Supplemental Material to "The real structure of a massless block: $SO(30)$ in the finite geometry"

Draft, 3 November 2026

This is the supplement of the paper *The real structure of a massless block: $SO(30)$ in the finite geometry*. Its parts keep the numbers of the full Supplemental Material, so that a citation of a part in the paper refers to the part of the same name here.

Parts contained. Lemma parts: 06, 18, 19, 21, 22, 23. Eleven-dimensional files: README, 01, 02, 06, 07, 08, 09, 11. Other parts: R4, R6, R7, R8, R9, R19, R21 (the letters H, L and R mark the history of the field, the literature check and the referee reports).

The full Supplemental Material, with all parts of the project, exists as `supplemental.pdf` and `supplemental_mobile.pdf` in the same folder as this file. A reference inside a part to a part not listed here points to the full version.

Contents

Part 06 **2**

Part 18 **8**

Part 19 **21**

Part 21 **23**

Part 22 **26**

Part 23 **32**

Part 11D, README, referee pending **39**

Part 11D, 01 interval and eleventh dimension, referee pending **48**

Part 11D, 02 witten spaces hypercharge quotient, referee pending **52**

Part 11D, 06 main point heterotic route, referee pending **56**

Part 11D, 07 dualities 9 11, referee pending **59**

Part 11D, 08 orientifold sign, referee pending **64**

Part 11D, 09 dilaton spectral and so32, referee pending **66**

Part 11D, 11 so30 fluctuations, referee pending **70**

Part R4 (referee report, lemma 18, two sheets, framework note) **74**

Part R6 (referee report, no-go results and lemma 19) **83**

Part R7 (referee report, lemmas 20 and 21, Proposition I) **89**

Part R8 (referee report, lemma 22) **96**

Part R9 (referee report, lemma 23) **101**

Part R19 (independent reading of 11D files 10 and 11) **107**

Part R21 (second check of the Terra referee of the 11D line) **110**

Part 06

The proposed uniqueness is false: **the same KO-6 triple can admit both reductions**. Moreover, under the stated definition, the strict heavy-limit triple is a doubling of KO-7, although it is not a doubling of KO-5. The distinction between the full real Hilbert space and its physical particle subspace is essential.

1. Lemma — Product convention and necessary correction of the "Clifford point."

Under ordinary spectral-triple conventions, a graded triple on `C^2` is not itself an odd KO-1 or KO-7 triple. The appropriate interpretation is an odd point

$$P_k = (\mathbb{C}, \mathbb{C}, 0, K), \qquad k = 1 \text{ or } 7,$$

together with the auxiliary `C^2` introduced by the odd-times-odd product. For real algebras, use the point algebra `R`, so that the product algebra remains `A`.

I use the following products, with the standard antilinear tensor product of real structures.

For even $\times$ even,

$$D = D_1 \otimes 1 + \gamma_1 \otimes D_2, \qquad \gamma = \gamma_1 \otimes \gamma_2,$$
$$J = J_1 \otimes (J_2\gamma_2^b), \qquad b = (1 - \epsilon_1'')/2.$$

For even $\times$ odd, use the same formula for `D`, no grading, and

$$J = (J_1\gamma_1^a) \otimes J_2, \qquad (-1)^a = \epsilon_1''\epsilon_2'.$$

Odd $\times$ even is obtained by transporting the even-first construction through the tensor flip.

For odd $\times$ odd,

$$H = H_1 \otimes H_2 \otimes \mathbb{C}^2, \quad D = D_1 \otimes 1 \otimes \sigma_1 + 1 \otimes D_2 \otimes \sigma_2, \quad \gamma = 1 \otimes 1 \otimes \sigma_3,$$
$$J = J_1 \otimes J_2 \otimes (BK),$$

where

$(\epsilon_1', \epsilon_2')$	$(+,+)$	$(+,-)$	$(-,+)$	$(-,-)$
B	σ_1	1	σ_3	$-i\sigma_2$.

This agrees, up to harmless phases, with Dąbrowski–Dossena, equation (25), Table 6.

Proof. In the odd-odd case,

$$B\sigma_1 = \epsilon_1'\sigma_1 B, \qquad B\sigma_2 = -\epsilon_2'\sigma_2 B.$$

The second minus sign compensates for complex conjugation of `sigma_2`. Consequently `JD=DJ`; computing `J^2` and `J gamma` gives the signs of the sum degree. The grading twists in the other two formulas follow by the same calculation. Order zero and first order pass to these products.

Thus the intended doubling has Hilbert space `H_0 ⊗ C^2`, with the displayed grading on the auxiliary factor. All subsequent statements use this corrected interpretation and the zero-Dirac point. ■

2. Theorem — Exact doubling criterion in KO-dimension six.

Let `(A,H,D,J,gamma)` be a finite KO-6 triple. For `s=±1`, the following are equivalent:

1. It is a product `T_n × P_k`, where
$$(s, n, k) = (+1, 7, 7) \quad \text{or} \quad (-1, 5, 1).$$
2. There is a complex-linear unitary `U` satisfying
$$U^2 = 1, \quad U\gamma = -\gamma U, \quad [U, a] = [U, a^o] = [U, D] = 0, \quad UJ = sJU,$$
where `a^o=Ja^\*J^{-1}`.

Write `H_L=H_{\gamma=+1}`, `H_R=H_{\gamma=-1}`, and
$$D = \begin{pmatrix} 0 & M^* \\ M & 0 \end{pmatrix}.$$
Equivalently, there is an `A`-bimodule unitary `V:H_L→H_R` such that
$$V^*M = (V^*M)^*, \qquad (V^*C)^2 = s, \quad C = J|_{H_L} : H_L \longrightarrow H_R.$$
Here `C` and `V^\*C` are antilinear.

Proof. Given `U`, identify
$$H_L \oplus H_L \longrightarrow H, \qquad (x, y) \longmapsto x + Uy.$$
Put
$$H_0 = H_L, \qquad d = (UD)|_{H_L}, \qquad j = (UJ)|_{H_L}.$$
The assumptions give
$$d = d^*, \qquad j^2 = s, \qquad jd = s\, dj.$$
Hence `(j^2,jd/dj)` is `(+,+)` for KO-7 and `(-,-)` for KO-5. In these coordinates,
$$a = \begin{pmatrix} a_0 & 0 \\ 0 & a_0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & d \\ d & 0 \end{pmatrix}, \quad \gamma = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$
$$J(x, y) = (sjy, jx).$$
These are precisely the products in Result 1: `B=σ_1` for `(7,7)` and `B=-iσ_2` for `(5,1)`.

The descended opposite action is
$$ja_0^* j^{-1} = a^o|_{H_L}.$$
Since `[d,a_0]=U[D,a]|_{H_L}`, order zero and first order descend. Conversely, the product has `U=1⊗σ_1`, with all the required properties.

Finally,
$$U = \begin{pmatrix} 0 & V^* \\ V & 0 \end{pmatrix}.$$
Its commutation with `D` is equivalent to self-adjointness of `V^\*M`, while `UJ=sJU` is equivalent to `(V^\*C)^2=s`.

If one prefers an anticommuting convention, replacing `U` by `i gamma U` exchanges `[U,D]=0` with `{U,D}=0` and preserves `s`. ■

In particular, ordinary equality of left and right module multiplicities is only part of the criterion; compatibility with `D` and `J` must also be checked.

3. Counterexample — J does not select a unique reduction; electrodynamics admits both.

Consider the electrodynamics triple in the basis `(e_L,e_R,bar e_L,bar e_R)`. After a phase change compatible with `A,J,gamma`, its matrices are

$$\rho(a,b) = \mathrm{diag}(a,a,b,b), \qquad \gamma = \mathrm{diag}(1,-1,-1,1),$$

$$J = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} K, \qquad D = m \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_1 \end{pmatrix}, \quad m \geq 0.$$

These are the input data of van den Dungen–van Suijlekom, §4.1, equations (4.1)–(4.2).

Define

$$U_+ = \mathrm{diag}(\sigma_1, \sigma_1), \qquad U_- = \mathrm{diag}(\sigma_1, -\sigma_1).$$

Proof. Both involutions reverse the grading and commute with the algebra, opposite algebra, and `D`. Direct multiplication gives

$$U_+ J = J U_+, \qquad U_- J = -J U_-.$$

Thus the same triple, with the same `J`, admits both factorizations.

On `H_0=span(e_L,bar e_R)`, the two descended triples have

$$d_+ = mI_2, \qquad j_+ = \sigma_1 K$$

and

$$d_- = \mathrm{diag}(m,-m), \qquad j_- = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} K.$$

They have KO-degrees seven and five, respectively. This also holds at `m=0`. ■

4. Lemma — Exact neutral-sector criterion, including Majorana masses.

Suppose a KO-6 sector has

$$H = H_+ \oplus H_-, \qquad \dim H_+ = \dim H_- = r,$$

and both algebra actions are the same scalar character throughout this sector. Then:

- a KO-7 reduction always exists;
- a KO-5 reduction exists exactly when **every singular value of `M`, including zero, has even multiplicity**.

Proof. Choose an orthonormal basis of `H_+` and its `J`-image as the basis of `H_-`. Then

$$J = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} K.$$

The equation `JD=DJ` says `M^T=M`. Takagi factorization, implemented by a basis change preserving this form of `J`, reduces `M` to

$$\Sigma = \mathrm{diag}(\mu_1, \ldots, \mu_r), \qquad \mu_i \geq 0.$$

Taking `V=I` satisfies Result 2 with `s=+1`.

For `s=-1`, the descended antiunitary `j` squares to `-1` and commutes with `d^2=D^2|_{H_+}`. Every eigenspace of `d^2` must therefore have even complex dimension. These dimensions are exactly the singular-value multiplicities.

Conversely, pair equal entries of `Sigma` and choose `V=σ_2` on each pair. Then `V` is a skew-symmetric Hermitian unitary commuting with `Sigma`; it satisfies the criterion with `s=-1`. ■

A single neutral Weyl state and its conjugate therefore **are** a KO-7 doubling, even without a Dirac partner. They cannot be a KO-5 doubling. The same statement holds for a single Majorana pair, whether its mass vanishes or not.

5. Proposition — The strict one-generation heavy limit is a KO-7 doubling, not a KO-5 doubling.

Here is an explicit convention for the Standard Model representation. Set

$$F = \mathbb{C} \oplus \mathbb{C}^3, \qquad H_p = (\mathbb{C}^2 \otimes F)_L \oplus (\mathbb{C}^2 \otimes F)_R, \qquad H = H_p \oplus H_{\bar{p}}.$$

For `a=(lambda,q,m)`, put

$$Q_\lambda = \mathrm{diag}(\lambda, \bar{\lambda}), \qquad B_{\lambda,m} = \mathrm{diag}(\lambda, m),$$
$$\rho_p(a) = (q \otimes I_4) \oplus (Q_\lambda \otimes I_4),$$
$$\rho_{\bar{p}}(a) = (I_2 \otimes B_{\lambda,m}) \oplus (I_2 \otimes B_{\lambda,m}).$$

Use

$$J(p, \bar{p}) = (K\bar{p}, Kp), \qquad \gamma = \mathrm{diag}(I_8, -I_8, -I_8, I_8).$$

On particles, the Dirac block is

$$S_p = \begin{pmatrix} 0 & Y^* \\ Y & 0 \end{pmatrix},$$

where `Y` has entries `y_nu,y_e,y_u I_3,y_d I_3` on the corresponding species. The antiparticle block is `bar S_p`; the additional Majorana block connects `nu_R` with `J nu_R`. This realizes the bimodule and mass-block structure in CCM, Proposition 2.4 and Theorem 2.21.

Assume precisely the strict projection described in canonical §3: remove `nu_R,J nu_R` and the complete up-type Dirac sector, and restrict to

$$B = \{(\lambda, Q_\lambda, m)\} \simeq \mathbb{C} \oplus M_3(\mathbb{C}).$$

The surviving space consists of `d_L,d_R,e_L,e_R,nu_L` and their conjugates, with `D_light=PSP`.

Proof. The surviving down-quark and electron masses pair identical `B`-bimodules. After making their masses nonnegative, each sector has both involutions from Result 3, with the identity on color.

On

$$N = \mathrm{span}(\nu_L, J\nu_L),$$

both algebra actions are the scalar character `lambda`, and `PSP|_N=0`. Thus

$$\gamma_N = \sigma_3, \qquad J_N = \sigma_1 K, \qquad U_N = \sigma_1$$

gives a KO-7 reduction. Combining this with `U_+` on the charged sectors gives a global KO-7 reduction.

A KO-5 reduction is impossible: the full space has dimension eighteen, so its descended space would have dimension nine, which cannot carry an antiunitary squaring to `-1`. More specifically, the neutral bimodule is inequivalent to the charged bimodules, so any permitted `U` preserves it; Result 4 already obstructs KO-5 there. ■

Thus the anticipated claim "the neutral sector prevents any doubling" is false under the definition in Question 1.

6. Proposition — Modifications and the distinction from physical vectorlikeness.

For the reduced algebra `B` above, the following conclusions hold.

Surviving neutral content	Odd reductions of the full light triple
No neutral sector	Both KO-5 and KO-7
`nu_L,nu_R` and conjugates, pure Dirac mass	Both KO-5 and KO-7
Only `nu_L,J nu_L`	KO-7 only
A single isolated Majorana pair	KO-7 only

Surviving neutral content	Odd reductions of the full light triple
General neutral Dirac–Majorana block	Always KO-7; KO-5 exactly under Result 4

Proof. The charged sectors admit both reductions by Result 3. Pure Dirac neutrinos have the same four-state construction, so adding or retaining `nu_R` restores both possibilities, including when its Dirac mass is zero.

For one generation with Dirac mass `m` and right Majorana mass `R`, the neutral symmetric matrix can be written

$$M_N = \begin{pmatrix} 0 & m \\ m & R \end{pmatrix}.$$

Phase changes make `m,R` nonnegative. Its singular values are

$$\mu_\pm = \frac{\sqrt{|R|^2 + 4|m|^2} \pm |R|}{2}.$$

They coincide exactly when `R=0`. Hence any nonzero right Majorana mass obstructs KO-5 in this one-generation neutral sector, but never KO-7. Treating that sector separately leaves a charged KO-5 reduction; it does not turn the complete direct sum into a KO-5 reduction. ■

There is an additional physically meaningful condition absent from Question 1. Let `Q` be `+1` on particles and `-1` on antiparticles.

No-go theorem under the extra hypothesis `[U,Q]=0`. An unpaired `nu_L` prevents doubling.

Proof. Such a `U` must map particle states of positive grading to particle states of negative grading within the same bimodule. No particle `nu_R` exists. The neutral involution used in Result 5 instead maps `nu_L` to its conjugate and violates `[U,Q]=0`. ■

This is why physical chirality and doubling of the complete real triple are different tests.

7. Proposition — Three generations and arbitrary flavor mixing.

Assume the reduced algebra acts trivially on generation indices.

If all three right-neutrino Majorana pairs are removed and only the top Dirac sector is made heavy, the light space has

$$96 - 6 - 12 = 78$$

complex dimensions. If all three up-type sectors are removed, it has dimension fifty-four. In either case, with three surviving left neutrinos and `D_light=PSP`, the light triple admits KO-7 and excludes KO-5.

Dropping the remaining neutral sector, or retaining three right neutrinos with purely Dirac neutrino masses, permits both reductions.

Proof. Each surviving charged species has equal left/right flavor multiplicities. Singular-value decomposition of its mass matrix uses bimodule unitaries; conjugate transformations on antiparticles preserve `J`. Consequently every charged sector reduces to copies of Result 3, irrespective of mass rank or flavor mixing.

The three unpaired neutral states give `r=3`, `M_N=0` in Result 4, excluding KO-5 and admitting KO-7. Equivalently, the proposed descended spaces have odd dimensions thirty-nine and twenty-seven.

With three right neutrinos retained, the general neutral matrix is

$$M_N = \begin{pmatrix} 0 & Y_\nu^T \\ Y_\nu & R \end{pmatrix}, \qquad R^T = R.$$

For `R=0`, every singular value of `Y_nu` appears twice, so both reductions exist. For general `R`, KO-7 always exists, while KO-5 exists exactly when every singular value of this six-by-six matrix has even multiplicity. Thus

nonzero Majorana masses are not an unconditional obstruction in several generations; the exact obstruction is the stated multiplicity condition. ■

8. No-go theorem — The massless Standard Model triple is not a doubling.

Let the algebra remain the full

$$A = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$$

with the representation of Result 5, for `g≥1` generations. Then no involution satisfying Result 2 exists, for any admissible `D`. In particular this applies at `v=0`, when only the Majorana block remains.

Proof. Consider the central projection

$$e_{\mathbb{H}} = (0, 1_{\mathbb{H}}, 0).$$

Its represented range lies entirely in the particle left-handed sector:

$$\dim \rho(e_{\mathbb{H}})H_+ = 8g, \qquad \dim \rho(e_{\mathbb{H}})H_- = 0.$$

An algebra-linear grading-reversing unitary would identify these spaces, which is impossible.

In physical representation-ring notation,

$$V_L = (3,2)_{1/6} \oplus (1,2)_{-1/2},$$

$$V_R = (3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,1)_{-1} \oplus (1,1)_0.$$

Thus `I_G=[V_L]-[V_R]≠0`. For the complete real graded Hilbert space, the relevant gauge character is

$$[H_+] - [H_-] = I_G - \overline{I_G},$$

not merely `I_G`. It too is nonzero: the coefficient of `(3,2)\_{1/6}` is `g`. Either this full-character calculation or the central-projection argument proves the obstruction. ■

Consequence for the 10 -> 9 claim

9. Proposition — What dimension reduction is actually established.

The heavy projection itself preserves KO-degree six, as canonical §3 explicitly states. Hence its spacetime product still has degree `4+6=10 mod 8`.

A subsequent choice of a KO-5 reduction removes a degree-one Clifford factor and gives the formal representative `4+5=9`. Such a reduction exists for the charged truncation or the pure-Dirac completion, but **both also admit KO-7 reductions**. Their fixed `J` does not select nine.

For the strict light spectrum with one or three unpaired left neutrinos, **KO-5 is obstructed; only KO-7 is available among the proposed reductions**. This gives the formal representative eleven, not nine. If particle preservation is additionally required, that strict light spectrum admits neither reduction.

Proof. These are Results 2–7. For completeness, the spacetime real structures are

$$J_{4+5} = (J_M \gamma_M) \otimes j_5, \qquad J_{4+7} = J_M \otimes j_7.$$

Thus even the odd reductions require different product conventions. No finite-dimensional operation here changes heat dimension: a finite internal factor remains spectrally zero-dimensional. Consequently the claimed loss of one dimension is, at most, a chosen Clifford-factor removal—not a proved metric collapse, and not a consequence of vectorlike charged matter alone. ■ —

Independent numerical check (Claude, 2 Nov 2026)

We used the electrodynamics data of Result 3, with m = 0.7, J = [[0, I_2],[I_2, 0]] K and gamma = diag(1,-1,-1,1). The triple has J^2 = 1, JD = DJ and J gamma = -gamma J, so its KO-dimension is 6. Both U_+ = diag(sigma_1, sigma_1) and U_- = diag(sigma_1, -sigma_1) satisfy U^2 = 1 and U gamma = -gamma U. Both commute with random algebra elements, their opposites and D; and U_+ J = J U_+, U_- J = -J U_-. On H_0 = span(e_L, bar e_R), U_+ descends to j^2 = +1 with jd = dj (KO-7) and U_- to j^2 = -1 with jd = -dj (KO-5). For No-go 8 we also rechecked by hand the dimension count dim rho(e_H) H_+ = 8g, dim rho(e_H) H_- = 0 in the representation of Result 5.

Terminology. Here "doubling" means the factorization of an even triple as an odd triple times a Clifford point (the doubling of an odd triple in Dabrowski-Dossena's product construction). It is unrelated to the "fermion doubling" of the noncommutative Standard Model (Lizzi-Mangano-Miele-Sparano). Prior work on products and on J versus J gamma: ../LITERATURE.md, section D; the factorization criterion and the 9/11 results were not found in print.

Referee note

Terra 6.1 (3 Nov 2026; report) checked Result 3, Lemma 4, Proposition 5, Result 7, No-go theorem 8 and the no-go under [U, Q] = 0: all correct. One amendment to Result 7: "arbitrary flavor mixing" means arbitrary matrices within the displayed species blocks, with the heavy projection chosen onto the specified top Dirac sector; mixing between inequivalent species is outside the displayed Dirac operator. Escapes: for KO-5, neutral-sector pairing or a symmetry enforcing even multiplicities; for No-go 8, a change of algebra or representation, the even description, and boundary or defect realizations of chirality; for the no-go under [U, Q] = 0, a matching particle bimodule of opposite grading.

Part 18

T5b: vector masses in the spectral action at fixed cutoff

Prompt: prompts/18 (STRATEGY T5b). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5 with Python (numpy, scipy), by computations independent of the kernel derived here: a direct heat trace with constant a reproduces equation (2) for five choices of (m_1, m_2), to about 0.03%; a direct heat trace on Fourier modes with a_2(x_1) = a cos(q x_1), including the F_{12} and [T, V] vertices, reproduces the kernel (37) at three values of (m, q), to 0.1-0.3%, the size of the discretization error. The closed forms (16)-(19), the bound (28), the limits (27), the neutral matrix (30) recomputed from Standard Model charges, and the identities behind (31) also check. The largest value of the Taylor mass is M_W^2 = 0.275 Lambda^2, at m_t^2 = 2.02 Lambda^2. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): all ten numbered results are correct, with small amendments incorporated below. The referee added Propositions 11 to 14 and Remark 15, which widen the scope in both directions; the ratio in Proposition 14 was added by Claude.

Scope. Within the three-generation, zero-Majorana, top-only family at fixed cutoff, these results refute an infinitely heavy W, Z, t limit for the stated quadratic mass definitions. They do not rule out other finite data (Proposition 14), a different action or constraint, a joint limit with Lambda -> infinity, or a quantum effective theory. They do not show unbroken SU(2) x U(1) on the light space (Remark 15).

For the heat cutoff, on $\mathbb{R}^4$, we obtain the exact quadratic vector form without expanding in m/Λ. In the three-generation Standard Model finite representation with only the top Dirac mass nonzero, both the mass defined by the q^0/q^2 Taylor ratio and the unique positive timelike pole mass satisfy $M_W \sim \Lambda^2/m_t$ as $m_t/\Lambda \to \infty$ at fixed Λ. The broken vectors consequently do not decouple by becoming heavy, although the heavy fermion contribution disappears from the gauge coefficients. These statements concern the classical bosonic quadratic action about a prescribed constant background; moreover, the positive Laplace class considered below does not itself select a finite nonzero stationary background when radial variations of that background are allowed.

1. Theorem (exact quadratic kernel).

Let

$$D_A = -i\gamma^\mu(\partial_\mu + iA_\mu) + \gamma_5 V, \qquad V = V_0 = V_0^\dagger,$$

on $\mathbb{R}^4$, with Hermitian Euclidean gamma matrices, $\{\gamma^\mu, \gamma^\nu\} = 2\delta^{\mu\nu}$, $\gamma_5^2 = 1$, and four spin components. Let the finite trace be the literal trace on H_F, including every stated multiplicity, without a subsequent projection; this agrees with the bosonic trace of Chamseddine-Connes-Marcolli, which counts particles, antiparticles and four spin components. Take A smooth and compactly supported.

We use the vacuum-subtracted action, or equivalently its quadratic variation, since the constant-background trace on $\mathbb{R}^4$ has infinite volume. Write

$$A_\mu(q) = a_\mu(q)T, \qquad T = T^\dagger, \qquad q^\mu a_\mu(q) = 0, \qquad \int_q = \int \frac{d^4q}{(2\pi)^4}.$$

Diagonalize V_0:

$$V_0 e_i = m_i e_i, \qquad m_i \in \mathbb{R}.$$

The signed eigenvalues are needed; their squares alone do not determine the answer.

For $S_t = \operatorname{Tr} e^{-tD_A^2}$, define

$$b(u) = u(1-u), \quad d_{ij} = m_i - m_j,$$

$$I_t(a, b; q^2) = \int_0^1 e^{-t[(1-u)a + ub + u(1-u)q^2]}\, du.$$

Then

$$S_t^{(2)} = \int_q |a_T(q)|^2\, C_{t,T}(q^2),$$

where

$$\boxed{C_{t,T}(q^2) = \frac{1}{8\pi^2} \sum_{ij} |T_{ij}|^2 \left[\left(\frac{2}{t} + d_{ij}^2 + q^2 \right) I_t(m_i^2, m_j^2; q^2) - \frac{e^{-tm_i^2} + e^{-tm_j^2}}{t} \right].} \tag{1}$$

For several generators, replace $|T_{ij}|^2$ by $\operatorname{Re}((T_a)_{ij}(T_b)_{ji})$.

In particular, writing $C(q^2) = C_0 + q^2 C_1 + O(q^4)$, we obtain

$$\boxed{C_{0,t} = \frac{1}{8\pi^2} \sum_{ij} |T_{ij}|^2 \left[\left(\frac{2}{t} + d_{ij}^2 \right) I_t(m_i^2, m_j^2; 0) - \frac{e^{-tm_i^2} + e^{-tm_j^2}}{t} \right],} \tag{2}$$

and

$$\boxed{C_{1,t} = \frac{1}{8\pi^2} \sum_{ij} |T_{ij}|^2 \int_0^1 e^{-t[(1-u)m_i^2 + um_j^2]} \left[1 - 2b(u) - t d_{ij}^2 b(u) \right] du.} \tag{3}$$

These formulas also apply, after integration over $t = s/\Lambda^2$, to

$$f(z) = \int_{(0,\infty)} e^{-sz}\, d\nu(s), \quad 0 < f(0) = \nu((0,\infty)) < \infty, \quad \int (s^{-2} + s^2)\, d\nu(s) < \infty, \tag{4}$$

where ν is positive. These hypotheses suffice for the Euclidean action density and the Taylor coefficients used below. They do not, by themselves, guarantee analytic continuation to every negative q^2.

Proof. With $\nabla_\mu = \partial_\mu + iA_\mu$,

$$D_A^2 = -\nabla^2 + V_0^2 - \frac{i}{2}\gamma^{\mu\nu} F_{\mu\nu} + \gamma^\mu \gamma_5 [A_\mu, V_0].$$

Let $P_0 = -\partial^2 + V_0^2$ and write $D_A^2 = P_0 + P_1 + P_2$, according to degree in A. Duhamel expansion and cyclicity give

$$S_t^{(2)} = -t\,\mathrm{Tr}(e^{-tP_0}P_2) + \frac{t^2}{2}\int_0^1 \mathrm{Tr}\,\left(e^{-(1-u)tP_0}P_1 e^{-utP_0}P_1\right) du. \tag{5}$$

The linear vertex from j, p to $i, p+q$ is

$$T_{ij}\left[(2p+q)\cdot a + \gamma^{\mu\nu}q_\mu a_\nu + (m_j - m_i)\gamma^\mu\gamma_5 a_\mu\right].$$

Completing the momentum square produces

$$(4\pi t)^{-2} e^{-t[(1-u)m_i^2 + um_j^2 + b(u)q^2]}.$$

For transverse a, the averaged spin trace of the vertex product is

$$4\left(\frac{2}{t} + d_{ij}^2 + q^2\right)|a|^2.$$

The P_2 term contributes

$$-\frac{1}{4\pi^2 t}\sum_{ij}|T_{ij}|^2 e^{-tm_i^2}|a|^2.$$

Symmetrizing in i, j proves (1). Differentiation proves (2)–(3).

The divided-difference structure is explicit:

$$I_t(a,b;0) = \frac{e^{-ta} - e^{-tb}}{t(b-a)}, \qquad I_t(a,a;0) = e^{-ta}, \tag{6}$$

and

$$\int_0^1 u(1-u)e^{-t[(1-u)a+ub]}\,du = \frac{1}{t^2}\partial_a\partial_b I_t(a,b;0). \tag{7}$$

Thus (2) uses a first divided difference, while (3) also uses its mixed derivative. The full momentum dependence follows from the convergent series

$$I_t(a,b;q^2) = \sum_{n=0}^{\infty}\frac{(-q^2/t)^n}{n!}\partial_a^n\partial_b^n I_t(a,b;0). \tag{8}$$

Repeated-node values are obtained continuously.

For completeness, the mass coefficient for (4) can be written entirely in dimensionless variables. Put

$$r_i = m_i/\Lambda, \quad x_i = r_i^2, \quad z(u) = (1-u)x_i + ux_j, \quad \Phi_1(z) = \int s^{-1}e^{-sz}\,d\nu(s).$$

The bracket replacing that in (2) is

$$\Lambda^2\left[2\int_0^1 \Phi_1(z(u))\,du + (r_i - r_j)^2\int_0^1 f(z(u))\,du - \Phi_1(x_i) - \Phi_1(x_j)\right]. \tag{9}$$

The corresponding bracket in (3) is

$$\int_0^1 \left[(1-2b)f(z) + (r_i - r_j)^2 b f'(z)\right] du. \tag{10}$$

Each unweighted integral is a divided difference of an antiderivative. The moment assumptions justify the integrations and differentiations. □

On a finite flat torus, (5) remains exact, but the Gaussian momentum integral must be replaced by the spin-structure-dependent lattice sum. Explicitly, for a transverse polarization a, its two-vertex contribution per unit volume is

$$\frac{2t^2}{\mathrm{Vol}}\sum_{ij,p}|T_{ij}|^2\int_0^1 e^{-t[(1-u)(p^2+m_i^2)+u((p+q)^2+m_j^2)]}\left(|(2p+q)\cdot a|^2+(d_{ij}^2+q^2)|a|^2\right)du,$$

and the quadratic-vertex contribution is

$$-\frac{4t}{\mathrm{Vol}}\sum_{ij,p}|T_{ij}|^2e^{-t(p^2+m_i^2)}|a|^2.$$

Consequently, the continuum formulas below are statements on $\mathbb{R}^4$, or in the infinite-volume limit. A finite torus can have additional holonomy dependence, including for constant commuting connections.

2. Lemma (gauge invariance and the Goldstone direction).

Let scalar fluctuations include the gauge orbit of V_0. Then the nonzero vector coefficient C_0 in (2) is compatible with gauge invariance.

For one generator, parameterize the orbit by

$$V(x)=e^{i\pi(x)T}V_0e^{-i\pi(x)T}.$$

With our conventions,

$$a_\mu\longmapsto a_\mu-\partial_\mu\alpha,\qquad \pi\longmapsto\pi+\alpha.$$

The invariant combination is

$$B_\mu=a_\mu+\partial_\mu\pi. \tag{11}$$

Proof. Gauge covariance gives

$$D_{A^U,V^U}=UD_{A,V}U^{-1}.$$

Applying $U=e^{-i\pi T}$ transforms V to V_0 and a to B. Hence

$$S[a,e^{i\pi T}V_0e^{-i\pi T}]=S[B,V_0]. \tag{12}$$

This establishes the Goldstone completion directly, without a derivative expansion.

For comparison, the longitudinal vector kernel is obtained from (1) by replacing the q^2 in its prefactor by $(1-2u)^2q^2$ inside the integral. If $[T,V_0]=0$, its integral vanishes by

$$\int_0^1[2+tq^2(1-2u)^2]e^{-tq^2u(1-u)}\,du=2.$$

For a broken generator the longitudinal kernel need not vanish; it belongs to the invariant combination (11).

If V_0 is not stationary, a Cartesian scalar Hessian alone need not annihilate the linear gauge-orbit tangent. The quadratic term in $e^{i\pi T}V_0e^{-i\pi T}$, contracted with the scalar tadpole, must also be retained. Equation (12) includes that term. Transverse vectors do not mix quadratically with scalar fluctuations on this translation-invariant background. □

3. Corollary (the asymptotic normalization).

Let all finite eigenvalues be small relative to Λ, let f satisfy (4), and let Q below be positive definite on the chosen generator space (otherwise pass to the quotient by its null space). Define

$$Q_{ab}=\mathrm{Re}\,\mathrm{Tr}(T_aT_b),\qquad K_{ab}=\mathrm{Re}\,\mathrm{Tr}([T_a,V_0]^\dagger[T_b,V_0]).$$

At leading order,

$$C_0=\frac{f(0)}{8\pi^2}K,\qquad C_1=\frac{f(0)}{12\pi^2}Q,$$

and therefore

$$\boxed{\mathcal{M}^2_{\text{Taylor}} = \frac{3}{2} Q^{-1/2} K Q^{-1/2}.} \tag{13}$$

This reproduces Proposition 13.5.

For the full three-generation Standard Model representation,

$$Q_{11} = 12, \qquad K_{11} = \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2),$$

so

$$\boxed{8M_W^2 = \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2).} \tag{14}$$

Proof. In (2), the orbital expression

$$\frac{2}{t} I_t(a, b; 0) - \frac{e^{-ta} + e^{-tb}}{t}$$

starts at order $t(a-b)^2$. The leading term is consequently d_{ij}^2. In (3), the leading integral is $\int_0^1 (1 - 2u(1-u))du = 2/3$, proving (13).

For $T_1 = \sigma_1/2$ on each left doublet, one generation of particles contributes 2 to Q_{11}. Charge conjugates and three generations give 12. A doublet with Dirac masses m_u, m_d contributes $\frac{1}{2}(m_u^2 + m_d^2)$ to the particle commutator trace, before color multiplicity. This proves (14).

Equivalently, the gauge coefficient is

$$\frac{f(0) g_2^2}{2\pi^2} = \frac{1}{4}.$$

For comparison with CCM's $\operatorname{Tr} F(D/\Lambda)$, take $F(v) = f(v^2)$; then $F(0) = f(0)$. The normalization and mass constraint are precisely equations (4.10) and (4.16), with the latter stated in Lemma 4.6 of the checked preprint. CCM describes the relation as "valid at unification scale." Chamseddine–Connes–Marcolli, pp. 37–39. □

The leading bound $M^2 \le 6\|V_0\|^2$ in Lemma 14 follows from (13) and $K \le 4\|V_0\|^2 Q$. Formula (13), rather than an exact finite-cutoff identity, is its applicable approximation.

4. Proposition (the exact charged-vector Taylor mass).

Let the finite data be the full three-generation Standard Model representation, including charge conjugates. Let the Majorana matrix and every Dirac mass except one up-type color triplet be zero. Write

$$m = \mu\Lambda, \qquad x = \mu^2, \qquad r = e^{-x}.$$

Let $f(z) = e^{-z}$. Define

$$H(y) = \int_0^1 e^{-yu(1-u)}\, du,$$

$$g(y) = (2+y)H(y) - 2,$$

$$h_x(y) = (2 + x + y) \int_0^1 e^{-xu - yu(1-u)}\, du - 1 - e^{-x}. \tag{15}$$

Then the full charged-vector kernel is

$$\boxed{C_W(q^2) = \frac{\Lambda^2}{4\pi^2} \left[\frac{9}{2} g(y) + \frac{3}{2} h_x(y) \right], \qquad y = q^2/\Lambda^2.} \tag{16}$$

Define

$$A(x) = h_x(0) = \frac{2[1 - (1+x)e^{-x}]}{x},$$

$$w(x) = 3\int_0^1 u^2 e^{-xu}\,du = \frac{3[2-(x^2+2x+2)e^{-x}]}{x^3}. \tag{17}$$

Continuous values are understood at $x = 0$. The particle-trace kinetic coefficient is

$$K_W(x) = \frac{9}{2} + \frac{3}{2}w(x) > 0.$$

The Taylor mass, defined by $M^2_{W,\mathrm{Taylor}} = C_W(0)/C'_W(0)$, is

$$\boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{9A(x)}{4K_W(x)} = \frac{3[1-(1+x)e^{-x}]}{x[3+w(x)]}.} \tag{18}$$

Its limiting behavior is

$$\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{3}{8}x - \frac{23}{128}x^2 + O(x^3), \qquad \frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{1}{x} \quad (x\to\infty). \tag{19}$$

Consequently,

$$\frac{m}{M_{W,\mathrm{Taylor}}} \longrightarrow \sqrt{\frac{8}{3}} \quad (\mu\to 0), \qquad \frac{m}{M_{W,\mathrm{Taylor}}} \sim \mu^2 \quad (\mu\to\infty). \tag{20}$$

Proof. On the particle space, $Q_{11} = 6$. The selected quark doublet contributes $3/2$, while the remaining doublets contribute $9/2$. Diagonalizing the massive left-right block produces eigenvalues $+m, -m$. The charged generator connects these to the massless down states; every such pair has $d_{ij}^2 = m^2$. Formula (1) therefore gives h_x for the selected doublet and g for the remaining doublets. Charge conjugation supplies the factor two in (16).

Integration by parts gives

$$h_x(y) = 2\int_0^1 u\,[x + 2y(1-u)]e^{-xu-yu(1-u)}\,du, \tag{21}$$

and

$$\boxed{h'_x(y) = 2\int_0^1 u^2[1-y(1-u)]e^{-xu-yu(1-u)}\,du.} \tag{22}$$

The boundary term proving (22) is $\partial_u[u(1-u)e^{-xu-yu(1-u)}]$. Thus

$$h'_x(0) = \frac{2}{3}w(x), \qquad g'(0) = \frac{2}{3}.$$

This proves (18). Expanding the elementary expressions at zero and taking their large-x limits proves (19)–(20). □

The same Taylor result extends to every f in (4). Put

$$\ell = f(0), \qquad w_f(x) = \int w(sx)\,d\nu(s), \qquad A_f(x) = \int \frac{A(sx)}{s}\,d\nu(s).$$

Then

$$\boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{9A_f(x)}{4\left(\frac{9}{2}\ell + \frac{3}{2}w_f(x)\right)}.} \tag{23}$$

All denominators are positive, and dominated convergence gives

$$M^2_{W,\mathrm{Taylor}} \sim \frac{3}{8}m^2 \quad (x\to 0), \qquad \boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{\int s^{-2}\,d\nu(s)}{\ell x} \quad (x\to\infty).} \tag{24}$$

Indeed, $A(z)/z \to 1$, $w(z) \to 1$ at zero, while $zA(z) \to 2$, $w(z) \to 0$ at infinity.

If the other Dirac masses are merely small and nonzero, the answer depends on those additional parameters and is given by (1). In particular, the vanishing limit in (19) assumes they are zero: masses of surviving light doublets can leave a residual vector mass.

5. Theorem (the charged-vector pole).

Under the heat-cutoff hypotheses of Proposition 4, define a timelike pole by analytic continuation of the quadratic inverse propagator:

$$C_W(-M^2_{W,\mathrm{pole}}) = 0. \tag{25}$$

For every $x > 0$, this equation has exactly one solution with $M^2_{W,\mathrm{pole}} > 0$. It is simple and has positive residue in the convention that the Euclidean propagator near the pole is a positive multiple of $(q^2 + M^2_{W,\mathrm{pole}})^{-1}$.

Let $z(x) = M^2_{W,\mathrm{pole}}/\Lambda^2$. Its exact specification is the unique positive root

$$\boxed{\frac{9}{2}g(-z) + \frac{3}{2}h_x(-z) = 0.} \tag{26}$$

Furthermore,

$$z(x) \sim \frac{3}{8}x \quad (x \to 0), \qquad z(x) \sim \frac{1}{x} \quad (x \to \infty), \tag{27}$$

and both mass definitions obey

$$\boxed{0 < M^2_{W,\mathrm{Taylor}},\ M^2_{W,\mathrm{pole}} \le \frac{\Lambda^2}{e}.} \tag{28}$$

Proof. Let

$$F_x(y) = \frac{9}{2}g(y) + \frac{3}{2}h_x(y).$$

Equation (22) implies $h'_x(y) > 0$ for $y \le 0$, and also $g'(y) > 0$, since $g = h_0$. Hence F_x is strictly increasing on that half-axis. We have

$$F_x(0) = \frac{3}{2}A(x) > 0.$$

As $y \to -\infty$, the prefactors in (15) become negative and both integrals grow exponentially, so $F_x(y) \to -\infty$. This proves existence, uniqueness, simplicity, and the residue sign.

For $y \le 0$, equation (22) gives $g'(y) \ge 2/3$, and therefore

$$F'_x(y) \ge 3.$$

The mean-value theorem yields

$$z(x) \le \frac{1}{2}A(x).$$

The Taylor denominator gives the same bound for $M^2_{W,\mathrm{Taylor}}/\Lambda^2$. Finally,

$$A(x) = 2\int_0^1 xu\, e^{-xu}\, du \le \frac{2}{e},$$

proving (28).

As $x \to 0$, the bound implies $z(x) \to 0$, and $F'_x(y) \to 4$ near the root. As $x \to \infty$, $z(x) \to 0$, $h'_x(y) \to 0$ near the root, and $F'_x(y) \to 3$. Together with $F_x(0) \sim 3x/2$ and $F_x(0) \sim 3/x$, respectively, these prove (27). □

The pole mass governs the linearized dispersion relation of this analytically continued quadratic action. The Taylor mass governs its two-derivative truncation; they need not coincide near the cutoff. We assert no closed-form inversion of (26), no classification of complex poles, and no Lorentzian unitarity theorem.

The pole proof also extends to positive measures supported in a compact subset of $(0,\infty)$, using

$$F_{f,x}(y) = \int \frac{1}{s}\left[\frac{9}{2}g(sy) + \frac{3}{2}h_{sx}(sy)\right] d\nu(s).$$

It gives the analogues of (24) for the pole mass. For the larger class (4), only the Euclidean and Taylor conclusions have been established here; its moment assumptions alone do not justify that continuation.

6. Proposition (unbroken coefficients, neutral vectors, and weak angles).

Retain the heat-cutoff Standard Model family of Proposition 4. Normalize color generators by $\mathrm{tr}_{\mathbf{3}}\, t_a t_b = \frac{1}{2}\delta_{ab}$, and use $Q = T_3 + Y$. Define the full-trace coefficients by

$$S^{(2)}_{\text{unbroken}} = \int \left(\kappa_{\text{em}} F_{\text{em}}^2 + \kappa_s \sum_a G_a^2\right).$$

Then

$$\boxed{\kappa_{\text{em}} = \frac{1}{24\pi^2}\frac{16}{3}(5+r), \qquad \kappa_s = \frac{1}{24\pi^2}2(5+r), \qquad \frac{\kappa_{\text{em}}}{\kappa_s} = \frac{8}{3}.} \tag{29}$$

Both coefficients approach 5/6 of their all-massless values.

Let K denote the particle-trace neutral q^2 coefficient in coordinates (A_3, A_Y). Then

$$\boxed{\begin{aligned} K_{33} &= \frac{21}{4} + \left(\frac{3}{4} - \frac{3x}{8}\right) r, \\ K_{YY} &= \frac{103}{12} + \left(\frac{17}{12} - \frac{3x}{8}\right) r, \\ K_{3Y} &= -\frac{1}{4} + \left(\frac{1}{4} + \frac{3x}{8}\right) r. \end{aligned}} \tag{30}$$

The matrix K, the charged coefficient K_W, and the unbroken coefficients are positive for every $x \geq 0$.

Define the neutral-current angle from the current of the massive direction after diagonalizing the two-derivative neutral kinetic form:

$$J_Z \;\propto\; T_3 - \sin^2\theta_{\text{NC}}\, Q.$$

Then

$$\boxed{\cos^2\theta_{\text{NC}} = \frac{K_{YY} + K_{3Y}}{K_{33} + 2K_{3Y} + K_{YY}} = \frac{5}{8}.} \tag{31}$$

A different continuation, namely

$$\cos^2\theta_{\text{match}} := \frac{K_{YY}}{K_W + K_{YY}},$$

satisfies

$$\boxed{\cos^2\theta_{\text{match}}(0) = \frac{5}{8}, \qquad \cos^2\theta_{\text{match}}(\infty) = \frac{103}{157}.} \tag{32}$$

Thus these definitions cease to describe one common electroweak angle.

Proof. Photon and color generators commute with V_0. Their coefficients therefore follow from (3) with $d_{ij} = 0$ on each invariant mass block. The charge traces give (29).

A Dirac species of mass m, with neutral charges q_L^a, q_R^a, contributes to K

$$e^{-x}\left[q_L^a q_L^b + q_R^a q_R^b - \frac{x}{2}(q_L^a - q_R^a)(q_L^b - q_R^b)\right]. \tag{33}$$

Applying (33) to the three top colors, and retaining all other states, proves (30).

Write

$$v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad B_\infty = \begin{pmatrix} 21/4 & -1/4 \\ -1/4 & 103/12 \end{pmatrix}, \quad U = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 17/12 \end{pmatrix}.$$

Then

$$K = B_\infty + rU - \frac{3xr}{8} vv^T.$$

We have $B_\infty \geq 5I$, $U \geq 0$, and $xr \leq 1/e$. Hence

$$K \geq \left(5 - \frac{3}{4e}\right) I > 0.$$

Positivity of K_W follows from (17).

The neutral mass direction is v, while the photon direction is $(1, 1)$. The kinetic-orthogonal massive field is proportional to $K^{-1}v$, giving the first expression in (31). Directly,

$$K_{33} + K_{3Y} = 5 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(5 + r),$$

which proves (31). Taking limits proves (32). □

The neutral mass also tends to zero. More precisely,

$$\boxed{\frac{M^2_{Z,\mathrm{Taylor}}}{\Lambda^2} = \frac{9}{4} xr\, v^T K^{-1} v \sim \frac{2}{3} xe^{-x} \quad (x \to \infty).} \tag{34}$$

For $x > 0$ there is likewise one positive massive neutral pole, besides the photon pole. To prove this, put $B = B_\infty + rU$. The exact neutral kernel, apart from the common factor $\Lambda^2/(4\pi^2)$, is

$$B\, g(y) + \frac{3}{2} xr\, H(y) vv^T. \tag{35}$$

After congruence by $B^{-1/2}$, its massive eigenvalue is

$$g(y) + \beta H(y), \qquad \beta = \frac{3}{2} xr\, v^T B^{-1} v > 0.$$

Its timelike root obeys

$$z - 2 + \frac{2}{H(-z)} = \beta. \tag{36}$$

The left side increases from zero to infinity, because

$$\frac{d}{dz}\left(z - 2 + \frac{2}{H(-z)}\right) = 1 - \frac{2\,\mathbb{E}_z[b]}{H(-z)} \geq \frac{1}{2},$$

where $\mathbb{E}_z$ uses density proportional to $e^{zb(u)}$. The pole is therefore unique and has positive residue. Since $v^T B_\infty^{-1} v = 8/27$,

$$\frac{M^2_{Z,\mathrm{pole}}}{\Lambda^2} \sim \frac{2}{3} xe^{-x}.$$

In particular, M_W^2/M_Z^2 does not continue to equal $5/8$.

For every f in (4), equations (29)–(33) extend by replacing 1 by $\ell = f(0)$, r by $f(x)$, and xr by $-xf'(x)$. Positivity follows by averaging the positive heat-cutoff matrices. The ratio $8/3$, the neutral-current value $5/8$, and the matching limit $103/157$ persist. At the endpoint both neutral masses vanish, so the distinguished massive neutral direction—and hence its angle—requires continuation from finite x.

7. Counterexample (positive cutoff does not ensure positive broken kinetic coefficients).

Let $H_F = \mathbb{C}^2$,

$$V_0 = m\sigma_3, \qquad T = \sigma_1, \qquad x = m^2/\Lambda^2,$$

with the same spacetime and trace conventions. For the heat cutoff,

$$C_T(q^2) = \frac{\mathrm{Tr}(T^2)\Lambda^2}{8\pi^2} e^{-x} \left[(2 + 4x + y)H(y) - 2\right]. \tag{37}$$

Its q^2 Taylor coefficient is proportional to

$$e^{-x}(1 - x).$$

It vanishes at $x = 1$ and becomes negative for $x > 1$. Thus the two-derivative truncation has a wrong-sign vector kinetic term in this example.

Nevertheless, the full quadratic form has a unique positive real timelike pole with positive residue, satisfying

$$\boxed{4x < \frac{M^2_{\mathrm{pole}}}{\Lambda^2} < 4x + 2.} \tag{38}$$

Consequently $M_{\mathrm{pole}} \sim 2m$ diverges at fixed cutoff.

Proof. The generator connects the eigenvalues $m, -m$, so $d_{ij}^2 = 4m^2$ while both squared eigenvalues equal m^2. Substitution in (1) proves (37). Its Taylor ratio is

$$M^2_{\mathrm{Taylor}} = \frac{6m^2}{1 - x},$$

where this is a positive mass only for $x < 1$.

The pole equation is (36) with $\beta = 4x$. The monotonicity proof there gives uniqueness and the residue sign. Evaluating (37) at $y = -4x$ gives $2H(-4x) - 2 > 0$, whereas at $y = -4x - 2$ it gives -2; this proves (38).

For general f satisfying (4), the Taylor kinetic coefficient in this example is proportional to

$$f(x) + xf'(x) = \frac{d}{dx}[xf(x)]. \tag{39}$$

It must be negative somewhere: $xf(x) > 0$ for $x > 0$, but $xf(x) \to 0$ at infinity under (4). A nonnegative derivative everywhere would contradict these facts. □

This counterexample concerns arbitrary finite data, not the top-only Standard Model family. It also distinguishes a wrong-sign **Taylor truncation** from a negative-residue pole of the **full** form factor. We have not proved absence of complex-pole or other nonlocal problems in either model.

8. Counterexample (heavy fermions do not dynamically select the centralizer).

Let the data be those of Proposition 4, and let

$$\mathfrak{g}_{V_0} = \{X \in \mathfrak{g}_{\mathrm{SM}} : [X, V_0] = 0\}.$$

For nonzero top mass this is the represented unbroken gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{\mathrm{em}}$. The limit $x \to \infty$ at fixed Λ does **not** produce a theory containing only these vector directions by quadratic mass decoupling.

Proof. For fixed y, equation (21) implies

$$h_x(y) \longrightarrow 0.$$

Consequently

$$\boxed{C_W(q^2) \longrightarrow \frac{\Lambda^2}{4\pi^2}\frac{9}{2} g(q^2/\Lambda^2).} \tag{40}$$

For positive y,

$$g(y) = 4y \int_0^1 u(1 - u)e^{-yu(1-u)}\, du > 0.$$

Thus a charged generator outside $\mathfrak{g}_{V_0}$ retains a finite nonzero quadratic kinetic form and a nonvanishing propagator at every fixed positive momentum. Its mass approaches zero. Equation (35) similarly leaves the positive matrix $B_\infty g(y)$, including the additional neutral direction. These facts contradict decoupling of the broken vectors through diverging masses or vanishing propagators. □

A narrower decoupling statement is true. If we **restrict the connection from the outset** to the centralizer, the finite mass blocks are invariant and the heat trace factorizes. On the heavy block its contribution carries e^{-sm^2/Λ^2}, which tends to zero. The restricted action therefore converges to the light-block spectral action. This proves decoupling of a finite fermion block under that restriction; it does not prove dynamical elimination of the other gauge fields.

The amendment to Lemma 03 Part C is therefore:

- Its transverse kinetic weights, including $w_L(x,0) = w(x)$, and its two weak-angle formulas are correct under their stated conventions.
- They must be accompanied by the mass coefficient (2), including the quadratic-vertex subtraction.
- A centralizer-only endpoint does not follow from those weights. In the present exact quadratic family, the broken-vector masses vanish while their remaining kinetic coefficients are positive.

9. No-go theorem (finite radial vacuum for the positive Laplace class).

Let f satisfy (4), and suppose the allowed scalar configurations include

$$V = cV_*, \qquad c > 0, \qquad V_* \neq 0,$$

with $A = 0$. Let the action be the literal spectral action, without an added scalar potential or a constraint fixing c. Then no finite $c > 0$ is stationary along this radial direction.

Proof. On $\mathbb{R}^4$, the constant-background action density is

$$\mathcal{U}(c) = \frac{\Lambda^4}{4\pi^2} \int s^{-2} \operatorname{Tr}_{H_F} e^{-sc^2V_*^2/\Lambda^2}\, d\nu(s).$$

Differentiating gives

$$\boxed{\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \int s^{-1} \operatorname{Tr}_{H_F} \left(V_*^2 e^{-sc^2V_*^2/\Lambda^2}\right) d\nu(s) < 0.} \tag{41}$$

The inequality is strict because $V_* \neq 0$ and ν is nonzero and positive. The same argument applies on a finite flat torus before performing its positive momentum sum. □

Thus the masses above are conditional quadratic-background quantities. In the top-only family, the ordinary Higgs radial variation supplies the direction used in this theorem. A finite symmetry-breaking minimum of the polynomial asymptotic scalar potential cannot automatically be regarded as a minimum of this full positive-Laplace spectral action.

10. Corollary (comparison of heavy limits).

Let the background family and mass definitions be those above.

At fixed Λ, with $\mu \to \infty$:

quantity, heat cutoff	behavior
m_t	$\mu\Lambda \to \infty$
$M_{W,\mathrm{Taylor}}$, $M_{W,\mathrm{pole}}$	$\sim \Lambda/\mu \to 0$
$M_{Z,\mathrm{Taylor}}$, $M_{Z,\mathrm{pole}}$	$\sim \sqrt{2/3}\,\Lambda\mu e^{-\mu^2/2} \to 0$
m_t/M_W	$\sim \mu^2 \to \infty$

(42)

At fixed $0 < \mu < \infty$, with $\Lambda \to \infty$, all four vector masses in (42) instead grow linearly with Λ, as does m_t. A controlled asymptotic family can also satisfy

$$\Lambda \to \infty, \qquad \mu \to 0, \qquad \mu\Lambda \to \infty;$$

then $M_W, m_t \to \infty$ while

$$m_t/M_W \to \sqrt{8/3}.$$

Proof. The first assertions follow from (19), (27), and (34)–(36). At fixed positive μ, their dimensionless mass functions are fixed and positive, so multiplication by Λ proves the joint-limit assertions. The final assertion follows from the small-μ limits. □

Accordingly, the obstruction has three distinct scopes. The CCM identity cannot be extrapolated to $\mu \to \infty$ at fixed cutoff because it belongs to the asymptotic truncation. The exact top-only quadratic action obstructs a fixed-cutoff heavy-W limit for the heat cutoff under both mass definitions, and for class (4) under the Taylor definition. A family of actual nonzero stationary vacua of the unmodified positive-Laplace action encounters the separate obstruction (41). Counterexample 7 excludes extending the first of these exact mass conclusions to arbitrary finite representations.

11. Proposition (the radial derivative for general f; referee). On flat $\mathbb{R}^4$, for a sufficiently integrable real cutoff f, with λ_j^2 the eigenvalues of V_*^2,

$$\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \sum_j \lambda_j^2 \int_{c^2\lambda_j^2/\Lambda^2}^{\infty} f(z)\, dz. \tag{43}$$

For nonnegative f, a finite $c > 0$ is stationary exactly when every tail associated with a nonzero λ_j vanishes. So if f has nonzero support arbitrarily far out, there is no finite stationary point on this ray; if f has compact support ending at R, there is a stationary plateau once every nonzero $c^2\lambda_j^2/\Lambda^2 \geq R$. Sign-changing cutoffs can have cancellations. A nonconstant cutoff flat near zero lies outside (4), since a nonzero positive measure gives $f'(0) < 0$; (43) covers the nonnegative ones. A radial path with an additional fixed noncommuting mass matrix need not have the form cV_*. □

12. Proposition (the top-only Taylor mass for general f; referee). For nonnegative smooth compactly supported f with $f(0) > 0$, the formulas of Proposition 4 hold with

$$A_f(x) = \frac{2}{x}\int_0^x z f(z)\, dz, \qquad w_f(x) = \frac{3}{x^3}\int_0^x z^2 f(z)\, dz,$$

by approximating f with its derivatives by real combinations of e^{-nz}. Hence

$$\frac{M_{W,\mathrm{Taylor}}^2}{\Lambda^2} \sim \frac{\int_0^\infty z f(z)\, dz}{f(0)\, x}.$$

This includes cutoffs flat near zero. For a nonanalytic cutoff there need be no analytic continuation defining the pole. □

13. Proposition (several heavy doublets; referee). For the heat cutoff, let P be the total particle-trace weight of complete massless doublets, and let heavy doublet i, of weight $d_i > 0$, have one Dirac member of mass m_i and one massless member. Then

$$C_W(y) = \frac{\Lambda^2}{4\pi^2}\Big(Pg(y) + \sum_i d_i h_{x_i}(y)\Big), \qquad \frac{M_{W,\mathrm{Taylor}}^2}{\Lambda^2} = \frac{3}{2}\,\frac{\sum_i d_i A(x_i)}{P + \sum_i d_i w(x_i)}.$$

For finitely many doublets, if $P > 0$ and all $x_i \to \infty$, the Taylor mass tends to zero. If $P = 0$ and all $x_i = x$, then $M_{W,\mathrm{Taylor}}^2 \sim m^4/(2\Lambda^2)$, the form factor has a unique positive pole with $M_{W,\mathrm{pole}}^2 \sim m^2/2$, and the whole unnormalized kernel tends to zero at every fixed Euclidean momentum. The divergent Taylor and pole scales differ, and neither alone establishes a healthy limiting gauge theory. □

14. Proposition (the fully massive family; referee; a complete proof by Astra 6 is in the validation, item 11). Give both members of every Standard Model doublet the same Dirac mass m, with no Majorana masses, and let z_W, z_Z be the charged and massive neutral poles in units of Λ^2. The charged form factor is proportional to $e^{-x}[g(y) + 2xH(y)]$, and

$$2x < z_W < 2x + 2, \qquad \tfrac{16}{5}x < z_Z < \tfrac{16}{5}x + 2.$$

So both pole masses diverge with m at fixed cutoff, and

$$\frac{M^2_{W,\mathrm{pole}}}{M^2_{Z,\mathrm{pole}}} \longrightarrow \frac{5}{8} \qquad (x \to \infty),$$

as also for $x \to 0$, where $z_W \sim 3x$ and $z_Z \sim \frac{24}{5}x$. In between the ratio differs, reaching about 0.66 near $x = 3$ (Python, solving the pole equations). The kernel tends to zero at every fixed Euclidean momentum, and the charged Taylor kinetic coefficient is proportional to $e^{-x}(1 - x/2)$, negative for $x > 2$.

Proof. The species with equal mass have eigenvalues $\pm m$. The charged generator connects them with $d^2 = 0$ and $d^2 = 4m^2$ in equal weights, which gives the charged form. For the neutral generators the diagonal charge has $d = 0$ and the off-diagonal charge $(q_L - q_R)/2 = T_3 v/2$ has $d^2 = 4m^2$; on the particle trace the kernel is $e^{-x}[\mathrm{diag}(6, 10)g + 12xHvv^T]$, so $\beta = 12x\, v^T \mathrm{diag}(6, 10)^{-1} v = \frac{16}{5}x$, and for the charged direction $\beta = 2x$. Equation (36) gives the bounds, since $H(-z) > 1$. □

This is a counterexample to any general no-go for diverging pole locations at fixed cutoff. It has no light fermions, and the physical quality of that limit is open.

15. Remark (the light space does not carry unbroken $SU(2) \times U(1)$; referee). In the top-only limit the broken masses vanish, but this does not restore the electroweak group on the whole light space. The light space retains b_L after removing t_L. If P_{light} projects off the top mass block, then on b_L we have $P_{\mathrm{light}} T_1 P_{\mathrm{light}} = P_{\mathrm{light}} T_2 P_{\mathrm{light}} = 0$ and $P_{\mathrm{light}} T_3 P_{\mathrm{light}} = -\frac{1}{2}$, which do not obey the $SU(2)$ relations. Correspondingly $K_W = 9/2$, $K_{33} = 21/4$, $K_{3Y} = -1/4$ in the limit, while a linearly realized unbroken action would have equal weak coefficients and no mixing. The remaining complete massless doublets carry the ordinary representation, and their contribution is gauge invariant. If both members of every removed doublet are heavy, the retained light space can be invariant under the full group; a full limiting theory still needs control of the remaining fields, normalizations, interactions and vacuum. "The heavy fermion leaves the loop" is shorthand for the suppression in this classical trace.

The light-space statement should be read together with the separate role of the top (lemma 21): among single charged species only an up-type quark can be removed while keeping cos^2 theta_NC = 5/8 and the ratio 8/3, and removing it leaves the five other quarks and the leptons, on which SU(3) x U(1)_em acts with b_L paired with b_R.

16. Remark (the CCM relation as the small-mass form of a weighted sum rule; Claude). For the heat cutoff and doublets with one massive member, Proposition 13 reads

$$8\kappa M^2_{W,\mathrm{Taylor}} = \sum_f N_f m_f^2\, \rho(m_f^2/\Lambda^2), \qquad \rho(x) = \frac{A(x)}{x} = \frac{2[1 - (1 + x)e^{-x}]}{x^2},$$

with $N_f = 2d_f$ and $\kappa = (P + \sum_i d_i w(x_i))/6$. As $x \to 0$, $\rho, \kappa \to 1$, which is the CCM relation (14). The relation is therefore the leading term of an expansion, and its range is $m \ll \Lambda$. For the top alone, $\kappa = (3 + w)/4$; m/M_W departs from $\sqrt{8/3}$ by 5% at $m \approx 0.45\Lambda$, and M_W peaks at about 0.52Λ at $m \approx 1.42\Lambda$ (Python). The maximum is unique, at $x = 2.024$ with $M_W/\Lambda = 0.5242$ (referee, by a sign argument and Python). A massive diagonal contribution to the trace is exponentially suppressed above the cutoff, while the mixed loop with a massless partner has the algebraic $\rho \sim 2/x^2$ at large x; a vacuum with no other mass leaves the W mass with no source. Whether the classical action is meaningful for $m > \Lambda$ is a separate question, on which these formulas do not decide. In the top-only heat-cutoff family, the ratio differs from $\sqrt{8/3}$ for every $x > 0$, since $3 + w - 4\rho$ vanishes at zero and is increasing, so the limit with M_W and m_t diverging together and $m_t/M_W \to \sqrt{8/3}$ needs $\Lambda \to \infty$ faster than m_t (Corollary 10). This necessity is for that family: with one complete massless lepton doublet and one member of each other doublet of common mass m, the weighted sum equals the unweighted CCM sum at some $x_* > 0$ (referee).

As algebraic checks, we used **Python 3 with SymPy** to verify the integration-by-parts identities, the displayed small-x Taylor series, and $v^T B_\infty^{-1} v = 8/27$. The arguments above provide the proofs; these computations introduce no fitted constants.

Consequence for T5b. At fixed cutoff, the specified top-only Standard Model spectral action does not realize an infinitely heavy W, Z, t family: the top grows while both vector masses tend to zero, for the heat

cutoff and the stated quadratic mass definitions. A joint limit with $\Lambda \to \infty$ permits all three background mass scales to diverge, but neither centralizer-only decoupling nor a stationary Higgs vacuum follows from that scaling. These conclusions concern the classical bosonic action and contain no quantum corrections or claims about the physical Standard Model at those scales. Other finite data behave differently: with every doublet heavy and no light fermions, the pole masses diverge at fixed cutoff with M_W^2/M_Z^2 -> 5/8 (Proposition 14). With a massless complete doublet retained, the Taylor mass tends to zero (Proposition 13). So in the families treated here, a heavy W and Z at fixed cutoff is found only without a massless complete doublet. Light nonzero masses are not treated: they leave a residual vector mass that depends on them.

Part 19

Inner fluctuations of the two-point data without order one

Step 1 of the next steps in the background note, for STRATEGY T0b. Proved by Claude Opus 5.5 without Astra or Fable; every identity checked with numpy on random data (section "Numerical check"). Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): Theorem 19.1 and the comparison with arXiv:1304.7583 are correct; Corollary 19.2 item 3 and the Wilson phase after the table were wrong as first written, and these and the amendments are incorporated below.

Outcome. The truncated two-point data satisfy order zero and fail order one, for each natural real structure. So the inner fluctuations of Chamseddine-Connes-van Suijlekom (arXiv:1304.7583), with their quadratic term, are the applicable ones. The result factorizes. With the inherited pointwise real structure, the fluctuated operator is m |Phi|^2 sigma_1: a real gauge-neutral radial field, with trivial gauge group. With the reflection real structure, which has the KO-6 signs, the link is m Phi^2: all of C is reached, and the gauge group U(1) acts with charge 2. Along a gauge orbit the quadratic term keeps the modulus of the link fixed. This parallels the fixed singular value of the covariant projection of lemma 14, which the fixed compression of lemma 13 lacks. So the first-order failure of Counterexample 08.4 points to the framework of arXiv:1304.7583 for these truncated data.

Setting

Let A = C + C act diagonally on H = C^2, with D = m sigma_1, m > 0, and gamma = sigma_3. For a real structure J = C K put x-hat = J x J^-1. We use two choices.

- **J_0 = K** (or sigma_3 K, the other sign of lemma 08), inherited from pointwise reality on the interval. Then x-hat = conj(x) on A, and J_0 commutes with gamma.
- **J_r = sigma_1 K**, inherited from the reflection real structure sigma_1 R_l K of Counterexample 08.4. Then diag(x_1, x_2)-hat = diag(conj(x_2), conj(x_1)), J_r^2 = 1, J_r D = D J_r and J_r gamma = -gamma J_r: the KO-6 signs.

Both satisfy order zero, since A is commutative and x-hat lies in A. Both fail order one: for e = diag(1, 0), [[D, e], e-hat] = m sigma_1 for J_0 and -m sigma_1 for J_r.

For a universal one-form A = sum_j a_j delta(b_j), the fluctuated operator of arXiv:1304.7583 is

```
D' = D + A_(1) + A~_(1) + A_(2),
A_(1) = sum a_j [D, b_j],   A~_(1) = sum a_j-hat [D, b_j-hat] = eps J A_(1) J^-1,   A_(2) = sum
↪  a_j-hat [A_(1), b_j-hat],
```

with J D J^-1 = eps D. Gauge transformations are U = u J u J^-1 for unitary u in A, and U D(A) U* = D(gamma_u(A)), where gamma_u(A) = u delta(u*) + u A u*.

Put

```
phi_1 = sum a_{j,1} (b_{j,2} - b_{j,1}),     phi_2 = sum a_{j,2} (b_{j,1} - b_{j,2}),
```

so that A_(1) = m [[0, phi_1], [phi_2, 0]], which is self-adjoint exactly when phi_2 = conj(phi_1). The pair (phi_1, phi_2) takes every value in C^2, as a = diag(x, y) and b = diag(0, 1) show.

The fluctuated operator

Theorem 19.1. For every universal one-form,

```
J_0:   D' = m [[0, |1 + phi_1|^2], [|1 + phi_2|^2, 0]],
J_r:   D' = m [[0, (1 + phi_1)(1 + conj(phi_2))], [(1 + phi_2)(1 + conj(phi_1)), 0]].
```

For a self-adjoint form, with Phi = 1 + phi_1,

```
J_0:   D' = m |Phi|^2 sigma_1,          J_r:   D' = m [[0, Phi^2], [conj(Phi)^2, 0]].
```

Proof. For diagonal b, [X, b] for off-diagonal X = [[0, x], [y, 0]] is [[0, x (b_2 - b_1)], [y (b_1 - b_2), 0]].

For J_0: a-hat and b-hat are the conjugates. Then A~*(1) = m [[0, conj(phi_1)], [conj(phi_2), 0]], and the first row of* $A(2)$ is m phi_1 sum conj(a_{j,1}) (conj(b_{j,2}) - conj(b_{j,1})) = m phi_1 conj(phi_1); the second row is m phi_2 conj(phi_2). So the (1,2) entry of D' is m (1 + phi_1 + conj(phi_1) + |phi_1|^2) = m |1 + phi_1|^2, and likewise (2,1).

For J_r: a-hat = diag(conj(a_2), conj(a_1)) and b-hat = diag(conj(b_2), conj(b_1)). Then A~*(1) = sigma_1 conj(*$A(1)$) sigma_1 = m [[0, conj(phi_2)], [conj(phi_1), 0]], and the first row of A_(2) is m phi_1 sum conj(a_{j,2}) (conj(b_{j,1}) - conj(b_{j,2})) = m phi_1 conj(phi_2), and the second is m phi_2 conj(phi_1). This gives the stated factorization. ■

Corollary 19.2 (gauge action and image). Let u = diag(e^{i alpha}, e^{i beta}) and psi = alpha - beta. Under gamma_u, Phi -> e^{i psi} Phi.

1. For J_0: U = 1. The image of the self-adjoint forms is {t sigma_1 : t >= 0}, so the link is real, and t is a gauge-neutral field. The value t = 0 is reached at Phi = 0.
2. For J_r: U = diag(e^{i psi}, e^{-i psi}), and the link m Phi^2 goes to e^{2 i psi} m Phi^2. So U(1) acts with charge 2 in the variable of the link, and the image is all of C, with Phi -> Phi^2 two-to-one away from 0. The image statement is pointwise: a globally prescribed smooth link need not have a global smooth square root.
3. The gauge orbit of the zero form has Phi = e^{i psi}. It gives D' = D for J_0, whose operator orbit is the single point D, and link m e^{2 i psi} for J_r. Without A_(2), the link is m (2 cos psi - 1) for J_0, of modulus m (1 - psi^2 + O(psi^4)), although U = 1; and m (2 e^{i psi} - 1) for J_r, of modulus m (5 - 4 cos psi)^{1/2} = m (1 + psi^2 + O(psi^4)), off the orbit of D.

Proof. gamma_u(0) has phi_1 = u_1 conj(u_2) - 1 = e^{i psi} - 1. The rest is direct. ■

Item 3 shows, in the smallest example, why the quadratic term is needed: it restores gauge covariance.

Comparison with the parent

Proposition 19.3. Consider the chiral interval at M = 0, with m = pi/(2l), and the links of lemmas 13 and 14.

1. *With no real structure*, the finite fluctuation is m (1 + phi_1), all of C, while the fixed compression of a gauge field gives only the line m + i B (Proposition 13.2).
2. *With pointwise reality*, the parent has no scalar fluctuation at all: the algebra C(I) is commutative with J f J^-1 = conj(f), it satisfies order one, and A + J A J^-1 = 0 for a scalar one-form. The truncated data do have a fluctuation, the neutral radial field t. So the finite fluctuation has a gauge-neutral radial parameter, while the specified pointwise-real parent has no surviving self-adjoint scalar inner fluctuation. This does not establish a dynamical scalar field or its action, and it says nothing about parent fields in other models. Compression is not multiplicative, so there is no canonical identification of finite potentials with parent potentials.
3. *With the reflection real structure*, the parent satisfies order one, since J_r f J_r^-1 = conj(f) composed with the reflection commutes with [D, g]. One finds J_r D_{A_5} J_r^-1 = D_{A_5 o R}, with R the reflection, so that the symmetrized field A_5 + A_5 o R survives, and the fixed compression gives m + i B^s with B^s = 2 B, because the weight sin(pi x/l) is even about l/2. The modulus along a pure-gauge field behaves as follows.

object	modulus of the link along a gauge orbit
covariant projection (lemma 14, Theorem 3)	constant: lambda W_l^* has fixed singular value
finite data with A_(2) (Corollary 19.2)	constant: m
fixed compression (Proposition 13.2), with B the integral of the symmetrized coefficient for J_r	(m^2 + B^2)^{1/2}, growing at second order along paths where B has a linear term
finite data with J_r, linear terms only	m (5 - 4 cos psi)^{1/2}, growing at second order

Proof. Item 1 is lemma 13. For item 2, a represented self-adjoint parent one-form is E h(x) with h = i a and a real, and K (i E a) K^-1 = -i E a, so the real fluctuation cancels. An individual commutator need not change sign, since K [D, x] K^-1 = [D, x] = E: self-adjointness is the step that matters. For item 3, K changes the sign of i A_5, the reflection changes the sign of the derivative and composes A_5 with R, and sigma_1 E sigma_1 = -E for E = [[0, 1], [-1, 0]]. The table collects Theorem 3 of lemma 14, Corollary 19.2 and Proposition 13.2. ■

The table states a parallel and no identification. The variables differ: for constant unsymmetrized A_5 = a, the transported link is m e^{i a l} = m + i (pi a/2) + O(a^2), with Wilson phase a l, against m + i (2 a/pi) in the compression. For the reflection-real fluctuation from the same a, these become m e^{2 i a l} and m + i (4 a/pi). The phase of a link depends on the frame, while its modulus does not. In the parent, the covariance defect is repaired by moving the projection, and in the finite data by the quadratic term.

Reading

Spectral compression can break order one, and it can also break order zero: on the circle with J = K, compression to the zero mode preserves both, while compression to the modes -1, 0, 1 sends e^{ix} to a truncated shift S, with [S, S*] != 0, so that order zero fails. Where a represented algebra, a preserved real structure and order zero hold, as here, the framework of arXiv:1304.7583 applies to the truncated data as they stand. So the first-order failure of Counterexample 08.4 has an escape, and this calculation is a first use of it. Two points were open. Whether the parallel in the table extends to an exact statement for a fluctuated parent and its truncation: this holds on gauge orbits, by Corollary 22.6 of lemma 22, and remains open for the nonunitary part. And whether the neutral radial parameter t is related to the singlet of the grand-symmetry model (Devastato-Lizzi-Martinetti, where sigma comes from a Majorana-operator fluctuation in a larger algebra). This is an analogy only: in arXiv:1304.7583 the quadratic term does not generate new scalar fields, and here too it depends on the same two parameters. An identification would need a map of algebras, representations, fields and gauge actions.

Numerical check

Python (numpy), run on 3 Nov 2026 in a temporary directory, not kept. On random complex a_j, b_j, with m = 1.3: the three real structures have the stated signs; the formulas of Theorem 19.1 hold to about 10^-15, without assuming self-adjointness; D(gamma_u(0)) = U D U* holds to 10^-16 for both real structures, and fails by 4.5 for the sample when A_(2) is dropped; U D(A) U* = D(gamma_u(A)) holds for a general form; the values of [[D, e], e-hat] are as stated.

Part 21

The separate role of the top: removal identities and counting

Prompted by the observation that the top has a separate role, which matters more than the loss of an electroweak partner for b_L. Proposition 21.1 and the counts below are exact calculations by Claude Opus 5.5, in Python with exact fractions. The readings in the last section are proposals, and nothing there is derived. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): one error, in Proposition 21.2, and several amendments, incorporated below.

Outcome. The spectral action detects the top: in the exact quadratic action of lemma 18, among single charged species only an up-type quark can be removed while keeping both the neutral-current value cos^2 theta_NC = 5/8 and the ratio 8/3 of the photon and gluon coefficients. Removing it leaves 84 of the 96 states of the finite Hilbert space. Those 84 are the seed of an SO(30), inside SO(32), whose adjoint contains the charged scalar counts of three generations, and 84 is also the number of on-shell components of the three-form of eleven-dimensional supergravity.

Removal identities

For a Dirac species with charges q_L, q_R in the coordinates (T_3, Y), equation (33) of lemma 18 gives a contribution to the neutral kinetic matrix K with weight e^{-x}. The off-diagonal charge is (q_L - q_R)/2 = (T_3/2)(1, -1), which is orthogonal to the photon direction (1, 1). So the combinations K_33 + K_3Y and K_YY + K_3Y depend on x only through the factor e^{-x} of each massive species, and cos^2 theta_NC = (K_YY + K_3Y)/(K_33 + 2 K_3Y + K_YY) is determined by the sums of T_3 Q and Y Q.

Proposition 21.1. Per species, with colour multiplicity, on the particle trace for the first two columns and with charge conjugates and both chiralities for the last two:

species	sum T_3 Q	sum Y Q	Q^2 coefficient	colour coefficient
up-type quark	1	5/3	16/3	2
down-type quark	1/2	1/6	4/3	2
charged lepton	1/2	3/2	4	0
neutrino	0	0	0	0
three generations	6	10	32	12

Hence the ratios of the whole, 5/3 and 8/3, are those of an up-type quark alone. As one species is removed, that is, as its mass tends to infinity at fixed cutoff in the setting of lemma 18:

removed	cos^2 theta_NC	photon/gluon coefficient
up-type quark	5/8	8/3
down-type quark	59/92	46/15
charged lepton	17/28	7/3

So among single charged species only an up-type quark preserves both values. For an up-type quark they hold at every x, as in Proposition 6 of lemma 18. Within one generation, the down-type quark and charged lepton together contribute exactly what the up-type quark contributes: (1, 5/3; 16/3, 2).

Proof. Arithmetic with the Standard Model charges, Y = Q - T_3. The last statement is direct addition. ■

The scope is that of lemma 18: classical bosonic trace, with these definitions of the coefficients and of the angle, three generations, and the other species with the mass assignments of that calculation. Other definitions of the weak angle do not stay at 5/8 (Proposition 6 there). At the endpoint where the neutral mass vanishes, the massive neutral direction is no longer distinguished, so its angle is read as the limit from finite mass, as lemma 18 states.

The state count in the action

Proposition 21.2. With three generations and nu_R, dim H_F = 96 (Chamseddine-Connes-Marcolli). With only an up-type Dirac mass m and no Majorana mass, the constant-background density of lemma 18 has finite trace Tr exp(-V_0^2/Lambda^2) = 84 + 12 e^{-x} for the heat cutoff. So the coefficient of the vacuum density moves from 96 to 84 along the radial direction, with no finite nonzero stationary point on this free radial ray (No-go theorem 9 and Proposition 11 of lemma 18). This concerns the literal flat-space action with no added potential or constraint. Compact support can give a plateau; sign-changing cutoffs,

additional terms, constraints and paths with an additional fixed noncommuting mass lie outside it. At m = 0 the derivative is zero.

Proof. The top block has 3 colours, 2 chiralities and charge conjugates, so 12 states, each with V_0^2 = m^2. ■

This also gives the top-only family a place in the radial problem, as the endpoint of a descending direction. A radial direction giving other species masses descends further, so this is the endpoint for that direction only.

SO(30)

Let the five quarks other than the top, in three colours, with their conjugates, form the vector 30 = (5, 3) + (5-bar, 3-bar) of SU(5) x SU(3) inside SO(30). The adjoint, of dimension 435, is

```
Lambda^2(30) = (24, 8) + (24, 1) + (1, 8) + (1, 1) + (10, 6) + (15, 3-bar) + conjugates of the last
↪  two.
```

With Q = 2/3 for u, c and Q = -1/3 for d, s, b, that is, Q = diag(3/5, 3/5, -2/5, -2/5, -2/5) + (1/15) I_5, which includes the U(1) commuting with SU(5), the charges are:

representation	charge: number
(15, 3-bar)	-2/3: 6; +1/3: 6; +4/3: 3
(10, 6)	-2/3: 3; +1/3: 6; +4/3: 1
(24, 1) + (1, 1)	+1: 6; -1: 6; 0: 13

Three generations of scalar partners in a supersymmetric Standard Model with nu_R need, for antisquarks in the 3-bar, 6 of charge -2/3 and 6 of charge +1/3, and for sleptons 6 of charge +1, 6 of charge -1 and 12 neutral real states. So (15, 3-bar) and (24, 1) + (1, 1) contain exactly the charged counts. What is left over in them is three triplets of charge +4/3 and one neutral state. The counting of 15 diquarks and 25 mesons, with the charge +4/3 states, and the remark that the top is unable to form them, go back to an early preprint (hep-ph/0512065), which finds neither SO(30) nor a role for SO(32). What SO(30) adds is one group whose adjoint contains both families. It also sits in SO(32), with 496 -> (435, 1) + (1, 1) + (30, 2) under SO(30) x SO(2), which hints at a role for SO(32); we have not explored this. These are representation and charge counts: identifying the entries with physical scalar partners needs a field construction, spin and chirality assignments, interactions and dynamics.

The two families of 84

Proposition 21.3. Count four states per Dirac fermion and colour, and take Dirac neutrinos. Then

```
96 = 84 + 12:   five quarks and all leptons (60 + 12 + 12), and the top (12);
96 = 84 + 12:   all electrically charged fermions (72 + 12), and the neutrinos (12).
```

The two families of 84 overlap in 72 states, and each is the other with the top and the neutrinos exchanged. Without nu_R, the first family has 78 states and the second stays at 84. In eleven dimensions, the three-form C_3 has C(9, 3) = 84 on-shell components, and its dual C_6 has C(9, 6) = 84. The 84 in H_F is a complex finite-Hilbert-space dimension, and the three-form count is a real bosonic polarization count, so an identification needs a map of representations and degrees of freedom.

Readings, as proposals

These are proposals to test, with no derivation behind them.

1. **Seiberg duality.** For SU(3) with five flavours, N_f = 5 lies in the conformal window 3 N_c/2 < N_f < 3 N_c, and the magnetic dual is SU(N_f - N_c) = SU(2) with five flavours and a 5 x 5 gauge-singlet meson chiral superfield coupled by W = M q q-tilde (N. Seiberg, Nucl. Phys. B 435 (1995) 129,

arXiv:hep-th/9411149, checked by the referee). The removal of the top is what gives N_f = 5, and SU(2) is the magnetic group there. To become a statement this needs an N = 1 supersymmetric theory with the required massless matter and dynamics, for example an emergent one, with an operator and anomaly map. Five-flavour ordinary QCD alone does not supply the duality, and the magnetic SU(2) needs a separate identification with any electroweak group.

2. **Branes.** The first family of 84 could relate to C_3 or to an M2 or M5 brane, and the second family to the dual object. The exchange of the top and the neutrinos would then be the content of the duality on the Standard Model side. The counts match, and equality of the number is insufficient: a representation and dynamics must relate the finite fermionic states to the bosonic form or brane degrees of freedom. The pairing of item 1 is one candidate. The first family also needs Dirac neutrinos for its 84, and the second does not.
3. **Relation to the odd labels.** The removal of the top leaves the five quarks and the leptons, on which SU(3) x U(1)_em acts with b_L paired with b_R. So the loss of the electroweak partner of b_L has no role at the label 9, where SU(2) is absent. A statement here needs the required involution on this finite sector, including its action on D, J, the algebra and its opposite; the vectorlike charges alone do not establish it, and the neutral states still need treatment.

Numerical check

Python with exact fractions, run on 3 Nov 2026 in a temporary directory, not kept: all tables above, and the dimension 435 = 224 + 1 + 105 + 105 with 105 = 60 + 45 and 224 = 192 + 24 + 8.

Part 22

The finite Dirac operator as a boundary condition on an interval

For STRATEGY T0, T0b, T10 and T13, following lemma 12 and lemma 20. Proved by Claude Opus 5.5, with the identities checked in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): two errors and one gap as first written, and amendments; all are incorporated below.

Outcome. Let (H_F, D_F, J_F, Gamma_F) be finite data of signs (+, +, -), and let l ||D_F|| < pi/2. Then D_F, with J_F and Gamma_F, is exactly the lowest spectral window of the pure derivative on an interval of length l, with no bulk mass and no Yukawa term, on the domain fixed by the boundary coupling tan(l D_F). So all the eigenvalues of D_F, Yukawa and Majorana blocks alike, are quantized internal momenta k with tan(k l) equal to a singular value of the coupling. The input is the finite triple: the content is that a consistent parent exists, with these properties:

- Preservation of the domain forces the observables at the wall into C_F, the commutant of D_F in A_F, while A_F acts freely in the bulk.
- The parent satisfies order one for any finite data with order zero.
- The compression is the exact average Q_l(a) = (2l)^-1 int_{-l}^{l} e^{ixD_F} a e^{-ixD_F} dx, so it tends to the finite representation as l -> 0, with defects of order l^2.
- The scalar inner fluctuations of the parent are pure gauge under its allowed endpoint gauge group. On the moving window, they act by the gauge orbit of D_F through the transport along the interval. This reproduces the quadratic term of Chamseddine-Connes-van Suijlekom on the orbit, and it leaves the singular values fixed, so the radial mode of the Higgs field is absent from this parent.

With the family X(l) = tan(l M) and l -> 0, the couplings of the light fermions vanish as m l at fixed mass, the tower decouples, and the broken gauge directions go to pi/(2l). The parent algebra is matrix-valued, so T0 for a commutative parent remains open.

Setting

Let H_F = H_+ (+) H_- be finite-dimensional, Gamma_F = diag(1, -1), let c : H_+ -> H_- be antiunitary, and put J_F(a, b) = (c^-1 b, c a), so that J_F^2 = 1 and J_F Gamma_F = -Gamma_F J_F. Let D_F = [[0, M*], [M, 0]] with M : H_+ -> H_-. Then J_F D_F = D_F J_F is equivalent to M = c M* c. By

the Takagi factorization, in bases related by c, we may write M = V |M| with a unitary V satisfying V = c V* c.

Let l > 0 with l ||D_F|| < pi/2, and define X : H_- -> H_+ by

```
tan(l D_F) = [[0, X], [X\*, 0]],   that is,   X = tan(l |M|) V\*.
```

Since J_F commutes with real functions of D_F, c X = X* c^-1. Conversely, every X with c X = X* c^-1 arises from exactly one such D_F, namely D_F = l^-1 arctan of the block operator. In the limit, X = l M* + O(l^3).

The parent. On H_l = L^2([0, l]; C^2) (x) (H_+ (+) H_-), write psi = (Q_L, Q_R; U_L, U_R), with Q valued in H_+ and U in H_-. Let (D psi)_L = psi_R' and (D psi)_R = -psi_L' on both fields, the operator of lemma 12 with no bulk mass, on the domain

```
Q_R(0) = 0,   U_L(0) = 0,   Q_R(l) = X U_R(l),   U_L(l) = -X\* Q_L(l).
```

Call it D_X. Put Gamma^ psi = (Q_L, -Q_R; U_L, -U_R) and J^ psi = (c^-1 U_R, -c^-1 U_L; -c Q_R, c Q_L). Let (E psi)_L = psi_R and (E psi)_R = -psi_L, so that D_X = E d/dx on its domain.

In the Standard Model form of the classification, with E = Hom(V, W) and W = W_+ (+) W_-, one takes H_+ = E_+ (+) E_-* and H_- = E_- (+) E_+*, with c the adjoint map. At X = 0, the Q-fields have boundary type (a) and the U-fields type (b), in the terms of lemma 09; for X != 0 the modes continue from those zero modes, which generally lift. A single matrix X encodes the Yukawa blocks and the Majorana block together. Its entries need not keep the original block pattern, since the tangent brings in higher powers of D_F: for M = [[a, b], [b, 0]] real symmetric, X_22 = (l^3/3) a b^2 + O(l^5).

The window

Theorem 22.1. D_X is self-adjoint with compact resolvent, and J^^2 = 1, J^ D_X = D_X J^, J^ Gamma^ = -Gamma^ J^, Gamma^ D_X = -D_X Gamma^. Let R(x) = exp(x Gamma_F D_F), and define W : H_F -> H_l so that for phi = (a, b), the L-components (Q_L, U_L) of W phi at x are l^{-1/2} R(x)(a, 0), and the R-components (Q_R, U_R) are l^{-1/2} R(x)(0, b). Then W is an isometry and

```
D_X W = W D_F,    J^ W = W J_F,    Gamma^ W = W Gamma_F.
```

With sigma_k the singular values of M and theta_k = l sigma_k, the signed spectrum of D_X is (n pi +- theta_k)/l, n in Z, with multiplicities added when values coincide; a zero singular value gives a zero mode of multiplicity two. So the spectrum of |D_X| consists of the sigma_k and of (n pi +- theta_k)/l for n >= 1, and ran W is the spectral subspace of |D_X| <= ||D_F||, and the next level is at least (pi - l ||D_F||)/l.

Proof. Self-adjointness is Lemma 12.1. Since Gamma_F D_F is anti-Hermitian, R(x) is unitary, and with C(x) = cos(x |M|), C'(x) = cos(x |M*|), S(x) = V sin(x |M|),

```
R(x)(a, 0) = (C(x) a, -S(x) a),    R(x)(0, b) = (S(x)\* b, C'(x) b).
```

Isometry follows from the unitarity of R(x) and the orthogonality of the L- and R-components. The conditions at 0 hold since S(0) = 0. At l, using V* cos(l |M*|) = cos(l |M|) V*, we get X C'(l) = S(l)* and X* C(l) = S(l), which are the conditions at the wall. For the derivative, the L-part of D W phi is d/dx R(x)(0, b) = R(x) Gamma_F D_F (0, b) = R(x)(M* b, 0), and the R-part is -d/dx R(x)(a, 0) = R(x)(0, M a); together these are W D_F phi. The relations for Gamma^ are direct. For J^, use c |M| c^-1 = |M*| and V = c V* c, which give c^-1 C' = C c^-1 and c S(x)* = S(x) c^-1. The signs of the parent follow in the same way, using c X = X* c^-1 for the domain. For the spectrum, constant unitaries on Q and on U commute with E d/dx and bring X to diag(tan theta_k), which reduces to Example 12.5 in each channel. ■

Corollary 22.2 (conjugated form). D_X = R (D_0 + E Gamma_F D_F) R*, with R acting pointwise on the pair (Q, U) and D_0 the uncoupled operator, X = 0. So after this unitary change of domain, D_F is a constant matrix connection, odd for the grading, with relative endpoint rotation R(l). This is equation (21) of lemma 20, for general D_F.

Proof. D(R phi) = R(D phi + E R* R' phi), and R* R' = Gamma_F D_F. For phi in the uncoupled domain, the computation at the wall in the proof above shows that R phi satisfies the coupled conditions. ■

So in this family a fermion mass is a momentum: D_X has no mass term, and the light eigenvalue k = theta/l is fixed by tan(k l) = tan(theta), a singular value of X. In the finite description, the condition relating k to l is lost and the value appears as free data in D_F.

The parent as a real spectral triple

Let (A_F, H_F, D_F, J_F, Gamma_F) be finite data as above, with an even representation pi = pi_+ (+) pi_- of a unital finite-dimensional real or complex *-algebra A_F, faithful, unital and satisfying order zero, with pi°(b) = J_F pi(b)* J_F^-1. Order one is not assumed. Let

```
C_F = {a in A_F : [pi(a), D_F] = 0},    A_l = {f in C^infinity([0, l], A_F) : f(l) in C_F},
```

with f acting pointwise by pi_+(f(x)) on Q and pi_-(f(x)) on U.

Theorem 22.3.

1. A function f in C^infinity([0, l], A_F) preserves Dom D_X if and only if f(l) lies in C_F.
2. (A_l, H_l, D_X, J^, Gamma^) is a real even spectral triple, with [D_X, pi(f)] = E pi(f') and J^ pi(g)* J^^-1 = pi°(g) pointwise.
3. It satisfies order zero and order one, with signs (+, +, -).

So A_F acts freely in the bulk, and the boundary condition restricts the observables at the wall to the commutant of D_F. In the Standard Model data with nonzero Yukawa blocks, this is the algebra {(lambda, lambda, m)} of the photon and gluons; with vanishing Yukawa blocks the commutant is larger. Order one holds in this parent whatever the finite data, for its pointwise algebra: the finite condition appears only on the truncation, and arbitrary compressions need not preserve either order condition.

Proof. Item 1: at l, the traces Q_L(l) and U_R(l) are free, and applying f to the conditions gives pi_+(f(l)) X = X pi_-(f(l)) and pi_-(f(l)) X* = X* pi_+(f(l)), as in Result 3 of lemma 20. That is [pi(f(l)), tan(l D_F)] = 0, equivalent to [pi(f(l)), D_F] = 0 because l ||D_F|| < pi/2 makes D_F a function of tan(l D_F). At 0 the conditions are preserved since pi is even. Item 2: the commutator is the Leibniz rule. Conjugation by J^ acts on (Q, U) as J_F does, and pi(g) acts identically on the two spin components, which gives the second formula. This preserves the domain since J_F commutes with D_F. Item 3: pi(f(x)) and pi°(g(x)) commute at each x by finite order zero, and E commutes with both; so [E pi(f'), pi°(g)] = E [pi(f'), pi°(g)] = 0. ■

This is consistent with Counterexample 4 of lemma 20, whose finite data A_F = C + C, D_F = m sigma_1 fail order one while the parent satisfies it.

The compression

Let Q_l(T) = W* T W for a bounded operator T on H_l. For T(x) a bounded function with values in the even operators on H_F, acting pointwise and identically on both spin components, we write T for the multiplication operator.

Theorem 22.4.

```
Q_l(T) = l^-1 int_0^l cos(x ad_{D_F})(T(x)) dx,      Q_l(E T) = -l^-1 int_0^l sin(x ad_{D_F})(T(x))
↪ dx.
```

For constant a in A_F,

```
Q_l(a) = sinc(l ad_{D_F})(pi(a)) = (2l)^-1 int_{-l}^{l} e^{ix D_F} pi(a) e^{-ix D_F} dx,
```

so in an eigenbasis of D_F with eigenvalues mu_i, the matrix elements are pi(a)_{ij} sin(l (mu_i - mu_j))/(l (mu_i - mu_j)). Moreover:

1. Q_l(A_l) = Q_l(C^infinity([0, l], A_F)), so the image of the parent algebra contains Q_l(a) for every a in A_F.
2. ||Q_l(a) - pi(a)|| <= (l^2/6) |||[D_F, [D_F, pi(a)]]|| <= (2/3) l^2 ||D_F||^2 ||a||.
3. Q_l([D_X, f]) = [D_F, Q_l(f)] for f in A_l.

4. With a' = Q_l(a) and b' = J_F Q_l(b)* J_F^-1, |||[a', b']|| <= (8/3) l^2 ||D_F||^2 ||a|| ||b||, and |||[[D_F, a'], b'] - [[D_F, pi(a)], pi°(b)]|| <= (16/3) l^2 ||D_F||^3 ||a|| ||b||.

So as l -> 0 with D_F fixed, D_F, J_F and Gamma_F stay exact, and the compressed observables converge to pi(A_F) with rate l^2. The finite order conditions hold in the limit exactly when they hold for the finite data.

Proof. From the definition of W, the expectation of T splits into the L- and R-parts, so Q_l(T) is l^-1 int of the Gamma_F-even part of R(x)* T(x) R(x). Put K = Gamma_F D_F. For even Z, ad_K Z = Gamma_F [D_F, Z], which is odd, and ad_K (Gamma_F [D_F, Z]) = -[D_F, [D_F, Z]]. So the even part of e^{-xK} T e^{xK} is cos(x ad_{D_F})(T). Likewise, E exchanges the L- and R-parts with a sign, so Q_l(E T) is l^-1 int of Gamma_F times the odd part of R* T R, and that odd part is -Gamma_F sin(x ad_{D_F})(T). The formula for constant a follows from cos(x ad_D)(Z) = (e^{ixD} Z e^{-ixD} + e^{-ixD} Z e^{ixD})/2.

Item 1: the map f -> Lambda(Q_l(f)), for a linear functional Lambda, is integration against a smooth density. If it vanishes on A_l, which contains all f of compact support in [0, l), the density vanishes on [0, l), hence everywhere by continuity. Item 2: by Taylor's formula with integral remainder, ||cos(x ad_D)(Z) - Z|| <= (x^2/2) ||ad_D^2 Z||, since cos(x ad_D) has norm at most 1 by the average formula; averaging over x gives l^2/6. Item 3: P commutes with D_X. It also follows from the two formulas by integration by parts, where the boundary term at l vanishes because f(l) is in C_F. Item 4: J_F commutes with D_F, so b' is the same average of pi°(b). Expand both commutators with the bound of item 2 and |||[D_F, Z]|| <= 2 ||D_F|| ||Z||. ■

The convergence is a statement about the maps Q_l, on the operator systems, in the setting of Connes and van Suijlekom (arXiv:2004.14115, sections 1 and 3). The generated algebra is not continuous at l = 0. An analytic example, from the referee: let M = [[0, 0, 1], [0, 0, 1], [1, 1, 0]], D = [[0, M], [M, 0]], with A_F = C^3 acting diagonally and equally on the two blocks. With P_1, P_2 the first two channel projections, [Q_l(P_1), Q_l(P_2)]_{12} = -l^2/3 + O(l^4), so the compressions of commuting projections generate a noncommutative algebra for small nonzero l. On the one-generation Standard Model data, a numerical closure with tolerance 10^-7 gives complex dimension 15 for the algebra generated by pi(B), with B = C + H + M_3(C), and 30 for Q_l(B) at l = 0.3; these are numerical values with no certified rank.

Scalar fluctuations

Theorem 22.5. Let h be smooth with values in the anti-Hermitian elements of A_F. The self-adjoint one-forms of the parent are exactly the E pi(h). Let u' = -h u, u(0) = 1, and let w = u(l), a unitary of A_F. Then

```
D_X + E pi(h) = pi(u) D_{X_w} pi(u)\*,      X_w = pi_+(w)\* X pi_-(w),
```

and the window of D_{X_w} carries pi(w)* D_F pi(w). With the real structure, for A = E pi(h) we have J^ A J^^-1 = E J_F pi(h) J_F^-1, and D_X + A + J^ A J^^-1 has the window U_w* D_F U_w, with U_w = pi(w) J_F pi(w) J_F^-1, and this commutes with J_F. Moreover, with u~(x) = u(x) w*, which lies in A_l since u~(l) = 1, and U~ = pi(u~) J_F pi(u~) J_F^-1,

```
D_X + A + J^ A J^^-1 = U~ D_X U~\*.
```

So every scalar connection in this model is gauge equivalent to the original parent under the allowed endpoint gauge group (referee).

Proof. A one-form is sum f_1 [D_X, f_2] = E pi(sum f_1 f_2'), and E pi(h) is self-adjoint exactly when h is anti-Hermitian, since E is anti-Hermitian and commutes with pi(h) and pi is faithful. Every smooth such h arises: with b(x) = int_l^x h, we have b(l) = 0, so b is in A_l, and [D_X, b] = E pi(h). Next, pi(u)* D pi(u) = D + E pi(u* u'), and pi(u)* E pi(h) pi(u) = E pi(u* h u), so the sum is D with no connection term. At 0 the conditions are unchanged. At l, the substitution psi = pi(u) phi gives the conditions with X_w. Then tan(l pi(w)* D_F pi(w)) has the block X_w. For general w, this operator need not commute with J_F, so we use the spectrum and the isometry of Theorem 22.1 without its real-structure part. For the real fluctuation, J^ commutes with E, u(x) is replaced by pi(u) J_F pi(u) J_F^-1, using finite order zero,

and U_w commutes with J_F, so the real window follows. The last formula follows by differentiating, with the endpoint condition. ■

Corollary 22.6 (the quadratic term as transport). By Chamseddine-Connes-van Suijlekom (arXiv:1304.7583, section III), for unitary v in A_F, with U = v J v J^-1, one has U D U* = D + A_(1) + A~*(1)* + *A*(2), for the universal form v delta(v*). Taking v = w*, on the gauge orbit the fluctuation of the parent, which satisfies order one and has no quadratic term, gives on the moving window exactly the fluctuated operator of that framework (equations (10)-(11) of section III.2 there), quadratic term included, for finite data that fail order one. This answers the first open point of lemma 19 on these gauge orbits and with the moving window. For the two-point data with J = sigma_1 K and w = diag(e^{i alpha}, e^{i beta}), it gives the upper link m e^{-2 i (alpha - beta)}, which is m e^{2 i psi} of Corollary 19.2 with psi defined from v = w*. Theorem 22.5 and the covariant projection of lemma 14 share a transport principle. The fixed compression, as in lemma 13, is a different object: by Theorem 22.4, it gives Q_l(E pi(h)) = -[D_F, l^-1 int_0^l x pi(h) dx] + O(l^3 ||D_F||^3 ||h||) for h fixed, so that for constant h it is -(l/2)[D_F, h], while the first derivative of the transported operator is -l [D_F, h]. The difference is the contribution of the moving spectral projection.

No-go theorem 22.7 (the radial mode). *Hypotheses:* the parent above, on one interval, with fixed X, no bulk masses, and scalar inner fluctuations from the even algebra A_l, with the moving spectral window. *Conclusion:* the fluctuations leave all singular values of the finite Dirac operator fixed. In the usual Standard Model representation with nonzero Yukawa data, the normalized shifted Higgs quaternion runs through the unit quaternions, a three-dimensional gauge orbit, with no radial mode. Degenerate data can have a smaller orbit. A fixed compression can show changes in singular values at second order, so the window has to be named.

Proof. Theorem 22.5: D_F and pi(w)* D_F pi(w) are unitarily equivalent. The same holds with U_w. ■

Known escapes.

These lie outside the hypotheses, and each needs its own construction.

- The coupling X as a field on the wall, with its own kinetic term and potential.
- Connections odd for the grading, as in Corollary 22.2 and lemma 23: the fluctuation K -> K + delta K changes the singular values.
- Bulk masses that do not commute with the gauge action.
- Two or more internal dimensions, where a connection has curvature, although curvature alone does not prove a radial mode.
- An additional composite or other degree of freedom; a quantum effective action on the same configuration space does not by itself add a radial coordinate.
- A fluctuation of the finite data alone, as with the radial parameter of lemma 19, which is not the moving window.

Direction. On the interval, the even connection is pure gauge up to its boundary effect, so the finite orbit is all it can reach. The physical Higgs modulus must then be sought at the wall, in odd directions, or in more dimensions. At the 10 -> 9 endpoint this agrees with T10, where the Higgs is frozen.

The gauge sector and the 10 -> 9 family

Proposition 22.8. Consider the product of M^4 with this parent, with algebra of functions f(y, x) valued in A_F and f(y, l) in C_F. Then the gauge one-forms sum f_1 d_mu f_2 take values in C_F at x = l. Assume, in addition, a flat product, a positive five-dimensional Yang-Mills kinetic form, the trivial background, unrestricted endpoint variation in the allowed directions, and no additional boundary kinetic or mass terms. Then the directions in the Lie algebra of C_F have masses n pi/l, n >= 0, and the orthogonal directions have masses (n - 1/2) pi/l, n >= 1. The physical gauge algebra comes from the skew-adjoint elements through the real gauge representation, with its kernel and any unimodularity condition.

Proof. The first statement is direct evaluation at x = l. The rest is the interval spectrum in the gauge A_5 = 0, with Neumann conditions at 0, and with Neumann or Dirichlet conditions at l. ■

In the Standard Model data with nonzero Yukawas, M_W = M_Z = pi/(2l). The Lie algebra of C_F has dimension 10 before unimodularity; with the unimodularity condition of Chamseddine-Connes-Marcolli (arXiv:hep-th/0610241, section 2.5), the massless modes are the photon and the gluons. These are the boundary conditions of Higgsless models (Csaki, Grojean, Murayama, Pilo and Terning, Phys. Rev. D 69 (2004) 055006), and the fermion formula tan(m L) = M_D L is equation (5.6) of Csaki, Grojean, Hubisz, Shirman and Terning (Phys. Rev. D 70 (2004) 015012); LITERATURE.md, section C. Boundary spectral geometry for the Standard Model has been studied by Chamseddine and Connes (arXiv:1008.3980, section 5). What is added here, as far as our searches went, is the combination: the real structure, the finite triple as an exact window, the wall algebra C_F, the compression formula and the fluctuations. A wider claim of novelty would need a fuller comparison.

Corollary 22.9 (the 10 -> 9 family). Fix a block M_0 for the light fermions and a block Theta/l on an orthogonal channel, compatible with J_F and the grading, with 0 < Theta < pi/2, for the top and if wanted for the Majorana block. Let D_F(l) have M(l) = M_0 (+) Theta/l, and let l -> 0. Then:

1. X(l) has the light block tan(l |M_0|) V_0* = l M_0* + O(l^3) and a heavy block with singular values tan(Theta): the light couplings vanish as m l at fixed mass m.
2. The heavy channel has mass Theta/l, the broken gauge directions have mass pi/(2l), and the other levels diverge.
3. The massless gauge sector is C_F. Let P_L be the projection reducing D_F(l) to the fixed light block D_L. For constant a, P_L Q_l(a) P_L = sinc(l ad_{D_L})(P_L pi(a) P_L), with ||P_L Q_l(a) P_L - P_L pi(a) P_L|| <= (2/3) l^2 ||D_L||^2 ||pi(a)||, also when pi(a) mixes light and heavy channels. The limiting light compression need not be a representation, since products can pass through the heavy subspace. The full estimate of Theorem 22.4 does not apply, because l ||D_F(l)|| tends to Theta. For example, with D_F(l) = sigma_1 (x) diag(m, Theta/l) and pi(a) = 1 (x) sigma_1, one has Q_l(a) = sinc(Theta - l m) pi(a), which tends to sinc(Theta) pi(a).
4. With l = pi/(2 M_W), we get X_q ~ (pi/2) m_q/M_W, which has the form of y_q = sqrt(2) m_q/v -> 0 at fixed mass, as the coupling vanishes while the mass stays. Identifying X_q with the four-dimensional Yukawa coupling is a hypothesis.
5. The ratio is m_t/M_W = 2 Theta/pi < 1, and M_W/M_Z = 1.

So the 10 -> 9 proposal is realized at this level as a limit l -> 0, with W, Z and top at the tower scale and the light fermions keeping explicit masses. Item 5 does not reach m_t/M_W = sqrt(8/3), which belongs to the interior. This is a statement in the stated family: lowest level, no bulk mass, strict wall algebra. Bulk masses for the top, and a changed gauge wall condition, are directions beyond it.

A numerical comparison for the interior (Python). A separate model, with a finite wall scalar and Neumann conditions at 0, lies outside Proposition 22.8 and No-go theorem 22.7. In it, the mass of a broken direction satisfies t tan t = s, with t = M l and s = g^2 v^2 l^2/4 (for the Z, s/cos^2 theta_0), which has the form of M_k tan(M_k pi R) = -V in Csaki et al. (arXiv:hep-ph/0305237, section on mixed boundary conditions, read in the LaTeX source), with the sign convention of V there. If g'^2/g^2 = 3/5 is taken at all scales, M_W^2/M_Z^2 goes from 5/8 as s -> 0 to 1 as s -> infinity. It equals the measured 0.777 at M_W l = 1.03, with 1/l = 78 GeV and a next W level at 280 GeV; whether collider searches allow this would need the couplings, branching fractions and an applicable limit, which we have not analyzed. The top needs M_W l < 0.96 for m_t/M_W = sqrt(8/3), or M_W l < 0.73 for the measured ratio, in the lowest level with no bulk mass. So in the flat tree-level toy without bulk masses or running, the top and gauge requirements are incompatible. It does show the interior interpolating between the spectral-action value 5/8 and the endpoint value 1. The directions are bulk masses, warping and running.

What is input and what is derived

The finite data (A_F, H_F, D_F, J_F, Gamma_F) are all inputs, so the existence of the window is close to a reparametrization: X = tan(l D_F). The derived statements are the wall algebra C_F for that input, the order conditions of the parent, the compression formula with its rate, the form of the fluctuations, and the gauge spectrum under the stated assumption. The window identifies D_F, J_F and Gamma_F exactly, while the represented algebra is obtained through compression and is not an exact representation of A_F at

l > 0. In particular, this does not select A_F. T0 asks for a commutative parent: a natural next step is a two-stage parent, compressing functions on S^4 and CP^3 at level one to get A_F-valued functions on the interval, and then taking this window, where the difficulty is the same interface problem as after Result 7 of lemma 20.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^5, l ||M|| = 1.3: D u_+(a) = u_-(M a) and D u_-(b) = u_+(M* b) by finite differences, to about 10^-9; boundary conditions and J^ to about 10^-15; norm of W to 10^-13.
- Q_l(a) = sinc(l ad)(a), by quadrature, to 2 x 10^-8. The formula for Q_l(E T), with a nonconstant T, to 10^-14.
- One-generation Standard Model data with Yukawas and the rank-one Majorana block, with signs, order zero and order one exact on B = C + H + M_3(C), and commutant of D in B of real dimension 20. At l = 0.1 and 0.03, ||Q_l(a) - a + (l^2/6)[D, [D, a]]|| is 10^-5 and 4 x 10^-8, and the order-one defect of Q_l is 1.5 x 10^-4 and 1.2 x 10^-5, which scales as l^2.
- The dimensions of the generated algebras quoted above, 15 and 30, from a numerical closure with tolerance 10^-7, not retained and not certified; at smaller l the numerical rank depends on the tolerance.
- The roots of t tan t = s, and the values quoted in the comparison for the interior.
- On the two-point data with J = sigma_1 K: D + A_(1) + A~*(1)* + *A*(2) = U D U* to 4 x 10^-16, and the link of U_w* D U_w equal to m e^{-2 i psi}.
- The bounds of Theorem 22.4, items 2 and 4, on 30 random samples of the one-generation data: largest ratios to the bound 0.33, 10^-4 and 10^-3.

Part 23

Two gradings on an interval: masses as principal angles, the Higgs modulus, and exact W and Z masses

For STRATEGY T0b, T5b, T10 and T13, following lemma 22. Proved by Claude Opus 5.5, with the identities checked in Python (section "Numerical check"). Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): four errors and one gap as first written, and amendments, all incorporated below. The simpler form of the mass formulas, and the proof of item 3 of Theorem 23.6, are by the referee.

Outcome. A self-adjoint boundary condition for the derivative on an interval is a grading at each wall. The whole finite Dirac operator is then the relative position of two gradings: its singular values are the principal angles between them, divided by l, and

```
D_F = (1/21) Gamma_0 log(Gamma_1 Gamma_0).
```

With a bulk prealgebra, this gives four results.

1. Aligned walls compress a prealgebra stable under the grading to its even part, with the grading inherited, which is the step that Example 16.9 had to add by hand.
2. Odd scalar fluctuations move the second grading, so they change the singular values and supply the radial mode that No-go theorem 22.7 excludes for even fluctuations. For M_2(H) the field space is S^4, and the Higgs modulus is half the geodesic distance between the points chosen by the two walls.
3. With a wall connector on one channel and a five-dimensional Yang-Mills action with common normalization, the masses are exact: sin^2(M_W l) = sin^2(2 theta)/2 and sin^2(M_Z l) = (4/5) sin^2(2 theta), for universal fermions of mass theta/l. At small theta this gives M_W^2 = 2 m^2 and M_Z^2 = (16/5) m^2, the constants in the lower bounds of the fully massive family of lemma 18, with M_W^2/M_Z^2 tending to 5/8 at both ends.
4. The heat-cutoff spectral action on the interval is a Jacobi theta function of theta, decreasing on [0, pi/2]. This family lies outside the hypotheses of the radial no-go of lemma 18: the radial direction is

compact, with a stationary point at m = pi/(2l), the Connes-Lott value, where M_W = M_Z = 0.

Everything finite is still an input, and the parent algebra is matrix-valued. For a commutative parent, lemma 24 finds the window exactly, with spectral walls, while direct compression of scalar functions gives only R + R on the S^4 factor. The formulas for gauge masses depend on the assumed action. The relation to measured values fails in the flat toy, as stated at the end.

Two gradings

Let H be a finite-dimensional Hilbert space, and let Gamma_0, Gamma_1 be self-adjoint involutions on H. On L^2([0, l]; C^2) (x) H, with psi = (psi_L, psi_R), let D = E d/dx, where (E psi)_L = psi_R and (E psi)_R = -psi_L, on the domain

```
Gamma_0 psi_L(0) = psi_L(0),   Gamma_0 psi_R(0) = -psi_R(0),   Gamma_1 psi_L(l) = psi_L(l),
↪  Gamma_1 psi_R(l) = -psi_R(l).
```

Call it D(Gamma_0, Gamma_1), and let Gamma^ act as +1 on psi_L and -1 on psi_R.

Theorem 23.1.

1. D(Gamma_0, Gamma_1) is self-adjoint with compact resolvent, and it anticommutes with Gamma^.
2. Suppose Gamma_0 and Gamma_1 have eigenspaces of equal dimensions, and that -1 is not an eigenvalue of Gamma_1 Gamma_0. Let theta_k in [0, pi/2) be the principal angles between the +1 eigenspaces. Then K = (1/2l) log(Gamma_1 Gamma_0), with the principal branch, is anti-Hermitian, Gamma_0 K Gamma_0 = -K, and D_F = Gamma_0 K is self-adjoint, odd for Gamma_0, with singular values theta_k/l on the +1 eigenspace. Moreover Gamma_1 = e^{lK} Gamma_0 e^{-lK}.
3. With H_+, H_- the eigenspaces of Gamma_0, and X the H_- -> H_+ block of tan(l D_F), as in lemma 22, D(Gamma_0, Gamma_1) = D_X, and the spectral part of Theorem 22.1 gives the spectrum and the window, with no real structure needed. Conversely, every D_F odd for Gamma_0 with l ||D_F|| < pi/2 arises from exactly one such Gamma_1.
4. Let an antiunitary J_1 satisfy J_1^2 = -1, J_1 Gamma_i J_1^-1 = -Gamma_i for i = 0, 1. Then (J^ psi)_L = J_1 psi_R, (J^ psi)_R = -J_1 psi_L defines a real structure of signs (+, +, -) for D(Gamma_0, Gamma_1). With J_F = Gamma_0 J_1 up to sign, J_F^2 = 1, J_F Gamma_0 = -Gamma_0 J_F and J_F D_F = D_F J_F, which are the signs of lemma 22.

So J_F itself does not anticommute with Gamma_1 in general: J_F Gamma_1 J_F^-1 = -Gamma_0 Gamma_1 Gamma_0, the reflection of Gamma_1 through Gamma_0.

Proof. Item 1: the boundary form is [<psi_R, phi_L> - <psi_L, phi_R>] evaluated between 0 and l, which vanishes on the domain because L-values and R-values lie in orthogonal eigenspaces at each wall, and these are half of the boundary data.

Item 2: U = Gamma_1 Gamma_0 is unitary and Gamma_0 U Gamma_0 = Gamma_0 Gamma_1 = U^-1, so K is anti-Hermitian and anticommutes with Gamma_0. The eigenvalues of U are e^{+-2 i theta_k}, by the standard two-projection decomposition, and K then has eigenvalues +-i theta_k/l. Since e^{lK} Gamma_0 e^{-lK} = e^{2lK} Gamma_0 = U Gamma_0 = Gamma_1, the last formula follows.

Item 3: in the notation of Theorem 22.1, R(l) = e^{lK} carries H_+ to the graph {(C a, -S a)} = graph(-X*), and H_- to graph(X). These are the +1 and -1 eigenspaces of Gamma_1, which is the condition at the wall. The converse uses the same formula, with uniqueness from injectivity of the exponential on the stated range.

Item 4: J^ commutes with E by direct computation, and J^^2 = -J_1^2 = 1. The domain is preserved since J_1 exchanges the eigenspaces of each Gamma_i. For the last statement, J_1 U J_1^-1 = U, so as J_1 is antilinear, J_1 K J_1^-1 = K in the sense of the principal branch, and J_F = Gamma_0 J_1 then satisfies J_F K J_F^-1 = -K and commutes with D_F = Gamma_0 K. ■

When theta_k = pi/2 the channel has type (c) of lemma 09, with no zero mode and lowest level pi/(2l), the Connes-Lott mass of Theorem A in the paper. There the principal logarithm and the isolated window cease to be unique.

The bulk prealgebra

Let P be a *-subalgebra of End(H), which need not commute with Gamma_0. Let

```
A(Gamma_0, Gamma_1) = {f in C^infinity([0, 1], P) : [f(0), Gamma_0] = 0, [f(1), Gamma_1] = 0},
```

acting pointwise and identically on both spin components.

Theorem 23.2.

1. A function f in C^infinity([0, 1], P) preserves the domain exactly when it lies in A(Gamma_0, Gamma_1). The commutators are [D, f] = E f', and Gamma^ commutes with every f, including those with values odd for Gamma_0.
2. Under the hypotheses of Theorem 23.1, item 2, with W the isometry of Theorem 22.1, Q(T) = 1^-1 int_0^1 E_0(R(x)* T(x) R(x)) dx, where E_0(Z) = (Z + Gamma_0 Z Gamma_0)/2 is the conditional expectation onto the even part.
3. *Aligned walls.* If Gamma_1 = Gamma_0, then D_F = 0, the window consists of constant L-values in H_+ and constant R-values in H_-, and Gamma^ restricts on it to Gamma_0. For an ambient constant multiplier a in P, Q(a) = E_0(a), and the image of the boundary algebra A(Gamma_0, Gamma_0) is E_0(P), by compactly supported profiles. If Gamma_0 P Gamma_0 = P, then E_0(P) is the even part of P, so the compressions of the prealgebra are exactly its even part, with the grading inherited from the even parent operator Gamma^. Without that hypothesis this can fail: for H = C^2, Gamma_0 = sigma_z and P generated by sigma_x + sigma_z, E_0(P) = span{1, sigma_z}, while the even part of P is C.
4. *Order conditions.* Let J_1 be as in Theorem 23.1 and let P° = {J_1 b* J_1^-1 : b in P} commute with P. Then the parent has order zero and order one for A(Gamma_0, Gamma_1).

Proof. Item 1: as in Theorem 22.3, using that the conditions at a wall say that L- and R-values lie in the eigenspaces of Gamma_i. Item 2: as in the proof of Theorem 22.4. Item 3 is the case R = 1. Item 4: [D, f] = E f', and pointwise commutation gives both conditions. ■

Item 3 evades the inherited-grading obstruction of Result 11 of lemma 20, because hypothesis 4 there fails: for odd a, here Q(a) = 0, while there the coordinates are retained as Q(x_a) = Gamma_a/5. So at the level of a matrix-valued parent this addresses the question left in step 3 of the framework note: the prealgebra is the bulk algebra, the even algebra is what the window sees, and the grading is a boundary condition. The commutative-parent and retained-coordinate questions remain open, and Theorem 4 of lemma 24 shows a real difference: scalar functions compressed directly give R + R on the S^4 factor, and H + H needs the matrix algebra generated at the intermediate stage.

Remark 23.3 (the twist). On H = E (+) E*, with J(x, y) = (y*, x*), the operator J_1 = Gamma_0 J is the one that anticommutes with both gradings. For an odd anti-Hermitian h in P, J_1 h J_1^-1 is y -> +y h on E*, where J alone gives y -> -y h. So the opposite representation b° = J_1 b* J_1^-1 incorporates on the W-factor the grading automorphism sigma(b) = gamma_W b gamma_W. In this parent the twist comes from the requirement that J_1 preserve both boundary conditions. Twisted spectral triples for the grand symmetry (Devastato, Lizzi and Martinetti, JHEP 01 (2014) 042, arXiv:1304.0415; Devastato and Martinetti, Math. Phys. Anal. Geom. 20 (2017) 2, arXiv:1411.1320, Eqs. (1.8)-(1.11)) use a different automorphism: their rho exchanges left and right spinorial indices and algebra summands, while conjugation by gamma_W is inner on M_2(H) and fixes its centre. So this is a structural analogy between involutive automorphisms, and an identification needs a comparison of the representations, opposites and first-order conditions.

Odd fluctuations and the radial mode

Theorem 23.4. Let h be smooth with values in the anti-Hermitian elements of P, and let u' = -h u, u(0) = 1, w = u(1). Then

```
D(Gamma_0, Gamma_1) + E h = u D(Gamma_0, w\* Gamma_1 w) u\*.
```

So scalar fluctuations move the second grading along its orbit under the unitary group of P. If h has an odd part, the principal angles between Gamma_0 and w* Gamma_1 w change in general.

Proof. u* D u = D + E u* u', and u* E h u = E u* h u, so the connection cancels. Since u(0) = 1, the wall at 0 is unchanged. At l, psi = u phi gives the stated grading for phi. ■

This lies outside No-go theorem 22.7, whose hypotheses include the even algebra: an even connection gives a w commuting with Gamma_0, and the angles are kept, while an odd connection changes them. For a real fluctuation, add the conjugate one-form: D + E h + J^(E h)J^^-1, with generator of the form (L_F, R_F) in Example 23.5. The invariant is a double coset. With H_i = Stab(Gamma_i) in the connected unitary group G of P, a gauge transformation removing h has the form u(x) c, and it preserves both endpoints exactly when c is in H_0 and w c in H_1, that is, when w is in H_1 H_0. So the scalar moduli of the parent are the double cosets H_1 G / H_0, the relative position of the two gradings. This is with the moving window, as in lemma 22.

Example 23.5 (M_2(H) and S^4). Let W = H^2 = C^4, with the Clifford generators Gamma_a of lemma 16, and for a unit vector p in R^5 put Gamma_p = p_a Gamma_a. Each grading of W with one-dimensional quaternionic +1 and -1 eigenspaces is such a Gamma_p, so these gradings form S^4 = Sp(2)/(Sp(1) x Sp(1)), the quaternionic projective line. For two points, Gamma_q Gamma_p = p . q + (p wedge q), with eigenvalues e^{+-i d}, where d is the geodesic distance; so all the principal angles equal theta = d/2.

Take H = Hom(V, W) (+) Hom(W, V), J(x, y) = (y*, x*), Gamma_0(x, y) = (gamma x, -y gamma) with gamma = Gamma_{p_0}, and K(x, y) = (theta/l)(F x, y F), with F an odd anti-Hermitian element of M_2(H), F^2 = -1. Then:

- D_F = Gamma_0 K has all singular values equal to theta/l: a universal Dirac mass, for all colours.
- The second grading is Gamma_1(x, y) = (Gamma_p x, -y Gamma_{p'}), where p is at distance 2 theta from p_0 and p' is its mirror image through p_0. J_1 = Gamma_0 J satisfies Theorem 23.1, item 4, while J alone does not.
- The coordinates (theta, q) on S^4, with F determined by the unit quaternion q, are the polar coordinates of a Higgs quaternion: theta is the radial coordinate, bounded by pi/2, and q gives three angular directions, undefined at the poles. At theta = pi/2 the finite logarithm and the lowest window are degenerate. Quotienting by the stabilizer leaves the radial interval. A canonically normalized field needs the scalar kinetic term, which we have not computed.

With bulk masses, by Theorem 12.6, the light masses are about (tan(theta)/l) Phi_Q Phi_U when small compared with the tower thresholds, with a universal matrix before flavour data; general finite-angle eigenvalues need the secular equation. This field space is the coset of the minimal composite Higgs model (Agashe, Contino and Pomarol, Nucl. Phys. B 719 (2005) 165, sections 2-3, with their h/f_pi corresponding to 2 theta). Their construction uses a warped interval and additional fermion and boundary data, so the shared coset does not identify the actions or potentials. The appearance here of the S^4 of the twistor window is an observation, and no identification between the two roles of S^4 is made.

The weak bosons

Take P = M_2(H) (+) M_4(C), acting on E = Hom(V, W) on the left and on the right. Assume the following.

- A five-dimensional Yang-Mills field valued in sp(2) (+) su(4), with the trace form Tr_W + Tr_V, so with one coupling (the normalization of the spectral action, g_3 = g_2 = sqrt(5/3) g_Y).
- At x = 0: Neumann for the stabilizer of p_0, Dirichlet for its complement. At x = l: Neumann for the stabilizer of p_l and of a connector, that is, the a with [a_W, Gamma_p] = 0, a_W w = lambda w and a_V v = lambda v, as in Results 2 and 5 of lemma 20, for a line w in the +1 eigenspace of Gamma_p and a line v in V; Dirichlet for the complement.
- The connector is prescribed gauge-breaking wall data here. If it is instead a term in the fermion boundary operator, as for a unit vector e in the +1 eigenspace of Gamma_1, f = J_1 e, K_c = |f><e| - |e><f| and Gamma_{1,c} = e^{eta K_c} Gamma_1 e^{-eta K_c}, then J_1 still anticommutes with Gamma_{1,c}, the parent order conditions are kept, and the affected fermion tower changes, so the universal fermion formula must be recalculated for that channel.

Theorem 23.6. With theta the angle above, the modes that are massless for all theta are exactly nine, the photon and the eight gluons, with three additional electroweak zero modes at theta = 0 and theta = pi/2. The lowest charged and neutral massive modes satisfy

```
sin^2(M_W l) = sin^2(2 theta)/2,
sin^2(M_Z l) = (4/5) sin^2(2 theta),
```

equivalently tan^2(M_W l) = sin^2(2 theta)/(1 + cos^2(2 theta)) and tan^2(M_Z l) = 4 sin^2(2 theta)/(1 + 4 cos^2(2 theta)).

The photon has 1/e^2 = (8/3)/g^2. Moreover:

1. As theta -> 0, with m = theta/l: M_W^2 -> 2 m^2 and M_Z^2 -> (16/5) m^2, so M_W^2/M_Z^2 -> 5/8.
2. As theta -> pi/2: M_W, M_Z -> 0, with M_W^2/M_Z^2 -> 5/8, while m -> pi/(2l).
3. At theta = pi/4: M_W l = pi/4, tan(M_Z l) = 2, and M_W^2/M_Z^2 = ((pi/4)/arctan 2)^2 = 0.5032…, the global minimum. So M_W^2/M_Z^2 stays in [0.5032, 0.625), with 5/8 as a limit at the ends, where the ratio itself is undefined.
4. m/M_W goes from 1/sqrt(2) at theta -> 0 to infinity at theta -> pi/2. It equals sqrt(8/3) at theta = 1.0589.

Proof. In the gauge A_5 = 0 the field is free, with cos(k x) on the Neumann directions at 0 and sin(k x) on the Dirichlet ones, and the condition at l holds in the frame rotated by Ad exp(theta F). The generators split according to the frequency of Ad F. Write L, R for su(2) on W_- and W_+, S = (L + R)/sqrt(2), A = (L - R)/sqrt(2), and O for the odd generators; Ad F has frequency 0 on S, and frequency 2 on the plane (A, O). So S = s cos kx, A = alpha cos kx, O = beta sin kx.

At l, in the rotated frame, A' = cos(2 theta) A + sin(2 theta) O and O' = -sin(2 theta) A + cos(2 theta) O, with O' = 0 at l, which gives beta = alpha tan(2 theta) cot(k l), and so A'(l) = alpha cos(k l)/cos(2 theta).

Charged directions. The connector forces R^+- to vanish at l, with L^+- Neumann. So S(l) = A'(l), and d/dx (S + A') = 0 at l. Substituting gives sin^2(kl) (1 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Neutral directions. Add B = b cos kx for the generator T_15 = diag(1, 1, 1, -3)/(2 sqrt 6) of su(4). With Y = sqrt(3/5) R + sqrt(2/5) B and Z' = -sqrt(2/5) R + sqrt(3/5) B, the connector forces Z' to vanish at l, with L and Y Neumann. The Neumann conditions give b = -sqrt(3) s, and Z' = 0 gives s = alpha/(4 cos(2 theta)). Substituting in the condition for L gives sin^2(kl) (1/4 + cos^2(2 theta)) = sin^2(2 theta) cos^2(kl).

Massless modes and limits. At k = 0, a zero mode is a constant in the intersection of the two stabilizers. This consists of a_W = diag(q, q) on W_+ and W_-, identified by F, with q preserving the line of w, together with a_V = lambda (+) m, and with the trace condition this has dimension 1 + 8. In it, L = R and B/L = sqrt(2/3), which gives 1/e^2 = (1 + 1 + 2/3)/g^2. The limits follow by expanding. A second derivation, from the referee, uses the principal angles between the Neumann subspaces at the two walls: for each charged direction the squared sine of the angle is sin^2(2 theta)/2, and in the neutral sector the single nonzero value is (4/5) sin^2(2 theta); it also covers cos(2 theta) = 0, where the divisions above need a limit. For item 3, with a = sin(2 theta), the ratio is [arcsin(a/sqrt 2)/arcsin(2a/sqrt 5)]^2, and this decreases with a on (0, 1], since x/(sqrt(1 - x^2) arcsin x) increases. ■

The constants in item 1 are the values ||[D_F, a]||^2/||a||^2 in the Hilbert-Schmidt norm, for universal mass m, for a = L^+- and for the direction orthogonal to the photon and Z', which has A-component of squared length 4/5. These are the coefficients of the lower bounds for the poles of the fully massive family in Proposition 14 of lemma 18 (2 x < z_W, (16/5) x < z_Z); the small-x pole coefficients there are 3 and 24/5. The limit 5/8 is common to both, with different behaviour: the finite-action poles grow in the heavy limit of lemma 18, while the interval masses here tend to zero at the antipodal point. The spectral-action Taylor mass for universal m is M_W^2 = 3 m^2, which exceeds the value here by the factor 3/2 of Proposition F in the paper: there the mass matrix carries the relative heat coefficients of the action, and here we have assumed a five-dimensional Yang-Mills action, which gives the commutator norm alone. A direct matching claim needs the full parent spectral action, with its tower and boundary conditions.

Item 2 gives a geometric form of the behaviour that lemma 18 found: heavy fermions with light vectors. At antipodal points Gamma_p = -Gamma_{p_0}, so both walls have the same stabilizer in sp(2) again, and the fermion channels have type (c) with mass pi/(2l).

The spectral action on the interval

Proposition 23.7. For the heat cutoff and universal theta, with s = l Lambda, the trace over a channel is

```
Tr_channel exp(-D^2/Lambda^2) = 2 sum_{n in Z} exp(-(n pi + theta)^2/s^2).
```

This is proportional to the Jacobi function theta_3(theta, exp(-s^2)). It is strictly decreasing on [0, pi/2], with a maximum at theta = 0 and a minimum at theta = pi/2, where it is stationary, and it is symmetric under theta -> pi - theta. In the towerless regime s -> 0 at fixed x = theta^2/s^2, it tends to 2 e^{-x}, the form of the vacuum density in lemma 18 and Proposition 21.2.

Proof. The eigenvalues in a channel are +-(n pi + theta)/l, by Theorem 22.1, and the sum is over Z after combining the signs. By Poisson summation the sum equals (s/sqrt(pi)) sum_{k in Z} exp(-k^2 s^2) e^{2 i k theta}, and the Jacobi triple product writes this as a constant times the product over n >= 1 of (1 + 2 q^{2n-1} cos(2 theta) + q^{4n-2}), with q = exp(-s^2). Each factor is positive and strictly decreasing in theta on [0, pi/2]. ■

In the form of the Jacobi function, the trace is (2 s/sqrt(pi)) theta_3(theta, exp(-s^2)). This family lies outside the hypotheses of No-go theorem 9 of lemma 18, which concerns finite internal data and a free ray V = c V_*: here the operator family is different, the radial direction is a compact orbit of relative gradings, and the tower, which the finite description omits, supplies a second branch with a stationary point at theta = pi/2. In the present family the literal action has its minimum at antipodal gradings, m = pi/(2l), with M_W = M_Z = 0 by Theorem 23.6, where the lowest eigenspace is degenerate. This is a vacuum with massless vectors, so it is not the physical point. The sign of the term agrees with the fermion loop in the Hosotani mechanism, where gauge-boson loops enter with the opposite sign. So an interior minimum could arise from that competition, which the classical spectral action does not contain. We have not computed it.

The one-loop potential in the same family

From the exact spectra, the products over the towers are, relative to theta = 0 and in Euclidean momentum x = p l,

```
fermion channel:  1 + sin^2(theta)/sinh^2(x),
charged vectors:  1 + sin^2(2 theta)/(2 sinh^2(x)),
neutral vector:   1 + (4/5) sin^2(2 theta)/sinh^2(x).
```

So in the gauge-boson and fermion-loop form of the Hosotani mechanism, assuming the usual one-loop formula with three polarizations per vector, and with n_f = 2 x (number of physical Dirac species),

```
8 pi^2 l^4 V(theta) = int_0^infinity x^3 dx [ 3 log(1 + s^2/(2 sinh^2 x)) + (3/2) log(1 + (4/5)
↪  s^2/sinh^2 x) - n_f log(1 + sin^2(theta)/sinh^2 x) ],
```

with s = sin(2 theta). A Dirac top with three colours has n_f = 6, which needs a separately specified fermion sector, since the universal connection gives masses to all: one generation has n_f = 16 and three have n_f = 48. With I = int x^3/sinh^2 x dx = (3/2) zeta(3) and I' = int x^3/cosh^2 x dx = (9/8) zeta(3), near theta = 0 the bracket is (10.8 - n_f) I theta^2 + o(theta^2), and with epsilon = pi/2 - theta, near theta = pi/2 it is V(pi/2) + (10.8 I + n_f I') epsilon^2 + o(epsilon^2), where we have written V for the left side.

Proposition 23.8 (referee). In the massless flat family with this content, for every n_f > 0, theta = pi/2 is the unique global minimum on [0, pi/2], since the gauge logarithms are nonnegative and vanish there, and the negative fermion logarithm is minimized at sin^2(theta) = 1. For n_f >= 10.8, V is strictly decreasing on the interior. Indeed, with t = sin^2(theta) and J(z) = int_0^infinity x^3 dx/(sinh^2 x + z), the derivative with respect to t is, up to a positive factor,

```
(1 - 2t) [6 J(2t(1 - t)) + (24/5) J((16/5) t (1 - t))] - n_f J(t).
```

For t <= 1/2, both arguments in the bracket are at least t, and J decreases; for t > 1/2 both contributions have the required sign.

For 0 < n_f < 10.8, on grids of theta (Python), for n_f in {1, 2, 3, 4, 6, 8, 10}, V has one interior maximum, at about 0.33 for n_f = 8, and no interior minimum. Theorem 9 of lemma 24, by Astra 6, proves this for all n_f in that range: exactly one interior stationary point, a strict maximum, using a log-concavity property of the integral.

These are negative results in an exact scope: one interval, no bulk masses, fermions of frequency 1 under F, this gauge content, one loop, this condition at the connector wall. As a formal leading approximation, not uniform at x = 0, V has the form B sin^2(2 theta) - A sin^2(theta) with A, B > 0, whose interior stationary point cos(2 theta) = A/(4B) is a maximum. *Bulk masses.* With bulk masses, the fermion product becomes cos^2(theta) + sin^2(theta) G(x), with G = (w_Q coth w_Q - c_Q)(w_U coth w_U - c_U)/x^2 and w = (c^2 + x^2)^{1/2} in the secular form of lemma 12. For a = c_Q + c_U != 0, G = 1 - a/x + O(x^-2), so the potential has angle-dependent ultraviolet divergences, and a finite value depends on a regulator and subtraction: as the referee showed, an earlier scan of general pairs here did not establish a potential, and it is withdrawn. The exceptional case c_U = -c_Q = c has a common bulk mass on the two mixed components and G = 1 + w^2/(x^2 sinh^2 w), which decays exponentially. Keeping the flat gauge products as a separate tree-level assumption, a scan for c in {0, 1/2, 1, 3/2, 2, 3, 5}, n_f in {1, 3, 6, 8, 10, 16, 48}, on a grid of 721 values of theta, finds no interior minimum (Python). General bulk masses need a gauge-invariant completion, a regulator and wall counterterms.

So an interior minimum needs another change: a negative coefficient of sin^2(2 theta), as from fermions of frequency 2 under F, which the bimodule E = Hom(V, W) does not contain; general bulk masses or other gauge form factors, as with warping; a wall scalar with its own potential; or tree-level terms at the walls. With h/f_pi = 2 theta, the leading flat form has the angular form alpha cos(h/f_pi) - beta sin^2(h/f_pi) of Agashe, Contino and Pomarol (section 2.1, Eqs. (19)-(23)) with spinorial SO(5) fermions, with alpha = A/2 and beta = -B, up to a constant. Their form factors can give the sign and competition needed for an interior minimum, so the flat results here do not exclude that construction. This points away from a pure flat Wilson line, toward those changed form factors or a wall scalar with its own potential, for the physical interior.

Comparison with measured values in the flat toy

These are numerical comparisons (Python), with no running and no bulk masses, in lowest levels, using the Particle Data Group 2026 values quoted by the referee, m_t = 172.60 GeV as a direct-measurement mass, M_W = 80.3625 GeV and M_Z = 91.1879 GeV, so m_t/M_W = 2.1478 and M_W^2/M_Z^2 = 0.7767. The ratio m_t/M_W reaches sqrt(8/3) at theta = 1.0589 and the measured value at theta = 1.1626, where M_W l = 0.541, so 1/l is about 148.5 GeV. There, M_W^2/M_Z^2 = 0.581. The wall-scalar model of lemma 22 covers [0.625, 1], and this family covers [0.5032, 0.625); in the wall-scalar model, fitting the measured vector ratio gives M_W l = 1.028, where the lowest no-bulk-mass fermion channel has m/M_W < 1.528. So neither flat family fits both ratios on its own, and the flat one-loop potential above does not select an interior point. Whether a tower at 1/l of this size is allowed would need the couplings and an applicable experimental limit, which we have not analyzed. The directions are bulk masses for the light fermions, warping, running of the couplings, and families combining a Wilson line with a wall scalar. These negative statements hold in the flat tree-level toy only.

Open questions

1. The interior vacuum: the effective potential with a wall scalar, or with warping, or with general bulk masses, with a gauge-invariant completion, a regulator and wall counterterms. The flat cases above have none on the grids.
2. The relation between the two roles of S^4: the base of the twistor window, and the field space of the Higgs quaternion.
3. A commutative parent: functions on S^4 x I and CP^3 x I, with the gradings as local boundary conditions.

4. The factor 3/2 between the five-dimensional normalization and the spectral action: whether the spectral action on the parent itself, with its tower, reproduces the formula of Theorem 23.6.
5. The comparison with twisted spectral triples, Remark 23.3.
6. The contribution of a real connector to the fermion tower, the degenerate antipodal window, and the scalar kinetic normalization needed for a canonically normalized Higgs field.

Numerical check

By Claude Opus 5.5, in Python (numpy and scipy), 3 Nov 2026, in a temporary directory, not kept.

- Random complex symmetric M on C^4 with l ||M|| = 1.25: Gamma_1 is an involution, Gamma_1 = exp(2 l K) Gamma_0; D_F = (1/2l) Gamma_0 log(Gamma_1 Gamma_0) to 3 x 10^-15; the principal angles over l equal the singular values of M to five digits.
- Sp(2) x SU(4), of dimension 25, with the Neumann and Dirichlet subspaces built from the stated stabilizers: the two-point boundary problem was solved as a linear system on a grid in k l. For theta = 0.2, pi/4, 1.06 and 1.3, the two lowest nonzero roots are (0.279, 0.356), (0.785, 1.107), (0.647, 0.868), (0.373, 0.479), equal to the formulas of Theorem 23.6 to three digits, with other roots at pi/2 and above. The intersection of the two stabilizers has dimension 9 at each theta tested.
- Small theta: M_W^2/m^2 = 2.000 and M_Z^2/m^2 = 3.200.
- The universal connection on E (+) E*: J D = D J, Gamma_0 D = -D Gamma_0, all |eigenvalues| equal to theta/l, Gamma_1 = (left Gamma_p, right mirror), J_1^2 = -1, J_1 Gamma_1 = -Gamma_1 J_1, while J alone fails; J_1 h J_1^-1 is right multiplication by +h on E*, for odd h.
- The theta function: monotone on a grid for s = 0.3, 1, 2, with the towerless value at s = 0.3 equal to e^{-x} to five digits; Poisson summation and the triple product to 10^-16.
- The ratio M_W^2/M_Z^2 on a grid: range [0.5032, 0.6250], symmetric under theta -> pi/2 - theta to 10^-15, minimum at pi/4. At theta = 1.1625, M_W l = 0.541, and at theta = 1.0589, M_W l = 0.648.
- The one-loop integrals by adaptive quadrature (scipy), and by Gauss-Legendre quadrature on 400 nodes for the scan with c_U = -c_Q. The referee recomputed the derivative with scipy, found one sign change for each n_f in {1, 2, 3, 4, 6, 8}, with the stationary angle about 0.3382 at n_f = 8, and checked I = (3/2) zeta(3).

Part 11D, README, referee pending

Eleven dimensions from massless W and Z

Branch `11D`, opened 3 Nov 2026. This folder studies how the limit M_W, M_Z -> 0 of hypothesis H could give an eleven-dimensional theory, with proposed connections to Witten's seven-manifolds and the M2 and M5 readings of lemma 21 and STRATEGY T12. The branch also contains the independent referee report and related ledger and status updates.

Attempted by Claude (Anthropic) on 3 Nov 2026. The Astra 6 audit of the same day (reviews/2026-11-03_astra_audit_d11.md, of snapshot b2ddf98) supplies the joint-limit family and corrections to Proposition 1.4 and Proposal 1.3. GPT-6.1 Terra has addressed the requested items analytically in the referee report, with amendments and explicitly conditional source comparisons. The full EPJC paper and several cited primary sources were unavailable, and the original localized-family numerical table was not recomputed. The general third-charge extension of the referee has been checked by Claude Sonnet 5.5. The original scoped no-go results had two-model support after the Terra report; Claude Sonnet 5.5 has since checked the Terra sections on files 01 to 04 a second time (second check), so the no-go results of those files have three model identifiers, the ledger's general-consensus threshold, each in its stated scope. Files 05 to 09 are being checked in the same way. Opening statements in the numbered files that say "referee pending" are historical; their adjacent referee notes and the table below give the current status. prompts/terra_check_11d.md is the referee request. The conversation in which the folder was written is recorded at https://claude.ai/code/cshk_01BpscANdX3GG8D45SQ1BxsaHviSe7ufxxCLNiNaDv4fWT.

The picture

H reads the two sheets of the finite geometry as the minimal truncation of an interval with opposite chiral walls, and v -> 0 as the interval growing. The labels 9, 10 and 11 of H are KO labels; the heat dimension at the 11 endpoint is 5. To make 11 a metric count, the finite KO-6 factor is resolved into a six-dimensional base, B_6 = CP^2 x S^2, which has isometry SU(3) x SU(2) and is the base of Witten's spaces:

```
11 = 4 (spacetime) + 1 (the interval of H) + 6 (B_6).
```

The following table summarizes two proposed completions of 4 + 6 and their brane readings. The M^{111} identification uses the recalled fibration, and the physical dictionaries and the single-family construction remain open:

	electric (M2) completion	magnetic (M5) completion
eleven-dimensional vacuum	AdS4 x Y_{3,2}, Y_{3,2} = M^{111}, a circle bundle over B_6	AdS5 x CP^2 x S^2
the interval of H	the fibre circle, antiperiodic, two walls at the fixed points of a reflection	the radial direction of AdS5, infrared end at z_IR ~ 1/v
SU(3) x SU(2)	isometries	isometries
hypercharge	the vector of C_3 on the harmonic two-form (b_2 = 1)	one combination of the two C_3 vectors (b_2 = 2)
v -> 0	the M2-brane moves to the apex; AdS4 x Y_{3,2} with the Standard Model group unbroken	the infrared end moves to infinity

Results

#	result	status	file
1.1	The chiral interval of length l is unitarily the scalar antiperiodic circle of radius R = l/pi; the grading is reflection, the walls its fixed points, m = 1/(2R)	Proposition independently checked, with counting clarification	01; referee
audit §3, §4	Joint family: D_F = epsilon D_*, l ~ epsilon^{-alpha}, alpha < 1, gives a finite window and an opening tower; a localized variant uses a bulk mass and logarithmic length factor	Window, Weyl constant and localized asymptotics independently checked analytically	01; referee
1.3	The circle of H as the M-theory circle: v -> 0 along the joint family is strong coupling with the Wilson-line angles going to zero, the 9 endpoint has no bulk fermions as in type 0 strings	Proposal	01
1.4	The angle-dependent one-loop term of lemma 24 is -93 zeta(5) n_f/(512 pi^2 l^4) at its minimum, relative to theta = 0, where M_W = M_Z = 0 at fixed l	Proposition independently checked in the scope of lemma 24	01; referee

#	result	status	file
1.5	An isolated stationary angle of a scale-free potential with theta = l m gives l ~ 1/m; flat directions allow other stationary paths. The specified one-loop determinant has a small-angle obstruction when n_f != 54/5	Corrected conditional result; universal claim has a counterexample	01; referee
2.1	The hypercharge quotient of the 5-bar is the circle bundle of class ±(3,2), with effective group (U(3) x U(2))/U(1)_Y and a trivial-canonical cone; its identification with M^{111} uses the recalled fibration	Quotient checked; literature identification conditional, with metric and dual-action amendments	02; referee
2.2	Only the 5 and 5-bar of SU(5) give a seven-dimensional free hypercharge quotient; among the specified circle quotients only (3, 2) has the required trivial canonical character	Proposition independently checked	02; referee
2.3	Y_{a,b}: H^2 = Z (+ torsion), H^4 of order a^2 (Z/9 for (3,2)), spin iff a odd and b even; H^2(B_6) -> H^2(Y_{3,2}) is (x, y) -> 2x - 3y	Proposition independently checked; literature comparison pending	02; referee
2.4	The Z_6 is the kernel of the displayed SU(3) x SU(2) action on Y_{3,2}; tensor Kaluza-Klein modes on an invariant background contain no triplet or doublet	Proposition independently checked in the tensor-mode scope	02; referee
2.5	No generation from one abelian line bundle per field on CP^2 x CP^1 with fluxes along Y and B - L; the asserted equal images and unsigned linearity need common ten-dimensional chirality	Scoped no-go independently checked, with amendment	02; referee
2.5, third charge	The original bounded third-charge search is excluded analytically; the proof extends to any real third charge and independent source chiralities	Bounded claim independently checked; the extension to any real third charge checked by Claude Sonnet 5.5 (two model identifiers)	02; referee
2.6	A line-bundle index gives one generation using a universal charge, chi and B - L; the colour-action convention changes the quark bidegrees, and compatible global flux quantization remains to be specified	Conditional index result independently checked, with amendment	02; referee

#	result	status	file
3.1	Hypercharge is the Betti vector of Y_{3,2}; the N = 1 fivebrane baryons are d^c and L	Proposal	03
3.2, 3.3	One generation is Λ^odd(5-bar); its realization as a Fock space of wrapped fivebranes with 6Y as homology class is conjectural	Standard representation count independently checked; brane interpretation remains a conjecture	03; referee
3.4	The Higgs as an M2 position is a proposal; the FI quotient has residual Z_3 or Z_2 stabilizers, and the fixed-N tower remains finite	Coordinate stabilizer independently checked; orbifold and physical-identification amendments	03; referee
3.5	AdS5 x CP^2 x S^2 with the displayed magnetic G_4 solves the classical field equations for scaled Einstein factors iff abs(f_1) < 2 abs(f_2); quantization and stability remain open	Proposition independently checked	03; referee
3.6, 3.7	Two completions of one base; the M2/M5 exchange as the electric-magnetic duality of hypercharge, which does not permute the 96 states	Proposals	03
3.8, 3.9	The 84 of C_3 is not colour-equivariantly either family of 84, through the specified little-group action or as tensor Kaluza-Klein modes on Y_{3,2}	Scoped no-go and corollary independently checked	03; referee
4.1	The ratio-preserving 12-state blocks of whole species are one up-type quark, or the three neutrinos	Proposition independently checked	04; referee
4.2, 4.3	The colour contents of F1 and F2 differ, and Lambda^3 C^9 is neither family for any linear SU(3) action; Lambda^3 C^9 has only 1, 3, 3bar exactly for C^9 = 9 (1), 3 + 6 (1) or its conjugate	Scoped no-go independently checked, with amendments	04; referee
4.A, 4.4	Two commuting order-3 gradings of e8 place the left-handed 10 and H_u in the C_3 sector and (d^c, L), nu^c in the other sectors; top and antitop each have chiralities in both sectors	Branching and counts independently checked, with chirality amendment	04; referee
4.5	Up-type grade selection is within the C_3 sector and the other Yukawa triples use one field per sector; the physical top mass needs family-breaking or compactification data	Grade selection independently checked, with family-tensor amendment	04; referee

#	result	status	file
4.6	The Chevalley involution exchanges conjugate sectors and preserves both families; a grading-preserving E8 automorphism cannot exchange the specified top and neutrino blocks. An antiunitary J needs an extra conjugation; its physical duality identification is a proposal	Scoped no-go independently checked, with amendment	04; referee
4 §7, §8	nu^c as the gravitational state selects t = +1/2; a discrete Z_3 from the M-grading	Proposal; discrete-grade selection independently checked	04; referee
5.1	As configurations of aligned three-family E8 roots the two families of 84 have equal pair counts but are not isometric; no element of W(E8) exchanges them	Scoped no-go independently checked	05; referee
5.2	With four families, M = -P in W(E8) transposes colour with family and maps the up-type quark block of family 0 onto the neutrino blocks of families 1, 2, 3	Proposition independently checked, with setwise-invariance clarification	05; referee
5.3	With a unitary colour-family identification, the flip commutes with gamma and J and conjugates the heavy-top and three-heavy-neutrino limits; the displayed algebras are not flip-invariant	Proposition independently checked, with unitarity amendment	05; referee
5.4	No assignment of the four types to three of four family slots, aligned or misaligned, admits the exchange as an isometry of the specified E8 roots; the original universal neighbour counts need correction	Scoped no-go independently checked analytically, with amendment	05; referee
5.5	Families as zero modes of an SU(4) bundle escape the single-root-slot assumptions; the index counts net chirality and a compatible Weyl action remains open	Proposal, independently reviewed with amendments	05; referee
6.1	The SO(32) charged Wilson-line sector has 30 copies of the lemma-22 channel spectrum; its light mass is a/R for 0 <= a <= 1/2	Spectrum independently checked, with multiplicity and range amendments	06; referee
6 §2, §3, §5	The heterotic web and type-I' dictionary as a route to the target family	Proposals reviewed; Wilson-line map, winding duality and slope rule confirmed against the two sources in the second check	06; referee

#	result	status	file
7.1	A channel spectrum has fundamental domain [0, pi/2], with doubly degenerate periodic and antiperiodic endpoint spectra	Proposition independently checked, with endpoint and scope clarifications	07; referee
7.2	The SO(32) endpoint has centralizer S(O(30) x O(2)); the lemma-24 one-loop minimum is a = 1/2 under its stated hypotheses	Proposition independently checked, with disconnected-centralizer amendment; brane interpretation remains a proposal	07; referee
7.3	A doubled flux torus gives KO-6 signs, a 2p-dimensional kernel and a generated M_p(C) corner; adding a scalar summand gives a unital finite datum	Error in the stated unital triple; signs and matrix corner independently checked	07; referee
7 §2, §4, §5, §6	Readings of dualities for finite geometry; the torus flux and M2 wrapping need a constructed identification	Proposals reviewed; duality recalls conditional on unavailable primary sources	07; referee
8.1	The descended antiunitary gives O(n) or USp(n) on each eigenspace of d^2; preservation of the rest of the data requires further conditions	Proposition independently checked, with symmetry-scope clarification	08; referee
8.2	A signed channel eigenvalue has multiplicity k in the interior and 2k at the endpoints; the corresponding abs(D) counts differ	Proposition independently checked, with counting clarification	08; referee
8.3, 8 readings	Wall slopes and 2L/3 placement follow from the recalled rule; E-type enhancements require tuned coupling, and the electroweak identification remains a proposal	Arithmetic independently checked; slope rule and E-series confirmed against the sources in the second check, physical readings conditional	08; referee
9.1	The integrated profile and far-wall condition follow from the recalled slope rule; its spectral trace uses sixteen selected channels	Integration independently checked; slope rule confirmed against Polchinski-Witten in the second check, channel dictionary linear only at constant dilaton	09; referee
9.2	Antitriplet counts are (6, 6, 3) in the antisymmetric block and (3, 6, 1) in the symmetric block; the single-line antisymmetric component vanishes	Representation counts independently checked; orientifold projection and physical identification conditional	09; referee
9.3	The stated KO signs force a particle-antiparticle block into S^2 V, excluding nonzero Lambda^2 V blocks; the full symmetric spectrum and interval scalar modes require further construction	Scoped no-go independently checked, with amendment	09; referee

#	result	status	file
9 §1	SO(32) branching and the SU(16) alternative; the archived abstract also states an extra chiral +4/3 quark per generation	Algebra independently checked; detailed full-paper comparison pending	09; referee
9 §5	E-type wall and del Pezzo correspondence; the two degree-eight surfaces have distinct root lattices and engineering assignments	Error in the E1 surface alternatives; dP_2 comparison conditional	09; referee
9 §5(2)	The E8 Weyl transposition conjugates a Wilson line and acts on the Narain lattice; a physical exchange must preserve the full background	Gauge action independently checked, with background-scope clarification	09; referee
5 Readings	Three families as the obstruction; M exchanges the two heavy directions of the algebra chain; 128 = 44 + 84 under Spin(9)	Proposals	05

The author's numerical checks are in checks_11d.py (Python 3, exact fractions). The independent referee arguments are analytic.

Added on master, 4 Nov 2026, by Claude Opus 5.5, with one independent reading by Claude Sonnet 5.5 (review): 10 tests the dilaton as second scale (negative in scope; computes a_0; corrects the dictionary of 06 to 09: nonlinear angle map, T-dual lengths, Romans mass) and 11 computes the SO(30) fluctuations (the antisymmetric block needs M_15(C) at both walls, and the window sees its commutator with the masses).

How the massless limit gives eleven dimensions here

Vanishing W and Z masses open a dimension only along a specified path (audit §2.2): at fixed l, both theta -> 0 and theta -> pi/2 make them massless with the tower fixed. The criterion used here is the joint family of the audit, m_weak « E_epsilon « 1/R -> 0, with finite gauge couplings.

1. In the circle language of Proposition 1.1, the joint family is R -> infinity for an antiperiodic circle with two fixed points, with the Wilson-line angles of D_F going to zero. Read as the M-theory circle (Proposal 1.3), it is the strong-coupling limit, with the weak scale below the momentum tower of the eleventh dimension.
2. What selects the path: an isolated stationary angle of a potential with l and the angles as its only scales gives alpha = 1, the original l ~ 1/M_W of H, on which the finite window does not separate. A flat direction permits other paths without selecting one (referee of Proposition 1.5). A second scale could select a joint family; the localized family of the audit (bulk mass mu, mu l ~ (1 - alpha) log(1/epsilon)) is the first candidate, and the four-dimensional couplings must stay finite, which on the circle needs g_5^2 ~ R.
3. In the brane language of Proposal 3.4, v -> 0 moves the M2-brane to the apex of the cone over the Witten space with Killing spinors, and the near-horizon geometry is AdS4 x Y_{3,2}: 4 + 7 = 11 with SU(3) x SU(2) from isometries and U(1)_Y from C_3 (Proposal 3.1). This is the family of manifolds with the Standard Model group, singled out by the 5-bar (Theorem 2.1, Proposition 2.2). At fixed N it is the fixed-tower path; a joint family there needs N -> infinity, in tension with the N = 1 matter content.
4. On the magnetic side, AdS5 x CP^2 x S^2 carries the same base, and v -> 0 sends the infrared end of AdS5 to infinity (Proposal 3.6), which is the interval of H growing.

The angle-dependent one-loop term (Proposition 1.4) is minimized at the massless point at fixed l; the direction in which l moves depends on the angle-independent Casimir coefficient, which is not yet computed.

Next computations, ranked

0a. **The main point** (06, PLAN task B): one family realizing the three regimes; first steps listed in 06 §6 (type I' interval with the D8 configuration 15 + 1 as the parent of the lemma-22 window; Chan-Paton algebras against C + H + M_3(C); inner fluctuations against Λ^2(30)). 0b. **A nine-dimensional spectral triple on the type I' interval** (07 §7): sixteen D8 positions as the eigenvalues of D_F, and its strong-coupling limit with E8 walls. File 09 gives a conditional dilaton profile and representation counts; the physical string dictionary and whether the coupling can select the path remain open. 0. **A potential that selects alpha < 1** (Proposition 1.5, its referee amendment and audit §7.1): the one-loop potential of the localized family, G(l m, l mu), from the exact secular equations of lemma 09, and the angle-independent Casimir coefficient a_0 of the flat family; then the vector kinetic matrix and mass Hessian along the selected path (audit eq. (17)), with g_4 finite. 1. **Remaining referee work** (report, second check): compare the EPJC paper and cited primary sources when available, check the original localized numerical table if needed, and compare the EPJC paper when available; the second check of files 01 to 09 and of the third-charge extension is complete. 2. **Spinors on Y_{3,2}** (prompts/astra_11d_01.md): the lift of (U(3) x U(2))/U(1)_Y to the spin bundle, whether fermionic Kaluza-Klein modes can carry triality or isospin 1/2, and which spin structures give the antiperiodic fibre momenta (k + 1/2)/R of Proposition 1.1. 3. **W and Z masses from the resolution**: specify the partial orbifold resolution, any further smoothing, and the brane background; compute the broken SU(2) x U(1)_Y vector masses as functions of zeta and N and compare M_W/M_Z with the cos^2 theta = 5/8 of the spectral action. 4. **The 9D reduction of Proposition 2.6**: with SU(2)_R broken by wall data, derive the mass basis of u and d against the charge basis, and the vectorlike limit at the 9 label, where the left-handed interactions are off. 5. **Fermionic zero modes of wrapped fivebranes** (Conjecture 3.3) at N = 1, and three generations. 6. **The magnetic completion**: flux quantization, stability and the two C_3 vectors of AdS5 x CP^2 x S^2 against the fluxed pair (chi, B - L) of Proposition 2.6. 7. **A four-family completion with the fourth family decoupled** (PLAN, steps 4-5): a finite geometry or Hořava-Witten configuration in which the colour-family transposition M of Proposition 5.2 is a symmetry, so that the three-family statement of lemma 21 follows as a limit; Propositions 5.1 and 5.4 (one root per state) show that with three families the exchange is not an isometry.

The Astra 6 audit of 3 Nov 2026

The audit (reviews/2026-11-03_astra_audit_d11.md) reviews the master snapshot b2ddf98 and proposes a route from the massless electroweak limit to eleven dimensions with X = D(Λ^2_- T^*S^4) as principal candidate and M^{111} as comparison. Its use here:

audit item	use in this folder
§2.2, the massless limit must specify a path	adopted as the criterion; Proposal 1.3 revised; Proposition 1.5 shows what a scale-free action selects
§3, joint family	restated in circle language in 01, checked against Theorems 22.1 and 22.4; the constant 2 r l E/pi of its Weyl count confirmed (Python, section H)
§3.1, vector poles of the lemma-23 toy	quoted in 01: M_W^2/M_Z^2 -> 5/8 along the family
§4, localized family and tail identity	table of roots and the identity (8) reproduced (Python, section H, bisection), independently of the audit's script
§7.1, modulus target	In the stated flat one-loop family the electroweak term is m^2/l^2 (q = -2); a second scale could select the audit's q > 0 path. General scale-free potentials can have flat directions

audit item	use in this folder
§7.2, gauge normalization	test 2 of Proposal 1.3, and the large-N couplings in 03
Appendix A, M^{111}	agrees with Theorem 2.1 and Proposition 2.3 for (3, 2), and with the placement of hypercharge in the charged module (03)
§2.1, §5, §6, X and lemma 28	outside this folder; the 7-dimensional parent X is a third possible completion, M-theory on R^4 x X with the Bryant-Salamon comparison and the circle of solidification §7 as a candidate M-circle, listed as a direction only
§2.4, docs and distribution	outside this branch: refreshing the PDFs, adding lemmas 17 and 28 to paper/build_supplement.sh and repairing the supplement layout belong to master's build, and LaTeX is not available in this session

References used here

Read in this session (3 Nov 2026), from the arXiv abstract pages or the ar5iv text through a summarizing fetch: D. Fabbri, P. Fré, L. Gualtieri and P. Termonia, *M-theory on AdS4 x M^{111}: the complete Osp(2|4) x SU(3) x SU(2) spectrum from harmonic analysis*, Nucl. Phys. B 560 (1999) 617, hep-th/9903036 (coset, Killing spinors only for p = q, R-symmetry, Betti multiplet, Kaluza-Klein states neutral under it); D. Fabbri, P. Fré, L. Gualtieri, C. Reina, A. Tomasiello, A. Zaffaroni and A. Zampa, *3D superconformal theories from Sasakian seven-manifolds*, hep-th/9907219 (D-term 2|U|^2 = 3|V|^2, U(1) fibration over P^1 x P^2, baryons det U and det V as wrapped M5-branes, E_0 = 4N/9 and N/3, b_2 = b_5 = 1). M. Kreck and S. Stolz, *A diffeomorphism classification of 7-dimensional homogeneous Einstein manifolds with SU(3) x SU(2) x U(1)-symmetry*, Ann. of Math. 127 (1988) 373 (title and scope only).

Recalled and not reread, to be checked before any of them carries an argument: E. Witten, *String theory dynamics in various dimensions*, hep-th/9503124 (D0 states as momentum modes); O. Bergman and M. Gaberdiel, *Dualities of type 0 strings*, hep-th/9906055; M. Fabinger and P. Hořava, *Casimir forces between end-of-the-world branes*, hep-th/0002073; E. Witten, *SL(2, Z) action on three-dimensional conformal field theories with Abelian symmetry*, hep-th/0307041; C. N. Pope and P. van Nieuwenhuizen, *Compactifications of d = 11 supergravity on Kähler manifolds*, Commun. Math. Phys. 122 (1989) 281; J. Gauntlett, D. Martelli, J. Sparks and D. Waldram, *Supersymmetric AdS5 solutions of M-theory*, hep-th/0402153.

Already in the corpus: Witten 1981 and Castellani-D'Auria-Fré (LITERATURE.md §F); Hořava-Witten, Atiyah-Witten and Acharya-Witten-type cones (solidification §6).

Prior work for the observation of Theorem 2.1, that the weights of M^{111} are the hypercharges of the 5-bar: a thin web search of 4 Nov 2026 found no statement of it, as recorded in LITERATURE.md; this does not establish priority.

Files

- 01_interval_and_eleventh_dimension.md: the "1".
- 02_witten_spaces_hypercharge_quotient.md: the "6 + 1" of Witten's spaces, the Z_6, chirality from fluxes.
- 03_branes_hypercharge_and_the_84.md: C_3, M2, M5 and the 84.
- 04_two_families_m2_m5.md: the two 12-state blocks, two gradings of E8, Yukawa grades, and a proposed M2/M5 reading. An antiunitary J requires more than the complex-linear Chevalley involution.
- 05_colour_family_exchange.md: the two families as E8 root configurations, the colour-family transposition with four families.

- 06_main_point_heterotic_route.md: the main point: target, the heterotic web 9/10/11, dictionary, Proposition 6.1.
- 07_dualities_9_11.md: recalled dualities, the proposed D8-position dictionary, the conditional one-loop endpoint, and the doubled flux-torus matrix corner with a unitality amendment.
- 08_orientifold_sign.md: antiunitary commutants, a proposed orientifold interpretation, and conditional wall-enhancement arithmetic.
- 09_dilaton_spectral_and_so32.md: a conditional type I' coupling profile, SO(32) branching and diquark counts, and proposed string and del Pezzo readings; the full EPJC comparison remains pending.
- PLAN.md: the handover plan of the line.
- checks_11d.py: exact checks.
- ledger_11d.pl: ledger facts, included by ../ledger.pl.
- prompts/: the referee request and the first Astra job.
- ../reviews/2026-11-03_astra_audit_d11.md: the Astra 6 audit used in the second version.

Part 11D, 01 interval and eleventh dimension, referee pending

The interval of H as an eleventh dimension

Part of the 11D line of research (README). Propositions 1.1, 1.4 and 1.5 are proved here; Proposal 1.3 is a dictionary to be tested. The joint-limit family of the section of that name is from the Astra 6 audit of 3 Nov 2026 (reviews/2026-11-03_astra_audit_d11.md, §3 and §4), restated here in circle language and checked. Attempted by Claude (Anthropic), 3 Nov 2026; checks in Python, checks_11d.py, sections F, F2 and H. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

Referee note (GPT-6.1 Terra, 4 Nov 2026). Proposition 1.1 and the joint family's Weyl constant are independently checked; the unfolded circle is scalar, and even extensions of arbitrary smooth interval functions need not be smooth at the fixed points. See the referee report. Proposition 1.5 is amended at its statement below.

Referee note on Proposition 1.4 (GPT-6.1 Terra). The endpoint value, minimum and small-angle coefficient are independently checked under the stated one-loop hypotheses; see the report.

Correction (3 Nov 2026). There was an error in the first version of Proposition 1.4: it read the potential of lemma 24 as the whole energy of the walls, but that potential is taken relative to theta = 0 and omits the theta-independent Casimir energy. The corrected statement is below; the reading "the walls attract" is withdrawn. The first version of Proposal 1.3 also placed the W at the bottom of the tower, M_W ~ 1/(2R), along every path to v = 0; after the audit (its §2.2), that holds only on the scale-free path of Proposition 1.5.

Setting

Hypothesis H of STRATEGY.md reads the two sheets of the finite geometry as the minimal truncation of an interval I_l with opposite chiral boundary conditions, with l ~ 1/M_W. Its three regimes carry the labels 9, 10 and 11. These are KO labels. At the 11 endpoint (v -> 0, l -> infinity) the heat dimension is 4 + 1 = 5, since the finite factor F keeps heat dimension 0 for every v (canonical §2.1). To read 11 as a metric dimension, the KO-6 factor must itself be resolved into a six-dimensional space, which gives the count

```
11 = 4 (spacetime) + 1 (the interval) + 6 (a space resolving F).
```

The files of this folder take the six-dimensional space to be B_6 = CP^2 x S^2, the base of Witten's spaces (02), and study two ways of completing 4 + 6 to eleven dimensions (03). This file treats the "1", and the path along which v -> 0 opens it.

Proposition 1.1 (the chiral interval is the antiperiodic circle)

Statement. Let D_l act on H_l = L^2([0, l], C^2) as in lemma 05, eq. (8): D_l(L, R) = (R', -L'), with R(0) = 0 and L(l) = 0, grading Gamma = diag(1, -1) and the scalar functions C^infinity([0, l]). Let S_R be the circle of length 2l, that is of radius R = l/pi, with coordinate x in [-l, l] and antiperiodic functions u(x + 2l) = -u(x). The map

```
(W(L, R))(x) = (L(|x|) + i sgn(x) R(|x|)) / sqrt 2
```

is a unitary H_l -> L^2_antiperiodic(S_R) with

1. W D_l W^* = -i d/dx on its antiperiodic domain;
2. W Gamma W^* = P, the reflection (Pu)(x) = u(-x);
3. W f W^* = multiplication by f(|x|), so the algebra goes to the reflection-even functions;
4. the walls x = 0 and x = l go to the two fixed points x = 0 and x = l ~ -l of the reflection.

Hence spec D_l = {(k + 1/2)/R : k in Z}, each eigenvalue simple, which is (9) of lemma 05, and the Connes-Lott mass of the minimal truncation is m = pi/(2l) = 1/(2R).

Proof. The inverse map is L(x) = (u(x) + u(-x))/sqrt 2, R(x) = (u(x) - u(-x))/(i sqrt 2) for x in [0, l], so W is a bijection, and |u(x)|^2 + |u(-x)|^2 = |L(x)|^2 + |R(x)|^2 gives unitarity. For H^1 data, u is continuous at 0 exactly when R(0) = 0, and u(l) + u(-l) = sqrt 2 L(l), so the antiperiodic matching u(-l) = -u(l) is the condition L(l) = 0. On x > 0, -i (L + iR)' = R' + i(-L'), the image of (R', -L') = D_l(L, R). On x < 0, u(x) = (L(-x) - i R(-x))/sqrt 2 and -i u'(x) = (R'(-x) - i(-L'(-x)))/sqrt 2, which is again the image of (R', -L'). Gamma(L, R) = (L, -R) maps to L(|x|) - i sgn(x) R(|x|) = u(-x). Items 3 and 4 are immediate. The antiperiodic spectrum of -i d/dx on a circle of length 2l is pi(2k + 1)/(2l) = (k + 1/2)/R. ■

So the two sheets of H are the two fixed points of a reflection on a circle whose spinors are antiperiodic, and the grading of the finite geometry is that reflection. With finite data D_F supplied as in lemma 22, the parent unfolds to -i d/dx + D_F on the periodic circle of length 2l (lemma 25, Proposition 25.1), and the angle theta interpolates between that circle (theta = 0) and the antiperiodic circle of Proposition 1.1 (theta = pi/2): D_F is a constant connection, a Wilson line, and the wall angles theta_k = l mu_k (mu_k the eigenvalues of D_F) are its holonomy phases.

The joint family: weak masses below a descending tower

At fixed l, the vector masses of lemma 23 vanish as theta -> 0 and also as theta -> pi/2, while the tower spacing pi/l stays fixed; vanishing masses alone open no dimension (audit §2.2). The audit gives a family in which they do.

Proposition (Astra 6, audit §3; restated). Let D_F(epsilon) = epsilon D_* and l_epsilon = l_* epsilon^{-alpha}, 0 < alpha < 1, with wall data X = tan(l D_F) as in lemma 22. Then the exact window is W^* D W = epsilon D_, *the rest of the spectrum lies at |lambda| >= pi/l_epsilon - epsilon ||D_||*, the compression error is at most (2/3) l_^2 ||D_||^2 epsilon^{2(1-alpha)} ||a|| (Theorem 22.4, item 2), and for alpha < beta < 1 a cutoff E_epsilon = E_* epsilon^beta contains exactly the r = dim H_F finite states, while at every fixed E > 0 the number of states N_epsilon(E) grows like l_epsilon E, which is proportional to epsilon^{-alpha}.

In the circle language of lemma 25 (the periodic circle, on which D_F is a Wilson line; the window is its n = 0 mode) this reads: the radius R_epsilon = l_epsilon/pi goes to infinity, and the Wilson-line angles theta_epsilon = l_epsilon epsilon mu_k go to zero like epsilon^{1-alpha}. The hierarchy is

```
m_weak ~ epsilon  <<  E_epsilon ~ epsilon^beta  <<  1/R_epsilon ~ epsilon^alpha  ->  0.
```

In the five-dimensional Yang-Mills toy of lemma 23 the vector poles along this family are M_W^2 = 2 m^2 (1 + O(theta^2)) and M_Z^2 = (16/5) m^2 (1 + O(theta^2)), so M_W^2/M_Z^2 -> 5/8 (audit §3.1, from Theorem 23.6).

Check (Claude). The parent space of lemma 22 is L^2([0, l]; C^2) ⊗ H_F, of fibre dimension 2r. By Theorem 22.1 its signed spectrum is (n pi +- theta_k)/l, n in Z, two progressions of spacing pi/l for each of the r/2

singular values, so N(E) = 2 r l E/pi + O(r), the constant of the audit (the scalar circle of Proposition 1.1 gives 2 E l/pi; with fibre H_F the count is 2 r l E/pi). The window, gap and compression statements follow from Theorems 22.1 and 22.4 as stated. (Python, section H: exact spectra for random singular values.)

The audit also gives a localized family (its §4): a bulk mass mu_epsilon = mu_* epsilon^alpha and mu l = (1 - alpha) log(1/epsilon) + c_0, with light mass m ~ 2 mu_* e^{-c_0} epsilon, gap ~ mu_* epsilon^alpha, and the exact tail identity e^{-2 kappa l} = (m/(mu + kappa))^2. Its table of roots is reproduced in section H (Python, bisection on tanh s = s/c and tan y = y/c): the four light masses and next masses agree with the audit to the digits printed there.

Referee note (GPT-6.1 Terra, 4 Nov 2026). The localized asymptotics and tail identity are independently checked analytically; the length includes a logarithmic factor. The numerical table was not recomputed. See the referee report.

Proposal 1.3 (the circle of H as the M-theory circle)

Take the circle of Proposition 1.1 to be the eleventh direction of M-theory over a ten-dimensional type IIA background, so R = g_s l_s. Then, along the joint family:

H	circle of Proposition 1.1	M-theory / string side (recalled, to check)
v -> 0 along the joint family, label 11	R -> infinity, Wilson-line angles -> 0, m_weak R -> 0	strong coupling: D0-brane bound states of mass n/(g_s l_s) become light and are the momentum modes of the eleventh dimension (Witten 1995); the weak masses lie below this tower
v finite, label 10, two sheets	R finite, fixed points of a reflection	M-theory on an interval with two walls (Hořava-Witten); here with walls of opposite chirality
v -> infinity, label 9, sheets merge	R -> 0, antiperiodic spinors	a circle with antiperiodic fermions shrinking: bulk fermions get masses >= 1/(2R); compare type 0 strings, which have no bulk spacetime fermions (Bergman-Gaberdiel 1999), and the walls of opposite chirality of Fabinger-Hořava 2000

Each row matches a statement already in the corpus: the 11 endpoint needs the tower (canonical §2.1, lemma 05 Part 4), and at the 9 endpoint the bare interval keeps no light fermion, so light fermions must come from another sector (STRATEGY, T1 and T3). On the string side the same happens: the light fermions of a type 0 theory come from open strings on branes. The proposal therefore places the light fermions of the 9 endpoint on branes or walls, and 03 develops a brane origin for them.

Tests.

1. Along the joint family m_weak/M_KK ~ epsilon^{1 - alpha} -> 0, so the weak scale is far below 1/(g_s l_s); the scale-free path of Proposition 1.5, with M_W R fixed, is the other possibility and keeps no finite window.
2. Gauge couplings. A five-dimensional gauge field with constant profile on the interval has g_4^{-2} = l/g_5^2 = pi R/g_5^2 (audit §7.2 for the general volume formula), so at fixed g_5 the four-dimensional couplings vanish as R -> infinity, and M_W = g v/2 could vanish through g. The limit sought keeps g finite, which needs g_5^2 ~ R, and this must be shown for the vectors in question before the masses are read as restoration of the symmetry.

3. The path must be selected by an action: Proposition 1.5 shows what the flat interval alone selects.

Proposition 1.4 (the angle-dependent one-loop term at the vacuum angle)

Statement. Under the hypotheses of Theorem 9 of lemma 24 (flat interval, zero bulk masses, frequency-one fermions, the stated gauge towers and multiplicities, no wall potential), the angle-dependent part of the one-loop potential, taken relative to theta = 0 as in lemma 23, has its minimum at theta = pi/2 with value

```
V(pi/2) - V(0) = - 93 zeta(5) n_f / (512 pi^2 l^4),
```

and near theta = 0 it is (3 zeta(3)/(16 pi^2)) (54/5 - n_f) theta^2/l^4 + o(theta^2). At theta = pi/2 the exact vector poles of lemma 23 give M_W = M_Z = 0, with the fermion at the tower scale m = pi/(2l).

Proof. In the notation of lemma 24, V = $\mathcal{V}$/(8 pi^2 l^4), with F(0) = 0, and $\mathcal{V}$(1) = -n_f F(1) since the gauge terms vanish at t = 1. With 1 + 1/sinh^2 x = coth^2 x and log coth x = 2 sum over odd k of e^{-2kx}/k,

```
F(1) = 2 int_0^infinity x^3 log coth x dx = 4 sum_{k odd} (1/k) 3!/(2k)^4 = (3/2) sum_{k odd}
↪  k^{-5} = (3/2)(31/32) zeta(5) = 93 zeta(5)/64.
```

Near t = 0, $\mathcal{V}$'(0) = (6 + 24/5 - n_f) J(0) with J(0) = int x^3/sinh^2 x dx = (3/2) zeta(3), the coefficient already given in lemma 23. Python: F(1) = 1.506786, J(0) = 1.803085, series against quadrature to 10^-6. ■

Reading. The θ-independent Casimir energy of the towers, a_0/l^4, is not part of this potential and is not computed in lemmas 23 or 24; the direction in which l moves depends on the sign of a_0 - 93 zeta(5) n_f/(512 pi^2). The minimum over the angle is the point with massless W and Z at fixed l, which opens no dimension (lemma 23 already notes that it is not the physical point).

Proposition 1.5 (a scale-free potential with an isolated stationary angle fixes M_W l)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The derivative identity is correct, and an isolated stationary angle gives $l \propto 1/m$. The universal conclusion has a flat-direction counterexample $G(\theta) = \theta^4$; for the displayed one-loop determinant the small-angle obstruction is recovered when $n_f \neq 54/5$. See the referee report.

Statement. Let V(l, theta) = l^{-4} G(theta) with G of class C^1, the form of every one-loop potential of a flat interval whose only scales are l and the wall angles (lemmas 23 and 24, with a_0 included in G).

1. If the angle is free, V is monotone in l at every fixed theta with G(theta) != 0. At the minimizing angle theta_min the length runs away: to l -> infinity if G(theta_min) > 0, to l -> 0 if G(theta_min) < 0. Stationary points exist only on the set G = G' = 0.
2. If the angle is tied to a fixed mass, theta = l m (as for X = tan(l M) in lemma 22), the stationary lengths satisfy theta G'(theta) = 4 G(theta) at theta = l m. So l = theta_*/m for a root theta_* that does not depend on m, and M_W l, M_Z l (functions of theta alone by Theorem 23.6) are the same at every m.

Hence, when the roots of theta G' - 4 G are isolated, a stationary branch has l = theta_*/m: the path l ~ 1/M_W of H's original statement, alpha = 1 in the scaling of the joint family (Remark 10.7 of 10 for a small fixed ratio). A root interval, as for G = theta^4 near 0, makes every alpha < 1 stationary and selects none (referee amendment). The joint family with alpha < 1 needs a second scale in the action: a bulk mass (the localized family, G(l m, l mu)), the cutoff of the spectral action (lemma 25, G(theta, l Lambda)), curvature, flux or a wall term.

Proof. d/dl [l^{-4} G(l m)] = l^{-5} (theta G'(theta) - 4 G(theta)) at theta = l m. For item 1, d/dl V = -4 l^{-5} G(theta) and d/dtheta V = l^{-4} G'(theta). ■

Amendment (4 Nov 2026). The window can separate on the scale-free path, with a fixed small ratio, if the light mass is exponentially small in a stabilized shape parameter, and a_0 is now computed. See 10, Remark 10.7.

Consequence for the modulus target of the audit (§7.1). The audit proposes V = V_*(a r^{-p} + b epsilon^2 r^q) with a, b, p, q > 0, whose minimum r ~ epsilon^{-2/(p+q)} gives alpha = 2/(p + q) < 1 when p + q > 2. In the flat one-loop family the epsilon-dependent term near theta = 0 is (3 zeta(3)/(16 pi^2)) (54/5 - n_f) m^2/l^2, so q = -2, and with the Casimir term p = 4 the stationary point has l ~ 1/m, alpha = 1, in agreement with item 2. A term growing with l at fixed v, as the ansatz requires, is absent from the flat family and must come from one of the second scales listed above.

What remains open here

- Whether the identification of Proposal 1.3 survives the gauge sector: lemmas 13, 14 and 25 tie M_W to the tower with model-dependent coefficients, which here become a relation between M_W, g_s and l_s, and test 2 above requires g_5^2 ~ R.
- The theta-independent Casimir coefficient a_0 of the flat family, which decides the runaway direction of Proposition 1.5, item 1.
- A potential with a second scale that selects alpha < 1: the localized family of the audit is the first candidate, with G(l m, l mu) computable from the exact secular equations of lemma 09.
- A fermion with both a fibre momentum and a base twist: on a nontrivial circle bundle (Witten spaces, file 02) the antiperiodic shift corresponds to a spin structure whose fibre circle does not lift the circle action. Which Witten spaces carry such a structure is an open item for the next job.

Part 11D, 02 witten spaces hypercharge quotient, referee pending

Witten spaces as the hypercharge quotient of the five-dimensional representation

Part of the 11D line of research (README). Theorem 2.1 and Propositions 2.2 to 2.4 and 2.6 are proved here, Theorem 2.1 and Proposition 2.6 conditionally as the referee notes say; Proposition 2.5 is a negative result in a stated scope, refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5. Attempted by Claude (Anthropic), 3 Nov 2026, with a remark from the steer of the same day on the mixing of u and d. Checks: checks_11d.py, sections A to D, Python with exact fractions. Referee pending.

Referee note (GPT-6.1 Terra, 4 Nov 2026). The exclusion in Proposition 2.5 has an independent analytic proof. The required-bidegrees lemma and the general assertion that u^c, d^c must have the same flux image require a common ten-dimensional chirality; the exclusion itself also holds without that extra assumption. See the referee report; other reviewed items are noted separately.

Referee note on Theorem 2.1 and Propositions 2.2, 2.4 (GPT-6.1 Terra). The quotient and tensor-centre calculations are independently checked. The cited M^{111} identification is conditional on the recalled fibration; the original quotient metric needs rescaling to be Sasaki-Einstein, and the action on holomorphic functions is dual to the action on U. See the report.

Referee note on Proposition 2.3 (GPT-6.1 Terra). The Gysin, Smith normal form and spin calculations are independently checked; the cited Kreck-Stolz paper was unavailable. See the report.

Setting

Witten's spaces M^{pqr} = G/H, with G = SU(3) x SU(2) x U(1) and H = SU(2) x U(1) x U(1), are the seven-dimensional spaces of least dimension with the isometry group of the Standard Model, so 4 + 7 = 11 (Witten 1981; LITERATURE.md §F). Castellani, D'Auria and Fré gave them Einstein metrics; Killing spinors exist only for p = q, where the AdS4 vacuum has N = 2 and the space is M^{111} (Fabbri, Fré, Gualtieri, Termonia, hep-th/9903036). Fabbri et al. (hep-th/9907219, §4) write the cone over M^{111} as the Kähler quotient of C^5 by a circle acting with weights 2 on a triplet U and -3 on a doublet V, with D-term 2|U|^2 = 3|V|^2, and M^{111} as a circle bundle over P^2 x P^1. In later notation it is M^{3,2}.

These weights are six times the hypercharges of the 5-bar of SU(5). The rest of the file develops that observation.

Theorem 2.1 (M^{111} is the hypercharge quotient of the 5-bar)

Statement. Let 5bar = C^3 + C^2 carry the Standard Model action on the left-handed fields d^c in (3bar, 1)_{1/3} and L in (1, 2)_{-1/2}; write U for d^c and V for L, so 6Y = 2 on U and -3 on V. Let mu_Y = (2|U|^2 - 3|V|^2)/6 be the moment map of hypercharge, and

```
Y_{3,2} = (mu_Y^{-1}(0) ∩ S^9) / U(1)_Y.
```

1. Y_{3,2} is a smooth, simply connected seven-manifold, the circle bundle of O(3, 2) over CP^2 x CP^1 (up to orientation), and its cone is the Kähler quotient C^5 // U(1)_Y of Fabbri et al.
2. The cone is Calabi-Yau (Gorenstein, sum of weights 0) because tr Y = 0 on the 5-bar, that is, because U(1)_Y lies in SU(5).
3. The group acting effectively on Y_{3,2} through the quotient is (U(3) x U(2))/U(1)_Y. Its subgroup S(U(3) x U(2))/U(1)_Y is PSU(3) x SO(3), and the overall phase of U(5) supplies one further circle, the Reeb (R-symmetry) direction.
4. The functions on Y_{3,2} that are polynomial in U, V and their conjugates have U(1)_Y-charge zero; the holomorphic ones are spanned by the monomials of degree (3k, 2k) in (U, V), whose span as tensors is Sym^{3k}(3bar) x Sym^{2k}(2) and as functions its dual Sym^{3k}(3) x Sym^{2k}(2): for k = 1 the functions form the (10, 3) of Fabbri et al.

Proof. On mu_Y^{-1}(0) ∩ S^9 we have |U|^2 = 3/5 and |V|^2 = 2/5, a product of round spheres S^5 x S^3. The circle acts by (lambda^2 U, lambda^{-3} V); a fixed point would need lambda^2 = lambda^3 = 1, so lambda = 1 and the action is free. The quotient of S^5 x S^3 by the full torus T^2 of phases of U and V is CP^2 x CP^1, and Y_{3,2} -> CP^2 x CP^1 is a principal bundle for T^2/U(1)_Y. A character lambda_U^m lambda_V^n of T^2 is trivial on U(1)_Y exactly when 2m - 3n = 0, so the characters of T^2/U(1)_Y are generated by (m, n) = (3, 2), whose line bundle is O(3) ⊠ O(2) up to the sign convention; this gives item 1 and item 4. Simple connectivity follows from the exact homotopy sequence of U(1) -> S^5 x S^3 -> Y_{3,2}. Item 2: the sum of weights is 3 * 2 + 2 * (-3) = 0. Item 3: an element (g, h) of U(3) x U(2) acts trivially on the quotient exactly when (g, h) = (lambda^2, lambda^{-3}), because mu_Y^{-1}(0) spans C^5. The intersection of SU(3) x SU(2) with U(1)_Y consists of lambda^6 = 1, generated by (e^{2 pi i/3} I_3, -I_2), the centre Z_3 x Z_2. ■

So the isometry group of the Witten space with Killing spinors, SU(3) x SU(2) x U(1) (Fabbri et al.), is the centralizer of hypercharge in U(5), divided by hypercharge. The SU(3) x SU(2) part is the Standard Model one; the remaining circle is the R-symmetry, and hypercharge itself has been quotiented. Witten assigned the U(1) of G to hypercharge; in this realization hypercharge is a different U(1) (03, Proposal 3.1).

Proposition 2.2 (uniqueness of the count 7)

Statement. (a) Among the quotients Y_{a,b} = (S^5 x S^3)/U(1), with weights b on C^3 and -a on C^2 and gcd(a, b) = 1, a, b > 0, the cone is Calabi-Yau only for (a, b) = (3, 2). (b) For a complex representation of dimension n and a free circle action, (S^{2n-1} ∩ mu^{-1}(0))/U(1) has dimension 2n - 3; this is 7 only for n = 5. Among representations of SU(5), with hypercharge as the circle, only the 5 and the 5-bar give a seven-dimensional quotient.

Proof. (a) The Calabi-Yau condition is 3b - 2a = 0. (b) Dimension count: 2n - 1 for the sphere, minus one for the moment map, minus one for the circle. The irreducible representations of SU(5) of dimension 5 are the 5 and the 5-bar. ■

Witten's count reads: 11 = 4 + (2 * 5 - 3). The seven internal dimensions come from the smallest representation of SU(5), with hypercharge as the circle. For the 10 the quotient has dimension 17, and for the 5 + 10 it has 27.

Proposition 2.3 (topology of Y_{a,b})

Statement. Let Y_{a,b} -> B = CP^2 x CP^1 be the circle bundle with Euler class a h_1 + b h_2, a != 0. Then H^2(Y) = Z + Z/gcd(a, b), b_3(Y) = 0, H^4(Y) is a group of order a^2, cyclic when gcd(a, b) = 1, and Y is spin exactly when a is odd and b is even. For (a, b) = (3, 2): H^2 = Z, H^4 = Z/9, spin. The

pullback H^2(B) = Z^2 -> H^2(Y) = Z is (x, y) -> 2x - 3y up to sign: the two divisor classes h_1 = [{l(U) = 0}] and h_2 = [{m(V) = 0}] go to 2 and -3, the values of 6Y on d^c and L.

Proof. The Gysin sequence H^{k-2}(B) -> H^k(B) -> H^k(Y) -> H^{k-1}(B) -> H^{k+1}(B), with the first map the cup product with c = a h_1 + b h_2 and H^*(B) = Z[h_1, h_2]/(h_1^3, h_2^2). In degree 2 the cokernel of 1 -> (a, b) is Z^2/(a, b). In degree 4 the map H^2(B) -> H^4(B) has matrix [[a, 0], [b, a]] in the bases (h_1, h_2), (h_1^2, h_1 h_2), with determinant a^2 and Smith invariants (gcd(a, b), a^2/gcd(a, b)); since H^3(B) = 0 this gives H^4(Y), and the same matrix is injective, so H^3(Y) = 0. For spin: w_2(B) = h_1 mod 2, and the kernel of H^2(B; Z/2) -> H^2(Y; Z/2) is spanned by c mod 2; so w_2(Y) = 0 exactly when (1, 0) = (a, b) mod 2. For (3, 2), Z^2/(3, 2) -> Z by (x, y) -> 2x - 3y is an isomorphism. ■ (Python: Smith invariants and the spin condition, section C.)

The product member CP^2 x S^2 x S^1 of Witten's family (a = b = 0) is not spin.

Proposition 2.4 (the Z_6 of the Standard Model)

Statement. (a) Every monomial in U and V, and every state of the exterior algebra Λ^*(5bar), satisfies 2t + 3d + 6Y = 0 mod 6, with t the triality (1 for 3, -1 for 3bar) and d = 2j mod 2. (b) The U(1)_Y-neutral monomials have t = 0 and d = 0. (c) Every tensor field on Y_{3,2} carries a representation of (U(3) x U(2))/U(1)_Y, on which SU(3) x SU(2) acts through PSU(3) x SO(3). So no tensor Kaluza-Klein mode on Y_{3,2}, on a background invariant under the displayed action, is a colour triplet or a weak doublet; spinors are open.

Proof. (a) holds for U (t = -1, d = 0, 6Y = 2) and V (t = 0, d = 1, 6Y = -3) and is additive. (b) U^alpha V^beta is neutral when 2 alpha = 3 beta, so 3 | alpha and 2 | beta. (c) The effective group of Theorem 2.1 acts on the frame bundle and on every tensor bundle, and its intersection with SU(3) x SU(2) is the centre Z_6 (Theorem 2.1, item 3). ■

(a) is the statement that the Standard Model group is (SU(3) x SU(2) x U(1))/Z_6 = S(U(3) x U(2)). In the quotient realization the Z_6 is the kernel of SU(3) x SU(2) -> Isom(Y_{3,2}). States with fractional hypercharge are those charged under the quotient circle, and they cannot be functions or tensors on Y_{3,2}. For spinors the isometry group may lift to a double cover; whether fermionic Kaluza-Klein modes on Y_{3,2} can carry half-integer weak isospin or triality is open (next job, item 2 of the README).

Chiral zero modes on the base with abelian fluxes

Witten's argument makes Kaluza-Klein fermions on an odd-dimensional space vectorlike, for gauge fields acting by isometries. The even base B_6 = CP^2 x CP^1 escapes it: abelian fluxes on B_6 give chiral four-dimensional fermions through the index of the twisted Dirac operator. This section asks which fluxes give one generation.

Setting. Ten-dimensional Weyl fermions on M^4 x CP^2 x CP^1, each with definite charges under some U(1) factors with fluxes on B_6. Since CP^2 is not spin, a fermion is a spin^c spinor, Λ^{0,*} ⊗ K^{1/2} ⊗ N with N the flux line bundle of its charges, and the four-dimensional chiral content is the virtual representation H^*(O(alpha, beta)) with (alpha, beta) = (-3/2, -1) + pairing of the charges with the fluxes. The cohomology is H^0(O(alpha)) = Sym^alpha(3bar) for alpha >= 0, H^2(O(alpha)) = Sym^{-alpha-3}(3) for alpha <= -3, and zero otherwise on CP^2; on CP^1, H^0(O(beta)) = Sym^beta(2) for beta >= 0 and H^1(O(beta)) = Sym^{-beta-2}(2) for beta <= -2. Each Standard Model Weyl field must come from one ten-dimensional field, from the whole cohomology of one line bundle.

Lemma (required bidegrees). Writing P_s = (alpha + 3/2, beta + 1) for the pairing of the charges of field s (up to the common sign of charge and chirality):

field	P_s	note
Q (3, 2)	(-5/2, 2)	H^2(O(-4)) x H^0(O(1))
u^c, d^c (3bar, 1)	(5/2, 1)	H^0(O(1)) x H^0(O(0))

field	P_s	note
L (1, 2)	(+-3/2, 2)	
e^c, nu^c (1, 1)	(+-3/2, 1)	

The second entry is 2j + 1 and the first is +-5/2 on colour triplets and +-3/2 on singlets. (Python, section D, by enumeration of the line bundles with these cohomologies and the Riemann-Roch check.)

Proposition 2.5 (no generation from fluxes along hypercharge and B - L) *Scope note (4 Nov 2026, after the referee).* The statement on equal images and the linearity of 2j + 1 assume a common ten-dimensional chirality for the fields. The conclusion holds without it: in the span of 6Y and 3(B - L) the charges of L are -3 times those of Q, so their second pairings are P_2(L) = -3 P_2(Q), against |P_2| = 2 on both doublets.

Referee note on the third-charge search (GPT-6.1 Terra, 4 Nov 2026). The bounded search below has an independent analytic proof. The same argument extends to any real third charge and independent source chiralities; this stronger extension was checked by Claude Sonnet 5.5 (second check: the vectors y, b, p, w have rank 4 for every sign pattern), so it stands as a no-go in the same scope with two model identifiers. See the referee report.

Statement. In the setting above, with the canonical spin^c root and fluxes only in U(1) directions spanned by Y and B - L (any flux quanta), no choice reproduces one generation, with or without nu^c. More generally, the pairing must give u^c and d^c the same image, and 2j + 1 = (2, 1, 1, 2, 1, 1) on (Q, u^c, d^c, L, e^c, nu^c) must be a linear function of the fluxed charges.

Proof. The pairing M is linear in the charges, and M c_s = P_s with c_s the charges of the left-handed field s. With only Y and B - L fluxed, c_u - c_d = (-6, 0) in the coordinates (6Y, 3(B - L)), so M vanishes on the hypercharge direction and P_s = 3(B - L)_s m for one vector m. The second entries 2 on Q and 1 on u^c would then need 3(B - L) to take values in the ratio 2 : 1 on (Q, u^c), against 1 : -1. In general, 2j + 1 is not in the span of 6Y = (1, -4, 2, -3, 6, 0) and 3(B - L) = (1, -1, -1, -3, 3, 3): from Q and u^c the coefficients are -1 and 3, and then d^c gives -5 against 1. ■ An exhaustive search adding a third integer charge z in [-3, 3]^6 to Y and B - L finds no solution either (Python, section D).

Scope and escapes. Abelian fluxes on CP^2 x CP^1 with the Fubini-Study complex structure; one ten-dimensional Weyl field per Standard Model field; canonical spin^c root; Dolbeault index as the chiral content. Known escapes: nonabelian bundles (an SU(2)_R doublet bundle), several ten-dimensional fields per four-dimensional field, a changed common twist, orbifolds and walls, and non-Kaluza-Klein (brane) states. The result points to the next proposition.

Proposition 2.6 (one generation from a universal charge, the grading and B - L) **Referee note (GPT-6.1 Terra, 4 Nov 2026).** The index calculation works with the stated cohomology convention. For the colour action of the earlier quotient, reverse the first-component coefficients to obtain the labelled colour representations; global flux quantization remains to be specified. See the referee report.

Statement. Let the fluxed charges be F (one on every field: a change of the common spin^c twist), chi (+1 on Q and L, -1 on u^c, d^c, e^c, nu^c) and B - L. Then one generation with nu^c is the exact index content for the pairing

```
P^{(1)} = c_0 chi + c_1 (B - L),   P^{(2)} = (3F + chi)/2,
```

with (c_0, c_1) = (-9/4, -3/4) or (-3/2, -3). The bidegrees are Q: (-4, 1); u^c, d^c: (1, 0); L: (-3, 1) or (0, 1); e^c, nu^c: (0, 0) or (-3, 0). Without chi, or without B - L, there is no solution.

Proof. Substitution, and the two failures by the span test (Python, section D2). ■

Reading.

1. u^c and d^c see the same line bundle O(1, 0), and so do e^c and nu^c. The flux treats them as SU(2)_R doublets with a common internal profile; hypercharge Y = T_3R + (B - L)/2 enters through T_3R, which carries no flux. This is the left-right pattern of Pati-Salam, produced here by the index.
2. chi is the finite grading gamma_F on the particle basis. The flux therefore couples to the grading, an axial abelian connection on B_6; in the language of the finite geometry this is a fluctuation proportional to gamma_F, the structure of the two gradings of lemma 23.
3. **Mass and charge bases.** Because u^c and d^c share one zero-mode profile, overlap integrals on B_6 cannot separate them; their masses come from data that break SU(2)_R (Higgs or wall couplings), and the basis that diagonalizes the masses is set by those data, independently of the basis in which the charges are diagonal. The steer of 3 Nov 2026 remarks that u and d should still mix, since mass eigenvalues are not charge eigenvalues, and that at the 9 label, with the left-handed interactions off, charge eigenvalues can coincide. Proposition 2.6 puts the coincidence in the flux charges: equal flux charges for u^c and d^c, unequal hypercharges through T_3R. At the 9 label the electroweak part of the algebra is removed (lemma 21, STRATEGY T2), and what distinguishes u^c from d^c in the light theory is then the vectorlike charge alone. Making this precise needs the SU(2)_R-breaking wall data and the 9D reduction of this index; listed as item 4 of the README.

What this file establishes, and the direction

Referee note (GPT-6.1 Terra, 4 Nov 2026). The M^{111} identification is conditional on the recalled fibration and metric normalization. The flux examples establish a particular construction and scoped obstructions; a general necessity for the grading flux remains open. See the wording audit.

The Witten space with Killing spinors is the hypercharge quotient of the 5-bar (Theorem 2.1), the count 7 is fixed by SU(5) (Proposition 2.2), and the Z_6 of the Standard Model is the kernel of the isometry action (Proposition 2.4). The price is that hypercharge is no longer an isometry, and no tensor Kaluza-Klein mode on an invariant background carries triality, weak isospin 1/2 or fractional hypercharge. Kaluza-Klein chirality on the base needs fluxes that see the grading and leave T_3R free (Propositions 2.5 and 2.6). Both results point to states outside the Kaluza-Klein spectrum of Y_{3,2} for the quarks and leptons: 03 takes them to be wrapped branes.

Part 11D, 06 main point heterotic route, referee pending

9 and 11 as limits of the modern spectral triple: the target, and a route through the heterotic strings

Part of the 11D line of research (README, PLAN, task B). Attempted by Claude (Anthropic), 4 Nov 2026, prompted by two remarks in the steer of the same day: that SO(32) against E8 × E8 resembles non-chiral against chiral, and that the SO(32) of lemma 21 was mainly scalars, a way in which supersymmetry could be hidden in the finite geometry as scalar composites. Proposition 6.1 is proved; everything else is a proposal, and the string-theory facts used are recalled, not reread (marked). Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

1. The target

The modern spectral triple is the real even finite triple of KO-dimension six with order zero and order one, fermion doubling, dim H_F = 96 (three generations with ν_R), A_F = C + H + M_3(C), or its grand algebra M_2(H) + M_4(C), and D_F with Yukawa and Majorana blocks. The main point of the project asks for one family of spectral triples with towers, T_s, and three regimes, each described by the three records of canonical §1 (geometry, fermions, physics):

regime	geometry	fermions	physics
finite v (label 10)	towerless window at the weak scale	the modern finite triple, with J, gamma, D_F exact on the window	the Standard Model
v -> 0 (label 11)	eleven resolved dimensions along a joint family (01)	chirality on walls or defects	massless W, Z with finite couplings
v -> infinity (label 9)	the interval collapses	vectorlike; KO-5 or KO-7 by the sign s of lemma 06	colour and electromagnetism

What exists: the exact window for D_F, J and gamma on an interval parent (lemma 22) and its circle form (lemma 25, Proposition 1.1); the joint family that opens the tower while keeping the window (Astra audit, 01); the grand algebra from the seven-dimensional parent X, conditionally (lemma 28); the KO parity of the odd endpoint (lemma 06, T2c); eleven-dimensional vacua with the Standard Model isometry (02, 03); placements of H_F in E8 (04, 05). What is missing: one family doing all three; the six internal dimensions resolved in the same family; and an action selecting the path (Proposition 1.5).

2. The heterotic web has the shape 9, 10, 11

Referee note (GPT-6.1 Terra, 4 Nov 2026). The nine-dimensional duality involves specified Wilson lines and radii; the finite-triple dictionary and field identification remain proposals. The cited primary papers were unavailable to the referee; Polchinski-Witten and Bergman-Gaberdiel-Lifschytz were read later on master (10) and the second check confirmed the Wilson-line map, the winding duality and the slope rule against them. See the referee report.

Recalled facts (Polchinski and Witten, hep-th/9510169; Hořava and Witten, hep-th/9510209 and hep-th/9603142; Bergman, Gaberdiel and Lifschytz, hep-th/9711098; to be checked before use):

- **9:** the heterotic strings on a circle. SO(32) and E8 × E8 are T-dual there, joined through the moduli space of Wilson lines; the theory is non-chiral, nine being odd. Its type I' form is an interval with two orientifold walls (O8) and sixteen D8-branes, whose positions are the Wilson-line eigenvalues.
- **10:** the two heterotic strings, both chiral, distinct.
- **11:** M-theory on an interval (Hořava-Witten), with an E8 multiplet on each wall, chiral at the walls; the length of the interval grows with the string coupling.

This is the pattern of hypothesis H: vectorlike at 9, chiral and KO-6 at 10, chiral walls at 11, with an interval whose two ends are the two sheets. In this web the chiral/non-chiral split is a split between dimensions: both heterotic strings are chiral at 10, and at 9 both are the same non-chiral theory. The two groups differ by their descriptions: SO(32) has the open-string description (type I, type I') and no eleven-dimensional end, and E8 × E8 has the eleven-dimensional end (Hořava-Witten) and no perturbative open-string description. The steer's assignment of chirality to the groups therefore does not hold as stated, which was noted on 4 Nov 2026. The pair of supergravity limits of eleven dimensions, on a circle (type IIA, non-chiral) and on the interval S^1/Z_2 (Hořava-Witten, chiral), matches hypothesis H more closely than the heterotic pair, since by Proposition 1.1 of 01 the finite grading is the reflection of an antiperiodic circle, so the two cases differ by whether the reflection is quotiented; whether the Z_2 action C_3 -> −C_3 of Hořava-Witten has a counterpart on the finite data is open.

3. Dictionary (proposal)

corpus	heterotic web
the two sheets of the finite geometry	the two walls of the interval (O8 planes in type I', E8 walls in Hořava-Witten)

corpus	heterotic web
D_F as a constant connection on the circle (lemma 25)	a Wilson line on the heterotic circle, T-dual to D8 positions on the interval (the map from angles to positions is linear only at constant dilaton: 10, Proposition 10.1)
the chiral interval as the antiperiodic circle (Proposition 1.1)	the interval as the orbifold S^1/Z_2 of that circle
lemma 21: removing the top keeps 5/8 and 8/3 and leaves SO(30) x SO(2) ⊂ SO(32)	fifteen D8-branes at one wall and the sixteenth displaced: the gauge group on the D8-branes is SO(30) x U(1)
the 30 = (5, 3) + (5bar, 3bar): five light quarks in three colours	the Chan-Paton labels of the fifteen-brane stack, with images
the SO(30) adjoint, Λ^2(30), which contains the counts of squarks and sleptons (lemma 21)	open strings with both ends on the stack: bilinear in the Chan-Paton labels, so composite in exactly the steer's sense, and in the adjoint scalars of a theory with sixteen supercharges
the (30, 2) of SO(32) ⊃ SO(30) x SO(2), states that the top cannot form	strings stretched between the stack and the displaced brane, with masses set by the displacement
v -> 0 opens the eleventh dimension (Proposal 1.3)	strong coupling: the interval of Hořava-Witten grows

4. Proposition 6.1 (the top channel as an SO(32) Wilson line)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The momentum progressions agree; the SO(32) charged sector has 30 copies, and $m = a/R$ is the light mass on $0 \le a \le 1/2$. The field identification remains a proposal. See the referee report.

Statement. On a circle of radius R, take the SO(32) Wilson line W = 1_30 ⊕ R(2 pi a) (rotation by the angle 2 pi a in the last two directions). The adjoint 496 decomposes under SO(30) x SO(2) as (435, 1)_0 + (1, 1)_0 + (30, 2), where (30, 2) = (30)_{+1} + (30)_{-1}; the Kaluza-Klein masses are n/R on the first two pieces and |n ± a|/R on (30)_{±1}, n in Z. With l = pi R and theta = pi a this is the spectrum (n pi ± theta)/l of a channel of lemma 22 (Theorem 22.1) and of the circle form of lemma 25, with mass parameter m = theta/l = a/R; so the top channel of lemma 22 and the SO(32)-charged states of this Wilson line have the same free spectrum when m_t = a/R.

Proof. The Wilson line commutes with SO(30) x SO(2) and acts on the vector 2 of SO(2) by e^{±2 pi i a} in the complex basis; a field of charge q on the circle with holonomy e^{2 pi i q a} has momenta (n + q a)/R. Substitution of l = pi R and theta = pi a. ■

Prior work (search of 4 Nov 2026). The content of Proposition 6.1 is known. A field of charge q on a circle with holonomy e^{2 pi i q a} has momenta (n + q a)/R: the Hosotani mechanism (Y. Hosotani, Phys. Lett. B 126 (1983) 309), which also gives the fermion masses of gauge-Higgs unification, including top-driven models (for example arXiv:0806.0480). An SO(32) Wilson line with one nonzero eigenvalue breaks SO(32) to SO(30) x U(1) and is T-dual to fifteen D8-branes at one point and the sixteenth displaced (Bergman, Gaberdiel and Lifschytz, hep-th/9711098; Kaplunovsky, Sonnenschein, Theisen and Yankielowicz, hep-th/0108135; seen at abstract level). The equality with the channel spectrum of lemma 22 is the circle form of lemma 25. Not found: the assignment of the top to the displaced D8-brane and of the five light quarks in three colours to the Chan-Paton labels of the fifteen-brane stack. That assignment is the proposal of section 3, and it has a gap: in the heterotic string the (30, ±1) are gauge bosons and gauginos, while the top is a quark, so the two spectra belong to fields of different spin unless the composite reading of section 5 supplies the map.

This identifies spectra, which is all it does. The physical identification needs the fermion content (in the heterotic string the (30, 2) are vectors and gauginos), the chirality at 10, and the map from the Yukawa

block to the Wilson line beyond the top channel.

5. Supersymmetry hidden as composite scalars (the steer's second remark)

In the type I' form the adjoint scalars of the D8 stack are open strings with two Chan-Paton ends, so they are bilinear in the labels that the dictionary assigns to the five light quarks in three colours: the composites of lemma 21 (the 15 diquarks and 25 mesons of five flavours recorded there, and the SO(30) adjoint containing the squark and slepton counts). The theory on the D8-branes has sixteen supercharges, which pair these adjoint scalars with gauge fields and gauginos. A finite spectral triple in which the scalars of the inner fluctuations are such bilinears would carry the supersymmetry in hidden form. The test is to compute the inner fluctuations of a finite triple whose algebra is the Chan-Paton algebra of the stack, and to compare them with Λ^2(30).

6. What would make this a theorem, and the first steps

Referee note (GPT-6.1 Terra, 4 Nov 2026). The proposed dictionary needs a physical construction. Proposition 1.5 allows flat directions; a second scale is a candidate for selecting a joint path, rather than a universal requirement. See the wording audit.

1. **One family, three regimes.** Realize T_s as the type I' interval with D8-branes in the configuration 15 + 1, with the corpus parent (lemma 22) as its low-energy window: the window must reproduce the modern D_F, J and gamma (lemma 22 does this for any finite data), the tower must open along a joint family as the interval grows (strong coupling, 11; but the length of H and the type I' length are T-dual, 10, Proposition 10.3), and the collapse must give the non-chiral nine-dimensional theory (9).
2. **The algebra.** The Chan-Paton algebras of orientifold stacks are of the types O(n), Sp(n), U(n); the Connes algebra C + H + M_3(C) has the form of stacks of one brane, two branes with a symplectic projection, and three branes. Decide whether the modern algebra is the Chan-Paton algebra of a stack configuration on the interval (for the colour factor, lemma 21 puts colour inside SO(30), on the same stack as the light flavours).
3. **The six dimensions and chirality.** Ten-dimensional chirality and the KO-6 class must come from the internal six dimensions (02: CP^2 x S^2 with fluxes, or a Calabi-Yau with an SU(4) bundle as in Proposal 5.5).
4. **The path.** Proposition 1.5 requires a second scale when the stationary angle is isolated; in the heterotic web the dilaton gradient of type I' (whose slope depends on the D8 positions) was a candidate, excluded in the scope of 10, Propositions 10.5 and 10.6; Proposition 9.1 of 09 writes it as a trace of a function of |D_F|.

References for this file

Recalled, not reread: J. Polchinski and E. Witten, "Evidence for heterotic-type I string duality", hep-th/9510169; P. Hořava and E. Witten, hep-th/9510209 and hep-th/9603142; O. Bergman, M. Gaberdiel and G. Lifschytz, "String creation and heterotic-type I' duality", hep-th/9711098; L. Ibáñez, F. Marchesano and R. Rabadán, "Getting just the Standard Model at intersecting branes", hep-th/0105155, for stacks with symplectic projections.

Part 11D, 07 dualities 9 11, referee pending

Dualities between nine and eleven dimensions, adapted to the spectral triple

Part of the 11D line of research (README, PLAN, task C). Attempted by Claude (Anthropic), 4 Nov 2026. Propositions 7.1 to 7.3 are proved here by hand; the string-theory statements of section 1 are recalled, not reread; the readings are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

1. The dualities (recalled)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The duality descriptions are conditional on the recalled sources. The M2 wrapping number and the auxiliary torus flux are separately specified; a map identifying them remains open. See the referee report.

1. **M-theory on T^2 and type IIB on S^1** (Schwarz, hep-th/9508143; Aspinwall, hep-th/9508154). The modulus tau of the torus is the IIB coupling and SL(2, Z) is IIB S-duality; the IIB radius scales as a negative power of the area A of the torus. M2-branes wrapped on T^2 are IIB momentum states, and momenta on T^2 are (p, q)-strings wound on S^1. Eleven dimensions at A -> infinity, ten (IIB) at A -> 0, nine in between.
2. **Sixteen supercharges, rank 17** (Polchinski and Witten, hep-th/9510169; Hořava and Witten, hep-th/9510209). The two heterotic strings on S^1 are T-dual through Wilson lines; their type I' form is an interval with two orientifold planes O8 and sixteen D8-branes whose positions are the Wilson-line eigenvalues; at strong coupling, M-theory on an interval times a circle, with E8 walls.
3. **Sixteen supercharges, lower rank.** The CHL string (rank 9) and M-theory on the Möbius strip; a rank-1 component related to the Klein bottle (Aharony, Komargodski and Patir, hep-th/0702195); further components with discrete theta angles (Montero and Parra de Freitas, arXiv:2209.03361).
4. **Light-cone M-theory.** On a light-like circle with momentum N, M-theory is the U(N) quantum mechanics of nine Hermitian matrices (Banks, Fischler, Shenker and Susskind, hep-th/9610043; Susskind). On T^2 with a light-like component of C_3, the torus becomes noncommutative and SL(2, Z) acts by Morita equivalence (Connes, Douglas and Schwarz, hep-th/9711162).

2. What each one offers the finite geometry

duality	9	11	the finite triple
1	the nine-dimensional theory	large torus	the lowest Landau level of a torus with flux p: algebra M_p(C), J from charge conjugation with doubling (Proposition 7.3); towers = Landau levels; p = wrapped M2-branes = IIB momentum
2	type I' interval, or heterotic on S^1	Hořava-Witten on S^1	two sheets = two orientifold walls; D_F = Wilson lines = D8 positions (Propositions 6.1, 7.1, 7.2); A_F from Chan-Paton algebras; J from the orientifold
3	rank-reduced nine-dimensional theories	M-theory on non-orientable surfaces	J as an orientation reversal; the reflection with antiperiodicity of Proposition 1.1 is the Möbius-type ingredient

duality	9	11	the finite triple
4	nine transverse matrices, Spin(9)	large N	A_F as the commutant of block-diagonal matrices: stacks of 1, 2, 3 give C + M_2(C) + M_3(C); H_F in off-diagonal blocks; D_F = Γ^i ad(X^i), masses = separations

Duality 2 is the one with an exact meeting point with the corpus, through the angle of the top channel.

3. Proposition 7.1 (the angle of a channel lives on an interval)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The spectral quotient and endpoint multiplicities are independently checked. Theorem 22.1 uses the strict interior; the antiperiodic endpoint is a limit, with a degenerate window. Equality of one channel spectrum does not by itself identify full physical backgrounds. See the referee report.

Statement. In the parent of lemma 22, a channel with singular value σ of the Yukawa block and angle θ = l σ has the signed spectrum {(n pi + θ)/l, (n pi − θ)/l : n in Z} (Theorem 22.1). As a set, this depends only on the class of θ in R modulo the group generated by θ -> −θ and θ -> θ + pi, whose fundamental domain is [0, pi/2]. At the endpoints:

1. θ = 0: the spectrum is n pi/l with a zero mode of multiplicity two (periodic);
2. θ = pi/2: the spectrum is (n + 1/2) pi/l, each value twice (antiperiodic), two copies of the spectrum of the chiral interval of Proposition 1.1; it is the limit X -> infinity of the wall data X = tan(θ), where the lowest eigenspace is degenerate, and in lemma 23 the point of antipodal gradings, where M_W = M_Z = 0 at fixed l with the fermion at pi/(2l).

Proof. The set is invariant under θ -> −θ (it exchanges the two progressions) and θ -> θ + pi (it shifts n). The orbit of θ meets [0, pi/2] in one point. The endpoint values are substitutions; for θ -> pi/2 the wall conditions Q_R(l) = X U_R(l), U_L(l) = −X^* Q_L(l) tend to U_R(l) = Q_L(l) = 0. The statements at θ = pi/2 are those of lemma 23 (Theorem 23.6 and the reading after Proposition 23.7). ■

So the parameter space of one channel of the finite Dirac operator, for a given parent, is an interval with two distinguished ends, and lemma 22's restriction l ||D_F|| < pi/2 is the choice of the fundamental domain.

4. Proposition 7.2 (the far end is the displaced D8-brane on the other wall)

Referee note (GPT-6.1 Terra, 4 Nov 2026). At $a = 1/2$ the full $SO(32)$ centralizer is $S(O(30) \times O(2))$, whose identity component is $SO(30) \times SO(2)$. The stated one-loop minimum uses the hypotheses of lemma 24 and $n_f > 0$; the D8 dictionary remains a proposal. (Second check.) In Spin(32)/Z_2 the component of determinant -1 sends the lift of W_{1/2} to its negative, so the heterotic centralizer is connected; a -> a + 1 holds for SO(32) matrices, and the lift has period 2. See the referee report.

Statement. Put a = θ/pi in [0, 1/2]. By Proposition 6.1 the channel spectrum is that of the states (30)_{±1} of SO(32) on a circle of radius R = l/pi with the Wilson line W_a = 1_30 ⊕ R(2 pi a). The interval [0, 1/2] is the moduli interval of one Wilson-line eigenvalue modulo a -> −a and a -> a + 1; the unbroken group is SO(32) at a = 0 and SO(30) x SO(2) for 0 < a <= 1/2; at a = 1/2 the holonomy is −1 on the plane of SO(2), W_{1/2} = diag(1_30, −1_2), and the (30)_{±1} tower is (n + 1/2)/R, the antiperiodic tower of Proposition 1.1. The one-loop minimum of lemma 24, Theorem 9, θ = pi/2, is the point a = 1/2.

Proof. Proposition 6.1 and Proposition 7.1 with θ = pi a; the commutant of W_a in SO(32) for 0 < a < 1/2 is SO(30) x SO(2), and at a = 1/2 its identity component is still SO(30) x SO(2), since only two eigenvalues

are −1. Lemma 24, Theorem 9, item 1, places the minimum at t = sin^2 θ = 1. ■

Reading (proposal). In type I' the Wilson line W_a is fifteen D8-branes, with images, on one orientifold plane and the sixteenth at distance proportional to a from it (proportional only at constant dilaton; see 10, Proposition 10.1); a = 1/2 puts it on the other orientifold plane (recalled: hep-th/9711098, hep-th/0108135). The dictionary of file 06 then reads: the top angle is the position of the top's D8-brane on the type I' interval, the two ends of the fundamental domain are the two orientifold walls, and the one-loop vacuum of the flat family of lemmas 23 and 24 carries the top brane to the far wall, where the tower of the stack-brane strings is antiperiodic. In supersymmetric type I' the D8 positions are flat directions; the potential of lemma 24 belongs to the non-supersymmetric flat family, the setting in which a force on the brane is expected.

Under the strong-coupling duality of item 2 of section 1, the D8 position is a Wilson-line eigenvalue of the heterotic string on S^1, and in eleven dimensions an E8 holonomy around the circle on a Hořava-Witten wall. So the same datum, the top angle, is a brane position at 9 and a holonomy at 11.

5. Proposition 7.3 (a torus with flux, doubled, has the colour form)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The signs, kernel dimensions and generated matrix corner check out, with a spin-line twist for nontrivial spin structures. The displayed action is degenerate and nonunital on the doubled Hilbert space, so the claimed unital finite triple needs an extra scalar summand. See the referee report.

Statement. Let T^2 be a flat torus of area A with a spin structure, L a line bundle of degree p >= 1 with a connection of constant curvature, and Lbar = L^{-1} with the conjugate connection. On H = L^2(S ⊗ L) ⊕ L^2(S ⊗ Lbar) put D = D_L ⊕ D_Lbar, gamma = gamma_S ⊕ gamma_S, and J(ψ, χ) = (Cχ, −Cψ), where C is the charge conjugation of the spin structure (antiunitary, C^2 = −1, C D = D C, C gamma_S = −gamma_S C, mapping S ⊗ Lbar to S ⊗ L). Let C^infinity(T^2) act by multiplication on the first summand only. Then:

1. J^2 = 1, J D = D J and J gamma = −gamma J, the signs of KO-dimension six, and the algebra satisfies order zero;
2. ker D = H^0(L) ⊕ H^1(Lbar) ≅ C^p ⊕ conj(C^p), with gamma = +1 on the first summand and −1 on the second; the other eigenvalues are the Landau levels, of order (p/A)^{1/2}, which go to infinity as A -> 0 at fixed p;
3. the compressions of the functions to the kernel generate M_p(C) on C^p ⊕ 0, so the window is the finite triple (M_p(C), C^p ⊕ conj(C^p), 0, J, gamma), with KO-6 signs and order zero: M_p(C) acts on the first summand and, through J, by its opposite on the second.

Proof. (1) J^2(ψ, χ) = (C(−Cψ), −C(Cχ)) = (ψ, χ) since C^2 = −1; JD = DJ and J gamma = −gamma J follow from the properties of C; order zero holds because the algebra acts on the first summand and J b J^{-1} on the second. (2) On a torus the spinor bundle is Λ^{0,*} up to a flat factor, so ker D_L = H^0(L) ⊕ H^1(L) = C^p ⊕ 0 by Riemann-Roch in genus one, and ker D_Lbar = 0 ⊕ H^1(Lbar) ≅ conj(C^p) by Serre duality; the Landau levels of a constant field of total flux p on area A scale as (p/A)^{1/2}. (3) The compression of a character e_{m,n} to H^0(L) is a nonzero multiple of a clock-and-shift Weyl operator U^m V^n with UV = e^{2 pi i/p} VU (Berezin-Toeplitz quantization of the torus; recalled; in the theta-function basis the compression is a Gaussian factor times the Weyl word, checked for p = 2 to 5 in the second check), and these operators span M_p(C). ■

The construction is the torus analogue of lemma 16 (a doubled parent with restricted algebra and a J covering the flip; Example 16.9 for the twistor case), and its window at p = 3 has the form of the colour sector of the Standard Model triple: M_3(C) on the quarks and its opposite on the antiquarks. Berezin-Toeplitz quantization of the torus is known; the KO-6 doubling is the corpus device.

Reading for duality 1 (proposal). The area A -> 0 at fixed flux keeps the window and sends the tower to infinity, the towerless limit; A -> infinity brings the Landau levels down. In M-theory on T^2, flux and wrapping number of M2-branes on T^2 are IIB momentum states, so the dimension p of the window counts IIB momentum on the nine-dimensional circle, and the SL(2, Z) of the torus acts on the family by Morita

equivalence of the noncommutative tori of Connes, Douglas and Schwarz when a C_3 background is on. Whether p = 3 for colour has a reason in this picture is open.

6. A remark on duality 4 (known)

For U(N) matrices X^i (i = 1, ..., 9) that are block-diagonal, X^i = $\oplus a$ *x_a^i 1*{n_a} with distinct points x_a in R^9, the commutant of all X^i is $\oplus a$ *M*{n_a}(C), and the fermion operator Γ^i ad(X^i) on the adjoint, with the sixteen-component spinor of Spin(9), vanishes on the diagonal blocks and has eigenvalues $\pm|x_a - x_b|$, each with multiplicity 8 n_a n_b per ordered block (a, b), that is 16 n_a n_b per unordered pair. With stacks of sizes 1, 2, 3 the commutant is C + M_2(C) + M_3(C), the complex form of A_F, the off-diagonal blocks are bifundamentals as in the Krajewski diagrams of the finite geometry, and the masses are the distances between stacks; coincident stacks restore U(N), the analogue of massless W and Z, and large N is the eleven-dimensional limit. A quaternionic factor H needs an orientifold projection on the two-stack, and chirality needs more than the nine Hermitian matrices.

6b. The three labels in type I' variables (proposal)

Correction (4 Nov 2026). The length l of H and the type I' length are T-dual, and the limit at label 11 below ends at ten-dimensional SO(32). See 10, Proposition 10.3.

Hypothesis H has two parameters for the top channel: the interval length l and the angle θ = l m_t in [0, pi/2] (Proposition 7.1). In the type I' reading of Proposition 7.2 they are the length of the interval and the position of the top's D8-brane as a fraction of it. Then:

- **label 9:** θ = pi/2 at fixed l, the brane on the far orientifold wall: the top at the tower scale pi/(2l) (lemma 22, Corollary 22.9: W, Z and top at the tower scale in the 10 -> 9 family), the antiperiodic tower, and the one-loop minimum of the flat family (lemma 24);
- **label 10:** 0 < θ < pi/2, the brane between the walls, a window below the tower;
- **label 11:** θ -> 0 together with l -> infinity, the joint family of 01 (θ_epsilon ~ epsilon^{1 − alpha}): the brane returns to the stack, SO(32) is restored on that channel, and the interval opens.

In this reading the 9 and 11 endpoints are the two ends of the brane's path, the far wall and the stack, and the second scale required by Proposition 1.5 must fix how l grows as the brane returns. The flat one-loop potential pushes the brane to the far wall (label 9); reaching label 11 needs a term that pulls it back to the stack while the interval grows.

7. What this adds, and what remains open

Referee note (GPT-6.1 Terra, 4 Nov 2026). The brane and E_8-holonomy readings below are proposals. Proposition 7.3's displayed algebra action is nonunital; the corrected finite datum is in the referee report. Proposition 1.5 also permits flat directions.

Added: the angle of the top channel lives on an interval whose ends are the two orientifold walls of type I' (7.1, 7.2); the corpus's one-loop vacuum puts the top brane on the far wall, the SO(32) Wilson line diag(1_30, −1_2); the same datum is an E8 holonomy in the eleven-dimensional frame; a torus with flux gives the colour form of the finite triple with KO-6 signs (7.3).

Open: - A nine-dimensional spectral triple with sixteen D8 positions as the eigenvalues of D_F on the type I' interval, and its strong-coupling limit with E8 walls; the configuration 15 + 1 lies at the SO(32) end, while E8 needs seven D8-branes at an orientifold plane with divergent coupling. - The quaternionic factor H from a torus with even flux and a real structure of quaternionic type, and the full A_F from several tori or from a rank-three bundle with flux. - The lower-rank nine-dimensional components (duality 3), whose J-like orientation reversal may supply the real structure of the finite triple.

Part 11D, 08 orientifold sign, referee pending

The sign of lemma 06 as an orientifold type, and E8 walls from the configuration 15 + 1

Part of the 11D line of research (README, PLAN, task C, next step of 07). Attempted by Claude (Anthropic), 4 Nov 2026. Proposition 8.1 is proved (it adds a commutant statement to lemma 06); Proposition 8.2 is proved from Proposition 7.1; Remark 8.3 rests on recalled string-theory rules and on the dictionary of 06; readings are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

1. Setting

In type I' the two ends of the interval are orientifold planes. A stack of n Chan-Paton labels (k D8-branes with their images, n = 2k) on an orientifold plane of type O^- carries SO(n), and on one of type O^+ carries USp(n) (recalled; the physical O8 planes of the SO(32) web are O8^-). Lemma 06 factors a KO-6 finite triple into an odd one through a grading-reversing unitary involution U, commuting with the algebra and with D, with U J = s J U; s = −1 gives KO-5 (label 9) and s = +1 gives KO-7 (label 11), and the antiunitary j induced on H_+ satisfies j^2 = s and commutes with d^2 = D^2|{$H+$} (lemma 06, proof of Result 2 and the check of the electrodynamics data). STRATEGY T2c: with a realistic seesaw sector only s = +1 occurs.

2. Proposition 8.1 (the factorization sign fixes orthogonal or symplectic symmetry)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The antiunitary commutants are independently checked. They need not preserve d or the remaining data, and the Chan-Paton identification is a proposal. See the referee report.

Statement. Let E be an eigenspace of d^2 in H_+, of dimension n. The unitaries of E that commute with j form O(n) if s = +1, and USp(n) if s = −1, in which case n is even.

Proof. j preserves E because it commutes with d^2. An antiunitary with j^2 = +1 is a real structure: E = E_R ⊗ C with E_R its fixed points, and a unitary commutes with j exactly when it preserves E_R, which gives O(n). An antiunitary with j^2 = −1 is a quaternionic structure: E is a right H-module of quaternionic dimension n/2, and the unitaries commuting with j form USp(n). ■

The even dimension for s = −1 is lemma 06's own statement (Kramers degeneracy of the singular values). The proposition adds the symmetry group.

Reading (proposal). The groups are those of Chan-Paton labels on an orientifold plane: s = +1 with O^- (SO), s = −1 with O^+ (USp). So the two odd factorizations of the modern triple, which carry the labels 11 and 9, have the form of the two kinds of orientifold planes. The seesaw sector selects s = +1 (T2c), which is the O^- type of the SO(32) web used in 06 and 07: the physical neutrino sector and the type I' walls of the dictionary agree in type. Two refinements from the second check: the symmetries of the pair (j, d) are smaller, O(n_+) x O(n_-) for s = +1 and U(n/2) for s = -1, since for s = -1 the antiunitary j swaps the +-mu eigenspaces of d; and there is a tension of labels, since here label 9 goes with O^+ while section 6b of 07 puts label 9 on a far O8^- wall (the two are the KO sign and the position of the top's brane, see 10).

3. Proposition 8.2 (wall channels come with doubled multiplicity)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The stated count is for one signed eigenvalue. The lowest $|D|$ multiplicity is $2k$ in the interior and at zero, and $4k$ at $\pi/2$; interpreting the signed count as Chan-Paton labels requires further data. See the referee report.

Statement. In the parent of lemma 22, k channels at a common angle θ contribute, at the eigenvalue θ/l of the window, an eigenspace of dimension k if 0 < θ < pi/2 and of dimension 2k at the ends θ = 0 and θ = pi/2.

Proof. By Proposition 7.1 each channel has the spectrum {(n pi ± θ)/l}; the value θ/l occurs once (n = 0, sign +) for 0 < θ < pi/2, and twice at the ends: at θ = 0 as the zero mode of multiplicity two, at θ = pi/2 as (0 + pi/2)/l = (pi − pi/2)/l. ■

Reading (proposal). k D8-branes in the bulk carry k Chan-Paton labels and k D8-branes on an orientifold plane carry 2k, with their images; the parent reproduces this count, with the ends of the angle interval as the walls. With Proposition 8.1, a stack of channels at a wall has a symmetry of type SO(2k) or USp(2k), as a stack of D8-branes on an O8^- or O8^+ plane. The rank-reduced nine-dimensional theory with one O8^- and one O8^+ (duality 3 of file 07) would then be a parent whose two walls carry opposite signs s: one sheet of type 11 and the other of type 9.

4. Remark 8.3 (E8 walls from 15 + 1; conditional)

Correction (4 Nov 2026). The far-wall group is reached at the string-scale radius R = R_*, with the tower fixed, and only its charged states are D0-branes. See 10.

Referee note (GPT-6.1 Terra, 4 Nov 2026). The brane and slope arithmetic is independently checked conditional on the recalled rule. Wall enhancement requires tuning $z = 0$; for $7 + 7$ the two bulk positions must sum to L. The E_2 identification with the physical electroweak sector needs further data. The cited primary sources were unavailable to the referee; the slope rule and the E_{n+1} table were confirmed against Polchinski-Witten and Bergman-Gaberdiel-Lifschytz in the second check. See the referee report.

Conditions: the dictionary of 06 (fifteen D8-branes for the five light quarks in three colours, the sixteenth for the top) and the recalled rule that an O8^- plane with n <= 7 D8-branes on it, at infinite coupling, carries E_{n+1} (E_4 = SU(5), E_5 = SO(10), E_6, E_7, E_8), while n >= 8 admits no such point (Seiberg, hep-th/9608111; Bergman, Gaberdiel and Lifschytz, hep-th/9711098; recalled).

1. Dilaton bookkeeping: the dilaton is e^{phi} = z^{−5/6} with z piecewise linear along the interval (conformal coordinate; Polchinski and Witten, hep-th/9510169) and slope proportional to 8 − n_left, where n_left counts the D8-branes to the left (O8^- carrying −8 units), so the coupling grows toward the wall with fewer branes; this reproduces the constant dilaton of 8 + 8, the SU(2) at the empty wall of the SO(32) point (the self-dual radius of the heterotic string) and 7 + 7 for E8 x E8 (extrapolated from the single-wall rule; neither Polchinski-Witten nor Bergman-Gaberdiel-Lifschytz states it). In the configuration 15 + 1 the wall with fifteen branes has no E point (15 > 7), and the coupling can diverge only at the far wall. With the top's brane on the far wall (the label-9 configuration of Proposition 7.2, n = 1) the far wall enhances to E_2 = SU(2) x U(1), the electroweak group; with the top's brane in the bulk it is empty and enhances to E_1 = SU(2). The light-quark wall keeps SO(30).
2. If branes move in colour triples (a light flavour is three D8-branes), a wall holds 3j branes, or 3j + 1 with the top's brane. The reachable wall groups are E_1, E_2 (the top's brane alone), E_4 = SU(5) (one flavour), E_5 = SO(10) (one flavour and the top's brane), E_7 (two flavours) and E_8 (two flavours and the top's brane); E_6 (five branes) is not reachable.
3. Hence an E8 wall needs the top's brane together with two light flavours, and two E8 walls are impossible with one top brane: E8 x E8 requires breaking a colour triple.

In this reading the top is what completes E8 at a wall (7 = 2 x 3 + 1), and with colour triples the proposed E8 x E7 arrangement has seven branes at one wall, six at the other and a bulk triple at x_b, while two E8 walls require relaxing the colour-triple rule (referee replacement): with that arrangement, the slopes are proportional to +1 on (0, x_b) and −2 on (x_b, L), so z vanishes at both walls when x_b = 2L/3. The E_n series also contains E_3 = SU(3) x SU(2) (two branes at a wall), which colour triples do not reach.

Reading (proposal). At strong coupling the wall that holds the top's brane in the label-9 configuration carries SU(2) x U(1), and the wall of the five light quarks in three colours carries SO(30). The electroweak group then appears at the top's wall, where the M-theory circle opens locally, and the quarks' flavour group at the other wall.

5. What remains open

- The bulk symmetry: k D8-branes in the bulk carry U(k), while Proposition 8.1 gives O or USp on every eigenspace of d^2; the relation between the window's degenerate levels in the bulk and U(k) needs the parent's own structures, not only j.
- The dilaton: a spectral-triple counterpart of the type I' dilaton profile (a coupling varying along the interval, with slope set by the brane charge to one side) and the infinite-coupling points of Remark 8.3. Proposition 9.1 of 09 writes the profile as a trace of a function of |D_F|, only implicitly after 10, Proposition 10.1, and the gradient is not the second scale in the scope of Propositions 10.5 and 10.6 there.
- The physical status of E8 x E7 with colour triples, and whether the top's brane at an E8 wall has a counterpart in the finite geometry.

Part 11D, 09 dilaton spectral and so32, referee pending

The type I' dilaton as a spectral function, the SO(32) of the EPJC paper, and two more dualities

Part of the 11D line of research (README, PLAN, task C). Attempted by Claude (Anthropic), 4 Nov 2026, after the steer of the same day (the coincident-brane route is overused; the type I' dilaton and the colour-flavour idea are worth pursuing; compare with the SO(32) of the EPJC paper). Proposition 9.1 is proved from the recalled type I' slope rule; Proposition 9.2 is proved from representation theory and the recalled orientifold projections; Proposition 9.3 is proved (linear algebra and Connes's sign table); the comparison of section 1 is read from the arXiv text of the paper through a summarizing fetch; the rest are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

Correction (same day): the first version stated the slope rule for e^{−phi}. The piecewise-linear function is z, with e^{phi} = z^{−5/6}; Proposition 9.1 holds as stated with z in place of e^{−phi}, and infinite coupling is z = 0.

1. The SO(32) of A. Rivero, Eur. Phys. J. C 84 (2024) 1058 (arXiv:2407.05397)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The branching algebra is independently checked. The archived original abstract also states an additional chiral charge-+4/3 quark per generation. The full paper was unavailable for direct comparison, so its detailed attribution remains pending. See the referee report.

That paper uses the adjoint of SO(32) to classify the scalars of a supersymmetric Standard Model with three generations, as composites: pairs of quarks as colour triplets and quark-antiquark pairs as singlets and octets, with three extra scalars of charge +4/3. Its pattern is SO(2n) ⊃ SO(n_1) x SU(n_2) x SU(3) x U(1) with 2n = n_1 + 6 n_2, which for SO(32) and n_2 = 5 gives SO(2) x SU(5) x SU(3) x U(1): the 30 = 6 n_2 are the five light quarks u, c, d, s, b ("turtles") in three colours with conjugates, and the top is the non-participating "elephant". Its explicit route also passes through SO(32) ⊃ SU(16) x U(1), with 496 = 1 + 120 + 120bar + 255 (quarks from the 120s, leptons from the 255), and it mentions open strings, orientifolds and an SU(9) for M2 and M5.

Reading of the LaTeX source (Claude Sonnet 5.5, 4 Nov 2026; the earlier comparison used a summarizing fetch). The paper uses colour singlets for mesons and colour antitriplets for diquarks, and keeps only (1, 24, 1), (1, 15, 3bar) and (1, 15bar, 3) of its decomposition; colour octets and sextets lie outside it. The top is the favoured non-binding quark ("we have not a concrete argument"), and the postulates force the non-binding quark to be up-type. The chain is SU(16) x U(1) to SU(15) to SU(5) x SU(3), with SO(30) and SO(32) reached by "adding singlets". The identification of the top with the SO(2) factor, the O8^- plane and the Chan-Paton reading belong to these files, and "orientifold" in the paper means planar orientifold QCD. The footnote on the 84 + 12 split and SU(9) for M2 and M5 anticipates Proposition 21.3 and 04. The abstract specifies a chiral +4/3 quark per generation; the body has three scalars of charge +4/3 and leaves their fermion partner open. The paper gives (15, 3bar) as the antisymmetric square of one fundamental

without stating chirality, so the question of Proposition 9.3 stays open; its remark on odd-parity diquarks is consistent with that proposition. The paper's selection of an up-type quark by its postulates and lemma 21's by the spectral-action values are independent arguments with the same outcome.

Coincidence. The chain SO(32) ⊃ SO(2) x SO(30) ⊃ SO(2) x SU(15) x U(1) ⊃ SO(2) x SU(5) x SU(3) x U(1), the 30 as five light quarks in three colours, and the top as the factor SO(2), are the same as lemma 21 and the dictionary of 06. Lemma 21 records the branching 496 -> (435, 1) + (1, 1) + (30, 2) and leaves SO(32) unexplored; the EPJC paper explores it through composites, and the files 06 to 09 through type I' branes. What the 11D files add: the type I' realization of the SO(2) as the top's D8-brane and of the 30 as the Chan-Paton labels of fifteen D8-branes on an O8^- wall (06); the top angle as the brane's position and the one-loop vacuum on the far wall (Propositions 7.1, 7.2); the orientifold type O^- matched by the seesaw sign (Proposition 8.1); the far wall's enhancement to SU(2) x U(1) at infinite coupling (Remark 8.3); the dilaton (Proposition 9.1); the diquark counts as strings between the stack and its image, which need the O^- type (Proposition 9.2); and the real-structure condition under which these counts arise, as connection components along the interval and not as finite inner fluctuations (Proposition 9.3). In type I' language the paper's other chain, SU(16) x U(1), is the configuration with all sixteen D8-branes coincident in the bulk, and the chain with SO(2) x SO(30) is the configuration with fifteen on a wall and one apart; the second is the one the corpus selects. The SU(9) of M2 and M5 in the paper is the grading of e8 used in 04.

2. Proposition 9.1 (the type I' coupling function is a trace of a function of the finite Dirac operator)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The integration is independently checked conditional on the recalled slope rule. The trace is over sixteen selected channels, not automatically the full physical D_F Hilbert space; direct comparison with the cited source was unavailable to the referee. (Second check, against Polchinski-Witten.) The formula z(L) = z(0) - c (8L - sum x_k) is exact in the positions; the dictionary x_k = (2L/pi) l m_k holds only for constant z, and in general theta_k = (pi/2) W(x_k)/W(L) with W = int z^{-1/3} (10, Proposition 10.1). In the profile 15 + 1 the endpoint values 7cL and 8cL stand, and theta_t = pi/4 gives 7.358 cL in place of 7.5 cL. See the referee report.

Setting. Type I' on an interval [0, L], in the conformal coordinate x, with sixteen D8-branes at x_1, ..., x_16 and O8^- planes at the ends. Recalled rule (Polchinski and Witten, hep-th/9510169): the dilaton is e^{phi} = z^{−5/6} for a function z that is continuous and piecewise linear in x, with slope c (8 − N(x)), where N(x) is the number of D8-branes to the left of x and c > 0; the coupling diverges where z vanishes. Dictionary (06, 07): x_k = (2L/pi) θ_k with θ_k = l m_k the angles of sixteen channels of the finite Dirac operator (fifteen light-quark channels with colour, and the top).

Statement.

1. z(x) = z(0) + c Σ_k min(x/2, x_k − x/2) for x in [0, L], so z is z(0) plus c Tr g_x(l |D_F|) over the sixteen channels, with g_x(θ) = min(x/2, (2L/pi) θ − x/2): a trace of a function of the finite Dirac operator, as the spectral action is.
2. At the far wall, z(L) = z(0) − c (8L − Σ_k x_k).
3. Infinite coupling at the far wall (z(L) = 0) holds exactly when z(0) = (2cL/pi)(4 pi − l Σ_k m_k): the coupling at the light-quark wall is fixed by the sum of the sixteen channel masses, the trace of |D_F| on these channels.

Proof. *∫0^x N(y) dy = Σ_k (x − x_k)+*, so z(x) = z(0) + c (8x − Σ_k (x − x_k)+) = *z(0) + c Σ_k (x/2 − (x − x_k)+)*, and x/2 − (x − x_k)_+ = min(x/2, x_k − x/2). Item 2 is x = L with Σ_k (L − x_k) = 16L − Σ x_k. Item 3 sets z(L) = 0 and substitutes x_k = (2L/pi) l m_k. ■

Correction (4 Nov 2026). The dictionary x_k = (2L/pi) theta_k is exact only for constant z; the map from angles to positions is nonlinear. See 10, Proposition 10.1.

Reading (proposal). With the light quarks near θ = 0, Σ_k m_k ≈ m_t and the condition is z(0) ≈ (2cL/pi)(4 pi − θ_t): 7cL at label 9 (θ_t = pi/2) and 8cL at label 11 (θ_t -> 0). The coupling therefore varies along the interval on a scale set by the spectrum of D_F, which is the kind of second scale that Proposition 1.5 requires to select a joint family; whether the type I' coupling can play that role in the

corpus's flat family is the next question. Since the slope decreases with x, z is concave, so z > 0 on [0, L) once z(0) > 0 and z(L) = 0; positivity of z(0) gives l Σ_k m_k < 4 pi, a bound on the trace of |D_F| in units of the tower scale, met when only the top is heavy (θ_t <= pi/2).

3. Proposition 9.2 (diquarks as strings between the stack and its image)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The representation and charge counts are independently checked, conditional on the recalled projections. The shifts describe the specified gauge-field channels; Λ^2 of a line excludes that component, not all open-string states or strings to other branes. See the referee report.

Setting. V = C^5 ⊗ C^3, five light flavours in three colours, with Q = 2/3 on u, c and Q = −1/3 on d, s, b; the 30 of lemma 21 is V ⊕ Vbar. Fifteen D8-branes on an O8 plane carry the Chan-Paton space 30 (branes and images), with group SO(30) on an O8^- and USp(30) on an O8^+, and U(15) when the stack is moved off the plane (recalled).

Statement.

1. Under U(15), the adjoint of SO(30) is Λ^2V ⊕ Λ^2Vbar ⊕ (V ⊗ Vbar) and the adjoint of USp(30) is S^2V ⊕ S^2Vbar ⊕ (V ⊗ Vbar). The summands Λ^2V or S^2V and their conjugates are the strings between the stack and its image; V ⊗ Vbar are the strings with both ends on the stack.
2. Λ^2V = ($S^2$5, 3bar) ⊕ ($\Lambda^2$5, 6) and S^2V = ($S^2$5, 6) ⊕ ($\Lambda^2$5, 3bar).
3. In the colour antitriplet, Λ^2V has the charge counts (−2/3: 6, +1/3: 6, +4/3: 3) of lemma 21, and S^2V has (−2/3: 3, +1/3: 6, +4/3: 1). So the antisquark counts of three generations recorded in lemma 21 (6 of charge −2/3 and 6 of charge +1/3 in the 3bar) occur among the stack-image strings of an O8^- plane and not among those of an O8^+ plane.
4. With the angles θ_k = l m_k of 06, a string from channel i to the image of channel j has the spectrum (n pi ± (θ_i + θ_j))/l, and a string from channel i to channel j has (n pi ± (θ_i − θ_j))/l, n in Z.
5. A single brane has no string to its own image on an O8^- plane (Λ^2 of a line vanishes) and one on an O8^+ plane.

Proof. 1: 30 = V ⊕ Vbar as a U(15) module, Λ^2(V ⊕ Vbar) = Λ^2V ⊕ Λ^2Vbar ⊕ (V ⊗ Vbar) and likewise for S^2; the orientifold projection keeps the antisymmetric part for SO and the symmetric part for USp, and the mixed summand V ⊗ Vbar is the U(15) adjoint. 2: Λ^2(A ⊗ B) = (S^2A ⊗ Λ^2B) ⊕ (Λ^2A ⊗ S^2B) and S^2(A ⊗ B) = (S^2A ⊗ S^2B) ⊕ (Λ^2A ⊗ Λ^2B), with $\Lambda^2$3 = 3bar. 3: split 5 = 2 ⊕ 3 by charge; $S^2$5 = $S^2$2 ⊕ (2 ⊗ 3) ⊕ $S^2$3 has dimensions 3, 6, 6 with charges 4/3, 1/3, −2/3, and $\Lambda^2$5 = $\Lambda^2$2 ⊕ (2 ⊗ 3) ⊕ $\Lambda^2$3 has dimensions 1, 6, 3 with the same charges. 4: Proposition 6.1 with charges (+1, +1) and (+1, −1) under U(1)_i x U(1)_j, whose holonomy is e^{2 pi i (a_i ± a_j)}, and θ = pi a. 5: dimensions. ■

Reading (proposal). (11, Theorem 11.3 and Proposition 11.4, qualify the selection below: the window sees the counts (3, 6, 1) also at s = +1, through the commutator with the masses, and parent scalars with the counts (6, 6, 3) need M_15(C) at both walls.) The diquark counts of lemma 21 and of the EPJC paper are the stack-image strings of an O8^- plane, the type that Proposition 8.1 associates with the factorization sign s = +1 selected by the seesaw sector (T2c); the O8^+ alternative gives the wrong number of up-type antisquarks. So the seesaw sign and the scalar counts select the same orientifold type. Two features follow from the projection. Diquark masses go with the sum θ_i + θ_j and meson masses with the difference θ_i − θ_j, so when the stack leaves the wall the diquarks become massive and the mesons stay massless: the diquarks are tied to the orientifold. And the top's single brane has no string to its own image, while its strings to the stack, the (30, 2), have masses near m_t (Proposition 6.1); in the EPJC paper the top's exclusion is an input, and here the O8^- projection with one brane for the top removes the top-top channel only (item 5); the top-to-stack strings (30, 2) remain, with masses near m_t, so the rest of the exclusion is a mass statement, as in the paper.

4. Proposition 9.3 (the real structure decides between the two diquark counts)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The stated sign constraint excludes a nonzero antisymmetric block in the three KO labels. The signs permit all of S^2 V (second check: B ranges over exactly

S^2 V for eps eps' = +1, with the diagonal block free); the algebra, grading and order conditions of a triple then select a subspace. Distinct-multiplet blocks remain related to their transposes. See the referee report.

Setting. E = V ⊕ Vbar with V as in Proposition 9.2, an antiunitary J on E that maps V onto Vbar and Vbar onto V (as particles and antiparticles are exchanged), $J^2 = \varepsilon$, and a Hermitian X on E with JX = ε' XJ, the condition that inner fluctuations obey in a KO-dimension with signs (ε, ε'). Let B be the block of X from Vbar to V, an element of V ⊗ V.

Statement.

1. B is symmetric (B in S^2V) if εε' = +1 and antisymmetric (B in Λ^2V) if εε' = −1.
2. In the KO-dimensions 5, 6, 7 of the finite part for the labels 9, 10, 11, (ε, ε') = (−1, −1), (+1, +1), (+1, +1), so εε' = +1: the diquark block of every J-compatible fluctuation lies in S^2V, whose colour antitriplet ($\Lambda^2$5, 3bar) has the counts (−2/3: 3, +1/3: 6, +4/3: 1).
3. For a connection on a circle whose real structure is complex conjugation (KO-dimension 1, (ε, ε') = (+1, −1)), the block lies in Λ^2V, whose colour antitriplet ($S^2$5, 3bar) has the counts (−2/3: 6, +1/3: 6, +4/3: 3) of lemma 21. The SO(32) Wilson line of Proposition 6.1 is of this type.

Proof. Choose a basis of V and let K: V -> Vbar be coordinatewise conjugation; J(v, w) = (ε K^{-1} w, K v) has $J^2 = \varepsilon$. Write X = [[A, B], [B*, D]]. The component of JX = ε' XJ that maps V to V gives ε K^{-1} B* = ε' B K, so B = εε' K^{-1} B* K^{-1} = εε' B^T in coordinates. Under the U(15)-equivariant identification of Hom(Vbar, V) with V ⊗ V, symmetric matrices form S^2V and antisymmetric ones Λ^2V. Items 2 and 3 use Connes's sign table, ε = (+, +, −, −, −, −, +, +) and ε' = (+, −, +, +, +, −, +, +) for KO-dimensions 0 to 7, and the counts of Proposition 9.2. On the circle, D = −i d/dθ and J = conjugation give JD = −DJ, and a Hermitian connection A with JA = −AJ is i times a real antisymmetric matrix, an element of so(n), where the generator of R(2 pi a) lies. ■

Item 2 is a negative statement with the following scope: the blocks of J-compatible Hermitian operators between particles and antiparticles, for one multiplet V, in the three labels. It is the finite-geometry form of the symmetry of Majorana mass matrices, and of the Fermi-statistics constraint on a Lorentz-scalar bilinear ψ_i ψ_j of one Weyl multiplet without derivatives (recalled), under which the (15, 3bar) vanishes. Known escapes: item 3 (connection components along a circle or an interval), blocks between two distinct multiplets, which J does not constrain, and derivative couplings.

Reading (proposal). The counts of lemma 21 and of the EPJC paper, (6, 6, 3), belong to the antisymmetric block. In the corpus's setting this block arises as the component of an SO(30) connection along the circle or interval: the Wilson-line moduli of the type I' stack, whose stack-image part is Proposition 9.2. The finite triple alone, in any of the three labels, gives the symmetric block and the counts (3, 6, 1). So the composite scalars with the counts of lemma 21 point to the interval parent of lemma 22 and its gauge field, and the selection of O8^- in Proposition 9.2 concerns these connection components. This narrows the spin gap of Propositions 6.1 and 9.2: the Λ^2V states are scalars as circle components of a nine-dimensional gauge field, and they come with that field's vector and gaugino partners.

5. Two more dualities between 9 and 11 (recalled; proposals)

Referee note (GPT-6.1 Terra, 4 Nov 2026). The E_1 alternatives need correction: the canonical-orthogonal root lattices distinguish $\mathbb{P}^1 \times \mathbb{P}^1$ from $dP_1 = \mathbb{F}_1$, and the usual local engineering assigns the latter to $\widetilde{E}_1$. The dP_2 and brane duality claims remain conditional on the unavailable primary sources. The Weyl action is a gauge relabelling of a holonomy, subject to the full background data. See the report on the del Pezzo comparison and Weyl action.

1. **E-type walls and del Pezzo surfaces.** The E_{n+1} symmetry of an O8^- plane with n D8-branes at infinite coupling is the symmetry of the five-dimensional theory on a D4-brane probe, which M-theory realizes on a Calabi-Yau threefold with a shrinking del Pezzo surface dP_{n+1} (Seiberg, hep-th/9608111; Morrison and Seiberg, hep-th/9609070; Intriligator, Morrison and Seiberg, hep-th/9702198). So the far wall with the top's brane (n = 1, E_2 = SU(2) x U(1)) has an eleven-dimensional description by a local dP_2, and the empty far wall (E_1 = SU(2)) by dP_1 or P^1 x

P^1; the light-quark wall (n = 15) has none. The Witten base CP^2 x CP^1 of 02 is built from the same pieces (CP^2 = dP_0, and P^1 factors).

2. **Colour-flavour exchange as a T-duality.** In the heterotic string on S^1, the Weyl group of the gauge group acts on the Wilson lines as part of the T-duality group O(17, 1; Z). The colour-family transposition M of 05 is an element of the Weyl group of E8, so it is a T-duality of the nine-dimensional E8 x E8 string, and through Hořava-Witten a symmetry of the E8 holonomy on a wall in eleven dimensions. As a Weyl element it acts as a global gauge transformation, so by itself it relabels gauge-equivalent vacua; it exchanges colour and family physically only relative to a structure that tells them apart, such as the placement of H_F in 05. The four-family requirement of Propositions 5.1 and 5.4 remains.

6. Open

Referee note (GPT-6.1 Terra, 4 Nov 2026). The finite KO signs restrict the symmetric block without producing the full representation. Light interval scalar modes, their boundary conditions and the physical string dictionary remain to be constructed; the detailed EPJC comparison is pending.

- Whether the type I' coupling, as a spectral function of D_F (Proposition 9.1), supplies the second scale of Proposition 1.5 in the flat family of lemmas 23 and 24.
- The local dP_2 of the top's wall and its relation to the Witten base CP^2 x CP^1.
- Spin: in type I' the strings of Proposition 9.2 belong to nine-dimensional vector multiplets, with one real scalar each (the relative position), the same gap as in Proposition 6.1. By Proposition 9.3 a finite triple in the labels 9 to 11 gives S^2V, so the Λ^2V scalars must come from the connection along the interval; the next computation is the inner fluctuations of the parent of lemma 22 with an SO(30) algebra, and whether its interval component carries Λ^2V with the counts of lemma 21.
- Whether the (15, 3bar) of the EPJC paper could be read as bilinears of two distinct multiplets (the paper itself gives one fundamental with itself) (for example left-handed quarks and conjugates of right-handed ones arranged into one 30), which Proposition 9.3 does not constrain.

Part 11D, 11 so30 fluctuations, referee pending

Inner fluctuations of the interval parent with an SO(30) algebra

Part of the 11D line (README, PLAN). Attempted by Claude Opus 5.5 on 4 November 2026, to answer the question left in 09: whether the gauge field along the interval of lemma 22 carries the antisymmetric block with the counts of lemma 21, given that Proposition 9.3 excludes it from the finite triple alone. Numbers in Python (numpy), scripts not kept by the rule of the project. The Codex quota is spent, so no Terra referee is available. An independent reading by Claude Sonnet 5.5, whose amendments are applied below, is in the review. The corrected text has not been read again.

Outcome

1. **Yes, under a condition on the walls.** With a real algebra M_30(R) on the light block, the fluctuation along the interval is an so(30) connection, and its block between particles and antiparticles lies in Λ^2 V, with the counts (6, 6, 3) in the colour antitriplet (Proposition 11.2). Massless scalars in exactly Λ^2 V + conjugate appear when both walls reduce the algebra to M_15(C), the commutant of quark number (Theorem 11.3).
2. **In lemma 22's own parent the massless scalars are absent.** Its algebra is free at x = 0, so the interval component has no zero mode, for any algebra (Theorem 11.3, item 2).
3. **A generator has a massless vector or a massless scalar, never both, on the interval**, in free five-dimensional Yang-Mills at trivial background. So the whole of Λ^2(30) of lemma 21 cannot be massless scalars with colour unbroken. The antisquark counts appear as scalars, while the slepton part (24, 1) + (1, 1) appears as vectors (Theorem 11.3, item 4).
4. **The finite window sees a symmetric shadow.** The connection acts on the window through its commutator with the masses, B_ij -> (m_i - m_j) B_ij, which maps the (S^2 5, 3bar) of Λ^2 V onto

the (Λ^2 5, 3bar) of S^2 V, with counts (3, 6, 1) (Proposition 11.4). This reconciles Propositions 9.2 and 9.3. The kernel is the same-flavour diquarks.

5. **SO(30) has a precise meaning in the finite geometry:** it is the commutant of the real structure j = J U of lemma 06 on the massless light block, for the sign s = +1, and USp(30) for s = -1 (Proposition 11.1).

Setting

Let V = C^5 ⊗ C^3, with Q = 2/3 on u, c and Q = -1/3 on d, s, b. In the finite Hilbert space of the five light quarks, with colour, the chirality gamma = +1 subspace is

```
H_+ = {q_L} ⊕ {conj(q_R)} = V ⊕ Vbar,
```

of dimension 30: the left-handed Weyl fields of fifteen Dirac fermions. H_- = {q_R} ⊕ {conj(q_L)}, and J_F exchanges H_+ and H_-, with c = J_F restricted to H_+, as in lemma 22. Let U be a unitary, grading-reversing and J-related as in lemma 06, with U q_L = q_R and U conj(q_R) = s conj(q_L), so U J_F = s J_F U. Then j = U^-1 c is an antiunitary on H_+, with j^2 = s, j q_L = s conj(q_R) and j conj(q_R) = q_L. Here U only defines j: the representation pi below does not commute with U, so this is not a doubling in the sense of lemma 06, in which U commutes with the algebra.

The interval parent is that of lemma 22: D_X = E d/dx on L^2([0, l]; C^2) ⊗ (H_+ ⊕ H_-), with Q in H_+, U-components in H_-, and the wall conditions Q_R(0) = U_L(0) = 0, Q_R(l) = X U_R(l), U_L(l) = -X^* Q_L(l).

The block

Proposition 11.1 (the commutant of j). Let s = +1. In the basis V ⊕ Vbar with j(v, w) = (K^-1 w, K v), the skew-Hermitian a with j a j^-1 = a form so(30), and

```
a = [[alpha, B], [conj(B), conj(alpha)]],    alpha^* = -alpha,   B^T = -B,
```

so so(30) = u(15) ⊕ Λ^2 V, with u(15) the commutant of the complex structure I = (i, -i), the quark number. The Hermitian generator H = i a satisfies j H j^-1 = -H. For s = -1 the same computation gives usp(30) = u(15) ⊕ S^2 V.

Proof. Linear algebra, with Proposition 8.1 for the groups. For s = +1, j a j^-1 = a gives the block form, and a^* = -a gives alpha^* = -alpha and B^* = -conj(B), that is B^T = -B. Then j (i a) j^-1 = -i a. ■

So the SO(30) of lemma 21 is the symmetry of the massless block for the sign s = +1, the sign that Proposition 8.1 and STRATEGY T2c associate with label 11 and with the seesaw; and the counts of lemma 21 select the same sign:

	colour antitriplet	its charges (-2/3, +1/3, +4/3)	colour sextet
s = +1, Λ^2 V	(S^2 5, 3bar)	(6, 6, 3)	(Λ^2 5, 6)
s = -1, S^2 V	(Λ^2 5, 3bar)	(3, 6, 1)	(S^2 5, 6)

Fluctuations along the interval

Take on the light block D_F = 0, so X = 0, and let C_0, C_l be subalgebras of A = M_30(R), the real elements of End H_+. Let

```
A_l = {f in C^infinity([0, l], A) : f(0) in C_0, f(l) in C_l},    pi(f) = f on Q, 0 on the H_-
↪ components.
```

Proposition 11.2. Assume 1 lies in C_0 and C_l.

1. pi(f) preserves the domain, for every choice of C_0, C_l. Order zero and order one hold on this block.

2. The self-adjoint one-forms sum pi(f_i) [D_X, pi(g_i)] are exactly E a_x on the Q components, with a_x in C^infinity([0, l], so(30)), with no condition at the walls. The fluctuated operator is

   ```
   D_A = E (d/dx + a_x ⊕ U a_x U^-1).
   ```

3. J-hat A J-hat^-1 = E (U a_x U^-1 on H_-), with sign +, as KO-dimension six requires. The block of a_x between Vbar and V is antisymmetric. The two statements agree since E is antisymmetric: the block of the Hermitian operator E a on C^2 ⊗ E is symmetric, as Proposition 9.3, item 2 requires, with B antisymmetric and the factor E carrying the other sign. So the interval component evades Proposition 9.3 through the escape in its item 3.

For D_F != 0 on this block, order one fails in the finite window: [[D_F, pi(a)], J pi(b) J^-1] has the blocks -U b U^-1 M a from H_+ to H_-, and -a M^* U b U^-1 from H_- to H_+, which vanish for all a, b only if M = 0. The parent still satisfies order one (Theorem 22.3), but its wall algebra shrinks to the commutant of D_F in pi, which is zero for pi = (a, 0) with M invertible. So this algebra describes the massless block, and the masses need a treatment without order one, as in lemma 19.

Proof. (1) The conditions Q_R = 0 and U_L = 0 at the walls are preserved by any operator acting on the H_+ index. [D_X, pi(g)] = E g' on Q, and J-hat pi(b) J-hat^-1 acts on the H_- components only, so both commutators vanish. (2) Given T, choose tau with support inside (0, l) and the same integral as T, which is a finite sum sum f_i g_i' of functions supported inside the interval. Take f = 1 and g = int_0^x (T - tau), which vanishes at both walls, so g lies in A_l and g' = T - tau. Self-adjointness of E T, with E skew, requires T skew. (3) J-hat E = E J-hat (direct from the formula for J-hat in Theorem 22.1), and c a c^-1 = U j a j^-1 U^-1 = U a U^-1. ■

Theorem 11.3 (zero modes of the interval component). Let k_0, k_l be the skew parts of C_0, C_l, and assume a five-dimensional Yang-Mills action for the fluctuation on a flat product with M^4, as in Proposition 22.8.

1. The gauge group is {g : I -> O(30), g(0) in K_0, g(l) in K_l}. The holonomy of a_x transforms as P -> g(l) P g(0)^-1, so the moduli are the double coset K_l SO(30) / K_0, and the massless four-dimensional scalars from a_x form

   ```
   the orthogonal complement of k_0 + k_l in so(30).
   ```

2. In lemma 22's algebra, C_0 = A, so k_0 = so(30), and there are no such scalars, for any algebra of this kind.

3. If C_0 = C_l = M_15(C), the massless scalars are exactly Λ^2 V ⊕ Λ^2 Vbar, with colour-antitriplet counts (6, 6, 3), and the massless vectors are u(15).

4. A generator in k_0 ∩ k_l has a massless vector and no massless scalar; a generator in the orthogonal complement of k_0 + k_l has a massless scalar and no massless vector; all others have neither. If the orthogonal projections onto k_0 and k_l commute (aligned walls), the towers are n pi/l in the first two cases, and (n - 1/2) pi/l for a generator in exactly one of k_0, k_l. For nonaligned walls the towers move: k_0 spanned by T_1 and k_l by cos(phi) T_1 + sin(phi) T_2 give k l = n pi +- phi, which is the mechanism of lemma 23.

5. With colour unbroken at both walls, the (1, 8) of Λ^2(30) is a vector, so Λ^2(30) cannot consist entirely of massless scalars. In item 3, the (24, 8) + (24, 1) + (1, 8) + (1, 1) of lemma 21 are vectors, and the scalars are (15, 3bar) + (10, 6) + conjugates.

Proof. (1) A gauge parameter xi(x) with xi(0) in k_0, xi(l) in k_l shifts int a_x by xi(l) - xi(0), so the linear moduli are so(30)/(k_0 + k_l), and a constant a_x in the orthogonal complement is a zero mode of the quadratic form. (2), (3): substitution, with Proposition 11.1. (4) A_mu has Neumann or Dirichlet conditions at each wall according to membership in k_wall, and a_x the opposite, so the mode functions are cos(n pi x/l), sin(n pi x/l) or their half-integer versions. (5) Item 4, and the branching of lemma 21. ■

Scope of item 4. Free five-dimensional Yang-Mills at trivial background. Escapes: a circle without walls, nontrivial holonomy, brane-localized terms, further internal dimensions.

Note that the choice in item 3 is an input: the domain does not force the wall algebra to M_15(C), since by Proposition 11.2, item 1, every f preserves it at D_F = 0. It says that the walls conserve quark number and the bulk does not. On a circle, without walls, every generator has both a massless vector and a massless scalar, and the whole of Λ^2(30) appears, which is the nine-dimensional adjoint scalar used in 09. A wall condition, or the reflection-even algebra of Proposition 1.1, removes half. So the two readings differ: for the stack off the O8 plane, with holonomy I, diquarks are massive and mesons have massless scalars; on the interval with parity Ad(I), diquarks have massless scalars and mesons have none.

The shadow in the finite window

Proposition 11.4. Let s = +1, M = U Lambda with Lambda real for j and symmetric, and let a_tot = a ⊕ U a U^-1 be constant. Then:

1. [D_F, a_tot] has the off-diagonal blocks U S and S U^-1, with S = [Lambda, a] real and symmetric, and [D_F, [D_F, a_tot]] = [Lambda, S] ⊕ U [Lambda, S] U^-1.
2. By Theorem 22.4, the compression of E a_tot to the window is -l^-1 int_0^l sin(x ad_{D_F})(a_tot) dx. For Lambda = diag(m_i) on flavours, on both V and Vbar, the block B_ij goes to

   ```
   - ((1 - cos(l (m_i - m_j))) / (l (m_i - m_j))) B_ij,    which is -(l/2)(m_i - m_j) B_ij to
   ↪ leading order,
   ```

   as a change of Lambda. The result is symmetric, as Proposition 9.3 requires of a finite J-compatible fluctuation.
3. For distinct masses, B -> (m_i - m_j) B_ij maps (S^2 5, 3bar) in Λ^2 V onto (Λ^2 5, 3bar) in S^2 V, with kernel the five same-flavour diquarks, and maps (Λ^2 5, 6) into (S^2 5, 6). In charges, (6, 6, 3) - (3, 0, 2) = (3, 6, 1). If the wall condition is imposed on the gauge unitaries g ⊕ U g U^-1, so that k_l is the commutant of Lambda in so(30), then with k_0 = u(15) the massless scalars of Theorem 11.3 are the same off-diagonal blocks. If it is imposed on pi(f) itself, as in the algebra of lemma 22, then k_l = 0 for invertible M.
4. For a universal mass, Lambda = m, the compression vanishes: S = 0, and M = m U is invariant under SO(30). Fifteen Dirac fermions of equal mass are thirty Majorana fermions of equal mass.

Proof. (1) Direct multiplication, using a_tot on H_- equal to U a U^-1. S is real and symmetric since a is real and antisymmetric and Lambda real and symmetric. (2) Item 1 gives sin(x ad)(a_tot) = (sin(x delta)/delta) ad(a_tot) on B_ij, with delta = m_i - m_j; integrate. (3) In the colour antitriplet, B is symmetric in flavour, and (m_i - m_j) B_ij is antisymmetric in flavour, hence total symmetric. Colour sextets are analogous. (4) S = 0. ■ Checked in Python on a small model, with V = C^3: reality, symmetry, the blocks of item 1 and their squares. Claude Sonnet 5.5 also checked the compression by direct quadrature of W^*(E a_tot)W, with five flavours in three colours, against Theorem 22.4 and the closed block (Python, 10^-15).

Reading. The commutator [D_F, a] is the form of the Higgs field in Connes' finite geometry, and E a_x is a Kaluza-Klein component along the interval. Theorem 22.4 (3), Q_l([D_X, f]) = [D_F, Q_l(f)], says that the second compresses to the first. Indeed E a_tot = [D_X, f_tot] with f(x) = (x - l) a, and f(l) = 0 lies in C_F. So in this parent, to first order and for the fixed compression, a finite Higgs one-form is the shadow of a gauge field along the interval, and it inherits the symmetry type of the finite triple by the commutator with masses. The moving window of Corollary 22.6 gives -l [D_F, h] and a quadratic term (lemma 19). This shadow is tangent to the O(30) orbit of Lambda, so it gives no radial mode, in agreement with No-go theorem 22.7; the odd generators of lemma 23 are absent here since M_30(R) maps H_+ to H_+. A generator commuting with the masses has no shadow. This explains the difference between the counts (6, 6, 3) of the parent and (3, 6, 1) of the finite triple, and it replaces the alternative O8^- against O8^+ of Proposition 9.2, for these blocks, by a parent and its window.

What remains open

- Whether M_15(C) at the walls can be derived from a boundary condition instead of chosen, for example by a wall term that conserves quark number.
- The treatment of nonzero masses, where order one fails on the block: the quadratic terms of lemma 19 for this algebra.
- The sleptons: on an interval the (24, 1) + (1, 1) are vectors. Scalars in these representations need components along further internal directions, such as the base of 02.
- The colour sextets (10, 6) come with the antitriplets, and nothing removes them in this construction.
- The relation to the top: the (30, 2) of SO(32), with the second factor as the top's SO(2).

Part R4 (referee report, lemma 18, two sheets, framework note)

Referee report on lemma 18, the two-sheet section of lemma 16, and the background note

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_18_16b.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemma 18, lemma 16, the note and the paper. Claude checked with Python the equations (R1) and (R3), the fully massive kernel and its poles, and read Remark 2.9 of arXiv:0706.3688.

We checked the numbered statements in lemma 18, the two-sheet section of lemma 16, and the claims and table in current_framework.md. We read the relevant passages of the cited papers, identified below. We did not run a computation.

The principal conclusions are:

- The fixed-cutoff quadratic calculation in lemma 18 is correct in its stated top-only family. Its conclusion extends to a broad class of nonnegative smooth cutoffs, including compactly supported cutoffs flat near zero.
- The limiting top-only quadratic action does **not** by itself have unbroken $SU(2) \times U(1)$ acting on the entire light space. It contains an incomplete top-bottom doublet.
- The two-sheet construction reproduces the classified bimodule and real structure as abstract finite data. Example 16.9 does not derive its nontrivial grading as an even spectral truncation of the parent it specifies.
- Corollary 16.8 incorrectly says that its Dirac operator commutes with the whole centre in every case. It always commutes with the two *sheet projections*.

1. Lemma 18, Theorem 1

Correct with amendment. State suitable regularity and decay assumptions on A, for example smooth compactly supported fields, and interpret the trace on $\mathbb{R}^4$ as the stated relative trace or quadratic variation. For several generators, contract the coefficient matrix with the corresponding polarizations.

The Duhamel expansion has the two terms displayed in (5). For transverse a, shifting the loop momentum gives the orbital contribution $2/t$. At the two opposite vertices, the spin traces of the curvature and commutator terms give q^2 and $(m_i - m_j)^2$, respectively; their mixed spin traces vanish. The A^2 vertex contributes

$$-\frac{1}{4\pi^2 t} \sum_{ij} |T_{ij}|^2 e^{-tm_i^2} |a|^2.$$

Symmetrizing this term gives the subtraction in (1), including its factor. Differentiation gives (2)–(3). The divided-difference identities (6)–(8) also follow as written.

The moment assumptions (4) are sufficient for the displayed Euclidean coefficients: $\int s^{-2}\,d\nu$ controls the action density, while the positive moments provide the required differentiability and domination. They do not supply arbitrary timelike analytic continuation.

The trace convention agrees with CCM's bosonic spectral action. Their calculation explicitly counts both particle and antiparticle sectors and four spin components; see CCM, Lemma 3.14 and §3.7.6. The fermionic chirality projection must not be inserted into that bosonic trace.

2. Lemma 18, Lemma 2

Correct. With the stated convention, $A_\mu \mapsto A_\mu - \partial_\mu \alpha$ and $\pi \mapsto \pi + \alpha$, so $A_\mu + \partial_\mu \pi$ is invariant. Conjugating by $e^{-i\pi T}$ proves (12), including the scalar contribution that is missed by using only the linear orbit tangent at a nonstationary background.

The stated longitudinal replacement follows from retaining the longitudinal part of the shifted orbital vertex. For a commuting generator, its integral vanishes by the displayed integration-by-parts identity. Translation invariance and rotation symmetry exclude quadratic mixing of a transverse vector with a scalar.

3. Lemma 18, Corollary 3

Correct with amendment. Formula (13) requires Q to be positive definite on the chosen generator space. Otherwise it applies after quotienting by its null space.

The orbital part of (2) starts at second order in the squared-mass difference, whereas the commutator term starts with d_{ij}^2. Also $\int_0^1 (1 - 2u(1-u))\,du = 2/3$. This gives the stated C_0, C_1, and factor $3/2$.

For the full Standard Model finite trace, one generation contributes 2 to the particle trace of T_1^2; three generations and charge conjugation give $Q_{11} = 12$. The commutator trace gives (14). This agrees with CCM, equations (4.10) and (4.16), with $F(v) = f(v^2)$. It is a leading expansion, not an identity at fixed m/Λ.

4. Lemma 18, Proposition 4

Correct. In one color of the massive block, let

$$|+\rangle = (t_L + t_R)/\sqrt{2}, \qquad |-\rangle = (t_L - t_R)/\sqrt{2}.$$

Each has squared charged-generator matrix element $1/8$ to b_L, and the reverse ordered entries have the same value. Their total is $1/2$ per color. Thus the massive color triplet contributes $3/2$ to the particle trace, and the other complete doublets contribute $9/2$. The charge-conjugate block supplies the factor two used in (16).

Integration by parts gives $h_x'(0) = 2w(x)/3$ and $g'(0) = 2/3$. Consequently (18) follows. Expanding A and w gives the small-x coefficient $-23/128$, and $A(x) \sim 2/x$, $w(x) \sim 6/x^3$ give the large-x limit. The positive-Laplace extension (23)–(24) follows by dominated convergence under (4).

5. Lemma 18, Theorem 5

Correct. Equation (22) makes h_x' and g' strictly positive for $y \le 0$. Since $F_x(0) > 0$ and $F_x(y) \to -\infty$ as $y \to -\infty$, there is exactly one positive timelike root, and its derivative has the sign giving the stated Euclidean residue.

For $y \le 0$, $g'(y) \ge 2/3$, hence $F_x'(y) \ge 3$. The mean-value theorem gives $z \le A(x)/2$; the Taylor ratio has the same bound. Since $xue^{-xu} \le e^{-1}$, $A(x) \le 2/e$, proving (28). The limiting derivatives are 4 and 3 at small and large x, respectively, yielding (27). The result says nothing about complex poles or Lorentzian unitarity.

6. Lemma 18, Proposition 6

Correct with amendment. The statement about one *positive massive* neutral pole requires $x > 0$. At $x = 0$ there is only the massless neutral system.

For one Dirac species, diagonalizing the left-right mass block gives diagonal neutral charge $(q_L + q_R)/2$ and off-diagonal charge $(q_L - q_R)/2$. In (3), the latter connects eigenvalues $+m$ and $-m$, giving exactly (33). Adding three top colors gives (30). The commuting photon and color traces give (29), including charge conjugates.

The angle in (31) is a sensible **neutral-current definition**: the kinetic-orthogonal massive direction is $K^{-1}(1,-1)^T$, and its current can be written $T_3 - \sin^2\theta_{\mathrm{NC}} Q$. Here

$$K_{33} + K_{3Y} = 5 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(5 + r),$$

so the value $5/8$ is constant. This is a charge-trace identity for this family; it does not make all finite-cutoff definitions of the weak angle coincide.

The neutral mass form is rank one in $v = (1,-1)^T$. Its generalized eigenvalue gives (34), using $v^T B_\infty^{-1} v = 8/27$. For the pole, the left side of (36) has derivative at least $1/2$, so the positive massive root is unique and has positive residue. The substitutions stated for (29)–(33) are valid for (4).

7. Lemma 18, Counterexample 7

Correct. The only nonzero entries of T connect m to $-m$, so (37) follows from (1). Its q^2 coefficient is proportional to $e^{-x}(1-x)$, while its Taylor ratio is $6m^2/(1-x)$.

The pole equation is (36) with $\beta = 4x$. Since $H(-z) > 1$ for $z > 0$, its solution lies strictly between $4x$ and $4x + 2$, and the derivative is positive.

For (39), positive Laplace mixtures give $f(x) + xf'(x)$. The assumptions imply $xf(x) \to 0$, while $xf(x) > 0$ for $x > 0$; therefore its derivative is negative somewhere. A negative Taylor kinetic coefficient does not itself establish a negative-residue pole of the full form factor.

8. Lemma 18, Counterexample 8

Correct. For fixed y, (21) gives $h_x(y) \to 0$, leaving (40), which is positive at fixed positive y. The neutral matrix likewise leaves nonzero kinetic terms. The broken vector directions therefore do not disappear by their masses diverging or their fixed-momentum propagators vanishing.

The narrower decoupling statement is also correct: after restricting the connection to the centralizer, the mass blocks are invariant and the heavy block carries its heat suppression. "Decoupling" here is a statement about the classical bosonic quadratic action, not a quantum effective-action theorem. The symmetry of the unrestricted limiting light space is addressed below.

9. Lemma 18, No-go theorem 9

Correct. Differentiation of the heat-trace density gives (41), which is strictly negative for $c > 0$, nonzero V_*, and the positive measure in (4). The theorem is fairly limited by the phrase "positive Laplace class."

The conclusion can nevertheless be sharpened on flat $\mathbb{R}^4$. For a sufficiently integrable real cutoff f, let λ_j^2 be the eigenvalues of V_*^2. Direct integration of the momentum variable gives

$$\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \sum_j \lambda_j^2 \int_{c^2\lambda_j^2/\Lambda^2}^{\infty} f(z)\, dz. \tag{R1}$$

For **nonnegative** f, a finite $c > 0$ is stationary exactly when every tail in (R1) associated with a nonzero λ_j vanishes. Thus:

- if f has nonzero support arbitrarily far out, there is no finite stationary point on this ray;
- if f has compact support ending at R, there is a stationary plateau once every nonzero $c^2\lambda_j^2/\Lambda^2 \geq R$.

This argument uses the continuum momentum integral; it is not the proof given for a finite torus. Sign-changing cutoffs can have cancellations in (R1), and a radial path with an additional fixed, noncommuting mass matrix need not have the form cV_*.

A nonconstant cutoff flat near zero cannot satisfy (4): a nonzero positive measure on $(0,\infty)$ gives $f'(0) = -\int s\, d\nu(s) < 0$. CCM use positive test functions and their asymptotic moments; profiles flat near zero are allowed and are outside the theorem's stated class. For nonnegative profiles, however, (R1) supplies the separate conclusion above. See CCM, Introduction and §6.2.

10. Lemma 18, Corollary 10

Correct. Equations (19), (27), and (34)–(36) give every entry of (42). At fixed positive μ, all dimensionless pole and Taylor masses are fixed and positive, so their dimensionful values grow with Λ. In the last joint limit, the small-μ expansion gives $M_W \sim \sqrt{3/8}\, m_t$; the neutral masses also diverge, with $M_Z^2 \sim (3/5)m_t^2$.

These are scaling statements for prescribed backgrounds. They do not establish stationary vacua.

11. Lemma 16, reducibility of Corollary 16.4's pair

Correct. The projection

$$P = 1 \otimes \frac{1+\sigma_1}{2}$$

commutes with the represented algebra and with $J_F \otimes \sigma_1 K$, and is neither zero nor one. It proves reducibility of the **pair** (A, J). The projection need not commute with the grading; irreducibility of the pair is the issue asserted in the text.

12. Lemma 16, Theorem 16.7

Correct. The graded flip is an antiunitary involution. The two direct-summand Dirac operators square to the corresponding sums of squares, and the graded-flip signs give $J\mathcal{D} = \mathcal{D}J$. The minus sign on the second grading gives $J\Gamma = -\Gamma J$.

For even a, b, the opposite action is on the other tensor factor, as displayed. Both order conditions follow from the tensor-factor separation. The product operators have compact resolvent under the spectral-triple hypotheses.

When the two finite spaces are ungraded and have equal dimension, the identification $\xi \otimes \bar{\eta} \leftrightarrow (v \mapsto \xi\langle \eta, v\rangle)$ gives left composition and $J(x,y) = (y^*, x^*)$, exactly as in Chamseddine–Connes, Proposition 2.8 and Remark 2.9.

13. Lemma 16, Corollary 16.8

Error. The KO signs, dimension, order conditions, real orientability formula, and the full-matrix special case are correct. The assertion that D_F commutes with the **whole centre** in every case is false. It always commutes with the two projections distinguishing the sheets.

For a counterexample, take

$$V_X = \mathbb{C}^2, \quad \gamma_X = \sigma_3, \quad d_X = \sigma_1, \quad A_X = \mathbb{C} \oplus \mathbb{C}$$

acting diagonally, and take $V_Y = \mathbb{C}$, $d_Y = 0$. The element $(\mathrm{diag}(1,0), 0)$ is central in $A_X \oplus A_Y$, but its commutator with D_F is nonzero.

The orientability formula is the usual **real** zero-dimensional relation $\Gamma = \sum_i \pi(a_i) J \pi(b_i) J^{-1}$, with opposite-algebra coefficients. It is not the stronger assertion $\Gamma \in \pi(A)$, which can fail. In the full-matrix, equal-dimension special case, an invertible matrix in each summand is a separating vector; the pair is irreducible because a projection commuting with both left and right full matrix actions is scalar on each summand, and J equates the two scalars.

14. Lemma 16, Example 16.9

Error. The two algebra-generation calculations and the abstract classified bimodule are correct. The claimed application of Corollary 16.8 with the subsequently chosen nontrivial grading is not justified by the specified even parent, and the compressed potential does not repair it.

For the generation calculations:

- The matrices $1, \Gamma_a, \Gamma_a\Gamma_b$ for $a < b$ are 16 real-linearly independent elements commuting with j. Since $\dim_{\mathbb{R}} \mathrm{End}_{\mathbb{H}}(W) = 16$, the real algebra generated by the five Γ_a is $M_2(\mathbb{H})$.
- For Hermitian B, the fourth moment gives $Q(w^*Bw/w^*w) = (B + \mathrm{tr}(B)1)/5$. This ranges over all Hermitian matrices. Their products generate $M_4(\mathbb{C})$ as a real algebra.

The use of two copies of the four-space is mathematically legitimate. It supplies independent direct summands and the 32-dimensional bimodule. For this particular grading of W and trivial grading of V, the graded flip is the ordinary adjoint flip, and

$$\Gamma_F(\xi, \eta) = (\Gamma_5 \xi, -\eta\Gamma_5),$$

exactly the graded representation in Chamseddine–Connes, Proposition 3.2. The full algebra is a **graded prealgebra**; its even part is

$$\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C}).$$

The obstruction is that an even parent has $[P\gamma_X P, PaP] = 0$ for every compressed parent function. But the proposed grading $\Gamma_5 = 5Q(x_5)$ anticommutes with $Q(x_1) = \Gamma_1/5$. Thus the full compressed $M_2(\mathbb{H})$ and that grading cannot both arise from the stated even parent. They can be chosen **after** truncation as abstract graded-algebra data, and the even subalgebra then gives the real zero-cycle orientability relation.

Similarly, $d = \mu\Gamma_1$ is the compression of a bounded real multiplication operator, but adding that operator to the original parent Dirac operator does not retain the original kernel as a spectral projection: multiplication by the nonconstant real coordinate does not preserve the holomorphic kernel. It also does not preserve the original parent grading as an odd perturbation.

Finally, d commutes with the centre of the full prealgebra, namely $\mathbb{R} \oplus \mathbb{C}$. It does **not** commute with the whole centre after passing to the even algebra: for $e_+ = (1 + \Gamma_5)/2$, $[\Gamma_1, e_+] \neq 0$. It still commutes with the two sheet projections.

15. Framework note: real structure and KO-dimension

Correct with amendment. The KO-degree and antiparticle grading statements are right. The Euclidean Pfaffian construction uses the fermionic action on the appropriate chiral subspace. Barrett's Lorentzian treatment removes the quadrupling through compatible reality and chirality conditions; it should not be presented as the same Euclidean Pfaffian argument. See Barrett, §§1–3 and CCM, §4.4.

16. Framework note: classification of (A, J)

Correct with amendment. The classification is under order zero, a separating vector, irreducibility, and $J^2 = 1$. "Complexified algebra" here means the represented complex span used in the paper. The exclusion of the one-centre case also uses compatibility of the grading with the algebra—normalization in

the graded-prealgebra formulation, or commutation in an even triple. Without compatibility, an arbitrary anticommuting grading is not excluded. See Chamseddine–Connes, §§2–3.

17. Framework note: real form and grading

Correct with amendment. The quaternionic grading and smallest choice $\dim_{\mathbb{H}} W = 2$ are hypotheses and choices in the source. The displayed full algebra is graded; $\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C})$ is its even part. On $\mathrm{Hom}(W, V)$, the grading acts by **right** composition, with the minus sign.

18. Framework note: order one

Correct. Under the hypotheses of Chamseddine–Connes, Theorem 4.1, the stated maximal even subalgebra is selected by the existence of a centre-connecting Dirac operator satisfying order one. The subsequent gauge group uses the represented unimodular lift; the statement is an algebraic selection, not a dynamical derivation of a vacuum.

19. Framework note: unbroken symmetry

Correct with amendment. CCM explicitly impose commutation with the nonscalar $\mathbb{C}_F = \{(\lambda, \lambda, 0)\}$ and identify its physical purpose with the massless photon; see CCM, equation (2.30) and Remark 2.19. The subalgebra is not unital in the full algebra, and the physical electromagnetic lift also uses the unimodularity condition. The nonscalar element in the kernel of $a \mapsto [D_F, a]$ explains the failure of the metric condition cited in the note; a treatment formulated only for complex algebras must specify its complexification.

20. Framework note: beyond first order

Correct with amendment. The quadratic fluctuation, gauge covariance, transitivity, and semigroup statements are supported by Chamseddine–Connes–van Suijlekom, §§I and III. The Pati–Salam construction is in the separate arXiv:1304.8050, §§III and VIII. Its discussion of a vacuum selecting the Standard Model subalgebra is model- and vacuum-dependent: the paper explicitly says that determining all vacua is difficult and checks a restricted Hessian. It should not be stated as a general theorem that every vacuum restores order one.

21. Framework note: quanta of geometry

Correct. In the conventions of Chamseddine–Connes–Mukhanov, equations (2)–(5), the two Clifford algebras in dimension four are $M_2(\mathbb{H})$ and $M_4(\mathbb{C})$, and the two-sided relation gives the sum of the pulled-back volume densities. This is a quantization condition with two maps, not a truncation result.

22. Framework note: twisted triples and grand symmetry

Correct. The larger algebra mixing spin and internal degrees of freedom and the extra scalar are described in Devastato–Lizzi–Martinetti, §§3–5. A twisted commutator replaces the ordinary bounded-commutator requirement in the later Devastato–Martinetti construction, §3. The note properly says that these constructions have not been studied here.

23. Table: noncommutative finite algebra

Correct. The interval and projective-space examples give the stated generated algebras, with algebra generation an additional operation. The distinction between PAP as an operator system and its generated C^*-algebra is also emphasized in Connes–van Suijlekom, §1.

24. Table: inherited pointwise reality

Correct. Under Proposition 16.1's pointwise-reality and invariant-window hypotheses, order zero for the generated algebra holds exactly when that algebra is commutative. This is scoped to that inherited correspondence.

25. Table: order zero, order one, centre $\mathbb{C}$

Error. Corollary 16.3 gives the order conditions and dimension k^2, but it does not generally give centre $\mathbb{C}$. For its interval example $A_F = \mathbb{C} \oplus \mathbb{C}$, the centre is $\mathbb{C} \oplus \mathbb{C}$. The stated centre requires an additional condition, such as a simple generated algebra.

26. Table: KO-6 signs

Correct. Both constructions give the signs. The first pair is reducible and fails the displayed real orientability relation; the two-sheet input is explicit.

27. Table: centre $\mathbb{C} \oplus \mathbb{C}$, dimension $2k^2$

Correct with amendment. Corollary 16.8 reaches this classified pair when the two spaces have equal dimension and the represented complex spans are full matrix algebras. Arbitrary generated algebras in that corollary can have larger centres and need not give an irreducible pair.

28. Table: orientability

Correct. The displayed zero-cycle works under the stated condition that both finite gradings lie in the even represented algebras. It is real zero-dimensional orientability, in the sense specified in section 13 of this report.

29. Table: $M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$

Correct with amendment. The real generated algebras and the 32-dimensional abstract bimodule are reached. The nontrivial grading is added after the stated truncation, and the full algebra with that grading is a graded prealgebra. It is not an even spectral-triple algebra, nor is the grading derived from the specified even parent. The construction requires a **real** generated algebra on the S^4 side; complex C^*-generation would also include $i1$.

30. Table: selection of A_F

Correct with amendment. An off-diagonal term is missing, but an arbitrary off-diagonal term alone is not enough. One must impose the required order-one condition and the maximal-subalgebra hypothesis of the classification.

31. Table: Yukawa block

Correct with amendment. Theorem 12.7 gives the texture in a towerless limit, up to grading-preserving unitary identifications. An algebra and its representation must also be specified before calling the result a finite spectral triple. It is not embedded in Example 16.9.

32. Table: $[D_F, \mathbb{C}_F] = 0$

Correct with amendment. Proposition 17.3 explains the consequence **when** a chosen subalgebra commutes with D_F; it does not make an arbitrary interval or Yukawa block commute with a Standard Model representation. The appropriate common action on the paired blocks is additional data.

33. Table: bosonic action

Correct with amendment. The leading-order results are correctly described, but "near-cutoff in progress" is stale: lemma 18 now gives the exact quadratic heat-cutoff calculation and the stated positive-Laplace extension. Its scope and limitations are those checked above.

34. Table: beyond first order

Correct. The inherited two-point data supply an order-zero, first-order-failing example to which the generalized-fluctuation question can be posed. The table correctly records that the calculation has not been done and that relaxing order one does not cure order-zero failure.

35. Table: quanta of geometry

Correct with amendment. The same five Clifford matrices occur, and the projector identity is valid. For the quaternionic rank-one projector $\Pi = P_w + P_{jw}$, expansion in the real basis $1, \Gamma_a$ gives $\Pi = (1 + \sum_a x_a \Gamma_a)/2$. This algebraic connection does not derive the two-sided quantization condition from truncation.

36. Table: twisted triples and grand symmetry

Correct. The table records no result and marks the subject untested.

Part 2: scope of the heavy-limit conclusion

Within the three-generation, zero-Majorana, top-only family of Proposition 4, at fixed cutoff and for the heat function, **both** the charged Taylor mass and its unique positive timelike pole tend to zero. The neutral Taylor and pole masses also tend to zero. The result refutes an infinitely heavy W, Z, t limit **for these background data and these quadratic mass definitions**. It also rules out a stationary nonzero radial vacuum in the positive-Laplace class. It does not rule out other finite data, a different action or constraint, a joint $\Lambda \to \infty$ limit, or a quantum effective theory.

The top-only Taylor conclusion is broader than (4). For a nonnegative smooth compactly supported f with $f(0) > 0$, define

$$M_1 = \int_0^\infty z f(z)\, dz, \qquad M_2 = \int_0^\infty z^2 f(z)\, dz.$$

The heat formulas extend by linearity and smooth functional calculus to

$$A_f(x) = \frac{2}{x} \int_0^x z f(z)\, dz, \qquad w_f(x) = \frac{3}{x^3} \int_0^x z^2 f(z)\, dz. \tag{R2}$$

One way to justify the extension is to approximate a smooth compactly supported $f(z)$, with the derivatives needed for the quadratic variation, by finite real linear combinations of e^{-nz}: write $f(z) = e^{-z} g(e^{-z})$ and approximate the smooth g and its derivatives by polynomials. The divided-difference formula for the variation permits passage to the limit. Thus

$$\frac{M_{W,\mathrm{Taylor}}^2}{\Lambda^2} \sim \frac{M_1}{f(0)x}.$$

This includes a nonnegative cutoff flat near zero. For a nonanalytic cutoff there need not be a prescribed analytic continuation defining the pole used in Theorem 5.

Several heavy doublets change the answer. Suppose, for the heat cutoff, that P is the total particle-trace weight of complete massless doublets, and heavy doublet i, of weight $d_i > 0$, has one Dirac member of mass m_i and one massless member. Then

$$C_W(y) = \frac{\Lambda^2}{4\pi^2} \left(P g(y) + \sum_i d_i h_{x_i}(y) \right), \qquad \frac{M_{W,\mathrm{Taylor}}^2}{\Lambda^2} = \frac{3}{2} \frac{\sum_i d_i A(x_i)}{P + \sum_i d_i w(x_i)}. \tag{R3}$$

For any fixed finite number of doublets, if $P > 0$ and all $x_i \to \infty$, the Taylor mass tends to zero. If $P = 0$ and all $x_i = x$, (R3) instead gives

$$M^2_{W,\mathrm{Taylor}} \sim \frac{m^4}{2\Lambda^2}.$$

The full heat form factor also has a unique positive pole in this equal-mass case. To find its scale, put $z = cx$, $0 < c < 1$, in (21). Concentration near $u = 0$ gives

$$h_x(-cx) \sim \frac{2(1-2c)}{x(1-c)^2}.$$

Monotonicity then places the root at $z/x \to 1/2$, so $M^2_{W,\mathrm{pole}} \sim m^2/2$. At every fixed Euclidean momentum, however, the whole unnormalized quadratic kernel tends to zero. The divergent Taylor and pole scales differ, and neither alone establishes a healthy limiting gauge theory.

There is also a fully massive example. Give **both** members of every Standard Model doublet the same Dirac mass m, with no Majorana masses. The charged form factor is proportional to

$$e^{-x}\big[g(y) + 2xH(y)\big].$$

Its positive pole obeys $2x < z_W < 2x+2$. The neutral massive pole obeys $(16/5)x < z_Z < (16/5)x+2$. Both pole masses therefore diverge with m. Yet the fixed-momentum action tends to zero, and the charged Taylor kinetic coefficient is proportional to $e^{-x}(1-x/2)$, which is negative for $x > 2$. This is a counterexample to a general fixed-cutoff no-go for diverging pole locations, while leaving the physical quality of that limit open.

The phrase "the heavy fermion leaves the loop" is only a shorthand for the suppression in this **classical spectral trace**. The light complete doublets supply the surviving charged kinetic form, but the top-only light space also retains b_L after removing t_L. If P_{light} is the projection off the top mass block, then on b_L

$$P_{\mathrm{light}}T_1P_{\mathrm{light}} = P_{\mathrm{light}}T_2P_{\mathrm{light}} = 0, \qquad P_{\mathrm{light}}T_3P_{\mathrm{light}} = -\tfrac{1}{2}.$$

These compressed operators do not obey the $SU(2)$ commutation relations. Correspondingly, the limiting particle-trace coefficients have $K_W = 9/2$, $K_{33} = 21/4$, and $K_{3Y} = -1/4$; a linearly realized unbroken electroweak gauge action would require equal weak coefficients and no $3Y$ mixing. Vanishing vector masses therefore do **not** prove unbroken $SU(2) \times U(1)$ on the entire light space.

The remaining **complete** massless doublets do carry the ordinary electroweak representation, and their separate contribution is gauge invariant. If finite data instead make both members of every removed doublet heavy, the retained light subspace can be invariant under the full group; a full limiting theory still requires control of the remaining fields, normalization, interactions, and vacuum.

Part 4: what the two-sheet construction establishes

Example 16.9 is substantive as a concrete compression calculation: real base functions generate a quaternionic matrix algebra, real total-space functions generate a complex matrix algebra, and two copies give the classified 32-dimensional bimodule. The calculation does not select those ingredients dynamically. The restriction to base functions on one sheet, the second independent sheet, real algebra generation, the four-dimensional window, and the grading axis are inputs.

At the finite algebraic level, the two-sheet bimodule is the standard construction from two representations:

$$(W \otimes \bar{V}) \oplus (V \otimes \bar{W}).$$

The graded flip supplies the real structure and the block sign supplies the KO-6 relation. The calculation's additional content is its realization from the particular compressed function systems and its explicit matching to the classified data. The abstract bimodule and its adjoint-flip real structure are already stated in Chamseddine–Connes, Remark 2.9; the report makes no broader novelty claim.

The note appropriately lists several choices and says that no ordinary Kaluza–Klein vacuum is derived. It overstates the reach where it treats the chosen nontrivial grading and nonzero compressed d as consequences of the stated even-parent truncation. It also labels the centre of Corollary 16.3 too generally. Lemma 16's later caveat that its generated algebras are complex and a quaternionic summand still needs additional real data is stale after Example 16.9's real-algebra construction.

For canonical section 4, the example closes the **abstract finite-module** problem after adding a second sheet: it turns the nested actions on one four-space into independent summands acting on two Hom spaces. It does not turn four global sections into rank-four fibres over S^4, derive the two sheets from the fibration, or produce the centre-connecting Dirac operator required for the subsequent algebra selection. The fibre-rank problem remains open.

Amendments and errors

- Add field regularity to Theorem 1, $Q > 0$ or a quotient by its null space to Corollary 3, and $x > 0$ to the massive neutral-pole assertion.
- Keep the neutral-current angle distinct from other finite-cutoff angle definitions.
- Replace Corollary 16.8's claim about the whole centre with commutation with the two sheet projections.
- In Example 16.9, distinguish the correct abstract graded prealgebra from an even spectral truncation; the proposed grading and nonzero compressed perturbation are not derived by the stated parent. The nonzero d also fails to commute with the whole centre of the even algebra.
- Correct the table's unrestricted centre-$\mathbb{C}$ claim for Corollary 16.3, and add the full-matrix and equal-dimension hypotheses to its classified two-sheet row.
- Update the table's near-cutoff entry and qualify the claims about the Yukawa representation, off-diagonal Dirac operator, real-algebra generation, and the vacuum discussion beyond first order.
- Distinguish the Euclidean Pfaffian and Lorentzian fermion reductions, and retain the grading-compatibility hypothesis in the classification.
- On flat $\mathbb{R}^4$, equation (R1) extends the radial analysis to nonnegative non-Laplace cutoffs; compact support permits a stationary plateau.
- The top-only limiting quadratic action does not establish an unbroken electroweak representation on the full light space.

Overall: 36 statements checked: 18 correct, 15 correct with amendment, 0 gaps, and 3 errors.

Part R6 (referee report, no-go results and lemma 19)

Referee report on the two-model no-go results and lemma 19

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_nogo_19.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemmas 05, 06 and 19.

We checked the statements below by direct mathematical reasoning. No numerical computation was run. Each numbered verdict contributes one count to the final line.

Part 1. Bare limits and odd factorizations

1. The limit $m \to 0$ in Lemma 05, Part 4 Verdict: Correct with amendment. The fixed-rank statement is correct. The phrase "exact continuum limit throughout a fixed positive energy window" should specify the mode of convergence.

At fixed N, $D_N = mT_N$, with T_N independent of m. Hence, for fixed $t > 0$,

$$K_N(t) = \operatorname{Tr} e^{-tm^2 T_N^2} \longrightarrow 2(N-1).$$

The ultraviolet heat dimension is zero for each finite N. The distances between distinct sites are proportional to $1/m$ and diverge.

For $l \to \infty$, the condition $N/l \to \infty$ gives uniform convergence of the finite-chain dispersion to the continuum dispersion on every fixed bounded energy window. It also gives convergence of normalized heat traces and counting densities. It does not by itself give convergence of the *unnormalized difference* $K_N(t) - K_l(t)$ to zero.

To see the distinction, put

$$b = \frac{l}{2N-1}, \qquad k_r = \frac{(r-\frac{1}{2})\pi}{l}, \qquad \theta = \frac{\pi}{4N-2}.$$

Then

$$\lambda_{N,r} = \frac{\theta}{\sin\theta}\frac{\sin(bk_r)}{b}.$$

For example, taking $N = \lfloor l^{5/4} \rfloor$, expansion of the Gaussian sums gives, for fixed $t > 0$,

$$K_N(t) - K_l(t) \sim \frac{l^3}{16\sqrt{\pi}\, t^{3/2} N^2},$$

which diverges, although $K_N(t)/l \to 1/\sqrt{\pi t}$.

Direction. The finite-rank obstruction concerns the stated bare chain and heat trace. A resolved decompactification requires growing rank; the rate should be chosen for the desired notion of spectral convergence.

2. The limit $m \to \infty$ in Lemma 05, Part 4 Verdict: Correct. Every eigenvalue of the massless chiral interval, and of the calibrated finite chain, has magnitude at least

$$m = \frac{\pi}{2l}.$$

Thus, at every fixed cutoff Λ, the light spectral projection vanishes once $m > \Lambda$. At fixed N,

$$\|(D_N - i)^{-1}\| = \frac{1}{\sqrt{m^2+1}} \longrightarrow 0, \qquad d_N(i,j) \longrightarrow 0.$$

The zero resolvent limit on the original nonzero Hilbert space is not the resolvent of a finite self-adjoint operator there. The metric collapse and the empty fixed-energy sector are consequently compatible.

Direction. A nonempty light sector in a collapse calls for altered data, such as a tuned bulk mass or a scaled boundary coupling. The conclusion above applies to the stated massless interval and its calibrated truncations.

3. Proposition A.1 Verdict: Correct. For the stated graded product

$$D_F = D_0 \otimes 1 + \Gamma_0 \otimes D_I, \qquad \{D_0, \Gamma_0\} = 0,$$

the cross terms cancel and

$$D_F^2 = D_0^2 \otimes 1 + 1 \otimes D_I^2 \geq \frac{\pi^2}{4l^2}.$$

This holds for any finite KO-6 factor with the stated product operator, including an l-dependent finite Dirac operator. The fixed-cutoff light sector is empty for sufficiently small l.

Direction. The proposition points to modifications of the interval operator, boundary data, product coupling, or limiting prescription. It does not exclude a light sector in those modified families.

4. Result 3: electrodynamics **Verdict: Correct.** With the displayed matrices, both

$$U_+ = \mathrm{diag}(\sigma_1, \sigma_1), \qquad U_- = \mathrm{diag}(\sigma_1, -\sigma_1)$$

are grading-reversing unitary involutions. They commute with the algebra, its opposite, and D, while

$$U_+ J = J U_+, \qquad U_- J = -J U_-.$$

Thus the same D, J, γ admits both reductions, including at $m = 0$. The displayed representation and Dirac matrix agree, after the stated phase choice, with the electrodynamics data in van den Dungen–van Suijlekom, Eqs. (4.1)–(4.2).

Direction. A selection of one factorization needs additional input beyond the displayed real structure and the listed doubling conditions.

5. Lemma 4: neutral-sector criterion **Verdict: Correct.** In bases related by J, the KO-6 relations make the off-diagonal block M complex symmetric. Takagi factorization gives $M = \Sigma$, with nonnegative diagonal entries.

For $s = +1$, $V = I$ satisfies the criterion. For $s = -1$, the descended antiunitary j has $j^2 = -1$ and commutes with $d^2 = M^* M$. Each eigenspace of d^2, including its kernel, therefore has even complex dimension. Conversely, on each pair of equal entries of Σ, $V = \sigma_2$ is a Hermitian unitary and gives

$$(V^* K)^2 = -1, \qquad V^* \Sigma = (V^* \Sigma)^*.$$

This proves the stated equivalence, including zero singular values.

Direction. A KO-5 reduction can be recovered by neutral-sector pairing or by a symmetry enforcing the required even singular-value multiplicities. The result assumes the same scalar algebra character on the whole sector.

6. Proposition 5: the strict one-generation heavy limit **Verdict: Correct.** In the specified reduced representation, the surviving electron and down-quark sectors have both involutions from Result 3. The neutral sector has basis $(\nu_L, J\nu_L)$, zero Dirac block, and

$$\gamma = \sigma_3, \qquad J = \sigma_1 K, \qquad U = \sigma_1.$$

It supplies a KO-7 reduction. The full light space has dimension 18; a KO-5 reduction would require an antiunitary squaring to -1 on a complex space of dimension 9, which is impossible. The neutral bimodule gives the same obstruction directly.

Direction. The conclusion concerns the stated projection and reduced algebra. Adding an appropriate neutral partner can change the result; imposing particle-number preservation changes the admissible reductions, as in item 9 below.

7. Result 7: three generations **Verdict: Correct with amendment.** State that "arbitrary flavor mixing" means arbitrary matrices within the displayed species blocks, with the heavy projection chosen as a projection onto the specified top Dirac sector. Mixing between inequivalent species is outside the displayed Dirac operator.

The dimension counts are

$$96 - 6 - 12 = 78, \qquad 96 - 6 - 36 = 54.$$

The surviving three left neutrinos give a zero neutral block of size 3, so Lemma 4 gives KO-7 and excludes KO-5. Charged square flavor blocks can be brought to singular-value form by bimodule unitaries.

With three right neutrinos and a purely Dirac matrix Y_ν, the symmetric neutral block

$$M_N = \begin{pmatrix} 0 & Y_\nu^T \\ Y_\nu & 0 \end{pmatrix}$$

has every singular value of Y_ν twice, including zero. For a general symmetric Majorana block, the even-multiplicity criterion of Lemma 4 applies.

Direction. The obstruction is controlled by the neutral multiplicities. Paired neutral content or protected degeneracies provide directions to investigate.

8. No-go theorem 8: the full chiral representation **Verdict: Correct.** For the full displayed representation with $g \geq 1$,

$$e_{\mathbb{H}} = (0, 1_{\mathbb{H}}, 0)$$

has range of dimension $8g$ in H_+ and zero range in H_-. An algebra-linear grading-reversing unitary would identify these spaces. No such unitary exists, independently of the admissible D and of the sign against J.

This conclusion concerns the full quaternionic algebra, the displayed representation and grading, and the doubling criterion of Result 2. A projection that removes only one member of a weak doublet is generally not invariant under the full quaternionic algebra.

Direction. Possible directions include changing the algebra or representation, retaining the even description, and studying boundary or defect realizations of chirality.

9. No-go theorem with $[U, Q] = 0$ **Verdict: Correct.** In the stated strict light representation, an algebra- and opposite-algebra-linear U preserving particle number must map the ν_L particle subspace to an opposite-grading particle subspace with the same bimodule. No such partner is present. The KO-7 involution in Proposition 5 instead maps ν_L to its conjugate and fails $[U, Q] = 0$.

Direction. A particle-preserving reduction requires a matching particle bimodule of opposite grading, or a change in the particle content or imposed conditions. The conclusion holds for either sign against J.

Part 2. Fluctuations and the parent comparison

10. Comparison with Chamseddine–Connes–van Suijlekom **Verdict: Correct.** The formulas in Lemma 19 agree with arXiv:1304.7583v2, §III.2, Eqs. (10)–(11). That paper takes a "self-adjoint, universal one-form" and gives

$$A_{(2)} = \sum_j \hat{a}_j [A_{(1)}, \hat{b}_j], \qquad U = uJuJ^{-1}, \qquad UD(\omega)U^* = D(u\delta(u^*) + u\omega u^*).$$

Here $\hat{x} = JxJ^{-1}$. The formula for $A_{(2)}$ uses the same universal one-form as $A_{(1)}$; replacing it by an independently chosen one-form would change the construction. The algebraic calculation in Theorem 19.1 also extends to universal forms that are not self-adjoint.

For $J = \sigma_3 K$, the alternative pointwise choice, $JDJ^{-1} = -D$. The factor $\epsilon = -1$ in

$$\widetilde{A}_{(1)} = \epsilon J A_{(1)} J^{-1}$$

then gives the same displayed two-point formulas.

11. Theorem 19.1 **Verdict: Correct.** Direct evaluation gives

$$\frac{A_{(2)}}{m} = \begin{cases} \begin{pmatrix} 0 & |\phi_1|^2 \\ |\phi_2|^2 & 0 \end{pmatrix}, & J_0 = K, \\ \begin{pmatrix} 0 & \phi_1\overline{\phi_2} \\ \phi_2\overline{\phi_1} & 0 \end{pmatrix}, & J_r = \sigma_1 K. \end{cases}$$

Adding the two linear terms and D gives both factorizations in the theorem.

The represented self-adjointness condition is $\phi_2 = \overline{\phi_1}$. In this two-point example the universal one-forms are spanned by $p_1\delta p_2$ and $p_2\delta p_1$, which are adjoints and are represented faithfully for $m > 0$. Thus the same condition characterizes self-adjoint universal forms. Setting $\Phi = 1 + \phi_1$ gives

$$D'_{J_0} = m|\Phi|^2\sigma_1, \qquad D'_{J_r} = m\begin{pmatrix} 0 & \Phi^2 \\ \overline{\Phi}^2 & 0 \end{pmatrix}.$$

12. Corollary 19.2 Verdict: Error. The gauge actions and full fluctuated images in items 1–2 are correct. The common linear-only formula in item 3 is incorrect for J_0.

Let $z = e^{i\psi}$. On the gauge orbit of the zero universal form,

$$A_{(1)} = m\begin{pmatrix} 0 & z-1 \\ \bar{z}-1 & 0 \end{pmatrix}.$$

Removing $A_{(2)}$ gives the upper links

	upper link	modulus near $\psi = 0$
J_0	$m(2\cos\psi - 1)$	$m(1 - \psi^2 + O(\psi^4))$
J_r	$m(2e^{i\psi} - 1)$	$m(1 + \psi^2 + O(\psi^4))$.

For example, at $\psi = \pi/3$ the J_0 linear-only operator is zero, while the stated J_r expression has link $i\sqrt{3}\,m$. Adding $A_{(2)}$ restores the stated full results: $D' = D$ for J_0, and the link is mz^2 for J_r.

Two wording amendments accompany the correction. The set $\Phi = e^{i\psi}$ is the orbit in the potential parameter; the J_0 operator orbit consists only of D. The map $\Phi \mapsto \Phi^2$ is two-to-one away from zero and one-to-one at zero. The image statement is pointwise; a globally prescribed smooth link need not have a global smooth square root.

13. Reflection real structure and compression Verdict: Correct. In Counterexample 08.4,

$$J_{\text{ref}} = \sigma_1 R_l K.$$

For the real lowest-pair basis

$$u_L = \sqrt{2/l}\,(\cos mx, 0), \qquad u_R = \sqrt{2/l}\,(0, \sin mx), \qquad m = \pi/(2l),$$

the identities $\cos(m(l-x)) = \sin(mx)$ and $\sin(m(l-x)) = \cos(mx)$ give

$$J_{\text{ref}}u_L = u_R, \qquad J_{\text{ref}}u_R = u_L.$$

The subspace is J_{ref}-invariant, and its compressed antiunitary is exactly $\sigma_1 K$. Its signs are $(J^2, JD/DJ, J\gamma/\gamma J) = (+,+,-)$.

For $e = \text{diag}(1,0)$, the two compressed real structures give

$$[[D,e],\hat{e}] = m\sigma_1 \quad (J_0), \qquad [[D,e],\hat{e}] = -m\sigma_1 \quad (J_r).$$

Both two-point data satisfy order zero and fail order one.

14. Proposition 19.3, item 1 Verdict: Correct. Without a real structure, self-adjoint finite one-forms give every upper link $m(1+\phi_1) \in \mathbb{C}$. For fixed compression of the real parent gauge coefficient A_5, the link is

$$m + iB, \qquad B = \frac{1}{l}\int_0^l \sin\frac{\pi x}{l}\,A_5(x)\,dx \in \mathbb{R}.$$

Every real B is reached, so the fixed-compression image is the stated vertical line.

15. Proposition 19.3, item 2 **Verdict: Correct with amendment.** "A neutral radial field with no parent scalar" is fair when it means: *the finite pointwise-real fluctuation has a gauge-neutral radial parameter, while the specified pointwise-real parent has no surviving self-adjoint scalar inner fluctuation.* It does not establish a dynamical scalar field, its action, or the absence of parent fields in other models.

Let $E = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. A represented parent one-form is $Eh(x)$. Self-adjointness requires $h = ia$ with a real, and

$$K(iEa)K^{-1} = -iEa.$$

Thus the parent real fluctuation cancels. By contrast, the finite gauge-invariant link is

$$t = m|\Phi|^2 \geq 0,$$

so t is a function of the radial variable $|\Phi|$, and every $t \geq 0$ occurs.

The proof at line 71 needs this self-adjointness step. An individual commutator does not generally change sign under K: for $f = x$, $K[D, f]K^{-1} = [D, f] = E$. The sign reversal holds for the self-adjoint represented sum iEa.

The parent does possess an A_5 one-form before real symmetrization; the cancellation concerns the resulting real fluctuation. Compression of the algebra is also nonmultiplicative, so the construction gives no canonical identification of all finite potentials with parent potentials.

16. Proposition 19.3, item 3 **Verdict: Correct.** The reflected opposite algebra consists of multiplication operators, so the parent satisfies order one. For real A_5,

$$J_r D_{A_5} J_r^{-1} = D_{A_5 \circ R}.$$

Consequently its real scalar fluctuation has coefficient

$$A_5^s = A_5 + A_5 \circ R.$$

Since the compression weight is reflection-even, $B^s = 2B$, where B is calculated from the unsymmetrized A_5.

17. The table in Proposition 19.3 **Verdict: Correct with amendment.** Label the real structure and the meaning of B in each row. The covariant projection has fixed singular value m, and the full finite J_r orbit has link modulus m. The linear-only row has the stated modulus for J_r, locally growing at second order.

The fixed-compression row

$$\sqrt{m^2 + B^2}$$

is the formula of Lemma 13 for the unsymmetrized coefficient. For the reflection-real parent of item 3, if B continues to denote the integral of the original A_5, the corresponding expression is

$$\sqrt{m^2 + (B^s)^2} = \sqrt{m^2 + 4B^2}.$$

Equivalently, the table may retain its formula after defining its B to be the integral of the *symmetrized* coefficient. The increase is quadratic along paths for which the relevant B has a nonzero linear term; a nontrivial gauge path can also have $B = 0$.

18. Wilson phase in the paragraph following the table **Verdict: Error.** For a constant unsymmetrized $A_5 = a$, the transported link and its expansion are

$$mW_l^* = me^{ial} = m + i\,mla + O(a^2) = m + i\frac{\pi a}{2} + O(a^2).$$

Thus the Wilson **phase** is al; $mla = \pi a/2$ is the coefficient of the first-order imaginary change in the link. The fixed-compression value is $m + i(2a/\pi)$.

For the reflection-real fluctuation starting from the same a, the symmetrized coefficient is $2a$: the transported link is me^{2ial}, and the fixed-compression link is $m + i(4a/\pi)$.

19. Reading at the end of Lemma 19 Verdict: Correct with amendment. Replace "truncation generically breaks order one" by the demonstrated statement that spectral compression *can* break it, and state the hypotheses needed to apply the cited framework: a represented algebra, a preserved real structure, and order zero.

For example, on the circle take $D = -i\,d/dx$, $J = K$, and the full multiplication algebra. The parent satisfies order zero and order one. Compression to the zero Fourier mode leaves only scalar operators and preserves both conditions. Compression instead to the modes $-1, 0, 1$ sends e^{ix} to a truncated shift S; its conjugate is S^*, and

$$[S, S^*] \neq 0.$$

Thus this compression fails order zero. It also illustrates why the compressed operator system and the algebra it generates require separate attention.

The proposed relation to a singlet remains an analogy. In the cited 1304.7583v2, Proposition 8 and §IV, the toy model uses three complex fields, and the authors state that its quadratic term "does not generate new scalar fields"; it is expressed in fields already present in $A_{(1)}$. The present $A_{(2)}$ likewise depends on the same two-point parameters, but that fact gives no identification of the models. Devastato–Lizzi–Martinetti, arXiv:1304.0415v3, §4.3, obtain their σ from a Majorana-operator fluctuation in a larger algebra. An identification would require a map of algebras, representations, fields and gauge actions, together with an action comparison.

Amendments and errors

- Specify the convergence asserted in the $m \to 0$ continuum limit; $N/l \to \infty$ controls dispersion and normalized densities.
- Restrict the three-generation mixing statement to the displayed species blocks and specified heavy projection.
- In the parent comparison, use self-adjointness to prove the pointwise-real cancellation, and distinguish the original A_5 from its reflection symmetrization.
- Correct Corollary 19.2's J_0 linear-only link and the Wilson phase following the table.
- Qualify the general reading of truncation and retain the singlet comparison as an open analogy.

Overall: 12 Correct; 5 Correct with amendment; 0 Gap; 2 Error.

Part R7 (referee report, lemmas 20 and 21, Proposition I)

Referee report on lemmas 20 and 21, the comparison in the steer note, and Proposition I

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_20_21.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemmas 20 and 21, the steer note and the paper.

The review finds errors in the marked-point reading, the radial statement, parts of the scale comparison, Proposition I, and the no-go table. The connecting-operator calculations and the three consensus checks otherwise hold in their stated scopes.

The checks below are analytic; no numerical code was run. Source references identify the arXiv versions read.

Part 1. Connecting operators 1. Correct. In Result 1, block multiplication gives $T = cT^*c$ and $hT + Tg = 0$. The lower off-diagonal block of order one is (2); the other follows using adjoints and the involutive algebra. In c-paired bases, the first condition is complex symmetry.

The averaging proof gives $T = R + \tau(R)$, with R intertwining the left actions and $\tau(R)$ the opposite actions. Conversely, such a decomposition gives order one on $\mathcal{A}(R)$, establishing (4). The final example has one arrow

preserving the right character and the other preserving the left character; it satisfies order one although B is not contained in $\mathcal{A}(T)$. No correction is needed.

2. Correct. For the displayed rank-one T_κ, with $\kappa \neq 0$, the two selected lines must reduce their respective actions and carry the same scalar. This gives (6), of real dimension $2 + 4 + 18 = 24$. The self-adjoint off-diagonal operator has complex rank 2.

Chamseddine–Connes, §4, equations (4.6)–(4.7) use an auxiliary intertwiner, also called T there. Its compression $K : \mathbb{C}^2 \to \mathbb{C}^4$ has rank one or two. Rank two gives dimension at most $4 + 4 + 8 = 16$; rank one gives dimension 24, uniquely up to automorphism. For the displayed block, $T_\kappa x = \kappa K x K$. For a general admissible connector, the paper's K is obtained from an auxiliary intertwiner and need not equal the Dirac block. No correction is needed.

3. Correct. Result 3 and Counterexample 4 have the stated domain-preservation equations. In the scalar example, the parent has signs $(+, +, -)$ and satisfies both order conditions. The two displayed chiral vectors form the window for

$$\theta/l < \Lambda_0 < (\pi - \theta)/l.$$

Their compressed algebra is the full diagonal $\mathbb{C}^2$, and for $p = \mathrm{diag}(1, 0)$,

$$[[m\sigma_1, p], p^\circ] = -m\sigma_1.$$

The negative conclusion concerns compression of the full smooth scalar gluing algebra in this window. Restricting the parent algebra, or changing the window or representation, changes the problem; Result 5 gives a positive constant-algebra construction.

4. Correct. In Result 5, (18) and (19) preserve the boundary domain. The equality $cX = X^* c^{-1}$ is the required compatibility for J. The window contains $2n - 2$ uncoupled zero modes and the pair $\pm m$, giving (20). Constant $B = \mathcal{A}(S)$ multipliers commute with the parent Dirac operator, so the stated order-one claim follows.

With the displayed conventions, $U(x) = e^{mxF}$ carries the uncoupled domain to the coupled one, and differentiation gives (21), including its sign. Its compression gives $m(S + S^*)$. The construction assumes the Hom fibres, grading, constant multipliers and coupled channel.

5. Correct with amendment. The rank bound in Result 6 follows from the factorization $T = \mathrm{ev}_2^* \, X \, \mathrm{ev}_1$. A point with N fibre channels can give rank N. An exact change of boundary domain does not itself establish that factorization.

The separation argument in Result 7 is valid for unchanged finite windows when independent test functions can be supported away from the gluing sets. It applies to the stated point and fibre examples, including S^4 pullbacks after fibre integration. For a general subset upstairs in the twistor space, add the hypothesis that its image in S^4 has empty interior, so that the required independent base test functions exist. A new coupled window, boundary degrees of freedom, or a changed observable representation lies outside the result. The direction is to construct and analyze such a coupled parent.

6. Correct. Hořava–Witten, §2, states $\Gamma_{11}\chi = \chi$ for the intrinsic boundary gaugino. With a common ten-dimensional chirality convention and the standard common orbifold lift, the same-chirality comparison uses two type-(a), or two type-(b), fields. Opposite localization masses can place them near opposite ends.

For unbroken $E_8 \times E_8$, the displayed Hom space vanishes, and an operator odd for the common chirality has no matrix elements between the two retained same-chirality modes. Compactification, symmetry breaking, or additional bulk fields may give different effective couplings; those require a separate construction.

7. Correct. The three checks in Results 9–11 hold:

- The assumed weighted mass relation gives $m_{\max}/(2\sqrt{2}) \leq M_W \leq \sqrt{N_g}\, m_{\max}$. It forces the maximum to grow, with a fixed species necessarily unbounded only along some subsequence. The relation belongs to the specified asymptotic matching action; a full finite-cutoff calculation is a direction beyond it.

- With pointwise reality and a J_0-invariant window, inherited order zero holds exactly when the algebra generated by the compressed functions is commutative. A separate opposite action, as in the coarse correspondence, changes that hypothesis.
- If $P\gamma P$ is an actual grading, its square forces $[P, \gamma] = 0$. Thus it commutes with compressed even observables, whereas Γ_5 fails to commute with $Q(x_1)$. An added finite grading and restriction to its even algebra evade that particular inheritance claim.

The cited repository reports record earlier checks by another model. This review is a further solo check of the displayed arguments.

Part 2. Readings 8. Correct with amendment. In the CCM preprint, §2.5–2.6, especially (2.34), the right-handed up lepton is ν_R, and the off-diagonal block maps it to its conjugate through a symmetric Majorana matrix. After the stated identification, and an overall grading convention if needed, the one-generation connector has that form.

The interval construction gives one mass $m = \arctan(t)/l < \pi/(2l)$, with next positive absolute level $(\pi - \arctan t)/l$. Three identical copies give three equal masses. A general three-generation CCM Majorana matrix requires additional flavour data; it is not supplied by the single rank-one construction.

9. Error. The projector identity and

$$2\Pi_p - 1 = p_a \Gamma_a = 5Q(p \cdot x)$$

are correct. The pair of projective points determines the two lines, but it does **not** determine the connector's magnitude. Rephasing w and v also changes KxK by a phase. At fixed nonzero $|\kappa|$, the line choices are conjugate under the stated groups; $|\kappa|$ remains a modulus, and other finite data can introduce relative moduli.

The exact correction is to specify the lines **and** a coupling, together with an evaluation and normalization prescription. Moreover, $5Q(p \cdot x)$ fails to commute with the other coordinate compressions. It can be chosen as additional finite grading data after restricting to its even algebra. That evades Result 11's inherited-grading hypothesis; it does not derive the grading from the even parent or construct the proposed point interface.

10. Correct with amendment. Equation (21) gives a constant matrix connection after a unitary change of domain, with relative endpoint rotation angle $\theta = ml$. On an interval, the open-path transport depends on endpoint frames; the physical quantity is the transport together with the specified boundary conditions. A closed Wilson loop or a dynamical gauge field needs further structure.

The gauge-Higgs comparison is an analogy at this stage: the interval calculation supplies a fixed matrix connection and no Higgs fluctuation or effective potential. The reference is Yutaka Hosotani, "Dynamical mass generation by compact extra dimensions," *Physics Letters B* **126** (1983), 309–313. For the dynamical Wilson-line and gauge-Higgs construction in an orbifold setting, see Hosotani, Noda and Takenaga, §1.

Part 3. Counting 11. Correct with amendment. In Proposition 21.1, $(q_L - q_R)/2 = (T_3/2)(1, -1)$, so the additional term in lemma 18's (33) is annihilated by the photon vector $(1, 1)$. The stated sums and removal entries reproduce:

removed	$\cos^2\theta_{\mathrm{NC}}$	$\kappa_{\mathrm{em}}/\kappa_s$
u-type	5/8	8/3
d-type	59/92	46/15
charged lepton	17/28	7/3

The table assumes the stated three-generation charges and that the other species have the mass assignments used in the removal calculation. At the endpoint where the neutral mass vanishes, the massive neutral direction is no longer distinguished: read its angle as the limit from finite mass, as lemma 18 states. Other weak-angle definitions need not have these values.

12. Error. The trace $84 + 12e^{-x}$ is correct for a top-only Dirac mass and no Majorana mass. With the positive heat cutoff and $m > 0$, its radial derivative is strictly negative. The text says there is "no finite stationary point"; at $m = 0$ the derivative is zero.

Replace that sentence by: **there is no finite nonzero stationary point on this free radial ray**. This concerns the literal flat-space action with no added potential or constraint. Compact support can give a plateau; sign-changing cutoffs, additional terms, constraints, and paths with an additional fixed noncommuting mass lie outside the strict conclusion. The result points to analyzing those changed actions or prescribed backgrounds.

13. Correct with amendment. Complexifying the real 30 gives

$$(\mathbf{5}, \mathbf{3}) \oplus (\bar{\mathbf{5}}, \bar{\mathbf{3}}),$$

and the displayed exterior-square branching has dimensions

$$192 + 24 + 8 + 1 + 60 + 45 + 60 + 45 = 435.$$

The charge tables and the charged scalar-partner counts follow. The charge operator includes the $U(1)$ commuting with $SU(5)$:

$$Q = \mathrm{diag}(3/5, 3/5, -2/5, -2/5, -2/5) + \tfrac{1}{15} I_5.$$

Also,

$$\mathbf{496} \to (\mathbf{435}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{1}) \oplus (\mathbf{30}, \mathbf{2})$$

under $SO(30) \times SO(2)$.

These are representation and charge counts. Identifying the entries with physical scalar partners requires a field construction, spin and chirality assignments, interactions and dynamics. Rivero's preprint, Tables 1–2, contains the stated diquark and meson counting; its proposed particle interpretation remains a proposal.

14. Correct with amendment. Both decompositions and the overlap of 72 follow by counting four internal states per Dirac species and colour. Without ν_R, the first family has 78, and the charged family remains 84. A massless three-form in eleven dimensions and its six-form dual each have $\binom{9}{3} = \binom{9}{6} = 84$ on-shell polarizations.

The 84 in H_F is a complex finite-Hilbert-space dimension; the three-form count is a real bosonic polarization count. A physical identification needs a map of representations and degrees of freedom.

15. Correct with amendment. Seiberg, "Electric-Magnetic Duality in Supersymmetric Non-Abelian Gauge Theories," §1–3, *Nuclear Physics B* **435** (1995), 129–146, gives the four-dimensional $\mathcal{N} = 1$ supersymmetric result. For $N_c = 3, N_f = 5$, $9/2 < 5 < 9$; the magnetic group is $SU(2)$, with five flavours and a 5×5 gauge-singlet meson chiral superfield coupled by $W = Mq\tilde{q}$.

To turn readings 1–3 into statements, one needs:

- An $\mathcal{N} = 1$ supersymmetric theory with the required massless matter and dynamics, plus an operator and anomaly map. Five-flavour ordinary QCD alone does not supply Seiberg duality, and the magnetic $SU(2)$ needs a separate identification with any electroweak group.
- A representation and dynamics relating the finite fermionic states to the bosonic form or brane degrees of freedom; equality of the number 84 is insufficient.
- For the KO-label reading, the required involution on the stated finite sector, including its action on D, J, the algebra and its opposite. The vectorlike $SU(3) \times U(1)_{\mathrm{em}}$ charges alone do not establish it, and neutral states still require treatment.

Part 4. Comparison with repository results **16. Correct with amendment.** Example 12.5 gives

$$\tan^2(kl) = |X|^2, \qquad k_{\mathrm{light}} = \frac{\arctan|X|}{l}.$$

For small X, $k_{\text{light}} = |X|/l + O(|X|^3/l)$; the half-integer spectrum occurs as $|X| \to \infty$. At $X = 0$, zero has multiplicity two and must be omitted from the positive spectrum.

With $X(l) = \tan(ml)$, the light mass is exactly m and the remaining levels diverge as $l \to 0$. Identifying X with a physical Yukawa coupling, or $1/l$ with v, remains an extra hypothesis. The Wilson-line wording needs the qualification in item 10.

17. Error. CCM, (4.12) gives

$$\mu_0^2 = 2\frac{f_2}{f_0}\Lambda^2 - \frac{e}{a}.$$

Top dominance in the Dirac Yukawa trace alone does not force $e/a = 0$: the Majorana–neutrino term can remain relevant. Under exact top-only data, or the additional assumption $e = 0$, the stated truncated-potential relations give

$$\frac{m_t^2}{\Lambda^2} = 2\frac{f_2}{f_0}.$$

For the heat profile $F(u) = e^{-u^2}$, $f_2/f_0 = 1/2$, giving $x = 1$. This is a stationary point of the **truncated** potential; lemma 18's literal positive heat action has no finite nonzero stationary point on the free top-only radial ray. Its exact quadratic-background calculation at $x = 1$ gives the stated departure from the small-mass ratio.

18. Error. As written, "finite low-energy electromagnetic coupling" includes a coupling tending to zero. The trajectory in the transcript provides that case: take $\Lambda = \mu_* = v$, keep $\Lambda_5 > 0$ fixed through one-loop matching, and choose

$$\frac{8\pi^2}{g_2^2} = \frac{23}{3}\log\frac{v}{\Lambda_5} - \frac{1}{3}\log\log\frac{v}{\Lambda_5} + O(1).$$

Then $g_2 \to 0$, $m_t/\Lambda \to 0$, the top-only Taylor ratio tends to $\sqrt{8/3}$, and the low-energy electromagnetic coupling tends to zero. The text itself lists that endpoint as an escape.

The intended no-go can be stated with **nonzero limiting** α_{em} and positive finite limiting Λ_5, a matching scale above the relevant thresholds, top dominance at matching, the stated one-loop running, and the top-only heat-cutoff Taylor mass. In a step-threshold regime with $\mu_0 < m_t < \mu_*$ and nonincreasing α_{em}^{-1} toward μ_*,

$$\alpha_{\text{em}}^{-1}(\mu_0) \geq \frac{28}{3\pi}\log\frac{\mu_*}{\Lambda_5}.$$

This follows from $b_6 = 7$, $b_5 = 23/3$, and $\alpha_{\text{em}}^{-1}(\mu_*) = \frac{8}{3}\alpha_3^{-1}(\mu_*)$. It bounds μ_*/Λ_5; matching then bounds g_2 away from zero, so $m_t \asymp g_2 v$ eventually conflicts with $m_t < \mu_*$.

The signs quoted in the note are correct in their respective active-field ranges: $\sum Q^2 = 20/3$ for five active quarks and three charged leptons, and $b_2 + b_Y = -11/3$ above electroweak thresholds. The thresholds are set by masses, rather than simply by v. The written route $m_t/\Lambda \to \infty$ cannot simultaneously use six-flavour matching at $\mu_* = \Lambda$ above the top. Extra thresholds or gauge mixing, altered matching, a different action or mass definition, and an $e \to 0$ endpoint are directions outside the amended no-go.

19. Error. The one-loop threshold relation

$$\Lambda_5 = \Lambda_6^{21/23} m_t^{2/23}$$

is correct under the transcript's matching assumptions. Its $n_f = 5$ and Seiberg's $N_f = 5$ count different theories; the equality of the numbers establishes no duality.

The transcript says pions *do not have to be* elementary KK modes or finite-algebra fluctuations. The comparison changes that into a stronger exclusion. Proposition 17.10 establishes absence of a kinetic term for the specified coordinate change of a fixed operator; it leaves a new operator or quantum effective action available. Proposition 17.8 gives the stated Banks–Casher order of limits under its measure and regularity assumptions.

For $T^3 = \operatorname{diag}(1/2, -1/2)$ and $Q = \operatorname{diag}(2/3, -1/3)$, the algebraic coefficient is $2N_c \operatorname{tr}(T^3 Q^2) = 1$. The reduced action supplies a starting point; a regulated fermion measure and anomaly calculation remain to be supplied.

Part 5. Paper 20. Error. Proposition I omits $\kappa \neq 0$ in item 2. At $\kappa = 0$, the block has rank zero and $\mathcal{A}(T_\kappa) = \mathcal{A}^{\mathrm{ev}}$, of real dimension 40. Add $\kappa \neq 0$.

Item 1 must also assume that the sheet-diagonal D_F already has the stated signs and order one. In item 3, specify a symmetric cutoff satisfying

$$m < \Lambda_0 < (\pi - \theta)/l.$$

The Majorana identification in the following paragraph needs the basis and flavour qualifications in item 8. The marked-pair paragraph needs the coupling modulus and grading qualification in item 9.

The abstract's wall-coupling sentence is supported for a **nonzero coupling on the chosen decomposable channel**, with the constant matrix observable algebra and the stated fibres and grading supplied as inputs.

21. Error. The table at lines 330–349 needs the following row-by-row corrections.

Row	Assessment and correction
Odd bulk	Say **no net chiral spectrum** under the specified standard smooth closed reduction. A periodic massless Dirac field on a circle has left- and right-handed zero modes. Boundaries and defects change the hypotheses. Baptista, §1, studies chiral interactions of massive non-Killing gauge fields; that is a distinct escape from an exact-isometry assumption.
Odd factorization	State the full chiral Standard Model representation and the particular involution criterion of Theorem D. Reducing to invariant vectorlike sectors changes the algebra and representation.
Only KO-7	State the scalar-character neutral sector and the odd multiplicity, including zero singular values, in Theorem D. Even multiplicity and appropriate Dirac pairing are the stated alternatives.
Empty light sector	Specify the **massless type-(c)** interval and fixed spectral cutoff. A massless type-(a) interval has a zero mode for every l. Tuned bulk mass and scaled boundary coupling give the stated alternative families.
Fixed compression	Specify the local gauge transformations and fixed light projection of Proposition F. A projection invariant under a given transformation does not have this defect. The covariant projection changes with the fluctuated operator.
First order	Specify the two-point compression and inherited pointwise J. Fluctuations without order one relax that axiom; the doubled parent changes the data.
Order zero	Add J_0-invariance of the window and the inherited pointwise reality. A doubled or otherwise changed opposite action lies outside those hypotheses.

Row	Assessment and correction
Bounded-fermion matching	Keep the leading spectral-action order, positive kinetic form, stated trace and common profile space. A tower-scale fermion removes the bound's obstruction; matching the full actions needs more.
$M_W \to 0$	Specify **massless complete doublet**, the stated heavy-doublet family, the Taylor definition, and the cutoff hypotheses including $f(0) > 0$. Keeping a doublet with a fixed nonzero mass can leave a positive residual mass even for the heat cutoff. Moreover, for the nonnegative $f(z) = ze^{-z}$, $f(0) = 0$, the top-only Taylor mass instead has $M_W^2/\Lambda^2 \sim x^2/3$. The heat-cutoff pole result has its own analytic-continuation hypotheses. A joint cutoff limit, altered light sector, or quantum effective action changes the problem.
Radial vacuum	Specify an allowed nonzero ray $V = cV_*$, $c > 0$, flat background, adequate integrability, and no added term or constraint. For nonnegative cutoff of unbounded essential support the derivative is negative. Compact support can give a plateau; other listed changes are possible directions.
Grading	Replace "grading not inherited" by "the proposed Γ_5 grading is not inherited" under the four hypotheses of Result 11. The Dolbeault window itself has an inherited scalar grading. A compressed height function can be added as finite data with a restricted even algebra.
Scalar gluing	Add closed empty-interior gluing sets and independent functions supported away from them, with unchanged separate windows. Matrix-valued multipliers alone need not evade the same separation argument; a restricted multiplier algebra, changed representation, or coupled window can.
Finite limits	Specify finite limits of **connected scalar algebras** in the stated spectral-propinquity class. Truncation, matrix-valued parent algebras, or another convergence notion changes the question and needs a separate construction.

The vector-mass row points to exact calculations with the desired light masses and cutoff, while the grading and gluing rows point to an explicitly constructed coupled parent.

Amendments and errors

- Amend the pullback scope in **5**, the flavour and grading conventions in **8**, and the interval transport terminology in **10**.
- Amend the endpoint and degree-of-freedom interpretations in **11, 13–16**.
- Correct the missing connector data in **9**, the stationary-origin statement in **12**, and the missing Majorana term in **17**.
- Replace **18** by the explicitly nonzero-coupling, threshold-matched no-go, with the $e \to 0$ trajectory retained as an escape.

- Correct the transcript attribution in **19**, the hypotheses and marked-point summary in **20**, and the individual no-go rows in **21**.

Overall: 6 Correct; 8 Correct with amendment; 0 Gap; 7 Error.

Part R8 (referee report, lemma 22)

Referee report on lemma 22

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_22.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 22.

We find two errors in the stated claims: the assignment of zero modes at nonzero wall coupling, and the identification of fixed compression with the first derivative of transport. The main interval construction and its compression formulas hold, subject to the amendments below. The numerical collider-exclusion claim remains unsupported.

Throughout, let $\ell > 0$, and write $D = D_F$, $\Gamma = \Gamma_F$, and $J = J_F$ when no confusion can arise.

1. Setting — Correct.

In bases related by c, the condition on M says that its matrix is complex symmetric. Let its Takagi factorization be

$$M = U\Sigma U^T.$$

Then

$$|M| = \overline{U}\,\Sigma U^T, \qquad V = UU^T$$

give $M = V|M|$, with V unitary and $V = cV^*c$. This construction includes the kernel: zero entries of Σ still have corresponding columns in the unitary U. The extension of the polar factor on the kernel need not be unique.

The formula

$$X = \tan(\ell|M|)V^*$$

and the reality condition $cX = X^*c^{-1}$ follow. Conversely, for every X satisfying that condition, the principal functional calculus gives the unique operator

$$D_F = \ell^{-1}\arctan\begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \qquad \ell\|D_F\| < \pi/2.$$

Thus the claimed bijection is valid. The expansion $X = \ell M^* + O(\ell^3)$ assumes fixed M.

2. Theorem 22.1 — Correct with amendment.

The boundary subspaces are Lagrangian for the first-order boundary form, giving self-adjointness; compactness of the H^1-to-L^2 embedding gives compact resolvent. Direct substitution verifies the stated signs and all three intertwining relations. The L- and R-components are orthogonal, while $R(x) = e^{x\Gamma_F D_F}$ is unitary, so W is an isometry.

For each singular value σ_k, with $\theta_k = \ell\sigma_k$, the signed spectrum is

$$\frac{n\pi \pm \theta_k}{\ell}, \qquad n \in \mathbb{Z},$$

with multiplicities added when values coincide. This gives the stated spectrum of $|D_X|$, the window, and the lower bound on the next level. The amendment is to state multiplicities, particularly when $\sigma_k = 0$: the zero mode then has multiplicity two per zero singular channel, and the two excited branches coincide.

For arbitrary signed momentum k, the scalar equation is

$$|\tan(k\ell)| = \sigma_k(X).$$

For the positive light mass $k = \sigma_k(D_F)$, its tangent is nonnegative and equals the corresponding singular value of X.

3. Corollary 22.2 — Correct.

On the uncoupled domain, multiplication by R maps the endpoint conditions to those of D_X, and

$$R^* D_X R = D_0 + E\Gamma_F D_F.$$

For one real channel, $\Gamma_F D_F = mF$, with F as defined in equation (21) of lemma 20. The sign and the placement of the spin matrix E agree.

4. Standard Model form and boundary types — Error.

The type assignment holds **at** $X = 0$: Q has type (a) and U has type (b). For $X \neq 0$, the wall conditions couple the fields, and the corresponding zero modes generally lift. For example, one channel with $X = t > 0$ has lowest masses $\pm \arctan(t)/\ell$ and no zero mode.

The correction is to say that the coupled modes continue from the type-(a) and type-(b) zero modes of the uncoupled problem.

One matrix X does encode the full finite operator, including Yukawa and Majorana data. Its entries need not have only the original block pattern: the tangent introduces higher powers of D_F. For example, with real symmetric

$$M = \begin{pmatrix} a & b \\ b & 0 \end{pmatrix},$$

the corresponding entry of $X = \tan(\ell M)$ is

$$X_{22} = \frac{\ell^3}{3} ab^2 + O(\ell^5).$$

Thus the original zero block can acquire an induced entry.

5. Theorem 22.3 — Correct with amendment.

Assume explicitly that the finite algebra and its faithful even representation are unital. The two endpoint intertwining equations are equivalent to

$$[\pi(f(\ell)), D_F] = 0.$$

The commutant of a self-adjoint operator is closed under products and adjoints. Hence C_F, and consequently A_ℓ, are unital $*$-algebras.

For every $f \in A_\ell$, multiplication preserves the whole H^1 domain and

$$[D_X, \pi(f)] = E\pi(f')$$

there. The right side extends to a bounded operator on all of H_ℓ. The opposite action also preserves the domain: this follows from domain preservation by g^* and $J^\wedge D_X = D_X J^\wedge$. Finite order zero then gives

$$[[D_X, \pi(f)], \pi^\circ(g)] = E[\pi(f'), \pi^\circ(g)] = 0.$$

No finite order-one hypothesis is needed.

This agrees with Counterexample 4 of lemma 20: its parent has order one, while its two-point compression fails finite order one. The statement concerns this parent and its pointwise algebra; it does not assert that arbitrary compressions preserve either order condition.

The Standard Model identification of C_F also needs the usual nonzero Yukawa data and the specified representation. For example, if all Yukawa blocks vanish and only an algebra-commuting Majorana block remains, the commutant is larger.

6. Theorem 22.4 — Correct with amendment.

Both compression formulas, including the minus sign in $Q_\ell(ET)$, and the symmetric exponential average for constant a are correct. The four items hold with the stated constants. In particular, interpreting $\|a\|$ as $\|\pi(a)\|$,

$$\|Q_\ell(a) - \pi(a)\| \le \frac{\ell^2}{6}\|[D,[D,\pi(a)]]\| \le \frac{2}{3}\ell^2\|D\|^2\|\pi(a)\|.$$

The constants in item 4 are respectively $8/3$ and $16/3$.

For item 1, compactly supported bulk functions suffice to test every smooth matrix-entry density. Finite-dimensional linear algebra then gives equality of the two images. The constant function a itself need not belong to A_ℓ; the theorem supplies a domain-preserving function with the same compression.

The general warning about generated algebras is sound. It also has a simple analytic example. Let

$$M = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & M \\ M & 0 \end{pmatrix}, \quad \pi(\mathbb{C}^3) = \{\mathrm{diag}(a,a) : a \text{ diagonal}\},$$

with $J = \left(\begin{smallmatrix} 0 & K \\ K & 0 \end{smallmatrix}\right)$. If P_1, P_2 are the first two diagonal channel projections, then on either graded block

$$[Q_\ell(P_1), Q_\ell(P_2)]_{12} = -\frac{\ell^2}{3} + O(\ell^4).$$

The original projections commute, whereas their compressed images generate a noncommutative algebra for sufficiently small nonzero ℓ.

The specific dimensions 15 and 30 remain **numerical claims**: the matrices and closure calculation were not retained, and the reported tolerance alone does not establish an exact rank. They should be accompanied by reproducible data and an exact or certified rank calculation. The general role of operator systems under spectral truncation is already developed in Connes and van Suijlekom, §1 and §3, arXiv:2004.14115v2.

7. Theorem 22.5 — Correct with amendment.

Every represented self-adjoint one-form is $E\pi(h)$, with $h^* = -h$. In fact **every smooth** such h arises, including those with nonzero boundary value: set

$$b(x) = \int_\ell^x h(t)\,dt.$$

Then $b(\ell) = 0$, so $b \in A_\ell$, and $[D_X, b] = E\pi(h)$.

Thus imposing compact support would exclude some connection profiles, although it loses no endpoint holonomy in the identity component: any smooth path from 1 to such a unitary can be made constant near both endpoints.

The transport formula and wall condition are correct. The proof should separate its two cases. For a general w, $\pi(w)^* D_F \pi(w)$ need not commute with J_F, so the non-real part uses the spectral and isometry proof of Theorem 22.1 **without** its real-structure hypothesis. For the real fluctuation, $U_w = \pi(w) J_F \pi(w) J_F^{-1}$ commutes with J_F, and the claimed real window follows.

There is a useful stronger statement. Put

$$\tilde{u}(x) = u(x)w^*.$$

Then $\tilde{u}(\ell) = 1$, so $\tilde{u} \in A_\ell$. With

$$\widetilde{U} = \pi(\tilde{u}) J_F \pi(\tilde{u}) J_F^{-1},$$

direct differentiation and the endpoint condition give

$$D_X + A + J^\wedge A (J^\wedge)^{-1} = \widetilde{U} D_X \widetilde{U}^*.$$

Thus every scalar connection in this interval model is gauge equivalent to the original parent under its allowed endpoint gauge group.

8. Corollary 22.6 — Error.

The gauge-orbit comparison itself is correct. Chamseddine–Connes–van Suijlekom, §III.2, equations (10)–(11), arXiv:1304.7583v2 give the quadratic fluctuation and its gauge transformation. Here the correct substitution is $v = w^*$, so the moving window carries

$$U_v D_F U_v^* = U_w^* D_F U_w.$$

For the two-point example, if $w = \mathrm{diag}(e^{i\alpha}, e^{i\beta})$, the upper link is $me^{-2i(\alpha-\beta)}$. It is $me^{2i\psi}$ in lemma 19 when ψ is defined using $v = w^*$.

The error is the assertion that fixed compression is the first derivative of the transported window for the **same** connection. Let h be constant and replace it by εh. At fixed D,

$$\frac{d}{d\varepsilon}\Big|_0 D_{\text{transported}} = -\ell[D, h],$$

$$Q_\ell(Eh) = -\frac{\ell}{2}[D, h] + O(\ell^3).$$

For example, take $D = m\sigma_1$ and $h = i\sigma_3$. Their leading terms differ by a factor of two. The real fluctuation doubles both expressions in this example and retains the discrepancy.

The correction is to state that the displayed fixed-compression expression is a tangent commutator with its own weighted generator. The moving spectral projection contributes to the derivative of the transported operator. The comparison with lemma 14 is a common transport principle, with its global path qualification; lemma 13 computes a distinct fixed compression. The first open point of lemma 19 is answered only on the specified gauge orbits and with the moving window.

9. No-go theorem 22.7 — Correct with amendment.

For the stated one-interval, zero-bulk-mass parent, fixed X, even scalar inner one-forms, and the **moving spectral window**, the finite operator is unitarily conjugate to D_F. Its singular values are therefore fixed. The author's argument and the direct endpoint-preserving gauge transformation in item 7 give the same conclusion in this scope. A fixed compression can show changes in singular values at second order, so the window must be named.

For the usual Standard Model representation with nonzero Yukawa data, the normalized shifted Higgs quaternion runs through unit quaternions, a three-real-dimensional gauge orbit. These are gauge-orbit directions. Calling them physical or eaten modes requires a spacetime gauge action and its boundary conditions; the interval calculation alone supplies no such mode count. Degenerate data, including vanishing Yukawas, can have a smaller orbit.

The listed changes of wall coupling, admissible odd bulk terms, bulk masses, or internal dimension lie outside the theorem's hypotheses and are directions for further constructions. Curvature in higher dimensions alone does not prove a radial mode. Likewise, a quantum effective action on the same fixed configuration space does not by itself add a radial coordinate; an additional composite or other degree of freedom would have to be identified.

10. Proposition 22.8 — Correct with amendment.

Evaluation shows that the unsymmetrized gauge one-form has boundary value in C_F. For the displayed masses, assume a flat product interval, a positive Yang–Mills kinetic form, the trivial background, unrestricted endpoint variation in the allowed directions, and no additional boundary kinetic or mass terms. Variation then gives Neumann conditions at 0, and Neumann on the wall Lie algebra and Dirichlet on its orthogonal complement at ℓ. The stated integer and half-integer towers follow.

The physical gauge Lie algebra is obtained from skew-adjoint algebra elements through the real gauge representation, with its kernel and any imposed unimodularity condition accounted for. For the usual Standard Model representation, $u(C_F)$ has dimension 10 before unimodularity, whereas photon plus gluons

have dimension 9. The extra abelian direction must be addressed. The role of unimodularity and the hypercharge embedding is explicit in Chamseddine–Connes–Marcolli, §2.5, arXiv:hep-th/0610241.

The literature comparison is valid at the level of the boundary mechanism. The Neumann/Dirichlet towers appear in Csáki et al., §5.1, arXiv:hep-ph/0305237v1, and the fermion equation $\tan(m_n L) = M_D L$ is equation (5.6) of Csáki et al., arXiv:hep-ph/0310355v1. Their full gauge models have additional structure.

11. Corollary 22.9 — Correct with amendment.

Items 1, 2, 4 and 5 hold for fixed, orthogonal, J_F- and grading-compatible light and heavy channels, with $0 < \Theta < \pi/2$. For a general non-normal light block, write the upper wall block as

$$X_0 = \tan(\ell|M_0|)V_0^*,$$

or state the formula in terms of singular values. The light-channel coupling is $\ell M_0^* + O(\ell^3)$. Identifying it with a four-dimensional Yukawa coupling remains an additional hypothesis.

For item 3, let P_L reduce $D_F(\ell)$, and suppose $D_L = D_F(\ell)|_{P_L H_F}$ is fixed. Then the precise statement for a constant observable is

$$P_L Q_\ell(a) P_L = \operatorname{sinc}(\ell \operatorname{ad}_{D_L})(P_L \pi(a) P_L),$$

and

$$\|P_L Q_\ell(a) P_L - P_L \pi(a) P_L\| \le \frac{2}{3}\ell^2 \|D_L\|^2 \|\pi(a)\|.$$

This remains true when $\pi(a)$ mixes light and heavy channels. The limiting light compression need not be a representation: products can pass through the heavy subspace.

The full observable estimate does **not** follow from Theorem 22.4, since $\ell\|D_F(\ell)\| \to \Theta$. An explicit counterexample uses

$$D_F(\ell) = \sigma_1 \otimes \operatorname{diag}(m, \Theta/\ell), \qquad \pi(a) = 1 \otimes \sigma_1,$$

with the commutative algebra generated by 1 and $\pi(a)$ and the compatible real structure $J = (\sigma_1 \otimes 1)K$. Then

$$Q_\ell(a) = \operatorname{sinc}(\Theta - \ell m)\, \pi(a),$$

which tends to $\operatorname{sinc}(\Theta)\pi(a)$, rather than $\pi(a)$.

The ratio $m_t/M_W = 2\Theta/\pi < 1$ is a result for the stated lowest-level, zero-bulk-mass fermion channel and strict-wall gauge spectrum. Bulk fermion masses or a changed gauge wall condition are possible directions beyond that family.

12. Numerical comparison for the interior — Gap.

For a separate flat model with Neumann conditions at 0 and a finite Robin wall mass,

$$t \tan t = s, \qquad s = \frac{g_4^2 v^2 \ell^2}{4},$$

where $g_4^2 = g_5^2/\ell$ in the usual flat normalization. The Z equation has $s/\cos^2\theta_0$. This agrees with the form and sign convention of Csáki et al., §5.2, equations (5.13) and (5.22).

For $g'^2/g^2 = 3/5$, the lowest-mode squared mass ratio increases from 5/8 to 1. A small **Python** calculation evaluated the equations at the stated rounded values: $M_W \ell = 1.03$ is consistent with the quoted ratio to the displayed precision, and the next W root lies between $3.58/\ell$ and $3.59/\ell$, about 280 GeV when $1/\ell \approx 78$ GeV. This is a numerical check; the monotonicity follows separately from the equation.

The top constraints are strict:

$$M_W \ell < \frac{\pi}{2\sqrt{8/3}} \simeq 0.962, \qquad M_W \ell < \frac{\pi}{2(m_t/M_W)} \simeq 0.732$$

for the measured ratio. The latter estimate is consistent with the 2025 PDG W, Z values and top-mass table.

The collider-exclusion sentence is a **gap**: a mass alone does not establish a search exclusion without the model's couplings, branching fractions, and an applicable experimental limit. It should be removed or supplied with that analysis. The finite wall scalar also changes the strict wall boundary problem; it is an extended model outside Proposition 22.8 and No-go theorem 22.7. Within the stated flat toy model, the incompatible top and gauge mass requirements support the limited conclusion; bulk masses, warping, and running are directions to investigate.

13. Input and derived statements — Correct with amendment.

The finite algebra, representation, operator, grading, and real structure are inputs. The construction derives the endpoint commutant for that input, the interval operator and its window, the stated compression map, and the transport formulas. The Yang–Mills spectrum additionally depends on the action and boundary assumptions in item 10. The exact window identifies D_F, J_F, Γ_F; the represented algebra is obtained through compression and need not be an exact representation of A_F.

Several ingredients have broader precedents. For any reducing finite spectral subspace, the identity

$$[D_F, W^*TW] = W^*[D_X, T]W$$

holds whenever T preserves the operator domain. Operator-system compression is treated generally in Connes–van Suijlekom, §3. Boundary spectral geometry for the Standard Model was also studied by Chamseddine and Connes, §5, arXiv:1008.3980v1, and the interval fermion and gauge mechanisms occur in the Higgsless papers cited in item 10. The particular combination and formulas of lemma 22 can be stated explicitly; a wider novelty claim would need a fuller literature comparison.

Amendments and errors

- **Errors:** correct the boundary-type zero-mode statement at nonzero X and the first-derivative comparison between fixed compression and transport.
- **Gap:** support or remove the collider-exclusion claim.
- **Amendments:** state spectral multiplicities and unitality; qualify the Standard Model data, gauge quotient, and unimodularity; distinguish profiles from holonomies and real from non-real transport; scope the radial result to the moving window; provide reproducible data for the numerical algebra dimensions; state the light-block estimate and the additional assumptions of the Yang–Mills and finite-wall-scalar models.

Overall: 2 Correct; 8 Correct with amendment; 1 Gap; 2 Error.

Part R9 (referee report, lemma 23)

Referee report on lemma 23

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_23.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemma 23 and the note.

The boundary-spectrum formulas are correct under the stated Yang–Mills boundary conditions. The principal corrections concern the aligned-wall compression, the comparison with twisted spectral triples, the order of the holonomy double coset, and the bulk-mass extension of the one-loop potential.

The verdicts below count the lettered entries. The numerical checks use Python 3 with SciPy. They supplement the analytic arguments and do not replace them.

1. Theorem 23.1 (statement and proof)

1a. Item 1 — Correct. Each wall condition is a maximal isotropic subspace for the boundary form. The resulting first-order operator is self-adjoint with compact resolvent, and the spin grading preserves its domain and anticommutes with it.

1b. Item 2 — Correct. Under the equal-dimension and no-−1-eigenvalue hypotheses, the principal logarithm gives an anti-Hermitian K satisfying $\Gamma_0 K \Gamma_0 = -K$. The stated $D_F = \Gamma_0 K$, its singular values, and

$$\Gamma_1 = e^{lK} \Gamma_0 e^{-lK}$$

follow from the two-projection decomposition.

1c. Item 3 — Correct with amendment. The graph boundary condition and the converse hold on the stated branch $l\|D_F\| < \pi/2$. Define X explicitly as the $H_- \to H_+$ block of $\tan(lD_F)$. The spectral part of Theorem 22.1 applies without a real structure; its real-structure conclusion additionally needs the hypothesis in item 4. At an angle $\pi/2$, the principal logarithm and the isolated window used here cease to be unique.

1d. Item 4 — Correct. The antiunitary J_1 is an additional hypothesis. With $\widehat{J} = EJ_1$, the signs are $(+,+,-)$, and $J_F = \Gamma_0 J_1$ has the stated finite signs. In particular,

$$J_F \Gamma_1 J_F^{-1} = \Gamma_0 (J_1 \Gamma_1 J_1^{-1}) \Gamma_0 = -\Gamma_0 \Gamma_1 \Gamma_0.$$

The antilinearity of J_1 is compatible with $J_1 K J_1^{-1} = K$ on the principal branch.

2. Theorem 23.2 (statement and proof)

2a. Items 1 and 2 — Correct with amendment. Domain preservation is equivalent to commuting with the respective wall grading, and $[D, f] = Ef'$. The compression formula in item 2 requires the equal-dimension and branch hypotheses under which the isometry W of Theorem 22.1 exists. State those hypotheses at the item.

2b. Item 3, aligned walls — Error as stated for an arbitrary $*$-subalgebra P. A constant odd $a \in P$ generally fails the endpoint conditions and is an ambient multiplier, rather than an element of the parent algebra. More seriously, $E_0(P)$ need not equal $P \cap \{\Gamma_0\}'$.

Let $H = \mathbb{C}^2$, $\Gamma_0 = \sigma_z$, and let P be the unital $*$-algebra generated by $\sigma_x + \sigma_z$. Then

$$E_0(P) = \operatorname{span}\{1, \sigma_z\}, \qquad P \cap \{\Gamma_0\}' = \mathbb{C}1.$$

The first space is the compression image, while the second is the even part of P. Compactly supported scalar profiles show that the image of the actual boundary algebra is $E_0(P)$.

The correction is to assume $\Gamma_0 P \Gamma_0 = P$. Then $E_0(P) = P^{\mathrm{ev}}$, and the aligned-wall conclusion holds. Continue to distinguish constant ambient multipliers from parent-algebra elements.

2c. Item 4 — Correct. Under the stated J_1 hypothesis and $[P, P^\circ] = 0$, the parent has a real structure and satisfies order zero and order one, even when P contains elements odd for Γ_0. The parent grading is $\widehat{\Gamma}$, which commutes with those elements. Pointwise,

$$[\pi(f), J\pi(g)^* J^{-1}] = 0, \qquad [[D, \pi(f)], J\pi(g)^* J^{-1}] = E[f', g^\circ] = 0.$$

2d. Comparison with Result 11 — Correct with amendment. The aligned-wall construction evades hypothesis 4 of Result 11: odd coordinate matrices are sent to zero rather than retained as $\Gamma_a/5$. Result 11 assumes an even parent, an inherited involutive grading, and retention of all those coordinate compressions simultaneously. Its stated escapes include restricting to the even algebra or changing the grading construction. The present construction points toward a matrix-valued parent with an inherited grading; it leaves the commutative-parent and retained-coordinate questions open.

3. Remark 23.3 (remark)

3a. Opposite action — Correct with amendment. For an odd **anti-Hermitian** h, the displayed signs for the unstarred conjugations $J_1 h J_1^{-1}$ and JhJ^{-1} are correct. The opposite representation used in order zero is $b^\circ = J_1 b^* J_1^{-1}$; state the star and the anti-Hermiticity assumption when giving signs. On the W-factor it incorporates $\sigma(b) = \gamma_W b \gamma_W$.

3b. Grand-symmetry twist — Error in the assertion that it is the same automorphism. The relevant references are:

- Agostino Devastato, Fedele Lizzi and Pierre Martinetti, "Grand symmetry, spectral action and the Higgs mass," *JHEP* 01 (2014) 042. The larger algebra and its reduction appear in §§3–5.
- Agostino Devastato and Pierre Martinetti, "Twisted spectral triple for the Standard Model and spontaneous breaking of the Grand Symmetry," *Mathematical Physics, Analysis and Geometry* **20** (2017), article 2. Its Eqs. (1.8)–(1.11) define the twisted commutator, first-order condition and automorphism; the publisher record confirms the publication details.

The latter authors call their ρ the "exchange of the left/right spinorial indices." It exchanges algebra summands. Conjugation by γ_W is inner on $M_2(\mathbb{H})$ and fixes its centre. Thus the full automorphisms have different algebraic actions. The correction is to describe a **structural analogy** between involutive automorphisms and compare the actual representations, opposites and first-order conditions before claiming an identification.

4. Theorem 23.4 and Example 23.5 (fluctuations and example)

4a. Connection removal — Correct with amendment. The displayed unitary identity follows from $u' = -hu$. It describes the perturbation $D + Eh$. For a fluctuation of a real triple, include the conjugate one-form:

$$D + Eh + \widehat{J}(Eh)\widehat{J}^{-1}.$$

In the example, the resulting generator has the form (L_F, R_F), as required for the stated action on $E \oplus E^*$. This also preserves the J_1 condition. The odd directions lie outside No-go theorem 22.7, whose hypotheses include an even algebra and a moving window.

4b. Holonomy and double cosets — Error in the order of the factors. With the convention $u(0) = 1$, $u(l) = w$, let $H_i = \mathrm{Stab}(\Gamma_i)$. A gauge transformation removing h has the form $u(x)c$, so it preserves both endpoints precisely when

$$c \in H_0, \qquad wc \in H_1, \qquad \text{equivalently} \qquad w \in H_1 H_0.$$

The holonomy quotient is $H_1 \backslash G / H_0$, or $H_0 \backslash G / H_1$ after replacing w by w^{-1}. The order matters for the stated w. Also restrict G to the unitary component reachable by a smooth path from the identity.

4c. Orbit, angles and polar coordinates — Correct with amendment. The orbit of quaternionic self-adjoint involutions with one-dimensional quaternionic $+1$ and -1 eigenspaces is

$$\mathbb{H}P^1 \simeq Sp(2)/(Sp(1) \times Sp(1)) \simeq S^4.$$

The wording "each grading with quaternionic eigenspaces" also includes ± 1; exclude those two cases. For $\Gamma_p = p_a \Gamma_a$, the eigenvalues of $\Gamma_q \Gamma_p$ are $e^{\pm i d(p,q)}$, so the principal angles are $d(p,q)/2$.

The formula on E^* is correct: p' is the reflection of p through p_0, because

$$\Gamma_{p'} = \gamma \Gamma_p \gamma.$$

The quaternion direction q supplies three angular coordinates away from the poles. It is undefined at both poles, and at $\theta = \pi/2$ the finite logarithm and lowest-window choice are degenerate. The unquotiented orbit is S^4; quotienting the relative orientation by the stabilizer leaves the radial interval.

4d. Bulk-mass texture — Correct with amendment. For $X = \tan\theta\, 1$, the light-mass approximation from Theorem 12.6 has the scale

$$M_{\text{light}} \simeq \frac{\tan\theta}{l}\, \Phi_Q \Phi_U$$

when the light eigenvalues are small compared with the relevant tower thresholds. In the convention of Theorem 12.7, $\widehat{Y} = X/l = (\tan\theta/l)1$; a dimensionless identity matrix can be stated after factoring out that common scale. Positive localization parameters suppress the wall profiles. General finite-angle eigenvalues require the secular equation.

4e. Minimal composite Higgs comparison — Correct with amendment. The coset and the four real Higgs coordinates agree with those in Agashe, Contino and Pomarol, "The Minimal Composite Higgs

Model," *Nuclear Physics B* **719** (2005), 165–187, §§2–3. Their Eq. (6) identifies h/f_π with the sphere angle, corresponding here to 2θ. Their construction uses a warped AdS interval and additional fermion and boundary data. The shared coset alone does not identify the actions or potentials.

5. Theorem 23.6 (boundary conditions and theorem)

5a. Connector at one wall — Correct with amendment. The stated gauge boundary conditions form a consistent boundary-value problem with the connector treated as prescribed wall data. A real fermion connector can also be placed at one wall. For example, choose a unit vector e in the $+1$ eigenspace of Γ_1, put $f = J_1 e$, and set

$$K_c = |f\rangle\langle e| - |e\rangle\langle f|, \qquad \Gamma_{1,c} = e^{\eta K_c}\Gamma_1 e^{-\eta K_c}.$$

Then K_c anticommutes with Γ_1, commutes with J_1, and J_1 anticommutes with $\Gamma_{1,c}$. Restricting the wall algebra to the common stabilizer gives the connector conditions and retains the parent order conditions.

The amendment is to specify whether the connector is prescribed gauge-breaking data or a term in the fermion boundary operator. In the latter case, $\Gamma_{1,c}$ changes the affected fermion tower; the universal fermion product later in the lemma must be recalculated for that channel.

5b. Charged and neutral masses — Correct. An independent solution uses principal angles between the Neumann subspaces at the two walls. Put $s = \sin(2\theta)$. For each charged direction the squared sine of the angle is $s^2/2$. In the neutral sector the two Neumann directions are L' and $\sqrt{3/5}\,R' + \sqrt{2/5}\,B$; their projection onto the Dirichlet complement at the first wall has one nonzero squared singular value $4s^2/5$. Consequently

$$\sin^2(M_W l) = \frac{s^2}{2}, \qquad \sin^2(M_Z l) = \frac{4s^2}{5},$$

which give the stated tangent equations. This derivation also covers $\cos(2\theta) = 0$, where the divisions in the written proof need a limiting interpretation.

5c. Massless modes — Correct with amendment. For $0 < \theta < \pi/2$, the intersection has dimension nine: the photon and eight gluons. At each endpoint it has dimension twelve, with three additional electroweak zero modes. The phrase "massless for all θ" is correct; state the endpoint enhancement explicitly.

5d. Normalization — Correct. With the stated fundamental trace forms, equal five-dimensional coefficients, flat profiles and no boundary kinetic terms,

$$g_3 = g_2 = \sqrt{\frac{5}{3}}\,g_Y, \qquad \frac{1}{e^2} = \frac{8}{3g_2^2}.$$

Indeed the hypercharge generator has squared trace norm $1 + 2/3 = 5/3$, while the photon generator has squared norm $1 + 1 + 2/3 = 8/3$. Here g_2 is the normalized four-dimensional zero-mode coupling.

5e. Items 1 and 2 — Correct. Expanding the sine equations gives $M_W^2 \sim 2m^2$, $M_Z^2 \sim (16/5)m^2$ as $\theta \to 0$. At $\theta \to \pi/2$, both displayed branches tend to zero and their squared ratio tends to $5/8$. The ratio itself is undefined at the endpoints.

5f. Item 3 — Correct. Let $a = \sin(2\theta)$. Then

$$\frac{M_W^2}{M_Z^2} = \left[\frac{\arcsin(a/\sqrt{2})}{\arcsin(2a/\sqrt{5})}\right]^2.$$

For $0 < a \le 1$, this decreases with a, since $x/(\sqrt{1-x^2}\arcsin x)$ increases. Hence the midpoint is the global minimum, with exact value

$$\left(\frac{\pi/4}{\arctan 2}\right)^2 = 0.5032315\ldots.$$

The upper value $5/8$ is approached at the endpoints. This supplies the proof missing from the grid check.

5g. Item 4 — Correct. The ratio m/M_W increases from $1/\sqrt{2}$ to infinity. Python 3 with SciPy gives $\theta = 1.05888846\ldots$ for $m/M_W = \sqrt{8/3}$.

6. Comparison with lemma 18 and Proposition F

6a. Constants 2 and 16/5 — Correct with amendment. These are the coefficients in the lower bounds for the fully massive finite-action pole masses in Proposition 14 of lemma 18. They are not the small-x pole coefficients there: those are 3 and 24/5. The common 5/8 limit is a valid comparison, with different limiting behaviour: the finite-action poles grow in the lemma 18 heavy limit, while the interval vector masses tend to zero at the antipodal endpoint.

Specify the trace or Hilbert–Schmidt norm in the commutator-norm claim. With the ordinary operator norm, the charged example gives m^2, rather than $2m^2$.

6b. Factor 3/2 — Correct with amendment. Proposition F gives the factor 3/2 for the stated leading finite spectral-action mass matrix. Comparing it with the assumed five-dimensional Yang–Mills quadratic form gives $3m^2$ versus $2m^2$ in the universal small-mass example. A calculation of the full parent spectral action, with its tower and boundary conditions, remains necessary for a direct matching claim.

7. Proposition 23.7 (heat trace)

7a. Heat trace and monotonicity — Correct. With $s = l\Lambda$,

$$\mathrm{Tr}_{\mathrm{channel}}\, e^{-D^2/\Lambda^2} = \frac{2s}{\sqrt{\pi}}\,\vartheta_3(\theta, e^{-s^2}).$$

The Jacobi product proves strict decrease on the interval, with zero derivative at the endpoints. At fixed $x = \theta^2/s^2$, its towerless limit is $2e^{-x}$. At $\theta = \pi/2$, the full lowest eigenspace is degenerate even though the heat trace is well defined.

7b. Escape from No-go theorem 9 — Correct with amendment. No-go theorem 9 assumes finite internal data, the positive-Laplace cutoff class, the literal flat-space spectral action, $A = 0$, and an allowed unconstrained ray $V = cV_*$, $c > 0$, without an added scalar potential. Its conclusion is strict decrease along that ray. The interval family changes the operator family and replaces the free ray by a compact relative-grading orbit with a tower. It therefore lies outside those hypotheses.

For the stated heat action, the antipodal point is a minimum of this family. Added potentials, constraints, other cutoff classes and quantum effects remain separate possibilities. The result points toward analysing the full tower and any additional scalar terms.

8. One-loop potential (products and potential)

8a. Three products — Correct. The product for a principal angle α, normalized at $\alpha = 0$, is

$$1 + \frac{\sin^2\alpha}{\sinh^2 x}.$$

Using the fermion angle θ and the two gauge angles from 5b gives all three displayed products.

8b. Fermion coefficient — Correct with amendment. With the conventional four-dimensional Dirac determinant and each physical species counted once, $n_f = 2N_D$; one colour-triplet Dirac top contributes six. The universal example gives masses to both members of each weak doublet. For one complete generation it contains eight physical Dirac species, hence $n_f = 16$, before any connector modification; three identical generations give $n_f = 48$. A top-only $n_f = 6$ potential therefore needs a separately specified fermion sector or a complete accounting of spectator channels.

8c. Potential and endpoint curvatures — Correct with amendment. The logarithmic integrand and the vector multiplicities are correct under the stated one-loop assumptions. The curvature sentence omits the prefactor on its left-hand side. Let

$$I = \int_0^\infty \frac{x^3\,dx}{\sinh^2 x} = \frac{3}{2}\zeta(3), \qquad J = \int_0^\infty \frac{x^3\,dx}{\cosh^2 x} = \frac{9}{8}\zeta(3).$$

The precise expansions are

$$8\pi^2 l^4[V(\theta) - V(0)] = (10.8 - n_f)I\,\theta^2 + o(\theta^2),$$

and, with $\epsilon = \pi/2 - \theta$,

$$8\pi^2 l^4[V(\tfrac{\pi}{2} - \epsilon) - V(\tfrac{\pi}{2})] = (10.8I + n_f J)\epsilon^2 + o(\epsilon^2).$$

For $n_f \geq 0$, the latter curvature is positive.

8d. Numerical absence of an interior minimum — Gap in any conclusion beyond the sampled grid. The recorded computation is explicitly a grid statement, and its original code was not retained. An independent Python 3/SciPy calculation of the derivative found one sign change for each tested $n_f = 1, 2, 3, 4, 6, 8$; for $n_f = 8$, the stationary angle is approximately 0.3382. The known integral $I = 3\zeta(3)/2$ agreed to 5×10^{-16} in absolute error, and a 1% perturbation failed the same check. A grid does not exclude additional stationary points between samples, so the claim about absence of an interior **local** minimum for all parameters in the displayed massless family remains open in the range $0 < n_f < 10.8$.

Two stronger statements follow analytically from the displayed potential:

- For every $n_f > 0$, $\theta = \pi/2$ is the unique global minimum on $[0, \pi/2]$. The gauge logarithms are nonnegative, and the negative fermion logarithm is minimized when $\sin^2\theta = 1$.
- For $n_f \geq 10.8$, the potential is strictly decreasing throughout the interior. To see this, let $t = \sin^2\theta$ and

  $$\mathcal{J}(z) = \int_0^\infty \frac{x^3\,dx}{\sinh^2 x + z}.$$

  The derivative, apart from the positive overall factor, is

  $$(1-2t)\Big[6\mathcal{J}(2t(1-t)) + \frac{24}{5}\mathcal{J}\Big(\frac{16}{5}t(1-t)\Big)\Big] - n_f\mathcal{J}(t).$$

  For $t \leq 1/2$, both arguments of $\mathcal{J}$ in the bracket are at least t, and $\mathcal{J}$ decreases. For $t > 1/2$, both contributions have the required sign.

These conclusions concern the displayed massless flat products, the stated gauge content and boundary conditions, and one-loop order. They direct attention to changed form factors or additional terms for an interior vacuum.

8e. Leading-logarithm form — Correct with amendment. The formal first term is $B\sin^2(2\theta) - A\sin^2\theta$, with $A, B > 0$, and any interior stationary point of that expression is a maximum. The logarithm expansion is not uniform at $x = 0$; the next term has an infrared-divergent integral. Present the expression as a formal leading approximation and use the full integral for conclusions about its global shape.

8f. Bulk-mass scan — Error as a finite one-loop potential for general scanned masses. Let $a = c_Q + c_U$. The stated fermion factor satisfies

$$G(x) = 1 - \frac{a}{x} + \frac{a^2}{2x^2} + O(x^{-3}),$$

so, for $t = \sin^2\theta$,

$$\log(1 + t(G-1)) = -\frac{ta}{x} + \frac{t(1-t)a^2}{2x^2} + O(x^{-3}).$$

Multiplication by x^3 gives angle-dependent ultraviolet divergences whenever $a \neq 0$. A finite quadrature value then depends on a regulator and subtraction prescription; the reported scan does not establish a regulator-independent potential.

The exceptional case $c_U = -c_Q$ has a common bulk mass on the two mixed components and

$$G(x) = 1 + \frac{w^2}{x^2\sinh^2 w},$$

which has the required exponential decay. Keeping the flat gauge products is consistent as a **separate tree-level gauge-sector assumption**. For a coupled $Sp(2)$ gauge theory, unequal masses on components

mixed by the gauge group require gauge-invariant completion or explicit breaking data. The correction is to specify that completion, the regulator and allowed wall counterterms, then recompute the potential.

8g. Escapes and spinor comparison — Correct with amendment. A frequency-two fermion contribution, modified gauge form factors, wall potentials and wall terms are possible changes to the stated flat calculation; each requires its own model and coefficients. For the comparison with Agashe–Contino–Pomarol, their §2.1, Eqs. (19)–(23), uses spinorial $SO(5)$ fermions and permits an approximate potential

$$\alpha\cos(h/f_\pi) - \beta\sin^2(h/f_\pi).$$

With $h/f_\pi = 2\theta$, the displayed leading flat expression has this angular form with $\alpha = A/2$ and $\beta = -B$, up to a constant. Their fermion form factors can also give the sign and competition needed for an interior minimum. The flat scan therefore has its stated scope; it does not exclude the spinorial composite-Higgs construction.

9. Measured values and open questions (comparison)

9a. Numerical comparison — Correct with amendment. Using the 2026 Particle Data Group values $m_t = 172.60$ GeV, $M_W = 80.3625$ GeV and $M_Z = 91.1879$ GeV gives $m_t/M_W \simeq 2.1478$ and $M_W^2/M_Z^2 \simeq 0.7767$. The top entry is a direct-measurement mass and its definition should be kept distinct from a specified pole-mass scheme. PDG quark table, p. 2; PDG gauge-boson table, pp. 1–2.

Python 3 with SciPy gives $\theta \simeq 1.1626$, $M_W l \simeq 0.5413$, $1/l \simeq 148.5$ GeV and the model ratio $M_W^2/M_Z^2 \simeq 0.5814$. For the separate wall-scalar model of lemma 22, fitting the measured vector ratio gives $M_W l \simeq 1.0282$; its lowest no-bulk-mass fermion channel then has $m/M_W < 1.528$. Thus the stated mismatch holds for those flat, tree-level families with the specified lowest-level and coupling assumptions. Bulk masses, warping, running and a combined wall/Wilson-line model remain directions to examine.

9b. Open questions — Correct with amendment. The listed questions remain appropriate. Add the real connector's contribution to the fermion tower, a gauge-invariant and renormalized bulk-mass potential, the degenerate antipodal window, and the scalar kinetic normalization needed to identify a canonically normalized Higgs field.

Amendments and errors

- Assume $\Gamma_0 P \Gamma_0 = P$ for the aligned-wall even-algebra claim, and distinguish ambient constant multipliers from elements of the boundary algebra.
- Replace the claimed equality with the grand-symmetry twist by a comparison of the involutions and their first-order conditions.
- Reverse the holonomy factors for the convention used: $w \in H_1 H_0$, with quotient $H_1 \backslash G / H_0$.
- Specify the real fluctuation, the connector's role, the altered fermion tower, endpoint degeneracies and trace norms.
- Add the missing $1/l$ scale and light-mode approximation to the bulk-mass texture; add $1/(8\pi^2 l^4)$ to the potential curvatures.
- Restrict the unproved absence of local minima to the sampled grid, and replace the generic bulk-mass scan by a calculation with a stated regulator, counterterms and gauge completion.

Overall: 11 Correct; 19 Correct with amendment; 1 Gap; 4 Error.

Part R19 (independent reading of 11D files 10 and 11)

Independent reading of 11D/10 and 11D/11, by Claude Sonnet 5.5

Model Claude Sonnet 5.5, effort max, read-only, run on 4 November 2026 because the Codex quota was spent. The subagent was refused a file write, so it returned the report as text, and the verdicts and corrections below are that text, verbatim, filed here by Claude Opus 5.5. Its scripts were not kept, by the rule of the project that checking scripts are not saved; the report names what each computed.

Assessment by Claude Opus 5.5, reading it against the text, as the user asked for reports of smaller models. Correction 1 was checked against lemma 23, line 180 ("Sp(2) x SU(4), of dimension 25") and its Theorem 23.6, and recomputed: the bracket is 169.3125 for three generations and 41.3125 for one, and tools/d11_needed_numbers.py, corrected, reproduces theta_* = 1.009, 1.031 and G(pi/2) = 23.88. The near-cancellation 64 - 63 in the first version of 11D/10 was an artifact and is withdrawn. Correction 2 was checked against the conclusions of Bergman, Gaberdiel and Lifschytz, and the Planck length formula was rederived. Correction 3: the rank argument for the Z was accepted on reasoning. Correction 5 is the mechanism of lemma 23. Correction 7: stationarity at c = 1 was recomputed (mpmath). Corrections 6, 9 to 12 were read and applied. Item 3's rank-17 statement stays marked as recalled, since neither source proves it. This reading is by a smaller model and no further independent reading followed the corrections.

A. Verdicts

1. Prop 10.1: CONFIRMED. AMENDMENT: z must have slope exactly 8 - N(x) on [0, 2 pi].
2. Prop 10.2: CONFIRMED.
3. Prop 10.3: AMENDMENT (metric of l). The 10D SO(32) endpoint is CONFIRMED. The rank-17 eleven-dimensional-limit statement: COULD NOT CHECK from the two sources (only BGL remarks support it).
4. Prop 10.4: linearity, C_L = 8 pi/5, bound, strong-coupling statement CONFIRMED. AMENDMENT on the inference "no eleven-dimensional lift".
5. Far wall: formula and limits CONFIRMED. AMENDMENT: only W^+ and W^- are D0 states; the closing sentence needs its scope.
6. Prop 10.5: CONFIRMED.
7. a_0: method, signs, normalization, F(1) check CONFIRMED. ERROR in the generator count: 21 integer towers against 16 integer + 9 half-integer in lemmas 23, 24, 26.
8. Prop 10.6: items 1, 2 CONFIRMED; item 3 CONFIRMED in scope (no proof given); n_s = 129 ERROR (inherits item 7).
9. Tied angle: classification CONFIRMED; the quoted numbers inherit the item 7 error.
10. Localized channel: CONFIRMED. AMENDMENT: dropped bulk and wall terms.
11. Remark 10.7: AMENDMENT (wording). Remark 10.8: heat-cutoff statement CONFIRMED, no conflict with lemma 25. AMENDMENT: q > -2, not q > 0.
12. Corrections list: AMENDMENT (the S - T sentence of Prop 5.4 is also false; 07 section 4 omitted).
13. Prop 11.1: CONFIRMED. AMENDMENT: factor s in j q_L; s = +1 implicit in 11.4.
14. Prop 11.2: CONFIRMED with three AMENDMENTS (proof choice of f; second order-one block; finite window against parent).
15. Thm 11.3: items 1, 2, 3, 5 CONFIRMED; item 4 AMENDMENT (aligned subalgebras); the rule "vector or scalar" holds in a stated scope.
16. Comparison paragraph: CONFIRMED.
17. Prop 11.4: CONFIRMED.
18. Reading after 11.4: AMENDMENT (first order, fixed compression); no conflict with lemmas 19, 22, 23.

B. Corrections in order of severity

1. ERROR (file 10: Outcome 2, a_0 section, Prop 10.6 last paragraph, tied-angle paragraph). Lemmas 23, 24, 26 use sp(2) + su(4), 25 generators: 16 integer-tower (8 gluons, 3 W_L, 4 O with (D,D), 1 hypercharge combination) and 9 half-integer-tower (2 R^+-, 6 X, 1 orthogonal neutral; (N,D)). File 10 counts 21 integer towers (the Neumann generators at x = 0). Correct: 8 pi^2 a_0 = (3 zeta(5)/16)(4 n_D' - 3(121/16) - n_s - 5 n_g). Brackets: 169.3125 (3 generations; file: 129), 41.3125 (1 generation; file: 1), 36.3125 (with graviton; file: -4), n_s = 169.3 for a_0 = 0 (file: 129). A_0 = 32.918 and 8.032. Tied-angle values: universal 3 generations theta_* = 1.009, M_W/m = 0.685 (file 0.877, 0.876); universal 1 generation theta_* = 1.031, M_W/m = 0.653 (file 0.202, 1.394, "small because A_0 is small", an artifact); top only G(pi/2) = 23.88, no stationary point. The sign a_0 > 0 and the

top-only classification survive. The single-weight formula is valid for integer towers only. Taking all 25 as integer would give 117 and -11. The script carries the same n_v = 21. The Outcome writes n_f where n_D' is meant.

2. AMENDMENT (Prop 10.4). "Known fact that massive IIA has no lift" has no reference. The source is BGL (introduction): type I' with N != 0 "has no simple M-theory interpretation"; conclusions: "we expect neither … applicable". The classical profile is not controlled at g of order 1, so the proposition establishes the statement for the extrapolated profile. Add: g(0) L_p <= C_L l_s/7 holds exactly for the total length (the "small change on the last segment" is unnecessary), and the region g >= g_0 has eleven-dimensional length (3/4)(C_L/|8-N|) g_0^{-4/3} l_11.
3. AMENDMENT (far wall). Only W^+ and W^- are D0 states. A neutral Z is a Cartan combination, and the rank of a 9D theory with 16 supercharges is fixed, so no massive neutral vector arises. Scope of the closing sentence: chamber 15 + 1, top's brane on the far wall, W^+- as D0 states. Escapes: other chambers, W as a Wilson-line D8-D8 string (lemma 23), top's brane in the bulk (far wall E_1). Add M_D0 R = R^2 - R_*^2 (alpha' = 2).
4. AMENDMENT (Prop 10.3). l L_p = pi^2 alpha' holds for l = pi R_I measured in the type I' metric. For the heterotic radius in the heterotic metric, l L_p = pi^2 alpha' sqrt(lambda_h) (BGL (hettypeI), (simpleSduality)). alpha' is the type I' alpha'. Mark paragraphs 2 and 3 as recalled; there is no proof block.
5. AMENDMENT (Thm 11.3 item 4, Outcome 3). The towers need commuting projections onto k_0 and k_l. Counterexample: k_0 = span T_1, k_l = span(cos phi T_1 + sin phi T_2) gives kl = n pi +- phi. List escapes: circle, nontrivial holonomy, brane-localized terms, extra dimensions, non-aligned subalgebras.
6. AMENDMENT (Prop 5.4 correction). The sentence "S - T has d = 16 on the neutrinos of one slot, 22 on the two others" fails in 96 of 192 configurations (pattern 16, 16, 22). Add 07 section 4 ("distance proportional to a") to the list.
7. AMENDMENT (localized channel). The bulk term (-(4/15) c^5 in dimensional regularization) and the wall term are dropped. "Prefers c near 1" refers to the interaction term only. The stationary point is c = 1 exactly. Large-c corrections are relative 1/c.
8. AMENDMENT (Remarks 10.7, 10.8). "Fixed ratio lambda_{M_*}/delta" instead of "realizes the hierarchy of lemma 30". A joint family with alpha < 1 needs p + q > 2 (q > -2 for p = 4, q != 0). Write "the potential" in the heat-cutoff sentence.
9. AMENDMENT (Prop 11.2). The choice f = T, g = x 1 gives only wall-compatible T; use f = 1 with g' = T - tau (tau interior-supported, same integral) plus tau = sum f_i g_i'. Add the second order-one block -a M^* U b U^-1. Order one fails in the finite window; the parent satisfies it (Thm 22.3), and its wall algebra shrinks to the commutant of D_F in pi (zero for pi = (a, 0), M invertible).
10. AMENDMENT (Prop 11.1, 11.4). j q_L = s conj(q_R). Prop 11.4 requires Lambda = s j Lambda j^-1, so it is stated for s = +1. pi is not U-invariant, so U only defines j (not a lemma 06 doubling). Move the sextet out of the "colour antitriplet" column.
11. AMENDMENT (Reading after 11.4). The statement is first order with the fixed compression (-(l/2)[D_F,h]). The moving window gives -l[D_F,h] and the quadratic term (Cor. 22.6, lemma 19).
12. Minor: normalization of z; define n (n = 1, N' = 7) and the reflection to BGL's x^9 = 0; Prop 10.6 item 3 needs its own proof; heading "what would work"; ellipsis in the PW quotation; the folded potential has stationary points near theta = 8.2, outside the window.

C. Derivations, in summary

The subagent's derivations are reproduced in its scripts: s1 (items 1, 2, 5, 6, mpmath), s2 (item 7, mpmath), s3 (items 8, 9, numpy, scipy), s4, s4b, s4c (item 10, mpmath), s5, s5b (item 12, exact integers), s6 (items 13, 14, 17, numpy, scipy), s7 (item 11, mpmath), all Python 3.14. It also reported: no first-person singular, possessive or "not Y; it is Z" constructions in the two files; Prop 10.3 mostly recalled with no proof; "known fact" without a reference; the evaluative heading of Remark 10.8; the PW quotation dropping e^{phi}. These are corrected in the files.

Part R21 (second check of the Terra referee of the 11D line)

Second check of the Terra referee of 11D files 01 to 09, by Claude Sonnet 5.5

The user asked, on 4 November 2026, for a second check of each evaluation in the Terra referee report, with Claude Sonnet 5.5, since no other model had quota left. The checks ran in five batches, one at a time, each redoing the step on which a verdict turns and checking the records. The subagents were refused file writes, so each returned its report as text; the reports are filed below verbatim, each followed by the assessment of Claude Opus 5.5, who rechecked the claims against the text and the source lemmas before applying them. Scripts are in the session scratchpad and were not kept in the repository.

Batch 1: files 01 and 02

Report A. Verdicts

1. Prop 1.1 and joint family: PARTLY. Map, domains, grading, spectrum (k+1/2)pi/l, r window states, gap and N(E)=2 r l E/pi redone and correct. Missing: the joint family lives on the periodic circle (B1).
2. Prop 1.4: AGREE. F(1)=93 zeta(5)/64, J(0)=3 zeta(3)/2 and slope (54/5-n_f)J(0) reproduced; normalization matches lemmas 23, 24.
3. Prop 1.5: AGREE. Counterexample G=theta^4 and small-angle claim hold; true-determinant roots of theta G'-4G are isolated (1.0094 at n_f=48, 1.0309 at n_f=16, A_0 from 11D/10). Remark 10.7 and Terra amend different sentences: no conflict.
4. Prop 2.5: AGREE. Cohomology table enumerated for common and independent chirality, both colour conventions: no flux in span(6Y, 3(B-L)) works, with or without nu^c; c_L=-3 c_Q forbids |P_2|=2 on both doublets.
5. Prop 2.6: AGREE. Exactly the two stated solutions; on P(3bar) the first components reverse; none without chi or B-L, for either chirality hypothesis.
6. Thm 2.1, Props 2.2, 2.4: AGREE. Freeness, Euler class (3,2), Z_6, congruence, transverse Einstein constants 10, 10, factor 5/4. M^{111} identification: COULD NOT CHECK (no repository source); label "conditional" is honest.
7. Prop 2.3: AGREE. Smith forms for nine pairs; (3,2): H^2=Z, H^3=0, H^4=Z/9, spin. Kreck-Stolz: COULD NOT CHECK.

Ledger and README rows match all seven verdicts (status and checkers). Recorded as sonnet_5_5, this check gives d11_flux_nogo a third identifier, in its stated scope.

B. Corrections, by severity

1. 11D/01 line 36 ("the parent unfolds to -i d/dx + D_F on this circle (lemma 25)") and line 44 ("In the circle language of Proposition 1.1"): Proposition 25.1 unfolds the lemma-22 parent to the periodic circle. Write "unfolds to -i d/dx + D_F on the periodic circle of length 2l; theta interpolates between it (theta=0) and the antiperiodic circle of Proposition 1.1 (theta=pi/2); the joint-family window is the n=0 mode".
2. 11D/01 Prop 1.5: line 5 says "amended at its statement below", yet heading (line 88) and lines 92-97 are unchanged. Replace line 97 by "If the roots of theta G'-4G are isolated, a stationary branch has l=theta_/m *(alpha=1; Remark 10.7 for small theta_)*. A root interval (G=theta^4 near 0) makes every alpha<1 stationary and selects none." The universal wording also stands in STRATEGY.md line 77, INDEX.md line 497, 11D/06 line 63, paper/connes_kk_dictionary_note.tex line 112.
3. 11D/02 lines 62-77: lemma table, Statement ("the same image", "2j+1 … linear") and first two proof sentences need "common ten-dimensional chirality"; add the chirality-free proof (|M_2(c)|=2 on Q and L, c_L=-3 c_Q). STRATEGY.md line 77 still reads "pending check".
4. Stale headers: 11D/01 line 3, 11D/02 line 3 ("Referee pending"; "pending an independent check"; "Theorem 2.1 … proved here" against ledger "conditional").
5. Prop 2.4(c): ledger d11_z6, 11D/02 lines 50, 52, 101 and STRATEGY.md line 77 say "no bosonic Kaluza-Klein mode" without the Terra scope; add "tensor modes on a background invariant under the displayed action; spinors open".

6. 11D/02 line 26 “Sym^{3k}(3bar) x Sym^{2k}(2): for k = 1 this is the (10, 3)”: Sym^3(3bar)=10bar; write Sym^{3k}(3) for functions. Line 60: add “convention for base P(3); for P(3bar) use (c_0,c_1)=(9/4,3/4) or (3/2,3)”.
7. 11D/01 line 50 “(on the circle of Proposition 1.1, fibre C^2, …)”: write “the scalar circle gives 2 l E/pi; fibre H_F gives 2 r l E/pi”. Prop 1.1 item 3: “even f(|x|), Lipschitz at the fixed points”.
8. 11D/10 Prop 10.6 item 3 (“no joint family results”) needs the Prop 1.5 scope (V=0 or tuned H leaves flat directions).

C. Derivations

B1: one channel, walls Q_R=0, U_L=0 at 0 and Q_R=X U_R, U_L=-X Q_L at l, X=tan(theta): the determinant vanishes at l*lambda = n pi +- theta, so n pi at X=0 (zero mode), (n+1/2) pi as X tends to infinity. B2: (theta G’-4G+4A_0)/theta^2 tends to -2c, c=(54/5-n_f)J(0). B3: left-handed fields come from left-handed modes (pairing +(alpha+3/2, beta+1)) or conjugated right-handed modes (pairing minus that); common chirality gives P_2=+(2j+1), independent chirality +-(2j+1). Third charge z: coefficients (-1,3), d^c gives -5 against 1; no vector of 117,649 passes both components. Scripts (Python): s1_oneloop.py, s2_flux.py, s3_topology_spectra.py, s4_periodicity.py, s5_third_charge.py, s6_unfolding_map.py.

Assessment Checked against the sources by Claude Opus 5.5. Correction 1: lemma 25, Proposition 25.1, is stated on the circle with periodic conditions, while Proposition 1.1 concerns the chiral interval, which is antiperiodic; applied in 11D/01 and in the dictionary note. Correction 2: G = theta^4 gives V = m^4, constant in l, so every l is stationary; applied, with the heading changed and the INDEX link updated. Correction 3: in the span of 6Y and 3(B - L), Q has charges (1, 1) and L has (-3, -3), so the chirality-free argument holds; added as a scope note. Corrections 4, 5, 7 (first part) and 8: applied. Correction 6: the tensor and function spans are now distinguished; the colour-convention point for Proposition 2.6 was already in the referee note of 11D/02. The Lipschitz point is in the referee note of 11D/01. The ledger records sonnet_5_5 as a second checker on the ten rows of files 01 and 02.

Batch 2: files 03 and 04

Report Second check of Terra sections on 11D files 03 and 04 (Claude Sonnet 5.5; Python 3 scripts). Ledger and README rows otherwise match Terra.

A. Verdicts 1. Prop 3.2: AGREE. Odd Lambda^k(5bar) = SM Weyl fields, even = conjugates, X = 3, -1, -5. 2. Prop 4.1: AGREE. Lemma 21 sums rebuilt from SM charges; among 12-state whole-species blocks only u, c, t and the three neutrinos keep (5/8, 8/3). 3. Prop 3.8, Cor 3.9: AGREE. All 17 nine-dimensional SU(3) representations decomposed; self-conjugate ones are 9(1), 3+3bar+3(1), 8+1; centre trivial on Y_{3,2}. 4. Props 4.2, 4.3: AGREE with both amendments. None of the 17 Lambda^3 equals F1 or F2; only 9(1), 3+6(1), 3bar+6(1) give {1,3,3bar}; 4.2(b) needs a factor-preserving map. 5. Lemma 4.A, Thm 4.4: PARTLY. Table, 45+45+6 and 33+33+30 reproduced from explicit E8 roots in two models; the chirality amendment covers only the top (B1). 6. Prop 4.5: AGREE. At root level Q u^c H_u, Q d^c H_d, L e^c H_d, L nu^c H_u all carry family tensor eps_ijk; Y = eps.h has singular values (|h|, |h|, 0). 7. Section 8: AGREE; 14 operator grades redone. 8. Prop 4.6: AGREE. Dimension and Schur arguments sound; the blocks are congruent in R^8, so the grading carries the obstruction; the orbit of the top block under <W(A2 x E6), -1> (810 sets) misses the neutrino block. 9. Prop 3.5: AGREE. Normalization R_MN = (1/12)(G_MPQR G_N^PQR - g_MN G^2/12) (action R - G^2/48, mostly plus), checked on AdS4 x S7 and AdS7 x S4; all components and |f1| < 2|f2| redone.

B. Corrections, by severity 1. 04 lines 74, 105, 111: the chirality amendment also applies to the neutrinos. For t = +1/2, nu_L is in C_6, nu^c and nu_R in gravity, conj(nu_L) in C_3. Complete blocks give (C_3, C_6, gravity) = (6,6,0) for the top, (3,3,6) for the neutrinos, both signs of t (conjugation-closed blocks always have equal C_3 and C_6 counts). Restrict “the top on the M2 side, the neutrinos on the M5 and gravity sides” and “The particle neutrinos lie outside the C_3 sector” to the a = 1 fields and add these counts. 2. 04 line 34: the false “exactly” clause stays in Proposition 4.3; write “exactly when C^9 is 9(1), 3+6(1) or 3bar+6(1)”. Ledger d11_colour_families and README row 4.2/4.3 (“With nonzero triplets and no other nontrivial irreducibles, C^9 is …”) apply the condition to C^9, which 3+3+3(1) satisfies; apply it

to Lambda^3 C^9. Line 28: add “tensor-factor-preserving” to 4.2(b). 3. 03 line 105, ledger d11_kk_triality, README row 3.8/3.9: add “tensor”, “on a background invariant under the displayed action”, “spinors open” (as 8b92f59 for d11_z6); likewise 03 line 25, 04 line 109. 4. 04 line 105: “does not preserve the family grading” -> “neither preserves nor inverts it”. The proved scope, phi(s_f) = s_f^{+-1}, contains every automorphism normalizing <s_f, s_M>. Add that Proposition 5.1 excludes any W(E8) element mapping F1 onto F2 for aligned families. STRATEGY.md T14 repeats the claim (“must come from a colour-family exchange”). 5. Stale: 03 and 04 line 3 (“Referee pending”, “pending an independent check”); STRATEGY.md T14 (“referee pending”, “(pending check)”); 11D/README.md lines 5, 92. 6. Section 8 (ledger d11_discrete_z3, plus a note in 04): selection rule of the single element s_M, a symmetry only if the E6 breaking commutes with it; (nu^c)^2, LLH_uH_u and LH_u, allowed by s_M, are forbidden by the family Z_3 (a = 2, 1, 2). 7. Wording: “closes” (03 line 105, 04 line 38) -> “excludes, in that scope”; two negation-first contrasts in 04 (lines 32, 105); 04 section 6 heading states a proposal as a result (fix INDEX.md line 527 with it); ledger d11_ads5: add flux quantization to “not checked”.

C. Scripts (Python): b2_ext5bar_blocks.py (A1, A2); b2_su3_cases.py (A3, A4, B2); b2_e8_gradings.py (A5, A7, B1, B6; 240 roots, grades from Bourbaki marks and from b = a + 2T_3); b2_yukawa_family.py (A6); b2_blocks_weyl.py and b2_z3_group.py (A8, B4); b2_ads5.py (A9). Recalled-fact labels are honest; no conflict with 11D/10 or 11D/11. After B2 and B4, recording sonnet_5_5 gives d11_colour84, d11_colour_families, d11_m2m5_is_j three identifiers.

Assessment Checked by Claude Opus 5.5. Correction 1: from Theorem 4.4 with b -> -b for conjugates, conj(nu_L) has b = -2 = 1 for t = +1/2, so it lies in C_3; the counts (6, 6, 0) and (3, 3, 6) follow per colour and per generation, for both signs of t. Correction 2: C^9 = 3 + 3 + 3 (1) has 3 x 3 = 6 + 3bar inside Lambda^2, so the condition belongs on Lambda^3 C^9; the trivial 9 (1) gives 84 singlets, as Terra noted. Corrections 3 to 7: applied as stated; the section 6 heading of 04 is now “The Chevalley involution and the real structure J”, with the INDEX link updated. The ledger records sonnet_5_5 on the eleven rows of files 03 and 04, and the 11D README now says that the checked no-go results have three model identifiers, each in its scope.

Batch 3: file 05

Report A. Verdicts

1. 5.1 AGREE. d = 22 on all 96 roots; F2: d = 20 (36 roots), 19 (48); F1: 16 (4), 22 (8), 20 (24), 19 (48); pair counts 42/816/1812/816; Terra 4x4 table and 12 slot/top choices reproduced. “Isometry of R^8” is the right notion (distance-preserving bijections extend; W(E8) is a subgroup). “Canonical up to W(E8)” holds for the chain E8 > SO(10) x SU(4) only (C2).
2. 5.2 AGREE. M = -P has 8 sign changes, lies in W(D8), M^2 = 1 on all 240 roots; T(0) <-> N(1,2,3), D(0) <-> E(1,2,3); nu_L, e_L of family 0 fixed; u^c, d^c, nu^c, e^c go to conjugate roots.
3. 5.3 PARTLY. Unitarity amendment and all stated properties confirmed (F^2 = 1, [F,gamma] = [F,J] = 0, rank 12, kernel 116). Missing: F differs from M (C1).
4. 5.4 AGREE. All 192 rerun; Terra formulas hold root by root; S - N: {19,20} in 48, {13,17,19,20} in 144 configurations; S - T pattern (16,16,22) in 96. The 11D/10 sentence is right.
5. 5.5 AGREE. No proved statement is contradicted; Props 5.1 and 5.4 concern E8-torus weights, which an irreducible SU(4) bundle lacks (argument). For aligned split bundles with n_i >= 0, sum 3, the obstruction persists (C3). Labels “recalled” (BHOP, Spin(9)) honest; 128 = 44 + 84 confirmed.

B. Corrections (11D/05 unless stated)

1. Line 66, “This is the finite-geometry form of Proposition 5.2.” -> “F equals M on left-handed states and M followed by particle-antiparticle exchange on right-handed ones; F exchanges T_top and T_nu but lies outside W(E8); M flips gamma on 64 states.” Reading 2 (line 91) likewise.
2. Line 72: replace “In every case S - N has d = 19 and d = 20 only, …” by “S - N has d in {13,17,19,20} ({19,20} iff the down-quark and charged-lepton slot sets coincide, 48 of 192); each neutrino root of S - T has d = 16 + 6(1_A(f) - 1_{0}(f)).” Line 74 (“scratch script not kept”): cite tools/check_11d_10_11/s5_e8_prop54.py and the referee proof.

3. Lines 84, 86: “their number” -> “their net number”; “fall under Proposition 5.1” -> “only for n = (1,1,1,0) up to order; for n_i >= 0 the obstruction persists (C3)”.
4. Line 46 item 3 -> “nu_L, e_L of family 0 fixed; nu^c, e^c to negatives; u^c, d^c to conjugates”. Line 58: “Fix an isomorphism” -> “Fix a unitary isomorphism”. Line 36: “canonical up to the Weyl group of E8” -> “unique up to W(E8) for the aligned (16,3)+(16bar,3bar)”.
5. Line 3: “pending an independent check”, “Referee pending.” -> “refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5”.
6. Line 78, Reading 1 (line 90), 11D/README.md line 99: “three families” results stated beyond the proved scope (single roots, three of four slots); cite 5.1 and 5.4.
7. 11D/ledger_11d.pl: add checked_by(_, sonnet_5_5, python(…)) to d11_no_isometry, d11_transposition, d11_flip, d11_misaligned (dry run: tests pass; both no-go rows reach three models); replace “not kept” in d11_misaligned by the kept script. d11_flip: conditional without a named hypothesis, README row 5.3 says independently checked. Referee notes and README rows 5.1-5.5 match Terra.

C. Derivations

C1 (b3_flip_vs_M.py): uR, dR are negatives of the u^c, d^c roots. F = M on the 64 left-handed states, F = -M on the 64 right-handed. F changes 3072 of 16384 ordered root inner products, M none. M sends gamma to -gamma on 64 states, so the T_top pair (uL, uR) goes to (nu_L, conjugate nu_R), violating gamma(s1) = -gamma(s2); F keeps it. C2 (b3_embedding.py, b3_classes.py): the 96 roots are the spinor-type roots among the 162 of (27,3)+(27bar,3bar); the 7560 A3 subsystems of E8 form one W(E8)-orbit with complement D5; the 192 placements give 3 invariant classes. C3 (b3_multiset.py): S has n_f copies of the 32 roots of slot f; degree multisets of S - T and S - N differ in all 40 (n, top slot) cases. C4 (b3_prop53.py, numpy): iota unitary, scaled unitary, diag(2,1,1,1) and a non-unitary GL(3) block; F unitary only in the first two; all other properties hold in all four. Others (Python 3): b3_prop51.py, b3_prop52.py, b3_prop54.py, b3_spin9.py.

Assessment Checked by Claude Fable 5.1 (the dialogue model from this batch on). Correction 1 is right and matters: M = -P sends t^c = (-c | ++ | 1) to the negative of a nu^c root, that is to an antiparticle state, while F of Proposition 5.3 acts inside H_f and keeps particles as particles; so F agrees with M only on the left-handed states, and “the finite-geometry form of Proposition 5.2” overstated it. Applied, with Reading 2 qualified. Correction 2: the values {13, 17, 19, 20} agree with the rerun recorded in 11D/10; applied, with the formula for the neutrino degrees. Corrections 3 to 6: applied. Correction 7: the four rows carry sonnet_5_5 with the method named, and the “not kept” entries now cite the kept script. The ledger row d11_flip stays conditional on the unitary identification, which the file now names.

Batch 4: files 06, 07 and 08

Report A. Verdicts

1. Prop 6.1: AGREE. Ad(W_a) on so(32): eigenvalue 1 on 436 = 435+1, e^{±2 pi i a} on 30 each; (n±a)/R = (n pi ± theta)/l; light mass a/R on [0,1/2].
2. Prop 7.1: AGREE. Wall determinant gives lambda l = n pi ± theta; nullity 2 at theta = 0 and, as X -> infinity, pi/2.
3. Prop 7.2: AGREE. Centralizer S(O(30) x O(2)), two components, Lie dimension 436; minimum at t = 1 (n_f = 1, 6, 16, 48). Scope gap: B2.
4. Prop 7.3: AGREE with Terra. h^0(L S_0) = h^1(L^-1 S_0) = p; C^2 = -1 gives (+,+,-); rho(1) has rank p of 2p. M_p + C is unital, order zero, and is lemma 16, Corollary 16.8 with (k_X, k_Y) = (p, 1). Berezin-Toeplitz step: true (Gaussian times Weyl word, exactly; p = 2..5, four spin structures, rank p^2), unproved in the file.
5. 07 §6 remark: AGREE (no Terra check). Commutant dimension 14 for stacks 1, 2, 3; spectrum 0, ±|x_a - x_b|.
6. Prop 8.1: AGREE (j-commutant dimensions n(n-1)/2, n(n+1)/2 for s = +1, -1).
7. Prop 8.2: AGREE (k, 2k; lowest |D|: 2k, 2k, 4k).

8. Remark 8.3: AGREE on arithmetic (15+1: z(0) = 7cL; 7+7: x_1 + x_2 = L; 7+6+triple: x_b = 2L/3). PARTLY on sources: B4.
9. Recalled dualities: AGREE, labels honest. PW and BGL confirm positions = Wilson lines (wilmap), heterotic KK = type I' winding, the slope rule, E_{n+1} (Table 1); others unchecked.

B. Corrections, by severity

1. 11D/10 is right on 10.1 to 10.3 (rechecked; 10.4 to 10.6 unchecked) and conflicts with 07 §4 Reading "distance proportional to a", 06 §3 last row, 06 §6 items 1 and 4, 08 §5 bullet 2. Only 07 §6b and 08 §4 point to 10. Append "(proportional only at constant dilaton; see 10, Proposition 10.1)" and equivalents.
2. 07 §4 note, append: "in SO(32). In Spin(32)/Z_2 the det -1 component sends the lift of W_{1/2} to its negative, so the heterotic centralizer is connected; a -> a+1 holds for SO(32) matrices, the lift has period 2."
3. 08 line 43 ("is not available with colour-respecting moves; E8 x E7 is:"): unapplied Terra replacement: "With colour triples, the proposed E8 x E7 arrangement has seven and six branes at the walls and a bulk triple at 2L/3. Two E8 walls require relaxing the colour-triple rule."
4. 08 §4 item 1: "7 + 7 for E8 x E8" is in neither PW (E8 x E8 left open) nor BGL (E8 x E8 at N = 0, R_h -> 0). Write "extrapolated from the single-wall rule".
5. 07 §5: cite lemma 16, Corollary 16.8; state the Gaussian formula in step (3).
6. 08 §2 Reading: add the (j, d) refinement and the label tension (label 9 with O^+ against the far O8^- wall).
7. 07 §6: "8 n_a n_b per ordered block, 16 n_a n_b per pair".
8. Stale: line 3 of 06, 07, 08 ends "Referee pending." (02, 03 too); README row 8.3 and the 06 §2, 08 §4 notes still call PW and BGL unavailable.
9. Records: add sonnet_5_5 checkers to d11_wilson_top, d11_angle_interval, d11_far_wall, d11_torus_flux, d11_orientifold_sign, d11_wall_multiplicity, d11_e_walls, d11_duality_readings. Notes, ledger rows and README rows otherwise match Terra. Also add the unitality amendment at paper/connes_kk_dictionary_note.tex lines 54, 104.

C. Derivations

B1: PW: lambda = W(x)/(2W(L)), W = int z^{-1/3}. For 15+1, z(L) = 0, rho = (z(x_t)/z(0))^{2/3}: theta = 4 pi (1-rho)/(8-rho), R^2 = 14(8-rho^2)/(8-rho)^2 (as 10.2); x_t/L = 1/2 gives theta = 0.5859 against pi/4. B2: lifts W~ = G_{2n-1}G_{2n}, g~ = G_1 G_{2n-1} (2n = 8, 12): g~ W~ g~^-1 = -W~. Only these enter, so 2n = 32 agrees; -1 differs from 1 and from the volume element. B5: rho(a, lambda) = a (+) lambda commutes with J rho(b, mu)^* J^-1 = mu (+) b^T; a unital M_3 on C^6 has commutant M_2, too small for M_3^op. B6: for s = -1, j swaps the ±mu eigenspaces of d, so the symmetries of (j, d) form U(n/2); s = +1 gives O(n_+) x O(n_-). Scripts (Python, not kept): b4_walls, b4_spin_lift, b4_so32_wilson, b4_unital, b4_torus, b4_commutant_j, b4_channel, b4_matrix_theory, b4_oneloop.

Assessment Checked by Claude Fable 5.1. Correction 1: the conflicts with 11D/10 are real, since Proposition 10.1 is the statement of the sources and the four readings predate it; pointers added at each place. Correction 2: the lift of diag(1_30, -1_2) to Spin(32) is Gamma_31 Gamma_32, which the lift of a reflection pair in the determinant -1 component conjugates to its negative, so the connectedness statement holds; added to the note of 07 §4. Correction 3: the Terra replacement had not been applied on the branch; applied. Correction 4: neither source read on master states the 7 + 7 configuration for E8 x E8; marked as extrapolated. Correction 5: Corollary 16.8 of lemma 16 gives the unital datum with (k_X, k_Y) = (p, 1), and the Gaussian form of the compressed characters is recorded as checked and unproved. Corrections 6 to 9: applied, with the dictionary note's table and text amended for the unital datum. Terra's error verdict on the unital triple of Proposition 7.3 stands, and the row keeps the status conditional.

Batch 5: file 09 and the sections added by the referee

Report A. Verdicts

1. Prop 9.1: PARTLY. Integration, concavity and far-wall formula are correct; the slope rule matches PW eq. (soln). Omitted: x_k = (2L/pi) theta_k holds only for constant z (11D/10 Prop 10.1 is right).
2. Prop 9.2: AGREE. Splits, dimensions and counts (6,6,3), (3,6,1) match lemma 21. PW confirms SO(2n) on O8^- and the (A_i +- A_j)/R strings; USp on O8^+ is outside the sources ("recalled" is honest).
3. Prop 9.3: PARTLY. Sign algebra is correct and consistent with 11.1 to 11.4; "the signs permit only a subspace of S^2 V" is wrong.
4. Section 1: AGREE on both branchings; the EPJC comparison COULD NOT CHECK (no source).
5. Del Pezzo: AGREE, the error is real. BGL Table 1 has E_1 (self-dual radius) at the empty wall; the 5d dictionary is outside the sources.
6. Weyl action: AGREE.
7. Audit section 4: AGREE; asymptotics redone from lemmas 9.4, 9.5; table recomputed.
8. Prop 2.5 third charge: AGREE; the argument holds for any real z and any source chiralities.
9. Prop 3.4: AGREE (Z_3 on U = 0, Z_2 on V = 0).
10. Wording audit: AGREE with all nine reclassifications and the 08 replacement; no first-person singular in 01 to 11.

B. Corrections, by severity (remaining notes, statuses and rows match Terra)

1. ledger d11_dilaton_spectral, README row 9.1, 09 note of section 2: z(L) = z(0) - c(8L - sum x_k) is exact in the positions; x_k = (2L/pi) l m_k holds only for constant z; in general theta_k = (pi/2) W(x_k)/W(L), W = int z^(-1/3) (Prop 10.1). In the 15 + 1 profile the endpoint values 7cL, 8cL stand; theta_t = pi/4 gives 7.358 cL, not 7.5 cL. Add depends(d11_dilaton_spectral, d11_angle_map).
2. Replace "permits only a subspace of the symmetric representation" (09 section 4 note), "The signs permit only a subspace of S^2 V" (ledger d11_real_structure_diquarks) and, by editorial annotation, the same phrase in the Terra audit by "The signs permit all of S^2 V; the algebra, grading and order conditions of a triple select a subspace."
3. paper/connes_kk_dictionary_note.tex, "X with JX = XJ, in the labels 9 to 11": at label 9, JX = -XJ and J^2 = -1; write "JX = eps' XJ with eps eps' = +1". The gauge block is antisymmetric only for s = +1 (usp(30): symmetric).
4. d11_flux_third_general: change to no_go, proved; add checked_by sonnet_5_5; delete referee_pending; update the 02 note, README (intro, row 2.5, next-step 1), PLAN and STRATEGY T14. The open-directions lists name "a changed common twist": that is z = F and is covered. The Prop 2.6 solution (fluxes F, chi, B - L, no Y) lies outside span(Y, B - L, z).
5. "Referee pending." ends line 3 of 09, 02, 03: replace by "Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5."
6. 09 section 3 Reading and section 6: add that 11D/11 Prop 11.4 and Theorem 11.3 qualify the O8^-/O8^+ selection (the window sees (3,6,1) also at s = +1; parent scalars need M_15(C) at both walls); 11D/11 is right.
7. Add checked_by(_, sonnet_5_5, by_hand) to a11_localized, d11_flux_third_bounded, d11_higgs_m2, d11_dilaton_spectral, d11_diquark_strings, d11_so32_algebra, d11_real_structure_diquarks, d11_dp_walls, d11_transposition_tduality; d11_so32_comparison stays pending. README and PLAN: the localized table is recomputed.

C. Derivations and runs

9.1: PW (mass), (wilmap): lambda_i = W(x_i)/(2W(L)). For 15 + 1 with z(L) = 0: z(0) = c(8L - x_t), theta_t = (pi/2) 8(1 - rho)/(8 - rho), rho = (z(x_t)/z(0))^(2/3) (Prop 10.2). 9.3: JX = eps' XJ on (v, w) is equivalent to B* = eps eps' K B K and D = eps' K A K^-1 with A Hermitian free, so B ranges over exactly S^2 V (eps eps' = +1) or Lambda^2 V (-1). A gauge generator H = ia has jH = -Hj, so (eps, eps') = (s, -1) (11.1). 2.5: phi(v) = 2v_Q + v_u + v_d kills y, b, p. Since w_L/w_Q = +-1, not -3, w lies outside span(y, b), so span(y, b, z) = span(y, b, w); phi(p) = 0 gives t(4s_Q + s_u + s_d) = 0, so t = 0, a = 0, b_0 = -5/2, p_L = 15/2, not +-3/2.

Runs (python3 -I, scripts not kept): exact fractions, rank [y, b, p, w] = 4 for 1024, 256, 64 sign patterns (six, five, four fields), the Prop 2.6 solution has rank 4, a rank-3 pattern would have refuted the extension;

bisection, the four audit rows agree to the printed digits; mpmath quadrature, the PW formula on 15 + 1 equals the Prop 10.2 closed form (theta_t = 0.5859 at x_t/L = 0.5; linear 0.7854).

Assessment Checked by Claude Fable 5.1. Correction 1 restates Proposition 10.1 at the places of file 09 that still assumed the linear dictionary; applied, with the dependency row. Correction 2: for eps eps' = +1 the condition on the block is B^T = B with the diagonal block free, so every symmetric B occurs, and Terra's report itself says that the algebra, grading and order conditions select the subspace; the note and the ledger row now say so, and the report needed no annotation. Correction 3: the KO table gives (eps, eps') = (-1, -1) at label 9, so JX = -XJ there; the note now writes eps' and scopes the antisymmetric block to s = +1. Correction 4: the two-vector steps were redone by hand (phi kills y and b and not p; every vector of span(y, b) has v_L = -3 v_Q, which w fails), and the rank-4 enumeration over all sign patterns is Sonnet's; the row is now a no-go with two model identifiers (Terra as author, Sonnet as checker), below the three-identifier threshold, which nogo_below_consensus reports. Corrections 5 to 7: applied. The EPJC comparison keeps its pending flag, since no reader had the paper.